ONGO WEIGHT LOSS · BOOK SEVEN

Ongo Weight Loss

Measure It Right.

Analytics and reporting · GTM, GA4, GSC, Clarity, BigQuery and Looker Studio, from zero

For every Ongo Weight Loss teammate who builds, reads or acts on our data: every step measured, every visitor protected, every decision backed by numbers.

PREPARED BY

Ashish Osten

Team Lead, Ongo Weight Loss

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ONGO WEIGHT LOSS · ONBOARDING SERIES

Analytics and Reporting

How to measure Ongo Weight Loss end to end, safely, and report on it every day
Book Seven of the Ongo Weight Loss onboarding series

Prepared by	Ashish Osten, Team Lead, Ongo Weight Loss
Edition	October 2026 (first edition)
Written for	The team lead and the whole team, from zero: developers, SEO, writers, designers, social and testers
Read first	Book One: The Foundation (Books Three and Four explain the content map and the first event names)
Inside	70 chapters · 700 MCQs · Event Dictionary v2 · Next.js code pack · GTM build sheet · 28 SQL queries · 5 dashboard specs · KPI ranges · privacy checklist · glossary · sources

Important

This book is internal orientation material. It is not legal, privacy, HIPAA or medical advice. Decisions about privacy zones, consent, business associate agreements and data sharing belong to Ongo Weight Loss compliance and counsel.

Product menus, limits, connectors, prices and privacy laws change often. Every claim carries a source and an evidence label so you can recheck it. When this book and a newer official source disagree, the official source wins.

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ABOUT THIS BOOK

What's inside

> **Before you start:** read **Book One: The Foundation** (what Ongo Weight Loss is and what the rules allow). Book Three (site architecture and the content map) defines the pages and funnel stages we measure; Book Four (on-page SEO) introduced our first event names; Book Five (social media) set the UTM convention; Book Six (writing) explains the content we judge here. Book Seven teaches the whole team to measure every important step, from the first search to the portal, safely, and to turn the numbers into decisions every day.

How every chapter is built

Part of chapter	What it gives you
Trivia hook	One surprising, sourced fact that shows why the topic matters
Plain explanation	The concept in very easy English, with click-by-click steps
Code and SQL	Copy-ready Next.js, GTM and BigQuery snippets for developers and analysts
Desi analogy (Hinglish)	Shop, godown, cricket, wedding, railway, kitchen and mohalla examples
Ongo Weight Loss example	How the idea applies to our sites, portal and team
Tables and templates	Settings, specs, checklists and report layouts at a glance
Evidence labels	Official, spokesperson/academic, industry, or expert/internal
Do's and Don'ts	Quick rules
Pros and Cons	Honest trade-offs
Where freshers go wrong	Common mistake, why it happens, how to avoid it, how to fix it
Thumb rules + Notes to remember	One-line rules to keep
Glossary	Every new term in the chapter

Part of chapter	What it gives you
10 MCQs	Answer key at the foot of the chapter, flipped upside down
Sources	Every claim linked to where it came from

The six levels

Level	Chapters	What you'll learn
Level 1 · Foundation	1–10	360° measurement, how tracking works, health privacy law, the HIPAA-safe three-zone architecture, consent, the tool stack, how GA4 thinks, the measurement plan, our baseline, roles and access
Level 2 · Setup and Collection	11–30	GA4 and GTM setup, installing GTM in Next.js, the data layer, page views, Event Dictionary v2, CTAs, internal links, contact actions, engagement, tools, forms and leads, portal exit and cross-domain, backend conversions, custom dimensions, key events, minimal UTMs, Clarity, consent, QA
Level 3 · Data Platforms	31–42	BigQuery from zero, GA4 export, schema, SQL, the query library (paths, entry/exit, funnels, attribution), Search Console, bulk export, joining GSC and GA4, Bing Webmaster Tools, Core Web Vitals and CrUX, social data, competitor benchmarking
Level 4 · Analysis	43–56	GA4 reports, explorations, funnels, paths, user explorer, segments and cohorts, attribution, content, internal link and CTA performance, search analysis, Clarity analysis, data quality, alerts
Level 5 · Looker Studio and Reporting	57–66	Looker basics, fast data sources, calculated fields and blends, daily, executive, SEO, social and technical dashboards, weekly and monthly reports, KPI dictionary and target ranges
Level 6 · Operating	67–70	Experiments, governance, the 90-day roadmap, ZETA for analytics

Your questions: where to find the answers

You asked about	Chapter(s)	You asked about	Chapter(s)
Which tools, what each does, alternatives and why	6	Fewest UTMs, more events	16, 27, Appendix G
Track each user entry → page → page → exit	15, 35, 46, 47	Funnels	35, 45
HIPAA-safe analytics with a BAA	3, 4, 70, Appendix L	Consent tool recommendation	5, 29
Developer instructions, complete	12–30, Appendices B, C, D, K	Payments and leads counted (backend)	22, 24, 35
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Daily supervision	56, 60, 65	Where to start (90 days)	69

What's new compared with Books One to Six

Topic	Book Three	Book Four	Book Five	Book Six	Book Seven
Funnel stages	✅ Defined	Mapped to pages	Social funnel	Writing by stage	✅ Measured step by step

Topic	Book Three	Book Four	Book Five	Book Six	Book Seven
Events	—	✅ First dictionary	UTM convention	—	✅ Event Dictionary v2, data layer, code
Privacy	—	Tag cautions	Social privacy	Content privacy	✅ Three zones, consent, ZETA for analytics
Reporting	—	GSC basics	Social metrics	Measuring content	✅ BigQuery, Looker dashboards, KPI ranges

BEFORE YOU BEGIN

Front Matter

A Letter to the Reader

Welcome to Book Seven.

You told us three things about how you want Ongo Weight Loss's website to work. You want to know **what every important visitor does**, from the page they land on to the page they leave from. You want **events and funnels, not a pile of UTMs**. And you want to **supervise the website every day**, with numbers you can trust.

This book makes that possible, and it starts from zero. You don't need to know what a data layer, a BigQuery table or a key event is. By the last chapter, the whole team should be able to set up tracking correctly, read the reports, build the dashboards and act on them, on their own.

On a health website, measurement comes with a special duty. Our visitors may be worried about their weight, their health and their money. Some of the pages they see, and everything in the patient portal, can involve protected health information. So this book teaches two things together: **how to measure everything that matters**, and **how to measure it safely.** You'll see a three-zone model, consent rules, aggregated backend counts and a Red List of analytics mistakes we never make.

Here is what we did before writing:

- We used your 20 answers: nothing is implemented yet, GTM is installed but not configured, there's no consent banner, you have a BAA, payments are the real conversion, the portal is separate and private, child sites like weightclinicsresults.com link to it, and you read reports daily.
- We kept the event names from Book Four and extended them into Event Dictionary v2, so nothing earlier is wasted.
- We read Google's, Microsoft's and HHS's own documentation for every setting, limit and rule we teach, and labelled every claim.
- We checked what changed recently. One example: the CrUX connector you asked about was retired after November 2025, so Chapter 40 teaches what replaced it.

Developers will find complete code and click-by-click steps. Analysts will find copy-ready SQL. Everyone will find plain explanations, analogies and examples.

Let's measure what matters, protect who matters, and use the numbers to help more people get the care they're looking for.

How to Read This Book

The six levels

Level	Who it's for	What you can do after
Level 1 – Foundation	Everyone, day one	Explain what we measure, why, and the privacy rules
Level 2 – Setup and Collection	Developers, analytics owner, tester	Build and QA the full tracking setup
Level 3 – Data Platforms	Analytics owner, SEO, developers	Run BigQuery, GSC export, Bing, CWV and benchmarks
Level 4 – Analysis	Team lead, SEO, writers, UX, social	Answer business questions from data
Level 5 – Looker Studio and Reporting	Analytics owner, team lead	Build dashboards and write reports
Level 6 – Operating	Team lead, everyone	Test, govern and roll out in 90 days

The boxes you'll see

> 💡 **Trivia** – a surprising fact that shows why the topic matters.
>
> 🍛 **Desi analogy** – the idea explained in Hinglish with an everyday example.
>
> 🟧 **Ongo Weight Loss example** – how it applies to our sites and team.
>
> ⚠️ **Where freshers go wrong** – the common mistake and how to avoid or fix it.
>
> 👍 **Thumb rule** – one line to remember forever.
>
> 📝 **Note to remember** – a key fact to keep.

Reading paths by role

Role	Start with	Then
Team lead	1–10, 60, 66, 69, 70	43–56, 61, 65
Front-end developer	1–4, 13–24, 28–30	40, 67, Appendices B, C, D, K
Back-end developer	1–4, 23, 24	31–32, 37–39, 58
On-page SEO / analytics owner	1–12, 16, 25–27	31–66, Appendices A, E–J
Off-page SEO	1–6, 27	36–39, 42, 53
Content writer	1–4, 8	50–53, 62
UI/UX designer	1–4, 17, 28	52, 54, 67
Social media	1–4, 27	41, 63
Tester	1–4, 30	55, 64, Appendix K

If you've never done analytics before

Start with Chapters 1, 2, 7 and 9. Then read Level 1 in order. Keep Appendix A (Event Dictionary), Appendix J (KPI dictionary) and Appendix L (privacy checklist) open while you work. Do the practice assignments in Appendix O at the end of each level.

Using the MCQs for onboarding

- Each chapter ends with 10 multiple-choice questions.
- Team leads can use them after each level; 8/10 or better means ready to move on.

The Evidence Label Legend

Source labels: where the claim comes from

Label	Meaning
🟢 **Official**	Google, Microsoft, Chrome, HHS and other official documentation, laws and regulators

Label	Meaning
🟡 **Spokesperson / academic / expert book**	Public statements, academic frameworks and recognised expert books
🟠 **Industry**	Vendor blogs, trade press, third-party tools' explanations and studies
🔵 **Expert / Ongo Weight Loss internal**	Expert judgment, our plans, decisions and recommended practice

How much to trust each label

Label	Trust	Before you rely on it
🟢	Highest	Check it's still current; product menus change
🟡	High	Remember it's one view or framework
🟠	Medium	Prefer the official source when possible
🔵	Varies	Confirm with the team lead, legal or the analytics owner

Changeable facts

Product menus, limits, prices, connectors and privacy laws change. Treat them as **changeable** and re-check before relying on them. When this book and a newer official source disagree, the official source wins.

Risk level: what can go wrong

Most analytics risk comes from five places: **personal or health data sent to non-BAA tools**, **tags on the portal (Zone C)**, **tracking Zone B pages without a legal decision**, **ignoring consent**, and **numbers presented without definitions**. Chapter 70 (ZETA) lists them all.

How the Answer Keys Work

Every chapter's answer key sits at the foot of the chapter, printed **upside down**. Try the quiz first, then turn the page around to check. All 70 keys are also collected in the back matter.

01

LEVEL 1

Foundation

360° measurement, how tracking works, health privacy law, the three-zone architecture, consent, tools, how GA4 thinks, the plan, the baseline and roles.

1 · What 360° Measurement Means at Ongo Weight Loss

2 · How Web Tracking Works (From Zero)

3 · Health Privacy Rules for Analytics

4 · Your BAA and a HIPAA-Safe Analytics Architecture

5 · Consent Management: Which Tool and How to Set It Up

6 · The Tracking Stack: Core Tools and Gap Tools

7 · How GA4 Thinks: Events, Parameters, Users and Sessions

8 · The Measurement Plan: Goals to Events to Reports

9 · Baseline Audit: Where We Are Today

10 · Roles, Access and Governance

LEVEL 1 · CHAPTER 1

Chapter 1: What 360° Measurement Means at Ongo Weight Loss

Most websites count visitors. We want to know what every important visitor did: where they came from, which page they entered on, which links moved them forward, and whether they became a paying patient. And we want to know all of it without ever touching a patient's health information.

> 💡 **Trivia**
>
> In GA4's user explorer, every visitor has an "Effective user ID": the User ID if one exists, otherwise the browser's client ID. Clicking a row opens a timeline of that visitor's events. [1]
> ● Official (Google)
>
> GA4 can show you one anonymous visitor's journey, page by page. This book teaches you how to use that power, and where it must stop.

1.1 From "how many?" to "what happened?"

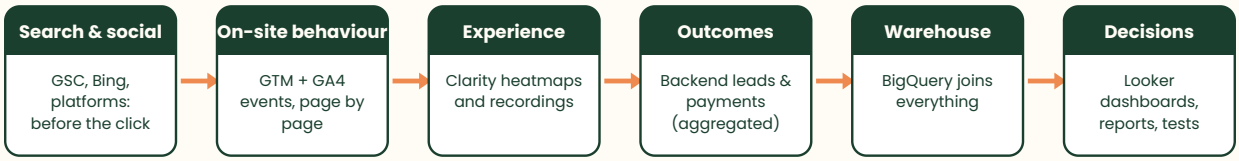
Old question	360° question
How many people visited?	Which pages do people **enter** on, and where do they go next?
How many clicks did the button get?	Which **CTA**, on which page, in which position, moved people toward a lead?
How much blog traffic?	Do blog readers move to **service and medication pages**?
How many sign-ups?	Which **first landing page and source** led to leads, and to **payments**?
Did the page get views?	Did the **internal link** from the GLP-1 article to the semaglutide page get clicked?

1.2 The Ongo Weight Loss journey we measure

```
Source (Google, social, email, referral, direct)
   ↓
Entry page (blog / service / medication / home)
   ↓
Page → page → page   (internal_link_click, page_view)
   ↓
CTA click            (cta_[pagegroup]_[action])
   ↓
Lead                 (eligibility completed / sign-up)       ← lead, not the goal
   ↓
exit_to_portal       (the last thing public analytics sees)
   ↓
Payment              (counted safely, without patient data) ← the real conversion
```

Important: you told us the **real conversion is payment**. Eligibility completed and sign-ups are **leads**. This book treats them differently everywhere: in GA4 key events, funnels, dashboards and KPIs. 🔵

The 360° Measurement Loop



Every tool answers a different part of the journey.

> 🪔 **Desi analogy: shaadi ka hall aur entry register**
>
> Shaadi ke hall mein ek register hota hai: kaun kis gate se aaya, kaun stage tak gaya, kaun khaana kha ke gaya, kaun shagun de ke gaya. Lekin register mein koi mehmaan ki bimaari ya tankhwaah nahi likhta. **360° measurement wahi register hai:** har kadam dikhta hai, kisi ki niji baat nahi.

1.3 The five tools and their jobs

Tool	Job in the 360° system
Google Tag Manager (GTM)	Decides what to measure and sends it
Google Analytics 4 (GA4)	Stores and reports behaviour: entry, paths, events, leads
Google Search Console (GSC)	What happened in Google **before** the click
Microsoft Clarity	**How** people use a page (heatmaps, recordings)
Looker Studio	Daily dashboards that combine everything

Plus the gap tools you approved: **BigQuery** (raw data, every path), **a consent manager**, **Bing Webmaster Tools**, Core Web Vitals sources, and a **HIPAA-safe route** for payment data (Chapters 4–6). 🔵

1.4 What "without exposing patients" means

✅ We measure	❌ We never measure
Page types, clusters, funnel stage	Names, emails, phone numbers, addresses
Which CTA was clicked, where	Answers to eligibility or medical questions
Which internal links move people	Weight, BMI, medications, conditions
Leads and payments **as counts**	Anything inside the patient portal, in GA4 or Clarity
Sources and landing pages	Personal data in URLs, events or user IDs

Chapters 3–5 explain why, and how to set up the BAA-covered route your team already has.

> 🟧 **Ongo Weight Loss example**
>
> You told us GTM is installed but not fully configured, nothing is implemented yet, and Looker Studio is not built. That is the best place to start: no bad data to clean up, no broken history. Everything in this book can be built correctly the first time. 🔵

1.5 Do's and Don'ts

✅ Do	❌ Don't
Measure journeys, not just totals	Report page views alone
Separate leads from payments	Call a form submit a "conversion"
Plan before tagging	Add tags ad hoc
Keep patients' data out of analytics	Track inside the portal with GA4

1.6 Pros and cons of 360° measurement

✅ Pros	❌ Cons
Know which content creates patients	Needs careful setup and upkeep
Spot drop-offs and fix them	Privacy rules limit what you can see
Daily supervision of site health	Requires team discipline (naming, QA)

1.7 Where freshers go wrong in Chapter 1 topics

Mistake	Why it happens	Avoid it by	Fix it by
Treating sign-ups as the goal	They're easy to count	1.2 journey	Separate lead and payment KPIs
Tracking everything "just in case"	Fear of missing data	Measurement plan (Chapter 8)	Removing unused tags
Putting GA4 on the portal	Wanting full journeys	1.4 table	Removing it; using the safe route

1.8 Thumb rules

👍 Every step visible, no person exposed.

👍 Payment is the conversion; leads are signals.

👍 Plan first, tag second, report third.

1.9 Notes to remember

📝 Five core tools: GTM, GA4, GSC, Clarity, Looker Studio.

📝 Public analytics stops at `exit_to_portal`.

Chapter 1 Glossary

Term	Simple meaning
360° measurement	Measuring the whole journey from source to conversion
Entry page	The first page of a visit (landing page)
Lead	Someone who signed up or completed eligibility
Conversion	For Ongo Weight Loss: a completed payment
Client ID	An anonymous ID GA4 stores in the browser

Chapter 1 Quiz

1. What is the real conversion for Ongo Weight Loss?

- A) A completed payment
- B) A blog visit
- C) A page view
- D) A scroll

2. What are eligibility completions and sign-ups?

- A) Payments
- B) Leads
- C) Errors
- D) Bounces

3. Which tool shows what happened in Google before the click?

- A) GTM
- B) Google Search Console
- C) Clarity
- D) Looker Studio

4. Where does public analytics stop?

- A) After payment
- B) At `exit_to_portal`
- C) Never
- D) Inside the portal

5. What does GA4's user explorer show?

- A) Only totals
- B) One anonymous visitor's event timeline
- C) Patient names
- D) Payment card numbers

6. Which data is never measured?

- A) Entry page
- B) Which CTA was clicked
- C) Answers to eligibility questions
- D) Source

7. Which tool shows how people use a page (heatmaps)?

- A) Microsoft Clarity
- B) Bing
- C) BigQuery
- D) GSC

8. What's the current state of tracking?

- A) GTM installed but not configured; nothing implemented
- B) Unknown forever
- C) No GTM at all
- D) Fully implemented

9. What does "360°" add over basic analytics?

- A) More colours in reports
- B) Bigger fonts
- C) More page views
- D) Journeys: entry, paths, CTAs, leads, payments

10. In the shaadi-register analogy, the register...

- A) Records nothing
- B) Records every step but no private details
- C) Records guests' salaries
- D) Records illnesses

Sources

1. Google Analytics Help, User explorer 🟢
2. Ongo Weight Loss, measurement goals and current setup (team lead, 30 October 2026), internal 🔵

Answer key, Chapter 1: 1-A · 2-B · 3-B · 4-B · 5-B · 6-C · 7-A · 8-A · 9-D · 10-B

LEVEL 1 · CHAPTER 2

Chapter 2: How Web Tracking Works (From Zero)

Before you set up anything, understand what actually happens when someone opens a page: what a tag is, what a cookie stores, how a "session" is counted, and why some data goes missing.

> 💡 **Trivia**
>
> In GA4 path explorations, a path can span multiple sessions, and a **new session starts after 30 minutes of inactivity**. [1] ● Official (Google)
>
> A "visit" isn't a fixed thing. It's a rule. Knowing the rules is how you read reports correctly.

2.1 What happens when a page loads

```
1. Browser requests the page        → Next.js server sends HTML
2. HTML loads the GTM container      → gtm.js runs in the browser
3. GTM reads the data layer          → page_type, funnel_stage, …
4. Triggers check conditions         → "Is this a page view? A CTA click?"
5. Tags fire                          → send an event to GA4 (and Clarity loads)
6. GA4 receives the event            → stores it with the client ID, time, page, source
7. Reports and BigQuery               → you analyse it later
```

2.2 The key words, simply

Word	Plain meaning	Example
Tag	A small piece of code that sends data somewhere	GA4 event tag
Trigger	The rule for when a tag fires	"Click on an element with data-event starting cta_"

Word	Plain meaning	Example
Variable	A value a tag or trigger reads	Page path, CTA position
Data layer	A JavaScript list where the website writes facts for GTM	`{ page_type: 'service' }`
Event	Something that happened	`page_view`, `internal_link_click`
Parameter	Detail attached to an event	`link_position: 'body'`
Cookie	A small file the browser keeps	`_ga` stores the client ID
Client ID	Anonymous ID for a browser	`123456.789012`
Session	A group of events from one visit	Ends after 30 minutes of inactivity
User	A browser (or a logged-in user ID; we don't use that)	One phone = one user

> 🍛 **Desi analogy: IT office ka access card**
>
> Office mein har card swipe register hota hai: kaunsa card, kaunsa gate, kab. Card pe naam nahi, sirf number hai. Agar card ghar bhool gaye aur naya liya, toh system aapko "naya employee" samjhega. **Client ID wahi card number hai:** browser badla ya cookie hati, toh naya user.

2.3 Why numbers are never perfect

Reason	Effect
Ad-blockers and privacy browsers	Some visits never reach GA4
Cookies cleared, new device	One person counted as several users
Safari limits cookies set by scripts	Returning visitors may look new (Chapter 6)
Consent declined	Less data (Chapter 5)
GA4 thresholds	Small numbers hidden in some reports
Bots and spam	Fake traffic (Chapter 55)

Rule: use analytics for **trends and comparisons**, not as an exact count of people. 🔵

2.4 First-party vs third-party

Type	Set by	Example	Status
First-party cookie	Our domain	GA4's `_ga` on ongoweightloss.com	Normal for analytics
Third-party cookie	Another domain	Ad networks	Increasingly blocked; we don't rely on them

2.5 Client-side vs server-side tracking

	Client-side (now)	Server-side (optional later, Chapter 6)
Where tags run	Visitor's browser	Your own tagging server
Pros	Simple, free	Cleaner data, strip risky fields, longer cookies
Cons	Blockers, browser limits	Hosting cost, setup work

> 🟧 **Ongo Weight Loss example**
>
> Our site is built with Next.js, which changes pages **without reloading** the whole page. A simple setup can miss page views or count them twice. Chapter 15 shows exactly how to handle this. 🔵

2.6 Do's and Don'ts

✅ Do	❌ Don't
Learn the tag-trigger-variable model	Paste random scripts into the site
Read numbers as trends	Treat analytics as an exact headcount
Use first-party analytics	Depend on third-party cookies

✅ Do	❌ Don't
Know why data goes missing	Panic at small gaps

2.7 Pros and cons of tag-based tracking

✅ Pros	❌ Cons
Flexible, no code change for many events	Can break silently after site changes
Free tools	Blocked by some browsers
Detailed behaviour data	Needs QA discipline

2.8 Where freshers go wrong in Chapter 2 topics

Mistake	Why it happens	Avoid it by	Fix it by
Thinking "users" = people	Plain meaning of the word	2.2 table	Explaining in reports
Expecting GA4 to match the CRM exactly	Different counting rules	2.3 table	Reporting trends
Not knowing Next.js navigation is special	Assuming page reloads	Chapter 15	Fixing page-view setup

2.9 Thumb rules

👍 Tag sends, trigger decides, variable describes.

👍 A user is a browser, not a person.

👍 Trends over totals.

2.10 Notes to remember

> 📝 Session ends after 30 minutes of inactivity.
>
> 📝 `_ga` cookie stores the client ID.

Chapter 2 Glossary

Term	Simple meaning
Tag	Code that sends data to a tool
Trigger	Rule for when a tag fires
Variable	A value used by tags and triggers
Data layer	Where the website writes facts for GTM
Session	One visit; ends after 30 minutes inactive

Chapter 2 Quiz

1. When does a GA4 session end by default?

- A) After 1 minute
- B) After 30 minutes of inactivity
- C) Never
- D) At midnight only

2. What decides when a tag fires?

- A) A report
- B) A trigger
- C) A cookie
- D) A dashboard

3. What is the data layer?

- A) Where the website writes facts for GTM
- B) A database of patients
- C) A server
- D) A Looker Studio chart

4. What does "user" usually mean in GA4?

- A) An employee
- B) A browser/device (client ID)
- C) A named person
- D) A patient record

5. Why are analytics numbers never perfect?

- A) Only because of fonts
- B) Numbers are invented
- C) GA4 is broken
- D) Blockers, cleared cookies, consent, thresholds, bots

6. Which cookie stores GA4's client ID?

- A) `_ga`
- B) `session_token`
- C) `cart`
- D) `lang`

7. What's a benefit of server-side tracking?

- A) It removes the need for consent
- B) It's always free
- C) No setup
- D) Cleaner data and the ability to strip risky fields

8. Why is Next.js special for page views?

- A) It blocks GTM
- B) It has no pages
- C) Pages change without full reloads
- D) It doesn't use URLs

9. What is a parameter?

- A) A browser
- B) Detail attached to an event
- C) A report name
- D) A type of cookie

10. In the access-card analogy, a new card means...

- A) A payment
- B) The system sees a "new user"
- C) A patient record
- D) The same user forever

Sources

1. Google Analytics Help, Path exploration 🟢
2. Ongo Weight Loss, Book Four (Next.js site), internal 🔵

Answer key, Chapter 2: 1-B · 2-B · 3-A · 4-B · 5-D · 6-A · 7-D · 8-C · 9-B · 10-B

LEVEL 1 · CHAPTER 3

Chapter 3: Health Privacy Rules for Analytics

A weight-care brand can't use analytics like a shoe store. Federal and state rules shape what we may track, where, and with which vendors. This chapter explains them in plain words. Legal makes the final decisions.

> 💡 **Trivia**
>
> Google's own help centre says it "does not offer Business Associate Agreements in connection with" Google Analytics, and that HIPAA-regulated customers "must not use Google Analytics in any way that implicates Google's access to, or collection of, PHI." [1] 🟢 Official (Google)
>
> The tool's maker is telling health companies exactly where not to use it.

3.1 HIPAA and tracking technologies (HHS)

The HHS Office for Civil Rights published a bulletin on online tracking (November 2022, updated March 2024). [2] 🟢

Point	What it says (plain words)
What a tracking technology is	"A script or code on a website or mobile app used to gather information about users or their actions": cookies, pixels, session replay, fingerprinting
Logged-in pages (patient portal, telehealth platform)	Tracking there "generally" has access to PHI
Public pages	Often no PHI, **but** HIPAA applies when the page collects PHI: e.g., appointment requests, symptom checkers, portal login/registration
Vendors	"Regulated entities must ensure that all tracking technology vendors have signed a BAA" when they receive PHI

Point	What it says (plain words)
Cookie banners	A banner accepting cookies is **not** a valid HIPAA authorisation
Court ruling (June 20, 2024)	A federal court in Texas (*AHA v. Becerra*) vacated part of the guidance: the part saying HIPAA is triggered when an IP address is linked to a visit to a public page about specific health conditions or providers
Legal force	The bulletin says it "does not have the force and effect of law"

What this means for us: the court ruling narrowed one part, but the rules for logged-in pages, forms that collect health information, and vendor BAAs still matter. **Legal decides** which pages are safe for which tools. 🔵

3.2 Google Analytics and HIPAA (Google's own guidance)

Google says	So we…
No BAA for Google Analytics	Never send anything that could be PHI to GA4
Don't tag **authenticated** pages	No GA4 or Clarity in the patient portal
Avoid unauthenticated pages tied to **providing health care services**	Legal classifies pages; some may need a BAA-covered tool instead
"Work with your legal team to identify pages … that do not relate to the provision of health care services"	We build a page-zone map (Chapter 4)

[1] 🟢

> 🍛 **Desi analogy: doctor ka clinic aur bahar ka chai-waala**
>
> Clinic ke reception par chai-waala aa sakta hai, lekin doctor ke cabin mein nahi, aur patient ki file kabhi nahi dekh sakta. **GA4 chai-waala hai:** public jagah theek, cabin (portal) aur file (health data) se door.

3.3 State consumer health privacy laws

Law	Key points for analytics
Washington My Health My Data Act	Main obligations from March 31, 2024 (small businesses June 30, 2024). "Consumer health data" can include **use or purchase of prescribed medication** and health **inferences** from non-health data. Requires a separate homepage link to a consumer health data privacy policy; violations are enforceable by the Attorney General **and private lawsuits** [3] 🟢
Other states	Several states have consumer health data or comprehensive privacy laws; legal tracks them

3.4 FTC rules (non-HIPAA health data)

Even where HIPAA doesn't apply, the FTC has acted against health companies that shared health data with advertising platforms, and the FTC's **Health Breach Notification Rule** covers many health apps and services that aren't HIPAA entities. Legal confirms how these apply to us. 🟠

3.5 Our privacy principles for analytics

#	Principle
1	**No PHI or personal data** in GA4, Clarity, GTM variables, URLs, event names or parameters
2	**No GA4 or Clarity on the patient portal** (authenticated)
3	**Legal classifies every page type** into zones (Chapter 4)
4	**BAA-covered tools only** wherever health information could be involved
5	**Consent** collected where legal requires it (Chapter 5)
6	**Google signals and ads features off** in GA4 (Chapter 11)
7	**Payments counted without identifiers** sent to Google (Chapter 24)
8	**Every new tag reviewed** before publishing (Chapter 30)

> 🟧 **Ongo Weight Loss example**
>
> Book Four listed portal events like `portal_payment_complete` as GA4 key events. Under these rules, **events from inside the portal must not be sent to GA4**. Book Seven replaces that with a safe design: payments are counted through your BAA-covered route or as aggregated daily totals (Chapter 24). 🔵

3.6 Do's and Don'ts

✅ Do	❌ Don't
Ask legal to classify pages	Assume "public page = always fine"
Keep GA4 and Clarity off the portal	Track portal steps with GA4
Use BAA-covered vendors for health data	Rely on a cookie banner as HIPAA consent
Review every new tag	Add vendor pixels "to test"

3.7 Pros and cons of privacy-first analytics

✅ Pros	❌ Cons
Protects patients and the company	Less granular data in some areas
Lower legal and reputational risk	More tools and setup (BAA route)
Builds trust	Requires legal time

3.8 Where freshers go wrong in Chapter 3 topics

Mistake	Why it happens	Avoid it by	Fix it by
Sending form answers to GA4 "to segment"	Wanting insight	Principle 1	Removing; using counts only

Mistake	Why it happens	Avoid it by	Fix it by
Clarity recording the eligibility form	Default install everywhere	Principle 2	Excluding pages
Thinking the 2024 court ruling allows anything	Misreading news	3.1 table	Asking legal

3.9 Thumb rules

👍 Public behaviour yes, private health never.

👍 No BAA, no health data.

👍 Legal classifies; analytics follows.

3.10 Notes to remember

📝 Google offers no BAA for GA4.

📝 HHS: all tracking vendors receiving PHI need a BAA; banners aren't HIPAA authorisation.

Chapter 3 Glossary

Term	Simple meaning
PHI	Protected health information
BAA	Business associate agreement: a HIPAA contract with a vendor
Authenticated page	A page you see only after logging in
Tracking technology	Code that collects information about users' actions
Consumer health data	Health-related data protected by state laws like Washington's

Chapter 3 Quiz

1. Does Google offer a BAA for Google Analytics?

- A) Yes
- B) Only for 360
- C) No
- D) On request

2. Should GA4 run on the patient portal?

- A) Yes
- B) Only on payment pages
- C) No
- D) Only on weekends

3. Is a cookie banner a valid HIPAA authorisation (HHS)?

- A) Yes
- B) If it's big
- C) No
- D) If it's green

4. What did the June 20, 2024 court ruling do?

- A) Required pixels
- B) Banned GA4
- C) Removed HIPAA
- D) Vacated part of the HHS guidance about IP addresses on public health pages

5. Which law covers purchases of prescribed medication as consumer health data?

- A) Plain Writing Act
- B) None
- C) Washington My Health My Data Act
- D) Copyright Act

6. What do HHS say about tracking vendors receiving PHI?

- A) They're always fine
- B) Nothing
- C) They need a logo

- D) They must have signed a BAA

7. Who classifies pages into zones?

- A) GA4 automatically
- B) Legal
- C) Visitors
- D) The graphic designer

8. Which GA4 features should be off?

- A) Events
- B) Reports
- C) Page views
- D) Google signals and ads features

9. What replaces portal events in GA4?

- A) Clarity recordings in the portal
- B) More portal tags
- C) Nothing
- D) Payments counted through a BAA route or aggregated totals

10. In the chai-waala analogy, GA4 belongs…

- A) In patient files
- B) In public areas, never in the cabin or files
- C) In the doctor's cabin
- D) Nowhere at all

Sources

1. Google Analytics Help, HIPAA and Google Analytics ●
2. HHS OCR, Use of Online Tracking Technologies by HIPAA Covered Entities and Business Associates ●
3. Washington Attorney General, Protecting Washingtonians' Personal Health Data and Privacy ●

Answer key, Chapter 3: 1-C · 2-C · 3-C · 4-D · 5-C · 6-D · 7-B · 8-D · 9-D · 10-B

LEVEL 1 · CHAPTER 4

Chapter 4: Your BAA and a HIPAA-Safe Analytics Architecture

You have a BAA. This chapter shows how to use it: which pages each tool may touch, where health-related data flows, and how payments get counted without exposing anyone.

> 💡 **Trivia**
>
> HHS says a tracking vendor "is a business associate if it meets the definition of a business associate, regardless of whether the required BAA is in place." [1] 🟢 Official (HHS)
>
> A BAA isn't paperwork you file once. It decides which tools may see which data.

4.1 First: which vendor does your BAA cover?

A BAA is signed **with a specific vendor** for specific services. Before building anything, confirm:

☐	Question for legal
☐	Which company signed the BAA (e.g., a HIPAA analytics/CDP vendor, the portal vendor, a cloud provider)?
☐	Which services does it cover (data collection, storage, forwarding)?
☐	May that vendor forward **de-identified** data to Google (GA4/BigQuery)?
☐	Which pages and data types are we allowed to send to it?
☐	Who is the contact for changes?

Important: a BAA with one vendor does **not** make GA4, Clarity or Looker Studio HIPAA-covered. Google states it offers no BAA for GA4. [2] 🟢

4.2 The three zones

Zone	Pages	GA4 + GTM	Clarity	BAA-covered tool
A · Public content	Blog, guides, glossary, about, medical team, policies, general home sections	✅ Allowed (no personal data)	✅ Strict masking	Optional
B · Service and decision pages	Service, medication, pricing, how it works, eligibility start pages, contact/forms	⚠️ **Legal decides** per page type: e.g., page views and CTA clicks only, no form fields	⚠️ Legal decides; never on forms	✅ Recommended for anything collecting information
C · Authenticated / health data	Patient portal, intake, eligibility questions and answers, payment, account	❌ Never	❌ Never	✅ Only BAA-covered tools

Rule: every page template gets a zone in the measurement plan, signed off by legal. 🔵



> 🪔 **Desi analogy: hospital ke zones**
>
> Hospital mein lobby sabke liye khuli hai, OPD mein kuch log, aur ICU mein sirf staff. Visitor pass lobby tak chalta hai, ICU mein nahi. **Zone A lobby hai, Zone B OPD, Zone C ICU.**

4.3 The architecture (how data flows)

```
ZONE A + allowed parts of ZONE B (public site)
  Next.js page → data layer (no personal data) → GTM
       ├─→ GA4  ──→ BigQuery (daily export) ──→ Looker Studio
       └─→ Clarity (Strict masking, excluded from forms)

exit_to_portal  (last public event; cross-domain _gl only if legal approves)

ZONE C (portal, intake, payment)
  Portal events → BAA-covered vendor only
       └─→ De-identified daily COUNTS (date, source group, leads, payments, revenue band)
              → Google Sheet / BigQuery table → Looker Studio
```

Key idea: Looker Studio can show **leads and payments next to GA4 data** using **aggregated** tables (daily totals by channel or landing-page group) without any identifier ever reaching Google. 🔵

4.4 Three options for counting payments

Option	How	Pros	Cons
1 · Aggregated daily totals (recommended start)	Backend or BAA vendor exports a daily table: date · channel group · landing page group · leads · payments	No identifiers; simple; safe	No per-user attribution
2 · BAA vendor attribution	BAA-covered tool joins the visit and the payment, then sends **de-identified, aggregated** results onward	Better attribution	Cost; vendor setup; legal sign-off
3 · Measurement Protocol to GA4 with client ID	Server sends a `purchase` event to GA4 with the visitor's client ID	Full GA4 attribution	**Highest risk:** links a browser ID to a health payment at Google; **only if legal explicitly approves**, usually not recommended

The full build is in Chapter 24. 🔵

4.5 Page-zone register (template)

Page template	Example URL	Zone	GA4 events allowed	Clarity	Notes / legal sign-off
Blog article	/blog/...	A	All public events	✅	
Medication page	/services/weight-loss/...	B	page_view, CTA clicks, internal links	Legal	
Eligibility start	/weightloss-onboard/	B/C	`exit_to_portal` only	❌	
Portal	portal domain	C	None	❌	BAA vendor

> 🟧 **Ongo Weight Loss example**
>
> Your portal is on a separate, non-public platform, and weightclinicsresults.com and other child sites also link to it. That makes the boundary clean: **everything on the public sites is Zone A/B, the portal is Zone C**. Public analytics records the click to the portal; the BAA-covered side counts what happens after. 🔵

4.6 Do's and Don'ts

✅ Do	❌ Don't
Confirm exactly what your BAA covers	Assume a BAA covers every tool
Give every template a zone	Tag new templates without review
Count payments as de-identified totals	Send payment events with client IDs to GA4 by default
Keep the zone register updated	Let it go stale after redesigns

4.7 Pros and cons of a zoned architecture

✅ Pros	❌ Cons
Clear, defensible rules	Less detailed attribution for payments
Easy to explain to developers	Needs legal input per template
Scales to child websites	Two data paths to maintain

4.8 Where freshers go wrong in Chapter 4 topics

Mistake	Why it happens	Avoid it by	Fix it by
"We have a BAA, so GA4 is fine"	Misunderstanding BAAs	4.1	Confirming the vendor
Clarity on the eligibility page	Site-wide install	Zone register	Excluding by URL
Sending payment events from the portal to GA4	Wanting attribution	Option table	Switching to Option 1 or 2

4.9 Thumb rules

👍 Lobby, OPD, ICU: every page has a zone.

👍 A BAA covers one vendor, not every tool.

👍 Count payments, don't track patients.

4.10 Notes to remember

📝 Zone C (portal, intake, payment): never GA4 or Clarity.

📝 Start with aggregated daily totals for payments.

Chapter 4 Glossary

Term	Simple meaning
Zone	A category of pages with its own tracking rules
Business associate	A vendor handling PHI for a covered entity
De-identified data	Data with identifiers removed
Aggregated data	Totals (e.g., per day), not individual records
Page-zone register	A list of templates with their zone and allowed tools

Chapter 4 Quiz

1. Does a BAA with one vendor cover GA4?

- A) Yes
- B) No
- C) Only on mobile
- D) Only for reports

2. Which zone is the patient portal?

- A) Zone B
- B) No zone
- C) Zone A
- D) Zone C

3. What's the recommended start for counting payments?

- A) GA4 purchase events with client IDs
- B) Clarity recordings
- C) Not counting them
- D) Aggregated daily totals with no identifiers

4. Who decides tracking on Zone B pages?

- A) Nobody

- B) The developer alone
- C) Legal
- D) GA4 automatically

5. What is de-identified data?

- A) Data with identifiers removed
- B) Data with names added
- C) Images
- D) Encrypted passwords

6. Which option is highest risk?

- A) Not tracking the portal
- B) Aggregated totals
- C) Measurement Protocol to GA4 with client ID for payments
- D) BAA vendor aggregation

7. Can Looker Studio show payments next to GA4 data safely?

- A) No, never
- B) Only with patient names
- C) Yes, using aggregated tables
- D) Only in Clarity

8. What is the last public event before the portal?

- A) `exit_to_portal`
- B) `scroll`
- C) `page_not_found`
- D) `purchase`

9. What should every page template have?

- A) Clarity
- B) A zone in the register
- C) A UTM
- D) A brand drug name

10. In the hospital analogy, Zone C is…

- A) The car park

- B) The lobby
- C) The canteen
- D) The ICU (staff only)

Sources

1. HHS OCR, Online tracking technologies bulletin
2. Google Analytics Help, HIPAA and Google Analytics
3. Ongo Weight Loss, portal and child-site structure (team lead, 30 October 2026), internal

Answer key, Chapter 4: 1-B · 2-D · 3-D · 4-C · 5-A · 6-C · 7-C · 8-A · 9-B · 10-D

LEVEL 1 · CHAPTER 5

Chapter 5: Consent Management: Which Tool and How to Set It Up

You have no consent banner today. This chapter explains what a consent manager does, which one to choose for a U.S. health brand, and how it connects to GTM through Google Consent Mode.

> 💡 **Trivia**
>
> Google Consent Mode has seven consent types: `ad_storage`, `ad_user_data`, `ad_personalization`, `analytics_storage`, `functionality_storage`, `personalization_storage` and `security_storage`. [1] ● Official (Google)
>
> A consent manager collects the visitor's choice; Consent Mode tells every Google tag what that choice was.

5.1 What a consent management platform (CMP) does

Job	Detail
Show a banner or notice	"We use cookies for analytics..." with choices
Store the choice	So the banner doesn't reappear every visit
Scan the site	Lists cookies and scripts it finds
Block or allow tags	Tells GTM which categories are allowed
Send Consent Mode signals	`analytics_storage: granted/denied`, etc.
Honour opt-out signals	e.g., Global Privacy Control (GPC) in states that require it
Keep records	Proof of what each visitor chose

5.2 Which CMP to choose

CMP	Best for	Why recommended	Flaws
CookieYes	Small to mid-size teams starting fast	Simple setup, GTM template, Google-certified CMP partner, U.S. state law support, affordable plans	Fewer enterprise workflows
Cookiebot (Usercentrics)	Teams wanting strong automatic scanning	Mature scanner, GTM integration, Google-certified	Pricing by number of pages/domains can grow
OneTrust	Large enterprises with legal teams	Very complete privacy suite	Expensive, complex
Termly / Osano	Alternatives	Policy + consent bundles	Check health-data features

Our recommendation: start with **CookieYes or Cookiebot** (both Google-certified CMP partners; certified CMPs update to the newest Consent Mode automatically). Choose with legal; check current pricing and plan limits (they change). [2] 🟢 / 🟠

> 🍛 **Desi analogy: shaadi mein photographer**
>
> Shaadi mein photographer har mehmaan ki photo nahi le sakta agar woh mana kare. Ek register hota hai: "photo haan / na". **CMP wahi register hai:** jo mana kare, uska data nahi.

5.3 Choosing the consent model (legal decides)

Model	How it works	When used
Opt-out (notice + choice)	Tags run by default; visitors can opt out; honour GPC	Common in many U.S. states for analytics
Opt-in	Tags wait until the visitor agrees	Required for some data types/regions; safest for health data
Hybrid by zone	Zone A: opt-out; Zone B: opt-in for non-essential tags	Practical for health sites

Note: Washington's My Health My Data Act and similar laws can require **consent before collecting consumer health data**, which is another reason to keep GA4 free of health data (Chapter 3). ●

5.4 Consent Mode: basic vs advanced

	Basic	Advanced
Before choice	Google tags **don't load**	Tags load with consent **denied** and send cookieless pings
After "deny"	Nothing sent	Cookieless pings only
Modelling	General model	Advertiser-specific modelling (needs enough data)
Our pick	**Basic is simpler and more conservative**; advanced only if legal approves	

[1] ●

5.5 Setup steps (developer + analytics)

Step	Action
1	Legal chooses the model per zone and approves banner wording
2	Create the CMP account; add ongoweightloss.com, weightclinicsresults.com and other child sites
3	Run the cookie scan; categorise cookies (necessary, analytics, functional, marketing)
4	In GTM: add the **CMP's Community Template** tag on the **Consent Initialization – All Pages** trigger
5	Set **default consent** in that tag (e.g., `analytics_storage: denied` until choice, if opt-in)
6	Turn on **Consent Overview** in GTM (Admin → Container settings → Enable consent overview)
7	For each tag, set **Additional consent checks** (GA4: `analytics_storage`; Clarity: analytics category)
8	Configure Clarity's consent behaviour (Clarity supports a consent signal; follow its docs)
9	Add the "Do not sell/share" or "Privacy choices" link if legal requires; enable GPC

Step	Action
10	Test: Preview mode → check consent states before and after choices (Chapter 30)

Developer code (if not using a template) — default consent before GTM loads:

```
<script>
  window.dataLayer = window.dataLayer || [];
  function gtag(){dataLayer.push(arguments);}
  gtag('consent', 'default', {
    analytics_storage: 'denied',
    ad_storage: 'denied',
    ad_user_data: 'denied',
    ad_personalization: 'denied',
    wait_for_update: 500
  });
</script>
```

The CMP then calls `gtag('consent', 'update', { analytics_storage: 'granted' })` when the visitor agrees. 🔵

> 🟧 **Ongo Weight Loss example**
>
> With no banner today, the fastest compliant path is: legal picks the model → CMP installed via GTM template → Consent Mode default set → GA4 and Clarity tags given consent checks → tested on all public sites. Ads tags are out of scope until the paid-media book. 🔵

5.6 Do's and Don'ts

✅ Do	❌ Don't
Let legal choose the consent model	Copy a European banner blindly
Use a Google-certified CMP	Hand-code a banner without records
Fire the CMP on Consent Initialization	Load GA4 before consent defaults are set
Test consent states in Preview	Assume the banner works

5.7 Pros and cons of a CMP

✅ Pros	❌ Cons
Legal protection and records	Some visitors decline: less data
Automatic scans and categories	Subscription cost
Clean integration with GTM	Banner design affects UX

5.8 Where freshers go wrong in Chapter 5 topics

Mistake	Why it happens	Avoid it by	Fix it by
GA4 fires before consent default	Wrong trigger	Consent Initialization trigger	Re-ordering
Banner on main site, not child sites	Forgetting domains	Step 2	Adding domains
Treating the banner as HIPAA consent	Confusion	Chapter 3	Keeping PHI out entirely

5.9 Thumb rules

👍 Choice collected by the CMP, respected by every tag.

👍 Defaults first, tags second.

👍 A banner is not a BAA.

5.10 Notes to remember

📝 Recommended CMPs: CookieYes or Cookiebot (Google-certified).

📝 GTM: CMP tag on Consent Initialization; consent checks on every tag.

Chapter 5 Glossary

Term	Simple meaning
CMP	Consent management platform
Consent Mode	Google's way to pass consent choices to its tags
Opt-in / opt-out	Agree first vs. allowed until refused
GPC	Global Privacy Control: a browser opt-out signal
Consent Initialization	The earliest GTM trigger, for consent defaults

Chapter 5 Quiz

1. What does a CMP do?

- A) Hosts videos
- B) Collects and stores consent choices and controls tags
- C) Builds dashboards
- D) Writes blogs

2. Which consent type controls analytics cookies?

- A) `ad_personalization`
- B) `security_storage`
- C) `functionality_storage`
- D) `analytics_storage`

3. Which GTM trigger should the CMP tag use?

- A) Scroll Depth
- B) Window Loaded
- C) Click – All Elements
- D) Consent Initialization – All Pages

4. In basic Consent Mode, what happens before a choice?

- A) Tags send full data

- B) Tags send names
- C) Tags crash
- D) Google tags don't load

5. Which CMPs are recommended to start?

- A) A spreadsheet
- B) CookieYes or Cookiebot
- C) Clarity
- D) None

6. Is a cookie banner HIPAA authorisation?

- A) Yes
- B) Sometimes
- C) No
- D) If opt-in

7. What is GPC?

- A) A cookie name
- B) A payment method
- C) A GA4 report
- D) A browser opt-out signal

8. Which sites need the CMP?

- A) Only the homepage
- B) Main site and child sites
- C) Only the portal
- D) None

9. Who chooses the consent model?

- A) Google
- B) The designer
- C) Visitors
- D) Legal

10. In the photographer analogy, the CMP is…

- A) The camera

- B) The food
- C) The wedding hall
- D) The yes/no register for photos

Sources

1. Google, Consent mode overview
2. Google Tag Manager Help, Consent mode and CMP partners
3. Washington Attorney General, My Health My Data Act

Answer key, Chapter 5: 1-B · 2-D · 3-D · 4-D · 5-B · 6-C · 7-D · 8-B · 9-D · 10-D

LEVEL 1 · CHAPTER 6

Chapter 6: The Tracking Stack: Core Tools and Gap Tools

Five core tools do most of the work. A few extra tools fill specific gaps the core can't. This chapter tells you what each one does, why it's here, and its flaws.

> 💡 **Trivia**
>
> GA4 standard properties can export up to **1 million events per day** to BigQuery as raw, unsampled data, and you can start in the free BigQuery sandbox. [1] ● Official (Google)
>
> That's how you see every visitor's full path, not just the top few.

6.1 The core five

Tool	Does	Can't do	Best practice
GTM	Manages tags, triggers, variables; reads the data layer; versions and rollbacks	Store or report data; collect consent by itself	Data-layer-driven tags; naming rules; Preview before publish
GA4	Events, users, sessions, sources, funnels, paths, user explorer, key events	Show every path (top 20 nodes max per step); keep explorations beyond 14 months; take PHI (no BAA)	Custom dimensions; 14-month retention; Google signals off; BigQuery export
GSC	Queries, impressions, clicks, CTR, position, indexing, Core Web Vitals	Show Bing; show anonymised queries; keep more than 16 months in the UI	Link to GA4; bulk export to BigQuery
Clarity	Heatmaps, recordings, rage/dead clicks, quick backs, scroll	Sign a BAA; run safely on forms/portal	Strict masking; public pages only; GA4 integration

Tool	Does	Can't do	Best practice
Looker Studio	Dashboards combining sources; scheduled emails	Store data; avoid GA4 API quota limits on big dashboards	Feed from BigQuery or extracts

6.2 Gap tools (only where the core can't)

Gap	Tool	Why	Flaws	Chapter
Every path, unsampled, beyond retention; custom attribution	**BigQuery**	Raw event-level data; free sandbox to start	SQL needed; costs above free tier; sandbox tables expire after 60 days	31–35
Consent collection	**CookieYes / Cookiebot**	GTM/GA4 can't ask for consent	Subscription; less data when declined	5, 29
Health-related data and payments	**Your BAA-covered vendor**	Google won't sign a BAA for GA4	Cost; setup	4, 24
Bing and Copilot search data	**Bing Webmaster Tools** (free)	GSC only covers Google	Smaller traffic	39
Real-user speed per page	**web-vitals → GA4**, **CrUX Vis**, **GSC CWV report**	GA4 doesn't measure Core Web Vitals by default; the old CrUX Looker connector was retired after November 2025	Small dev work	40
Blocked/short-lived browser data	**Server-side GTM** (optional)	Clean fields before sending; longer first-party cookies	Hosting cost; setup	12
Tag QA	**Tag Assistant, Omnibug, dataLayer inspectors** (free)	Faster debugging	Manual	30
Competitor benchmarking	**Semrush / Similarweb**	GA4/GSC only show our own site	Estimates, not real data; paid	42

> 🍛 **Desi analogy: rasoi ke bartan**
>
> Roz ka khaana kadhai, tawa, cooker, chakla-belan aur thali se ban jaata hai (core five). Biryani ke liye handi chahiye, aur mithai ke liye kadhai alag (gap tools). **Har bartan ka kaam hai; bina zaroorat naya bartan mat khareedo.**

6.3 Tools we don't recommend now

Tool	Why not now
Mixpanel, Amplitude, Heap	Strong journey tools, but GA4 + BigQuery covers our needs; HIPAA versions are enterprise-priced
Hotjar (instead of Clarity)	Clarity is free and covers heatmaps/recordings; consider only for on-site surveys
Ad pixels (Meta, TikTok)	No paid media yet; serious health-privacy risk (future paid book)

6.4 How the stack connects

```
Website (Next.js) ──data layer──> GTM ──> GA4 ──> BigQuery ──> Looker Studio
                                       └──> Clarity ──(GA4 integration)
GSC ──(link)──> GA4      GSC ──(bulk export)──> BigQuery
Bing Webmaster Tools ──(manual export / API)──> Sheets ──> Looker Studio
CMP ──(Consent Mode)──> GTM
BAA vendor / backend ──(daily aggregated counts)──> Sheets/BigQuery ──> Looker Studio
Social platforms ──(exports/connectors)──> Sheets ──> Looker Studio
```

> 🟧 **Ongo Weight Loss example**
>
> Our full stack costs very little: GTM, GA4, GSC, Clarity, Looker Studio, Bing Webmaster Tools and CrUX Vis are free; BigQuery starts free (sandbox) and stays cheap at our size; the CMP and BAA vendor are the main costs. 🔵

6.5 Do's and Don'ts

✅ Do	❌ Don't
Use core tools first	Buy overlapping tools
Add gap tools only for real gaps	Add tools "because others use them"
Document how tools connect	Leave integrations undocumented
Check vendor terms (BAA, pricing)	Assume free tools are HIPAA-safe

6.6 Pros and cons of a lean stack

✅ Pros	❌ Cons
Low cost, easy training	Some advanced features missing
Fewer privacy risks	More manual work in places
Google tools integrate well	Dependency on Google

6.7 Where freshers go wrong in Chapter 6 topics

Mistake	Why it happens	Avoid it by	Fix it by
Installing Hotjar and Clarity together	Overlap	6.3	Keeping Clarity
Expecting GA4 to show every path	Not knowing limits	6.1	Using BigQuery
Using the old CrUX dashboard	Outdated guides	6.2	CrUX Vis / web-vitals

6.8 Thumb rules

👍 Core five first, gap tools second.

👍 Every tool has one clear job.

👍 Free isn't the same as safe for health data.

6.9 Notes to remember

📝 BigQuery: every path; sandbox free; tables expire in 60 days on sandbox.

📝 CrUX Looker connector retired after November 2025.

Chapter 6 Glossary

Term	Simple meaning
Stack	The set of tools used together
Gap tool	A tool added for something the core can't do
Raw data	Every event, unsummarised
Server-side GTM	Tag manager running on your own server
Benchmarking	Comparing with competitors

Chapter 6 Quiz

1. What does BigQuery add?

- A) Consent banners
- B) Bing data
- C) Raw, unsampled event data for every path
- D) Heatmaps

2. How many events per day can standard GA4 export to BigQuery?

- A) Up to 1 million
- B) 100
- C) Unlimited always
- D) 10

3. Which tool collects consent?

- A) A CMP (CookieYes/Cookiebot)
- B) GA4
- C) GSC
- D) Looker Studio

4. Which tool shows Bing search data?

- A) GSC
- B) Clarity
- C) Bing Webmaster Tools
- D) GTM

5. What happened to the CrUX Looker connector?

- A) Nothing
- B) Moved to GA4
- C) Retired after November 2025
- D) Became paid

6. Why not use Meta/TikTok pixels now?

- A) They're required
- B) No paid media yet and serious health-privacy risk
- C) They're free
- D) They replace GA4

7. Can Clarity sign a BAA?

- A) On request
- B) Only for blogs
- C) No
- D) Yes

8. What does server-side GTM offer?

- A) Cleaner data and longer first-party cookies
- B) Automatic consent
- C) Patient tracking
- D) Free hosting

9. What does Semrush/Similarweb provide?

- A) Exact competitor data
- B) Competitor estimates
- C) Consent
- D) Our GA4 data

10. In the rasoi analogy, gap tools are…

- A) Special utensils for specific dishes
- B) Daily utensils
- C) The ingredients
- D) Unnecessary

Sources

1. Google Analytics Help, BigQuery Export
2. Chrome for Developers, CrUX Dashboard deprecation (September 9, 2025)
3. Google Analytics Help, HIPAA and Google Analytics

Answer key, Chapter 6: 1-C · 2-A · 3-A · 4-C · 5-C · 6-B · 7-C · 8-A · 9-B · 10-A

LEVEL 1 · CHAPTER 7

Chapter 7: How GA4 Thinks: Events, Parameters, Users and Sessions

GA4 is built on one idea: everything is an event. Once you understand events, parameters and scopes, every report, exploration and BigQuery table makes sense.

> 💡 **Trivia**
>
> GA4's data retention setting (2 or 14 months on standard properties) affects **only explorations and funnel reports**, not standard aggregated reports. [1] ● Official (Google)
>
> Set it to 14 months on day one. You can't get older exploration data back later.

7.1 Everything is an event

Event type	Who creates it	Examples
Automatically collected	GA4 itself	`first_visit`, `session_start`, `user_engagement`
Enhanced measurement	GA4 toggles	`page_view`, `scroll` (90%), `click` (outbound), `file_download`, `video_start`, `form_start`
Recommended	Google-defined names you send	`generate_lead`, `sign_up`, `purchase`
Custom	Our own names	`internal_link_click`, `cta_semaglutide_get_started`, `exit_to_portal`

7.2 Parameters and scopes

Concept	Meaning	Example
Parameter	Detail sent with an event	`link_position = body`
Event-scoped dimension	Describes one event	`cta_position` , `link_target`
User-scoped dimension (user property)	Describes the user over time	`first_landing_group` (set once, no personal data)
Item-scoped	E-commerce items	Not used now
Custom dimension	A parameter you register so it appears in reports	Chapter 25

Rule: a parameter not registered as a custom dimension appears in BigQuery and DebugView, but **not** in standard reports and explorations. 🔵

> 🍛 **Desi analogy: cricket scorebook**
>
> Scorebook mein har gend ek "event" hai: kaun bowler, kaun batsman, kitne run, kahan shot gaya (parameters). Match (session) bahut saari gendon se banta hai, aur player (user) poore series mein khelta hai. **GA4 bhi har gend likhta hai, phir jodta hai.**

7.3 Users, sessions and engagement

Metric	GA4 definition (plain)
Users (active users)	Browsers with an engaged session or first visit in the period
Sessions	Visits; a new session after 30 minutes inactive
Engaged session	Lasted 10+ seconds, or had a key event, or 2+ page views
Engagement rate	Engaged sessions ÷ sessions
Average engagement time	Time the page was in focus
Key event	An event you mark as important (Chapter 26)

7.4 Attribution basics in GA4

Scope	Dimension example	Meaning
User	First user source / medium	How the user first arrived
Session	Session source / medium	How this visit arrived
Event / key event	Source / medium (key-event attribution)	Credited source per your attribution model

Use the right scope for the question: "what brought new people?" (user) vs "what brought visits?" (session). 🔵

7.5 Identity

Identity	Use at Ongo Weight Loss
Device / client ID	✅ Default
User ID	❌ Not on public site (no logins); never an email or patient ID
Google signals	❌ Off (health site)
Modelling	Only if consent mode advanced + legal approval

> 🟧 **Ongo Weight Loss example**
>
> We will send our CTA events with custom names (Book Four style, e.g., `cta_semaglutide_get_started`) **and** shared parameters (`cta_name`, `cta_group`, `cta_location`). That gives you separate events per CTA, as you wanted, plus one combined view for comparisons. 🔵

7.6 Do's and Don'ts

✅ Do	❌ Don't
Plan events and parameters before tagging	Invent names on the fly
Register parameters you'll report on	Expect unregistered parameters in reports

✅ Do	❌ Don't
Set 14-month retention immediately	Leave 2-month default
Use client ID only	Set user IDs from emails

7.7 Pros and cons of GA4's event model

✅ Pros	❌ Cons
Very flexible	Easy to create messy names
Same model in BigQuery	Needs custom dimensions for reporting
Great for journeys	Learning curve for beginners

7.8 Where freshers go wrong in Chapter 7 topics

Mistake	Why it happens	Avoid it by	Fix it by
Parameters missing in reports	Not registered	Chapter 25	Registering (not retroactive)
Mixing user and session source	Similar names	7.4 table	Choosing scope per question
Retention left at 2 months	Default	Day-one checklist	Changing now (not retroactive)

7.9 Thumb rules

> 👍 Event = what happened; parameter = details.
>
> 👍 Register what you'll report.
>
> 👍 14 months, day one.

7.10 Notes to remember

📝 Engaged session: 10+ seconds, a key event, or 2+ page views.

📝 Retention affects explorations and funnels only.

Chapter 7 Glossary

Term	Simple meaning
Event	Something that happened
Parameter	Detail about an event
Scope	Whether a dimension describes an event, user or item
Engaged session	A meaningful visit by GA4's rules
User property	A user-scoped attribute

Chapter 7 Quiz

1. What is GA4's basic unit of data?

- A) The report
- B) The cookie banner
- C) The event
- D) The page only

2. What does the retention setting affect?

- A) Nothing
- B) BigQuery only
- C) All standard reports
- D) Explorations and funnel reports only

3. Which is a custom event?

- A) `user_engagement`
- B) `internal_link_click`
- C) `session_start`
- D) `first_visit`

4. When does a parameter appear in reports?

- A) Always automatically
- B) After it's registered as a custom dimension
- C) Only in Clarity
- D) Never

5. What is an engaged session?

- A) 10+ seconds, a key event, or 2+ page views
- B) Only purchases
- C) Only logins
- D) Any session

6. Should Google signals be on?

- A) No (health site)
- B) Yes
- C) Only on weekends
- D) Only for blogs

7. Which dimension answers "what brought new people"?

- A) Device model
- B) First user source / medium
- C) Session source / medium
- D) Page title

8. Can User ID be an email?

- A) If hashed, always fine
- B) Never
- C) Only for doctors
- D) Yes

9. What retention should be set on day one?

- A) 1 day
- B) 14 months
- C) None
- D) 2 months

10. In the scorebook analogy, each ball is…

- A) An event
- B) A cookie
- C) A user
- D) A report

Sources

1. Google Analytics Help, Data retention 🟢
2. Ongo Weight Loss, Book Four event dictionary (Appendix C), internal 🔵

Answer key, Chapter 7: 1-C · 2-D · 3-B · 4-B · 5-A · 6-A · 7-B · 8-B · 9-B · 10-A

LEVEL 1 · CHAPTER 8

Chapter 8: The Measurement Plan: Goals to Events to Reports

A measurement plan is the blueprint. It connects what the business wants to the exact events developers build and the reports you read every day.

> 💡 **Trivia**
>
> GA4 funnel explorations allow **up to 10 steps**, open or closed, with "directly" or "indirectly followed by" rules. [1] ● Official (Google)
>
> Ten steps is plenty, if you define the right ones first.

8.1 The plan structure

```
Business goal → Questions → KPIs → Events & parameters → Reports & dashboards → Owner
```

8.2 Ongo Weight Loss measurement plan (version 1)

Business goal	Questions	KPIs	Events / data	Report
Grow paying patients	How many payments? From which channels and landing-page groups?	Payments, payment rate per lead	Aggregated daily payments (Chapter 24)	Daily + executive dashboard
Generate qualified leads	How many eligibility completions and sign-ups?	Leads, lead rate per session	Lead counts (BAA/aggregated) + `exit_to_portal`	Daily dashboard

Business goal	Questions	KPIs	Events / data	Report
Move readers toward care	Do blog readers reach service/medication pages?	Blog → money-page rate	`internal_link_click`, `page_meta`	Content dashboard
Make CTAs work	Which CTAs, where, drive portal exits?	CTA click rate; CTA → portal rate	`cta_[group]_[action]` + params	CTA report
Earn search traffic	Which queries and pages grow?	Clicks, CTR, position	GSC + GSC export	SEO dashboard
Healthy site	Errors, speed, 404s?	CWV pass rate; 404 count	`page_not_found`, web-vitals	Technical dashboard
Grow social reach	Which platforms send engaged visits?	Organic social sessions, engagement	UTMs (off-site only) + platform data	Social dashboard
Understand behaviour	Where do people struggle?	Rage clicks, quick backs	Clarity	Clarity review

8.3 Leads vs conversions (your definition)

Stage	Definition	Where counted	GA4 key event?
Micro-conversion	CTA click, internal link to money page, tool complete	GA4	Some (Chapter 26)
Lead	Eligibility completed or sign-up	BAA route / aggregated	No GA4 event from portal
Portal entry	`exit_to_portal`	GA4	✅ Yes (proxy for leads)
Conversion	**Payment completed**	BAA route / aggregated	Reported in Looker, not GA4 by default

> 🪔 **Desi analogy: ghar ka naksha**
>
> Mistri seedha deewar nahi banata; pehle naksha banta hai: kitne kamre, kahan khidki, kahan bijli ke point. **Measurement plan naksha hai:** bina naksha ke tags = bina naksha ka ghar.

8.4 The core funnel (TOFU → payment)

Step	Definition	Source
1	Session starts on any page	GA4
2	Views a TOFU/MOFU content page (`funnel_stage` = tofu/mofu)	GA4
3	Views a BOFU page (service/medication/pricing)	GA4
4	Clicks a primary CTA	GA4
5	`exit_to_portal`	GA4
6	Lead (eligibility complete)	Aggregated
7	Payment	Aggregated

Steps 1–5 are a GA4 funnel exploration; steps 6–7 are joined in Looker Studio as daily totals by channel and landing-page group. 🔵

8.5 Plan ownership

Section	Owner
Goals and KPIs	Team lead
Events and data layer	On-page SEO (spec) + developer (build)
Privacy zones	Legal
Dashboards	On-page SEO / analytics
Payment data feed	Developer/backend + BAA vendor

> 🟧 **Ongo Weight Loss example**
>
> Your Book Three content map already gives every page a silo, parent, funnel stage and entity. Those become data layer fields (Chapter 14), so every event automatically knows "this happened on a MOFU semaglutide page". 🔵

8.6 Do's and Don'ts

✅ Do	❌ Don't
Start from business goals	Start from available reports
Define leads vs payments clearly	Mix them in one metric
Assign an owner to every line	Leave events ownerless
Version the plan (v1, v2...)	Change it silently

8.7 Pros and cons of a written plan

✅ Pros	❌ Cons
Developers know exactly what to build	Takes time up front
Reports answer real questions	Needs updates as the site grows
Easier audits	—

8.8 Where freshers go wrong in Chapter 8 topics

Mistake	Why it happens	Avoid it by	Fix it by
Tracking without questions	Excitement	Plan structure	Linking each event to a question
Calling sign-ups "conversions"	Habit	8.3 table	Renaming in reports

Mistake	Why it happens	Avoid it by	Fix it by
No owner for payment data	Cross-team gap	8.5	Assigning backend owner

8.9 Thumb rules

👍 No question, no event.

👍 Leads are signals; payments are the goal.

👍 Every line has an owner.

8.10 Notes to remember

📝 Funnel steps 1–5 in GA4; leads and payments joined in Looker.

📝 Book Three's content map feeds the data layer.

Chapter 8 Glossary

Term	Simple meaning
Measurement plan	Document linking goals to events and reports
KPI	Key performance indicator
Micro-conversion	A small step toward the goal
Funnel	Ordered steps toward a goal
Owner	Person responsible for a plan item

Chapter 8 Quiz

1. What comes first in a measurement plan?

- A) Business goals
- B) Colours
- C) Dashboards
- D) Tags

2. What is the conversion for Ongo Weight Loss?

- A) Payment completed
- B) Page view
- C) FAQ open
- D) Scroll

3. How many steps can a GA4 funnel exploration have?

- A) 1
- B) 3
- C) Up to 10
- D) 100

4. Where are leads and payments joined with GA4 data?

- A) Looker Studio (aggregated totals)
- B) Clarity
- C) GA4 portal tags
- D) GSC

5. Which event is a GA4 proxy for leads?

- A) `faq_open`
- B) `scroll`
- C) `exit_to_portal`
- D) `page_not_found`

6. Who owns privacy zones?

- A) Designer
- B) Legal
- C) Social team
- D) Nobody

7. Which source provides funnel_stage for every page?

- A) Book Three's content map via the data layer
- B) Clarity
- C) GSC
- D) Bing

8. What's a micro-conversion?

- A) A refund
- B) A server error
- C) A small step like a CTA click
- D) A payment

9. Should the plan be versioned?

- A) Only by Google
- B) Yes
- C) Only once
- D) No

10. In the naksha analogy, tags without a plan are like…

- A) Building a house without a map
- B) A wedding
- C) A finished house
- D) A cricket match

Sources

1. Google Analytics Help, Funnel exploration 🟢
2. Ongo Weight Loss, Book Three (content map) and conversion definitions (team lead, 30 October 2026), internal 🔵

Answer key, Chapter 8: 1-A · 2-A · 3-C · 4-A · 5-C · 6-B · 7-A · 8-C · 9-B · 10-A

LEVEL 1 · CHAPTER 9

Chapter 9: Baseline Audit: Where We Are Today

Before building, write down exactly what exists. The baseline is your "before" photo: it shows what's missing, what's risky and what must be fixed first.

> 💡 **Trivia**
>
> Clarity's default masking mode is **Balanced**, which masks only numbers and email addresses; **Strict** masks all content. Input boxes and dropdowns are masked in every mode. [1] 🟢 Official (Microsoft)
>
> If Clarity is already installed, its masking mode is the first thing to check.

9.1 Current state (team lead, 30 October 2026)

Area	Status	Risk
GTM	Installed, not fully configured	🟠 Unknown tags may exist
GA4	Access available; events not implemented	🟠 No journey data yet
GSC	Access available	🟢
Clarity	Present on site (Book Four audit); masking/scope to confirm	🔴 Until masking and exclusions verified
Looker Studio	Available, nothing built	🟢 Clean start
BigQuery	Not set up	🟠 No raw history
Consent banner	None	🔴 Legal review needed

Area	Status	Risk
BAA	Exists (vendor to confirm)	🟠 Not yet used for analytics
Payment data in reports	None	🟠
Child sites (weightclinicsresults.com etc.)	Link to portal; tracking unknown	🟠

9.2 Baseline audit checklist (do this in week 1)

#	Check	Where	Record
1	List every tag, trigger, variable in GTM	GTM → Workspace → Tags/Triggers/Variables	Export container JSON (Admin → Export)
2	Which GA4 property and data streams exist?	GA4 Admin → Data streams	Stream IDs, domains
3	Is GA4 loading via GTM, hard-coded, or both?	Browser DevTools → Network: `collect?v=2`	Note duplicates
4	Enhanced measurement toggles	GA4 → Data streams → Enhanced measurement	Screenshot
5	Data retention	GA4 Admin → Data retention	Set to 14 months
6	Google signals	GA4 Admin → Data collection	Must be off
7	Clarity masking mode and page scope	Clarity → Settings → Masking	Set Strict; exclude forms/portal
8	Any other scripts (pixels, chat)	DevTools → Network/Sources	List vendors
9	GSC properties	GSC → Property selector	Domain property?
10	Personal data leak check	GA4 → Explore → page_location contains "@" or "email="	Must be zero

> 🪔 **Desi analogy: naye ghar mein shift hone se pehle**
>
> Naye ghar mein jaane se pehle sab check karte hain: kaun si tap leak karti hai, kaun sa switch kaam nahi karta, purane kiraayedaar ka kya saamaan pada hai. **Baseline audit wahi walkthrough hai.**

9.3 Gap list (what this book builds)

Gap	Fix	Chapter
No measurement plan	Plan v1	8
No data layer	Spec + Next.js code	13–15
No events	Event Dictionary v2 build	16–23
No payment/lead data	Aggregated feed	24
No custom dimensions / key events	Register	25–26
No consent	CMP + Consent Mode	5, 29
Clarity risk	Strict masking + exclusions	28
No BigQuery	GA4 + GSC exports	31–37
No dashboards	Looker Studio set	57–65

9.4 Recording the baseline

Keep a one-page "Baseline v1" doc with screenshots, container export, and the date. Every later change is compared against it. 🔵

> 🟧 **Ongo Weight Loss example**
>
> We couldn't run a live browser check for this book, so this baseline uses your answers and Book Four's audit. Step one of the 90-day plan (Chapter 69) is to complete the 10-point checklist above with your access and save it as Baseline v1. 🔵

9.5 Do's and Don'ts

✅ Do	❌ Don't
Export the GTM container before changing anything	Delete tags you don't understand
Check for duplicate GA4 tags	Add a second GA4 install
Check Clarity masking first	Assume defaults are safe
Date the baseline	Overwrite it later

9.6 Pros and cons of a formal baseline

✅ Pros	❌ Cons
Clear before/after	Takes a day
Finds risks early	Needs screenshots and exports
Easier troubleshooting later	—

9.7 Where freshers go wrong in Chapter 9 topics

Mistake	Why it happens	Avoid it by	Fix it by
Two GA4 tags firing	Hard-coded + GTM	Check #3	Removing one
Clarity on forms	Default install	Check #7	Excluding pages
Emails in URLs	Form GET parameters	Check #10	Fixing forms (POST) and redacting

9.8 Thumb rules

👍 Photograph before you renovate.

👍 Export the container before editing.

👍 Zero personal data in URLs, always.

9.9 Notes to remember

📝 Baseline checklist: 10 checks in week 1.

📝 Clarity default is Balanced; we need Strict.

Chapter 9 Glossary

Term	Simple meaning
Baseline	The starting state before changes
Container export	A JSON backup of GTM
Data stream	A GA4 source (website or app)
Duplicate tag	The same tracking firing twice
Leak	Personal data accidentally collected

Chapter 9 Quiz

1. What is Clarity's default masking mode?

- A) None
- B) Strict
- C) Relaxed
- D) Balanced

2. What should be exported before changing GTM?

- A) Patient data
- B) Clarity recordings
- C) Nothing
- D) The container JSON

3. What does a `collect?v=2` request show?

- A) A payment
- B) A GSC query
- C) A GA4 hit being sent
- D) A Clarity recording

4. What should data retention be set to?

- A) 1 week
- B) 14 months
- C) Off
- D) 2 months

5. Which status is high risk today?

- A) No consent banner
- B) GSC access
- C) Clean start
- D) Looker Studio available

6. How do you check for emails in URLs?

- A) Explore page_location for "@" or "email="
- B) Ask visitors
- C) Check fonts
- D) Look at the logo

7. Why check for duplicate GA4 tags?

- A) They're required
- B) They improve SEO
- C) They add consent
- D) Duplicates double-count data

8. What's step one of the 90-day plan?

- A) Build dashboards
- B) Complete the baseline checklist
- C) Buy tools
- D) Delete GTM

9. What should Clarity masking be set to?

- A) Strict
- B) Relaxed
- C) Balanced
- D) Off

10. In the new-house analogy, the baseline is…

- A) The rent
- B) The walkthrough before moving in
- C) The party
- D) The furniture

Sources

1. Microsoft Learn, Clarity masking 🟢
2. Ongo Weight Loss, current setup (team lead, 30 October 2026) and Book Four audit, internal 🔵

Answer key, Chapter 9: 1-D · 2-D · 3-C · 4-B · 5-A · 6-A · 7-D · 8-B · 9-A · 10-B

LEVEL 1 · CHAPTER 10

Chapter 10: Roles, Access and Governance

Analytics breaks when everyone can change everything and nobody writes anything down. This chapter sets who does what, who gets which access, and how changes are controlled.

> 💡 **Trivia**
>
> To link GA4 to BigQuery you need **Editor** (or above) on the GA4 property and **Owner** access on the Google Cloud project. [1] ● Official (Google)
>
> Access levels aren't bureaucracy; some setup steps literally won't work without the right role.

10.1 Roles (RACI)

Task	Team lead	On-page SEO / analytics	Developer (front-end)	Developer (back-end)	Legal	Social	Content
Measurement plan	**A**	**R**	C	C	C	I	I
Privacy zones	A	C	I	I	**R**	I	I
Data layer build	I	C	**R**	C	I	—	—
GTM tags/triggers	A	**R**	C	—	I	—	—
GA4 admin settings	A	**R**	—	—	C	—	—

Task	Team lead	On-page SEO / analytics	Developer (front-end)	Developer (back-end)	Legal	Social	Content
Payment/lead feed	A	C	—	**R**	C	—	—
BigQuery setup	A	**R**	C	C	I	—	—
Dashboards	A	**R**	—	—	—	C	C
Daily review	**R**	R	I	I	—	I	I

R = responsible, A = accountable, C = consulted, I = informed.

10.2 Access levels

Tool	Admin	Edit/Publish	View
GA4	Team lead (+1 backup)	On-page SEO/analytics	Whole team
GTM	Team lead	On-page SEO publishes; developers **edit only** (no publish)	Others as needed
GSC	Team lead (owner)	SEO (full)	Content, social (restricted)
Clarity	Team lead	SEO	Content, design
Looker Studio	Owner: company account	SEO edits	Team views
BigQuery / Cloud	Owner: company account	SEO (BigQuery Data Editor/Job User)	Viewer roles

Rules: use company Google accounts, not personal ones; always two admins; remove access the day someone leaves. 🔵

> 🪔 **Desi analogy: ghar ki chaabiyan**
>
> Ghar ki master chaabi do logon ke paas hoti hai, kaam waali didi ke paas sirf rasoi ki. Jo ghar chhod de, uski chaabi usi din wapas. **Access bhi chaabiyon jaisa hai.**

10.3 Change control

Rule	How
Every GTM publish has a version name and note	"v12 – add internal_link_click (Ch 18)"
Workspaces per task	One workspace per change set
Preview + QA before publish	Chapter 30 checklist
Change log	Shared sheet: date, tool, change, who, why, version
Release coordination	Website releases noted; tracking QA after each deploy
Monthly access review	Remove unused accounts

10.4 Documentation set

Document	Owner	Where
Measurement plan	SEO	Shared drive
Event Dictionary v2	SEO	Appendix A + sheet
Data layer spec	Developer + SEO	Appendix B + repo
Page-zone register	Legal + SEO	Shared drive
Change log	SEO	Sheet
Dashboard catalogue	SEO	Sheet

> 🟧 **Ongo Weight Loss example**
>
> You said roles are flexible and anyone can be assigned by expertise. For analytics, keep two fixed rules anyway: only named publishers can publish GTM, and every publish goes in the change log. Flexibility in people, discipline in process. 🔵

10.5 Do's and Don'ts

✅ Do	❌ Don't
Use company accounts	Own analytics on a personal Gmail
Keep two admins per tool	Have one person who can unlock everything
Name every GTM version	Publish "Version 7" with no notes
Review access monthly	Leave ex-staff with access

10.6 Pros and cons of governance

✅ Pros	❌ Cons
Fewer silent breakages	Slower changes
Clear accountability	Some admin work
Safer for privacy	—

10.7 Where freshers go wrong in Chapter 10 topics

Mistake	Why it happens	Avoid it by	Fix it by
Developers publishing GTM	Convenience	Access table	Removing publish rights
No version notes	Hurry	Change control	Adding notes retroactively

Mistake	Why it happens	Avoid it by	Fix it by
Personal-account ownership	Quick setup	Company accounts	Transferring ownership

10.8 Thumb rules

> 👍 Two admins, company accounts, named publishers.
>
> 👍 No note, no publish.
>
> 👍 Access ends the day work ends.

10.9 Notes to remember

> 📝 BigQuery link: GA4 Editor + Cloud project Owner.
>
> 📝 Change log for every tool change and every site release.

Chapter 10 Glossary

Term	Simple meaning
RACI	Responsible, Accountable, Consulted, Informed
Workspace	A GTM draft area for one set of changes
Version	A saved, publishable snapshot of GTM
Change log	Record of all tracking changes
Access review	Regular check of who has access

Chapter 10 Quiz

1. What access is needed to link GA4 to BigQuery?

- A) Viewer only
- B) Clarity admin
- C) None
- D) GA4 Editor + Cloud project Owner

2. Who should publish GTM?

- A) Visitors
- B) Anyone
- C) Every developer
- D) Named publishers (on-page SEO/analytics)

3. How many admins per tool?

- A) None
- B) Ten
- C) One
- D) At least two

4. What goes with every GTM publish?

- A) Nothing
- B) A new GA4 property
- C) A version name and note
- D) A new container

5. Which accounts should own tools?

- A) Freelancer accounts
- B) Personal Gmail
- C) Company accounts
- D) Shared passwords on sticky notes

6. When should access be removed?

- A) Never
- B) The day someone leaves
- C) When convenient
- D) After a year

7. Who is responsible for the payment feed build?

- A) Back-end developer
- B) Graphic designer
- C) Social team
- D) Legal

8. What is a GTM workspace?

- A) A report
- B) A cookie
- C) A draft area for one set of changes
- D) A dashboard

9. How often should access be reviewed?

- A) Never
- B) Monthly
- C) Hourly
- D) Every 5 years

10. In the chaabi analogy, access is…

- A) Shared publicly
- B) Never taken back
- C) Given to everyone
- D) Given by need and taken back when work ends

Sources

1. Google Analytics Help, Set up BigQuery Export 🟢
2. Ongo Weight Loss, team roles (team lead, 30 October 2026), internal 🔵

Answer key, Chapter 10: 1-D · 2-D · 3-D · 4-C · 5-C · 6-B · 7-A · 8-C · 9-B · 10-D

02

LEVEL 2

Setup and Collection

GA4 and GTM, Next.js, the data layer, page views, the event dictionary, CTAs, links, contacts, engagement, tools, leads, portal exit, backend counts, dimensions, key events, UTMs, Clarity, consent and QA.

11 · GA4 Account, Property and Settings (Click by Click)

12 · GTM Container Setup and Conventions

13 · Installing GTM in Next.js (Developer Guide)

14 · The Data Layer Specification

15 · Page Views in Next.js and the `page_meta` Event

16 · Event Dictionary v2

17 · CTA Tracking (Developer + GTM)

18 · Internal Link Tracking

19 · Outbound, Phone, Email and Chat Tracking

20 · Engagement Events: Scroll, Video and FAQs

21 · Tracking Tools and Calculators (Without Their Inputs)

22 · Forms and Leads

23 · Portal Exit, Child Sites and Cross-Domain Tracking

24 · Counting Leads and Payments from the Backend

25 · Custom Dimensions and Metrics

26 · Key Events: What Counts as Success in GA4

27 · Minimal UTMs and Channel Groups

28 · Microsoft Clarity: Setup and What It Gives You

29 · Consent Implementation in GTM (Step by Step)

30 · QA, Debugging and the Release Checklist

LEVEL 2 · CHAPTER 11

Chapter 11: GA4 Account, Property and Settings (Click by Click)

Before a single event is sent, GA4 must be set up correctly: one account, one property, the right data streams, and a set of privacy and data settings that are hard or impossible to fix later.

> 💡 **Trivia**
>
> Standard GA4 properties offer only **2 or 14 months** of user-level data retention for explorations; longer options are for GA4 360. [1] 🟢 Official (Google)
>
> The default is 2 months. If you forget to change it, your funnels and paths will only ever look back eight weeks.

11.1 Account structure for Ongo Weight Loss

```
GA4 Account: Ongo Weight Loss (company)
 └─ Property: Ongo Weight Loss – Public Sites (time zone: America/New_York or company's reporting zone, currency: USD)
     ├─ Web stream: ongoweightloss.com          (G-XXXXXXX)  ← same G- ID used on child sites
     └─ (child sites use the SAME stream/tag for cross-domain journeys, Chapter 23)
 └─ Property: Ongo Weight Loss – Staging (separate, for testing)
```

Why one property for the main and child sites: journeys that cross from weightclinicsresults.com to the main site and to the portal can only be stitched if all public sites use the **same tag ID** (Chapter 23). Use a `site_id` dimension to separate them in reports. 🔵

11.2 Click-by-click setup

Step	Where	Setting
1	analytics.google.com → Admin → **Create → Account**	Name: "Ongo Weight Loss"; data-sharing settings: **untick** all optional sharing except what legal approves
2	**Create property**	Name; reporting **time zone**; **currency USD**
3	Business details	Industry: Healthcare; size
4	Business objectives	Choose "Generate leads" (affects default reports only)
5	**Data stream → Web**	URL ongoweightloss.com; stream name "Public sites"; copy the **G-measurement ID**
6	**Enhanced measurement** (stream settings)	ON: scrolls (we'll add our own depth too), outbound clicks, file downloads, form interactions **OFF** (we track forms ourselves, Zone rules), video engagement ON, site search ON if search exists. **Page views → Advanced → untick "Page changes based on browser history events"** (we send page views from GTM; Chapter 15)
7	Admin → **Data retention**	Event data and user data: **14 months**; "Reset user data on new activity": ON
8	Admin → **Data collection**	**Google signals: OFF**; granular location and device data: legal decides (recommended OFF for health)
9	Admin → **Data filters**	Create "Internal traffic" filter; set to **Active** after testing
10	Stream → Configure tag settings → **Define internal traffic**	Office IP ranges → `traffic_type = internal`
11	Configure tag settings → **List unwanted referrals**	Payment processor domains, portal domain if it bounces back
12	Configure tag settings → **Configure your domains**	ongoweightloss.com, weightclinicsresults.com, other child sites (Chapter 23)
13	Configure tag settings → **Adjust session timeout**	Keep 30 min; engaged session timer default 10 s

Step	Where	Setting
14	Admin → **Product links**	Link **Search Console** (Chapter 38) and **BigQuery** (Chapter 32); **no** Google Ads link (paid book)
15	Admin → **Property access management**	Team per Chapter 10
16	Admin → **Reporting identity**	**Device-based** (we don't use User ID)
17	Admin → **Ads personalization**	Disable for all regions

11.3 Settings that are hard to change later

Setting	Why
Time zone and currency	Reports and BigQuery dates follow it
Data retention	Not retroactive
BigQuery data location	Changing needs relink and dataset recreation
Event and parameter names	History stays under old names
Custom dimension creation	Not retroactive

> 🪔 **Desi analogy: naye ghar ki neev**
>
> Ghar ki neev ek baar padti hai. Baad mein kamre ka rang badal sakte ho, neev nahi. **Time zone, retention aur naam GA4 ki neev hain.**

11.4 Staging property

Why	How
Test events without polluting real data	Separate property "Staging"; staging site uses its G- ID; GTM environment variable picks the right ID (Chapter 12)

> 🟧 **Ongo Weight Loss example**
>
> Because nothing is implemented yet, you can set every item above today before any traffic is tracked: 14-month retention, Google signals off, form interactions off (forms handled by zone rules), history-based page views off (GTM sends them), internal traffic filter, and one property for all public sites with `site_id` to separate them. 🔵

11.5 Do's and Don'ts

✅ Do	❌ Don't
One property for all public sites	Separate properties that break journeys
14-month retention on day one	Keep the 2-month default
Google signals and ads personalisation off	Turn on ads features "for later"
Separate staging property	Test on production data

11.6 Pros and cons of a single property

✅ Pros	❌ Cons
Cross-site journeys	Need `site_id` filters in reports
One set of dimensions and key events	Mistakes affect all sites
Simpler dashboards	—

11.7 Where freshers go wrong in Chapter 11 topics

Mistake	Why it happens	Avoid it by	Fix it by
Duplicate page views on Next.js	History-change page views + GTM page views	Step 6	Untick history changes

Mistake	Why it happens	Avoid it by	Fix it by
Internal visits inflating data	No filter	Steps 9–10	Adding filter
Wrong time zone	Default	Step 2	Change early (affects future only)

11.8 Thumb rules

👍 Set the foundation before the first event.

👍 One tag ID across all public sites.

👍 Privacy settings off by default, on only by legal approval.

11.9 Notes to remember

📝 Untick "Page changes based on browser history events": GTM sends page views.

📝 Retention 14 months; Google signals off; reporting identity device-based.

Chapter 11 Glossary

Term	Simple meaning
Property	A GA4 container for one business's data
Data stream	A source of data (website) with a G- ID
Enhanced measurement	Automatic events GA4 can collect
Internal traffic filter	Removes your team's visits
Reporting identity	How GA4 recognises users

Chapter 11 Quiz

1. What data retention should be set?

- A) None
- B) 1 month
- C) 2 months
- D) 14 months

2. Why untick "Page changes based on browser history events"?

- A) To stop all tracking
- B) GTM will send page views; avoids duplicates
- C) To speed up the site
- D) It's required by law

3. Should Google signals be on?

- A) Yes
- B) No
- C) Only in staging
- D) Only for blogs

4. Why use one property for main and child sites?

- A) To stitch cross-site journeys with the same tag ID
- B) To save money
- C) To hide data
- D) No reason

5. Which setting is not retroactive?

- A) Bookmarks
- B) Report colours
- C) Data retention
- D) Dashboard names

6. Which enhanced measurement option is off?

- A) Form interactions
- B) Outbound clicks
- C) File downloads

- D) Video engagement

7. What reporting identity do we use?

- A) None
- B) Blended with signals
- C) Device-based
- D) User ID with emails

8. Why a staging property?

- A) To double traffic
- B) For ads
- C) For patients
- D) Test without polluting real data

9. Which link is not made now?

- A) Google Ads
- B) None
- C) BigQuery
- D) Search Console

10. In the neev analogy, which is foundation?

- A) Chart types
- B) Time zone, retention and names
- C) Dashboard layout
- D) Report colours

Sources

1. Google Analytics Help, Data retention
2. Google Analytics Help, Cross-domain measurement
3. Next.js Docs, Third-party libraries: tracking pageviews

Answer key, Chapter 11: 1-D · 2-B · 3-B · 4-A · 5-C · 6-A · 7-C · 8-D · 9-A · 10-B

LEVEL 2 · CHAPTER 12

Chapter 12: GTM Container Setup and Conventions

GTM is installed but not fully configured. This chapter rebuilds it properly: clean structure, environments, naming rules, folders, built-in variables and the base Google tag.

> 💡 **Trivia**
>
> Next.js's official `@next/third-parties` library supports a `gtmScriptUrl` option for server-side tagging, so the same component works if you later serve GTM from your own tagging server. [1] ● Official (Next.js)
>
> Set up the container cleanly now and it can grow into server-side later without a rebuild.

12.1 Container structure

Item	Setting
Container	One **web** container for all public sites (main + child sites)
Environments	**Live**, **Staging** (Admin → Environments → New), Latest (built-in)
Workspaces	One per change set: "ws-ch18-internal-links"
Folders	01 Consent · 02 Google tag · 03 Page views · 04 CTAs · 05 Links · 06 Engagement · 07 Tools & forms · 08 Portal · 09 Clarity · 10 Utilities
Permissions	Publish: team lead + on-page SEO; Edit: developers (Chapter 10)

12.2 Naming conventions

Type	Pattern	Example
Tag	`[Tool] – [Type] – [Name]`	`GA4 – Event – internal_link_click`

Type	Pattern	Example
Trigger	`[Type] — [Condition]`	`CE — page_meta`, `Click — data-event starts cta_`
Variable	`[Type] — [Name]`	`DLV — page_type`, `CJS — clean page_location`, `Const — GA4 ID`
Folder	Number + topic	`05 Links`
Version	`vNN — what — chapter/ticket`	`v14 — add exit_to_portal — Ch23`

(CE = Custom Event, DLV = Data Layer Variable, CJS = Custom JavaScript.)

> 🍛 **Desi analogy: kirana ki almaari**
>
> Achhi kirana dukaan mein har dabba label kiya hota hai: "Chana dal", "Moong dal". Bina label sab dabbe ek jaise lagte hain aur galat dal chali jaati hai. **GTM ke naam labels hain.**

12.3 Built-in variables to enable

Variables → Configure → enable: **Page URL, Page Path, Page Hostname, Referrer, Event, Click Element, Click Classes, Click ID, Click URL, Click Text, Form ID, Form Classes, Scroll Depth Threshold, Video Title, Video Percent, History Source, New History Fragment**.

12.4 Core variables to create

Variable	Type	Value
`Const — GA4 ID`	Lookup table on `{{Environment Name}}`	Live → G-LIVE..., Staging → G-STAGE...
`DLV — page_type`, `DLV — page_group`, `DLV — funnel_stage`, `DLV — content_cluster`, `DLV — primary_entity`, `DLV — hub_parent`, `DLV — privacy_zone`, `DLV — site_id`, `DLV — template`, `DLV — content_id`	Data Layer Variable (version 2)	Default value: `(not set)`

Variable	Type	Value
`CJS – clean page_location`	Custom JavaScript	Removes query parameters except `utm_*` (code below)
`CJS – clean page_referrer`	Custom JavaScript	Removes query strings
`DOM – data-event` etc.	Auto-Event Variable → Element attribute	`data-event`, `data-cta-location`, `data-link-position`

```
// CJS – clean page_location : keep only utm_* parameters, drop everything else
function () {
  try {
    var u = new URL({{Page URL}});
    var keep = new URLSearchParams();
    u.searchParams.forEach(function (v, k) {
      if (k.indexOf('utm_') === 0) keep.append(k, v);
    });
    var qs = keep.toString();
    return u.origin + u.pathname + (qs ? '?' + qs : '');
  } catch (e) { return {{Page Hostname}} + {{Page Path}}; }
}
```

Why: URLs can accidentally carry emails, names or form values (`?email=…`). Sending only the clean URL keeps them out of GA4. 🔵

12.5 The base Google tag

Field	Value
Tag type	**Google Tag**
Tag ID	`{{Const – GA4 ID}}`
Configuration parameters	`send_page_view` = **false** (Chapter 15); `page_location` = `{{CJS – clean page_location}}`; `page_referrer` = `{{CJS – clean page_referrer}}`; `site_id` = `{{DLV – site_id}}`

Field	Value
Shared event settings	Create a **Google tag: Event Settings** variable with `page_location`, `page_referrer`, `site_id`, `privacy_zone` used by every GA4 event tag
Trigger	**Initialization – All Pages** (after the consent tag on Consent Initialization)
Consent	Additional consent check: `analytics_storage`

12.6 Cleaning up the existing container

Step	Action
1	Export the current container (Admin → Export)
2	List tags and owners; pause unknown tags (don't delete yet)
3	Remove duplicate GA4/Google tags
4	Build the new structure in a new workspace
5	Preview, QA (Chapter 30), publish with notes
6	Delete paused tags after 30 days if nobody claims them

12.7 Server-side GTM (optional, later)

When	Why	Cost
Traffic grows; ad-blocker loss noticeable; need to strip fields before vendors	Cleaner, longer first-party cookies; central data control	Hosting (Cloud Run or hosts like Stape) + setup

Not needed for launch; revisit at the 90-day review. 🔵

> 🟧 **Ongo Weight Loss example**
>
> Today's container is "installed but not fully configured". Don't patch it piece by piece: export it, build the clean structure above in one workspace, test, and publish as "v1 – clean rebuild – Book Seven Ch12". 🔵

12.8 Do's and Don'ts

✅ Do	❌ Don't
One container for all public sites	Separate containers per child site without reason
Use environments for staging	Test on live
Strip query strings from URLs	Send raw URLs
Pause before deleting	Delete unknown tags immediately

12.9 Pros and cons of a structured container

✅ Pros	❌ Cons
Easy to read and debug	Takes a day to rebuild
Safe handoffs	Naming discipline needed
Privacy controls in one place	—

12.10 Where freshers go wrong in Chapter 12 topics

Mistake	Why it happens	Avoid it by	Fix it by
Hard-coding the G- ID in every tag	Speed	Constant/lookup variable	Replacing with variable
`send_page_view` left true	Default	12.5	Setting false
No default values for DLVs	Not knowing	12.4	`(not set)` defaults

12.11 Thumb rules

👍 Label every dabba.

👍 Clean URL before sending.

👍 Pause, wait, then delete.

12.12 Notes to remember

📝 Google tag: `send_page_view` false; consent check `analytics_storage`.

📝 Environments: Live and Staging with a GA4 ID lookup.

Chapter 12 Glossary

Term	Simple meaning
Container	A GTM installation for a site
Environment	Live or staging version of GTM
Data Layer Variable	Reads a value from the data layer
Custom JavaScript variable	Small function returning a value
Google tag	The base tag that loads GA4

Chapter 12 Quiz

1. What should `send_page_view` be in the Google tag?

- A) blank
- B) false
- C) true
- D) random

2. Why clean `page_location`?

- A) To improve speed
- B) To shorten reports
- C) For SEO
- D) To keep personal data in query strings out of GA4

3. What's the tag naming pattern?

- A) Emojis
- B) Random names
- C) `[Tool] – [Type] – [Name]`
- D) Numbers only

4. How should a staging GA4 ID be chosen?

- A) Same as live
- B) None
- C) Lookup on environment name
- D) Manual edits each time

5. What should DLVs have?

- A) No default
- B) Random values
- C) Personal data
- D) A default value like `(not set)`

6. What do you do with unknown existing tags?

- A) Publish more
- B) Delete immediately
- C) Pause first, delete after 30 days if unclaimed
- D) Duplicate them

7. What trigger does the Google tag use?

- A) Scroll
- B) Initialization – All Pages
- C) Click – All Elements
- D) Timer

8. When is server-side GTM recommended?

- A) Later, as traffic and data needs grow
- B) Only for portals
- C) Day one required
- D) Never

9. How many containers for public sites?

- A) One per page
- B) One
- C) None
- D) Three

10. In the kirana analogy, names are…

- A) Prices
- B) The shop sign
- C) Customers
- D) Labels on every container

Sources

1. Next.js Docs, Third-party libraries (GoogleTagManager, gtmScriptUrl) 🟢
2. Google Analytics Help, Data retention 🟢
3. Ongo Weight Loss, current GTM state (team lead, 30 October 2026), internal 🔵

Answer key, Chapter 12: 1-B · 2-D · 3-C · 4-C · 5-D · 6-C · 7-B · 8-A · 9-B · 10-D

LEVEL 2 · CHAPTER 13

Chapter 13: Installing GTM in Next.js (Developer Guide)

This chapter is for the front-end developer. It shows exactly where GTM, consent defaults and the data layer go in a Next.js App Router site, with copy-ready code.

> 💡 **Trivia**
>
> Next.js's `@next/third-parties` package provides a `GoogleTagManager` component for the root layout and a `sendGTMEvent` helper that pushes to the data layer. The library is labelled **experimental**. [1] ● Official (Next.js)
>
> You can use it, or a plain `next/script` setup. This chapter shows both, so the developer can choose.

13.1 Load order (must be exactly this)

```
1. dataLayer created + consent defaults   (inline, before GTM)
2. GTM container loads                     (CMP tag fires on Consent Initialization)
3. Page metadata pushed (page_meta)        (on every route, Chapter 15)
4. Interaction events pushed               (CTA, links, tools)
```

13.2 Option A: `@next/third-parties` (recommended)

```
npm install @next/third-parties@latest
```

```
// app/layout.tsx
import Script from 'next/script'
import { GoogleTagManager } from '@next/third-parties/google'

const GTM_ID = process.env.NEXT_PUBLIC_GTM_ID! // e.g. GTM-XXXXXXX

export default function RootLayout({ children }: { children: React.ReactNode }) {
  return (
    <html lang="en">
      <head>
        {/* 1. dataLayer + consent defaults BEFORE GTM */}
        <Script id="consent-default" strategy="beforeInteractive">
          {`
            window.dataLayer = window.dataLayer || [];
            function gtag(){dataLayer.push(arguments);}
            gtag('consent','default',{
              analytics_storage:'denied', ad_storage:'denied',
              ad_user_data:'denied', ad_personalization:'denied',
              wait_for_update:500
            });
          `}
        </Script>
      </head>
      {/* 2. GTM */}
      <GoogleTagManager gtmId={GTM_ID} />
      <body>{children}</body>
    </html>
  )
}
```

Notes: the consent defaults shown are for an **opt-in** model; legal may choose different defaults (Chapter 5). If the CMP's GTM template sets defaults itself, remove the inline block to avoid conflicts. ●

13.3 Option B: plain `next/script`

```
// app/layout.tsx (alternative)
<Script id="gtm" strategy="afterInteractive">
{`(function(w,d,s,l,i){w[l]=w[l]||[];w[l].push({'gtm.start':
new Date().getTime(),event:'gtm.js'});var f=d.getElementsByTagName(s)[0],
j=d.createElement(s),dl=l!='dataLayer'?'&l='+l:'';j.async=true;j.src=
'https://www.googletagmanager.com/gtm.js?id='+i+dl;f.parentNode.insertBefore(j,f);
})(window,document,'script','dataLayer','${GTM_ID}');`}
</Script>
```

Add the `<noscript>` iframe right after `<body>` if required by your setup.

> **Desi analogy: shaadi mein pehle pandit, phir band**
>
> Phere se pehle pandit ji aate hain, mantra ka kram tay hota hai, phir band baaja. Ulta ho jaaye toh rasm bigad jaati hai. **Pehle consent default, phir GTM, phir events: kram mat todo.**

13.4 Environments (staging vs live)

Variable	Live	Staging
`NEXT_PUBLIC_GTM_ID`	GTM-LIVE	Same container
`NEXT_PUBLIC_GTM_AUTH`, `NEXT_PUBLIC_GTM_PREVIEW`	—	From GTM → Admin → Environments → Staging → Get snippet

```
<GoogleTagManager
  gtmId={GTM_ID}
  auth={process.env.NEXT_PUBLIC_GTM_AUTH}
  preview={process.env.NEXT_PUBLIC_GTM_PREVIEW}
/>
```

13.5 Type safety for the data layer

```
// types/analytics.d.ts
export {}
declare global {
  interface Window {
    dataLayer: Record<string, unknown>[]
  }
}
```

```
// lib/analytics.ts
export function track(event: string, params: Record<string, string | number | boolean> =
{}) {
  if (typeof window === 'undefined') return
  window.dataLayer = window.dataLayer || []
  window.dataLayer.push({ event, ...params })
}
```

Rule for `track()`: never pass names, emails, phone numbers, form answers, weights, medications or any free text typed by a user. Code review checks every call. 🔵

13.6 Content Security Policy (if used)

If the site uses a CSP, allow: `https://www.googletagmanager.com`, `https://*.google-analytics.com`, `https://*.analytics.google.com`, `https://www.clarity.ms`, `https://*.clarity.ms`, and your CMP's domains. Use a **nonce** with `next/script` if inline scripts are restricted (Next.js supports nonces). 🔵

13.7 Child websites

Every public child site (weightclinicsresults.com and others) uses the **same GTM container** (or the same GA4 tag ID if built differently) and pushes `site_id` in its data layer. 🔵

13.8 Developer checklist

☐	Item
☐	Consent default runs before GTM

☐	Item
☐	GTM loads once per page (no duplicates)
☐	No hard-coded GA4 `gtag('config')` anywhere
☐	`track()` helper used for all custom events
☐	Staging uses GTM environment snippet
☐	CSP allows required domains
☐	Portal and Zone C apps do **not** include this layout

> **🟧 Ongo Weight Loss example**
>
> The portal is a separate, non-public platform, so it won't inherit this layout. Still add a check: search the portal codebase for `googletagmanager` and `clarity.ms` and confirm zero results. 🔵

13.9 Do's and Don'ts

✅ Do	❌ Don't
Set consent defaults before GTM	Load GTM first
Use one `track()` helper	Push ad-hoc objects everywhere
Use environment snippets for staging	Point staging at live data
Keep GA4 config only in GTM	Hard-code `gtag('config')` too

13.10 Pros and cons of `@next/third-parties`

✅ Pros	❌ Cons
Official, simple, performance-minded	Labelled experimental
`sendGTMEvent` helper	Fewer options than custom scripts

✅ Pros	❌ Cons
Supports server-side script URL	Must check version updates

13.11 Where freshers go wrong in Chapter 13 topics

Mistake	Why it happens	Avoid it by	Fix it by
GA4 hard-coded and in GTM	Old snippets left	Checklist	Removing hard-coded tag
Consent default after GTM	Script order	13.1	`beforeInteractive`
Personal data in `track()`	Debugging habit	Code review rule	Removing params

13.12 Thumb rules

> 👍 Consent, GTM, page_meta, events: in that order.
>
> 👍 One helper for every push.
>
> 👍 The portal never gets this layout.

13.13 Notes to remember

> 📝 `GoogleTagManager gtmId` in the root layout.
>
> 📝 Staging uses `auth` and `preview` from GTM environments.

Chapter 13 Glossary

Term	Simple meaning
App Router	Next.js's routing system using the `app/` folder

Term	Simple meaning
`next/script`	Next.js component for loading scripts
`beforeInteractive`	Load a script before the page becomes interactive
CSP	Content Security Policy: rules for allowed scripts
Nonce	A one-time value allowing an inline script under CSP

Chapter 13 Quiz

1. What must load before GTM?

- A) Fonts
- B) Clarity
- C) Images
- D) dataLayer + consent defaults

2. Which component loads GTM in Next.js?

- A) `Head` only
- B) `Image`
- C) `GoogleTagManager` from `@next/third-parties/google`
- D) `Link`

3. What does `sendGTMEvent` do?

- A) Pushes to the data layer
- B) Sends email
- C) Creates a GA4 property
- D) Loads Clarity

4. Should GA4 also be hard-coded?

- A) No; only via GTM
- B) Yes, twice
- C) Only in the portal
- D) Only on blogs

5. How does staging load GTM?

- A) With live data
- B) With a new container per page
- C) Without GTM
- D) With environment `auth` and `preview` values

6. What must never go into `track()`?

- A) CTA positions
- B) Personal or health data
- C) Event names
- D) Page types

7. Does the portal use this layout?

- A) No
- B) Only for payments
- C) Yes
- D) Only for doctors

8. How is `@next/third-parties` labelled?

- A) Experimental
- B) Deprecated
- C) Paid
- D) Mandatory

9. What do child sites push in the data layer?

- A) `site_id`
- B) Emails
- C) Nothing
- D) Patient IDs

10. In the shaadi analogy, the order is…

- A) Random
- B) Pandit first, then band (consent first, then GTM)
- C) No order
- D) Band first

Sources

1. Next.js Docs, Third-party libraries (updated June 1, 2026)
2. Google, Consent mode overview

Answer key, Chapter 13: 1-D · 2-C · 3-A · 4-A · 5-D · 6-B · 7-A · 8-A · 9-A · 10-B

LEVEL 2 · CHAPTER 14

Chapter 14: The Data Layer Specification

The data layer is the single most important part of the setup. It's where the website tells GTM "what kind of page this is" in a clean, consistent way, so every event can be grouped by silo, funnel stage and cluster without guessing from URLs.

> 💡 **Trivia**
>
> Next.js's `GoogleTagManager` component accepts a `dataLayer` option to initialise the container with data, and `dataLayerName` (default `dataLayer`). [1] 🟢 Official (Next.js)
>
> The data layer isn't a GTM trick; frameworks support it directly.

14.1 The page metadata fields

Field	Values (examples)	Source	Required
`site_id`	`ongo_main`, `wcr` (weightclinicsresults.com), `child_[name]`	Site config	✅
`page_type`	`home`, `service`, `medication`, `pricing`, `how_it_works`, `blog_article`, `blog_hub`, `tool`, `eligibility_start`, `policy`, `about`, `medical_team`, `contributor`, `reviews`, `contact`, `not_found`	Template	✅
`page_group` (silo)	`weight_loss`, `longevity`, `mens_health`, `womens_health`, `general`	Content map (Book Three)	✅
`funnel_stage`	`tofu`, `mofu`, `bofu`, `retention`, `none`	Content map	✅

Field	Values (examples)	Source	Required
`content_cluster`	`glp1`, `semaglutide`, `tirzepatide`, `nutrition`, `lifestyle`, `none`	Content map	✅
`primary_entity`	`semaglutide`, `tirzepatide`, `glp1`, `none`	Content map	✅
`hub_parent`	Parent hub slug, e.g., `glp1-medications`	Content map	✅
`template`	Next.js template name: `article_v2`	Code	✅
`content_id`	Stable CMS/content ID (not the title)	CMS	✅
`privacy_zone`	`A`, `B`	Page-zone register	✅
`author_role`	`writer`, `clinician`, `none` (role only, never a name)	CMS	Optional
`publish_month`	`2026-10`	CMS	Optional
`index_state`	`index`, `noindex`	Book Four	Optional

Never in the data layer: names, emails, phone numbers, IP addresses, form values, eligibility answers, BMI, weight, medications a person takes, payment details, or any free text typed by a visitor. 🔵

14.2 Where the values come from

```
Book Three content map (sheet / CMS fields)
   → page data in Next.js (generateMetadata / page props)
   → <PageMeta …/> component (client)
   → dataLayer.push({ event: 'page_meta', … })
```

Best practice: store these fields **with the content** (CMS fields or a JSON map keyed by route), not derived from URL patterns, so moving a URL doesn't break reporting. 🔵

> 🍛 **Desi analogy: tiffin ke dabbe pe naam ki parchi**
>
> Mumbai ke dabbawaale har dabbe pe code likhte hain: kahan se, kahan jaana, kaunsa station. Isi se laakhon dabbe sahi pahunchte hain. **Data layer har page ki parchi hai:** GTM usi se samajhta hai page kya hai.

14.3 TypeScript definition

```
// lib/page-meta.ts
export type PageMeta = {
  site_id: 'ongo_main' | 'wcr' | `child_${string}`
  page_type:
    | 'home' | 'service' | 'medication' | 'pricing' | 'how_it_works'
    | 'blog_article' | 'blog_hub' | 'tool' | 'eligibility_start' | 'policy'
    | 'about' | 'medical_team' | 'contributor' | 'reviews' | 'contact' | 'not_found'
  page_group: 'weight_loss' | 'longevity' | 'mens_health' | 'womens_health' | 'general'
  funnel_stage: 'tofu' | 'mofu' | 'bofu' | 'retention' | 'none'
  content_cluster: string        // from content map, lowercase snake_case
  primary_entity: 'semaglutide' | 'tirzepatide' | 'glp1' | 'none'
  hub_parent: string
  template: string
  content_id: string
  privacy_zone: 'A' | 'B'
  author_role?: 'writer' | 'clinician' | 'none'
  publish_month?: string         // YYYY-MM
  index_state?: 'index' | 'noindex'
}
```

14.4 Example map entry (JSON)

```
{
  "/blog/semaglutide-nausea/": {
    "site_id": "ongo_main",
    "page_type": "blog_article",
    "page_group": "weight_loss",
    "funnel_stage": "mofu",
    "content_cluster": "semaglutide",
    "primary_entity": "semaglutide",
    "hub_parent": "semaglutide",
    "template": "article_v2",
    "content_id": "blog-0142",
    "privacy_zone": "A",
    "author_role": "writer",
    "publish_month": "2026-11",
    "index_state": "index"
  }
}
```

14.5 Validation rules (developer + SEO)

Rule	Check
Every route has a map entry	Build-time test fails if missing
Values from allowed lists only	TypeScript types + unit test
`privacy_zone` matches the register	Legal-approved list
No personal data	Code review + automated scan for `@`, digits patterns
Same keys on every push	Prevents stale values between pages

> 🟧 **Ongo Weight Loss example**
>
> The project's master content map already defines URL, parent, entity, intent and funnel stage for each page. Export it to the JSON map above and the data layer is 80% done: developers don't invent values, they read them. 🔵

14.6 Do's and Don'ts

✅ Do	❌ Don't
Store metadata with content	Derive everything from URL regex
Push every key on every page	Push only changed keys
Use fixed value lists	Free-text values
Test the map at build time	Discover gaps in reports months later

14.7 Pros and cons of a rich data layer

✅ Pros	❌ Cons
Every event grouped by silo/funnel/cluster	Upfront dev work
Reports survive URL changes	Map must stay updated
One source of truth with the content map	—

14.8 Where freshers go wrong in Chapter 14 topics

Mistake	Why it happens	Avoid it by	Fix it by
Stale values on the next page	Not pushing all keys	14.5 rule	Full push each time
Page title as content ID	Easy	`content_id` rule	Stable IDs
Author names in data layer	Thought harmless	Role-only rule	Replacing with role

14.9 Thumb rules

👍 Every page wears its label.

👍 Values from lists, never free text.

👍 Metadata lives with content, not in URL guesses.

14.10 Notes to remember

📝 Required fields: site_id, page_type, page_group, funnel_stage, content_cluster, primary_entity, hub_parent, template, content_id, privacy_zone.

📝 Source: Book Three content map.

Chapter 14 Glossary

Term	Simple meaning
Data layer	JavaScript array where the site shares facts with GTM
Page metadata	Facts about a page (type, stage, cluster)
Content ID	A stable identifier for a piece of content
Stale value	A value left over from a previous page
Build-time test	A check that runs when the site is built

Chapter 14 Quiz

1. Where should page metadata come from?

- A) Visitors
- B) Guessing from URLs only
- C) Clarity

- D) The content map/CMS fields

2. Which field marks the privacy zone?

- A) `publish_month`
- B) `template`
- C) `site_id`
- D) `privacy_zone`

3. Can author names go in the data layer?

- A) Only doctors
- B) Only writers
- C) No; role only
- D) Yes

4. Why push every key on every page?

- A) For fun
- B) For SEO
- C) To prevent stale values
- D) To slow the site

5. What is `content_id`?

- A) A user ID
- B) A stable ID for the content
- C) A cookie
- D) The page title

6. Which `funnel_stage` values are allowed?

- A) tofu, mofu, bofu, retention, none
- B) Emojis
- C) Any text
- D) Numbers only

7. What fails the build?

- A) Nothing
- B) A new blog post with a map entry
- C) A route without a map entry

- D) A correct push

8. Which `site_id` represents weightclinicsresults.com?

- A) `ongo_main`
- B) `wcr`
- C) `none`
- D) `portal`

9. Which value is never allowed?

- A) `semaglutide`
- B) `mofu`
- C) A visitor's email
- D) `blog_article`

10. In the dabbawaala analogy, the data layer is...

- A) The train
- B) The lunch
- C) The code on each tiffin
- D) The office

Sources

1. Next.js Docs, Third-party libraries (GoogleTagManager options) 🟢
2. Ongo Weight Loss, Book Three master content map, internal 🔵

Answer key, Chapter 14: 1-D · 2-D · 3-C · 4-C · 5-B · 6-A · 7-C · 8-B · 9-C · 10-C

LEVEL 2 · CHAPTER 15

Chapter 15: Page Views in Next.js and the `page_meta` Event

Next.js changes pages without a full reload. If page views aren't handled deliberately, GA4 either misses pages or counts them twice. This chapter gives one clean design.

> 💡 **Trivia**
>
> Next.js documentation notes that GA4 tracks page views automatically when browser history changes, and warns: "If you decide to manually send pageview events, make sure to disable the default pageview measurement to avoid having duplicate data." [1] 🟢 Official (Next.js)
>
> We send page views manually, with metadata, so we turn the default off.

15.1 The design

Piece	Setting
GA4 enhanced measurement	Page views ON for first load is irrelevant because the Google tag has `send_page_view: false`; **"Page changes based on browser history events": OFF**
Next.js	A `PageMeta` client component pushes `{ event: 'page_meta', …fields }` on first load **and** every route change
GTM trigger	`CE – page_meta` (Custom Event = `page_meta`)
GTM tag	`GA4 – Event – page_view` with all metadata parameters

Result: **exactly one** `page_view` per page, each carrying `page_type`, `funnel_stage`, `content_cluster`, etc. 🔵

15.2 The `PageMeta` component

```
// components/PageMeta.tsx
'use client'
import { useEffect } from 'react'
import { usePathname } from 'next/navigation'
import type { PageMeta as Meta } from '@/lib/page-meta'

export function PageMeta(meta: Meta) {
  const pathname = usePathname()
  useEffect(() => {
    window.dataLayer = window.dataLayer || []
    window.dataLayer.push({
      event: 'page_meta',
      site_id: meta.site_id,
      page_type: meta.page_type,
      page_group: meta.page_group,
      funnel_stage: meta.funnel_stage,
      content_cluster: meta.content_cluster,
      primary_entity: meta.primary_entity,
      hub_parent: meta.hub_parent,
      template: meta.template,
      content_id: meta.content_id,
      privacy_zone: meta.privacy_zone,
      author_role: meta.author_role ?? 'none',
      publish_month: meta.publish_month ?? 'none',
      index_state: meta.index_state ?? 'index',
    })
  }, [pathname, meta.content_id])
  return null
}
```

```
// app/blog/[slug]/page.tsx (server component)
import { PageMeta } from '@/components/PageMeta'
import { getPageMeta } from '@/lib/content-map'

export default async function Page({ params }: { params: Promise<{ slug: string }> }) {
  const { slug } = await params
  const meta = getPageMeta(`/blog/${slug}/`)
  return (
    <>
      <PageMeta {...meta} />
      {/* article content */}
    </>
  )
}
```

Why `useEffect` on `pathname`: it runs after the new route renders, so GTM reads the new page's URL and title. ●

One Page View, the Right Way



History-change page views stay OFF.

> **Desi analogy: railway station ki announcement**
>
> Har station aane par announcement hoti hai: "Agla station Dadar." Agar announcement do baar ho, ya bilkul na ho, toh yatri confuse. **Har page pe ek hi announcement: page_meta.**

15.3 GTM build

Item	Setting
Trigger	Custom Event, event name `page_meta`
Tag	Google Analytics: GA4 Event · Measurement ID `{{Const – GA4 ID}}` · Event name `page_view`
Parameters	`page_location` = `{{CJS – clean page_location}}`, `page_referrer` = `{{CJS – clean page_referrer}}`, `page_title` = `{{Page Title}}` (built-in or DOM), and all DLVs: `site_id`, `page_type`, `page_group`, `funnel_stage`, `content_cluster`, `primary_entity`, `hub_parent`, `template`, `content_id`, `privacy_zone`, `author_role`, `publish_month`, `index_state`
Consent	`analytics_storage`
Firing	Once per event

Page referrer on route changes: for client-side navigations the browser's `document.referrer` doesn't update. Store the previous clean URL in a GTM variable (or push `previous_path` from `PageMeta`) so internal page-to-page flows are visible in BigQuery. 🔵

15.4 404 pages

`app/not-found.tsx` renders `PageMeta` with `page_type: 'not_found'` **and** pushes:

```
track('page_not_found', { requested_path: window.location.pathname })
```

(Path only, never the query string.)

15.5 Testing

Test	Expected
Load the homepage	1 `page_view` with `page_type=home`
Click to a blog article	1 new `page_view` with `page_type=blog_article`, `funnel_stage=mofu`

Test	Expected
Browser back button	1 `page_view` for the previous page
Refresh	1 `page_view`
Visit `/does-not-exist`	`page_view` (not_found) + `page_not_found`

Check in GTM Preview and GA4 DebugView (Chapter 30). 🔵

> 🟧 **Ongo Weight Loss example**
>
> This design also fixes a common report problem: GA4's "Landing page" dimension and our `page_type` / `funnel_stage` dimensions now come from the same `page_view` events, so a dashboard can show "sessions that **entered** on a MOFU semaglutide article" directly. 🔵

15.6 Do's and Don'ts

✅ Do	❌ Don't
Send one page view per route from `page_meta`	Keep history-change page views on too
Attach metadata to page views	Send bare page views
Store previous path for flows	Rely on `document.referrer` in SPAs
Test back/forward/refresh	Test only first load

15.7 Pros and cons of manual page views

✅ Pros	❌ Cons
No duplicates; rich metadata	Requires the component on every page
Consistent across sites	Forgotten component = missing page views
Works with clean URLs	Developer discipline needed

15.8 Where freshers go wrong in Chapter 15 topics

Mistake	Why it happens	Avoid it by	Fix it by
Two page views per navigation	Both methods on	15.1	Turning history setting off
Missing page views on some templates	Component not added	Build test (Chapter 14)	Adding `PageMeta`
Query strings in 404 events	Copying full URL	15.4	Path only

15.9 Thumb rules

> 👍 One station, one announcement.
>
> 👍 Page view = page_meta, with labels.
>
> 👍 Test back, forward and refresh.

15.10 Notes to remember

> 📝 Google tag `send_page_view: false` + history setting off + page_meta trigger.
>
> 📝 404: page_view (not_found) + page_not_found (path only).

Chapter 15 Glossary

Term	Simple meaning
SPA navigation	Changing pages without a full reload
Route change	Moving to a new URL inside the app
`usePathname`	Next.js hook returning the current path
Custom Event trigger	GTM trigger on a data layer event name
DebugView	GA4's real-time debugging screen

Chapter 15 Quiz

1. What does Next.js warn about manual page views?

- A) Disable default pageview measurement to avoid duplicates
- B) They need cookies
- C) They're impossible
- D) Nothing

2. Which data layer event triggers GA4 page_view?

- A) `page_meta`
- B) `gtm.click`
- C) `purchase`
- D) `scroll`

3. Which hook detects route changes?

- A) `useRef`
- B) `useMemo`
- C) `useState`
- D) `usePathname`

4. Why store the previous path?

- A) `document.referrer` doesn't update on client navigations
- B) For decoration
- C) To speed up pages
- D) To track names

5. What does a 404 page push?

- A) Nothing
- B) Visitor email
- C) page_view (not_found) and page_not_found with path only
- D) Full URL with query string

6. How many page_views per navigation?

- A) Zero
- B) Two
- C) Exactly one
- D) Random

7. Which setting must be off?

- A) Scrolls
- B) Outbound clicks
- C) Page changes based on browser history events
- D) Video engagement

8. What should you test besides first load?

- A) Only colours
- B) Nothing
- C) Only mobile
- D) Back, forward and refresh

9. What does attaching metadata to page views enable?

- A) Ads
- B) Faster site
- C) Patient tracking
- D) Reports by page type, stage, cluster from the same events

10. In the station analogy, page_meta is...

- A) The train driver
- B) Two announcements
- C) No announcement
- D) One announcement per station

Sources

1. Next.js Docs, Third-party libraries: tracking pageviews 🟢
2. Ongo Weight Loss, Book Four (page_meta), internal 🔵

Answer key, Chapter 15: 1-A · 2-A · 3-D · 4-A · 5-C · 6-C · 7-C · 8-D · 9-D · 10-D

LEVEL 2 · CHAPTER 16

Chapter 16: Event Dictionary v2

This is the master list of every event we send: its name, when it fires, its parameters, whether it's a key event, and what must never be included. It builds on Book Four's dictionary and updates it for privacy and the 360° plan.

> 💡 **Trivia**
>
> GA4 Measurement Protocol can't send event or parameter names that are "reserved" for automatic collection. [1] ● Official (Google)
>
> Naming isn't only style; some names are off-limits. A dictionary keeps everyone inside the rules.

16.1 Naming rules (unchanged from Book Four)

Rule	✅	❌
snake_case, lowercase	`internal_link_click`	`InternalLinkClick`
Start with a letter; ≤ 40 characters	`tool_complete_bmi_calculator`	60-character names
Outcome at the end	`form_submit_contact`	`contact_form`
Page group, not page title, in CTA names	`cta_semaglutide_get_started`	`cta_compounded_semaglutide_page_button_1`
No brand drug names	`cta_tirzepatide_get_started`	`cta_zepbound_get_started`
No personal or health values	—	`bmi=34`

16.2 The dictionary

Event	Fires when	Parameters	Key event?	Never include
`page_view` (from `page_meta`)	Every page/route	page_location (clean), page_referrer, page_title, site_id, page_type, page_group, funnel_stage, content_cluster, primary_entity, hub_parent, template, content_id, privacy_zone, author_role, publish_month	—	Query strings (except utm_*)
`cta_[group]_[action]`	CTA click	cta_name (= action), cta_group (= group), cta_location (hero/body/sticky/nav/footer/inline), cta_text (≤100 chars), link_url_path	`cta_*_get_started` optional	—
`internal_link_click`	Click on a link to our own domain(s)	link_source, link_target, link_position (body/nav/footer/related/breadcrumb/toc/card), link_text (≤100), source_stage, target_stage, target_page_type	—	Query strings

Event	Fires when	Parameters	Key event?	Never include
`exit_to_portal`	Click that leaves for the portal	source_page, cta_name, site_id, page_type, funnel_stage	✅ **Primary**	Any form data
`outbound_click_source` / `_social` / `_partner`	Click to another domain	link_domain, link_url_path, page_type	—	Query strings
`phone_click` / `email_click` / `chat_open`	Contact actions	contact_location, page_type	✅ (phone, chat)	Numbers/emails typed
`tool_start_[tool]` / `tool_complete_[tool]`	Public tools (e.g., BMI calculator)	tool_name, page_type	`tool_complete_*` optional	Inputs and results
`form_start_[form]` / `form_submit_[form]`	Public non-health forms (contact, newsletter)	form_name, page_type	`form_submit_contact` ✅	Field contents
`faq_open`	FAQ expanded	faq_id, page_type	—	Question text over 100 chars
`video_start` / `video_progress` / `video_complete`	Video plays	video_title, video_percent, page_type	—	—
`scroll_depth`	25/50/75/90% scrolled	percent_scrolled, page_type	—	—
`page_not_found`	404	requested_path	—	Query strings
`web_vitals`	Each Core Web Vital measured	metric_name (LCP/INP/CLS), metric_value, metric_rating	—	—
`consent_update` (optional)	Consent changed	consent_state (granted/denied)	—	—

16.3 Changes from Book Four

Book Four	Book Seven	Why
GA4 event `page_meta`	Data-layer event `page_meta` → GA4 `page_view` with metadata	One page view, rich dimensions
`portal_[step]` events to GA4 (signup, payment)	**Removed from GA4**; counted via BAA route/aggregated (Chapter 24)	Portal is authenticated (Zone C)
`tool_complete_eligibility_checker` key event	Only if the tool is public and collects **no** health answers; otherwise counted in the portal route	Health data rules
CTA params cta_position, cta_text, page_path	Added cta_name, cta_group, cta_location, link_url_path	Combined CTA reporting

> 🍛 **Desi analogy: restaurant ka menu card**
>
> Menu mein har dish ka naam, kya milega aur keemat fixed hoti hai. Waiter apni marzi se naam nahi badalta. **Event dictionary menu card hai:** sab wahi bolte hain jo likha hai.

16.4 Parameter length and value rules

Rule	Detail
Text parameters	Keep ≤ 100 characters (GA4 truncates long values)
Allowed values	Use fixed lists where possible (positions, stages, page types)
Paths	Path only (`/blog/x/`), never full URL with query
Text from the page	Only visible, non-personal UI text (button labels, link text)

16.5 Governance

Rule	How
New event	Proposed in a ticket → added to dictionary → built → QA → published
Rename	Avoid; if necessary, run old and new in parallel for 30 days
Retire	Mark "retired" with date; keep history note
Source of truth	Appendix A + a shared sheet

> 🟧 **Ongo Weight Loss example**
>
> You wanted separate CTA tracking (Call Us, Chat With Us, Email Us, Get Started, Check Eligibility, Book Appointment, Weight Loss CTA, etc.), not one generic `button_click` . The dictionary does both: distinct events like `cta_weight_loss_get_started` , `phone_click` , `chat_open` , plus shared parameters so you can still compare all CTAs in one table. 🔵

16.6 Do's and Don'ts

✅ Do	❌ Don't
Use the dictionary for every event	Invent names in GTM
Keep values in fixed lists	Free-text values
Retire, don't delete history	Rename casually
Review new events for privacy	Add "debug" events with personal data

16.7 Pros and cons of a strict dictionary

✅ Pros	❌ Cons
Clean, comparable reports	Slower to add new events
Easy developer handoff	Needs an owner
Privacy built in	—

16.8 Where freshers go wrong in Chapter 16 topics

Mistake	Why it happens	Avoid it by	Fix it by
`cta_get_started` with no group	Short names	16.1	Adding page group
Full URLs in parameters	Copying `Click URL`	16.4	Path-only variables
Portal events sent to GA4	Book Four habit	16.3	Removing; using Chapter 24

16.9 Thumb rules

> 👍 If it's not in the dictionary, it doesn't fire.
>
> 👍 Distinct events, shared parameters.
>
> 👍 Paths, not URLs; labels, not inputs.

16.10 Notes to remember

> 📝 Primary key event: `exit_to_portal`.
>
> 📝 Portal events never go to GA4.

Chapter 16 Glossary

Term	Simple meaning
Event dictionary	The master list of events and parameters
Reserved name	A name GA4 keeps for its own events
Retire	Stop using an event but keep its history notes
Truncation	Cutting text that's too long
Key event	An event marked as important for conversions

Chapter 16 Quiz

1. What's the primary key event?

- A) `exit_to_portal`
- B) `page_view`
- C) `scroll_depth`
- D) `faq_open`

2. Where do portal events go?

- A) Clarity
- B) GA4 key events
- C) BAA route / aggregated, never GA4
- D) GSC

3. Which CTA name follows the rules?

- A) `button1`
- B) `CTA-GetStarted`
- C) `cta_zepbound_get_started`
- D) `cta_semaglutide_get_started`

4. What parameter shows where a link was clicked?

- A) `video_percent`
- B) `metric_value`
- C) `page_title`
- D) `link_position`

5. What's the rule for paths in parameters?

- A) Path only, never full URL with query
- B) Include emails
- C) Full URL always
- D) Include form values

6. What happens when renaming an event?

- A) Rename instantly
- B) Never document
- C) Run old and new in parallel for 30 days
- D) Delete history

7. Which is never included in tool events?

- A) event name
- B) tool_name
- C) Inputs and results
- D) page_type

8. Why add shared CTA parameters?

- A) To compare all CTAs in one table
- B) For ads

- C) To slow GA4
- D) To track names

9. Which event records Core Web Vitals?

- A) `web_vitals`
- B) `chat_open`
- C) `faq_open`
- D) `page_not_found`

10. In the menu-card analogy, the dictionary is…

- A) The kitchen
- B) The bill
- C) The waiter's mood
- D) The fixed menu everyone follows

Sources

1. Google Developers, Measurement Protocol (GA4) 🟢
2. Ongo Weight Loss, Book Four Appendix C (event dictionary), internal 🔵

Answer key, Chapter 16: 1-A · 2-C · 3-D · 4-D · 5-A · 6-C · 7-C · 8-A · 9-A · 10-D

LEVEL 2 · CHAPTER 17

Chapter 17: CTA Tracking (Developer + GTM)

Every call to action gets its own event, tells you where it sat on the page, and can still be compared with every other CTA. This chapter shows the HTML attributes, the GTM build and the tests.

> 💡 **Trivia**
>
> GA4's funnel exploration "next action" card shows the **top 5** actions after each step, and "(no next action)" can indicate a drop-off. [1] 🟢 Official (Google)
>
> Clear CTA events make that card readable: "cta_semaglutide_get_started" tells a story; "click" doesn't.

17.1 The CTA catalogue

CTA	Event name pattern	Example
Get Started	`cta_[group]_get_started`	`cta_weight_loss_get_started`
Check Eligibility	`cta_[group]_check_eligibility`	`cta_semaglutide_check_eligibility`
Book Appointment	`cta_[group]_book_appointment`	`cta_weight_loss_book_appointment`
See Pricing	`cta_[group]_see_pricing`	`cta_tirzepatide_see_pricing`
How It Works	`cta_[group]_how_it_works`	`cta_weight_loss_how_it_works`
Learn More (to a money page)	`cta_[group]_learn_more`	`cta_glp1_learn_more`
Call Us	`phone_click`	—
Chat With Us	`chat_open`	—
Email Us	`email_click`	—
Longevity / Men's / Women's Health CTAs	`cta_longevity_get_started`, `cta_mens_health_get_started`, `cta_womens_health_get_started`	When those silos launch

`[group]` = the page group or entity the CTA promotes (`weight_loss`, `semaglutide`, `tirzepatide`, `glp1`, `longevity`, `mens_health`, `womens_health`). 🔵

17.2 HTML attributes (developer)

```
<Link
  href="/weightloss-onboard/"
  data-event="cta_semaglutide_get_started"
  data-cta-name="get_started"
  data-cta-group="semaglutide"
  data-cta-location="hero"
  data-exit-portal="true"
  className="btn-primary"
>
  See if I'm eligible
</Link>
```

Attribute	Values
`data-event`	Exact dictionary name
`data-cta-name`	get_started, check_eligibility, book_appointment, see_pricing, how_it_works, learn_more
`data-cta-group`	page group/entity
`data-cta-location`	hero, body, inline, sticky, nav, footer, card, modal
`data-exit-portal`	`true` only if the link goes to the portal

Make it a component (`<CtaButton …/>`) so attributes are never forgotten. 🔵

> 🪔 **Desi analogy: shaadi ke darwaaze**
>
> Hall ke chaar darwaaze: mukhya gate, side gate, kitchen gate, VIP gate. Register mein likho kaun kis darwaaze se aaya. **CTA location wahi darwaaza hai:** hero, body, sticky, footer.

17.3 GTM build

Item	Setting
Variables	Auto-Event Variable → Element Attribute: `data-event` , `data-cta-name` , `data-cta-group` , `data-cta-location` , `data-exit-portal` (use the **closest** CTA element; see JS below)
Trigger	Click – All Elements; fires on **Some Clicks**: `Click Element` matches CSS selector `[data-event^="cta_"], [data-event^="cta_"] *`
Tag	GA4 Event · Event name `{{CJS – cta event}}` · params `cta_name` , `cta_group` , `cta_location` , `cta_text` = `{{Click Text}}` (trimmed, ≤100), `link_url_path` = `{{CJS – click path}}` , plus `page_type` , `funnel_stage`

Because a click may land on an icon inside the button, read attributes from the closest CTA element:

```
// CJS – cta event
function () {
  var el = {{Click Element}};
  var cta = el && el.closest ? el.closest('[data-event^="cta_"]') : null;
  return cta ? cta.getAttribute('data-event') : undefined;
}
```

Create similar variables for `data-cta-name` , `data-cta-group` , `data-cta-location` , `data-exit-portal` .

```
// CJS — click path (path only, no query string)
function () {
  try { return new URL({{Click URL}}).pathname; } catch (e) { return undefined; }
}
```

17.4 Alternative: push from code

```
onClick={() => track('cta_semaglutide_get_started', {
  cta_name: 'get_started', cta_group: 'semaglutide', cta_location: 'hero'
})}
```

Then use a Custom Event trigger matching `^cta_` (regex). **Choose one method per site** to avoid double counting. 🔵

17.5 Tests

Test	Expected
Click hero "Get started" on semaglutide page	`cta_semaglutide_get_started`, cta_location=hero
Click the icon inside the button	Same event (closest() works)
Click footer CTA	cta_location=footer
CTA to portal	Also `exit_to_portal` (Chapter 23)
Middle-click/open in new tab	Event still fires (Click trigger, not Link Click with wait)

> 🟧 **Ongo Weight Loss example**
>
> The homepage uses several labels for the same action ("See If I'm Eligible", "Check My Eligibility", "Take free assessment"). With `cta_name` fixed to `check_eligibility` and the visible text in `cta_text`, you can compare which **label** and **location** perform best without breaking the event structure. 🔵

17.6 Do's and Don'ts

✅ Do	❌ Don't
Use a CTA component with all attributes	Hand-add attributes inconsistently
Read attributes with `closest()`	Rely on CSS classes that designers change
Keep `cta_text` short and non-personal	Send long or user-typed text
Pick one tracking method	Track the same click twice

17.7 Pros and cons of attribute-based tracking

✅ Pros	❌ Cons
One trigger handles every CTA	Developers must add attributes
Survives design changes	Needs code review
Clear, readable events	—

17.8 Where freshers go wrong in Chapter 17 topics

Mistake	Why it happens	Avoid it by	Fix it by
Event missing when clicking the icon	Click Element is the `<svg>`	`closest()`	Updating variable
Double events	Both code push and GTM click	One method	Removing one
CTA names by label text	Labels change	`cta_name` list	Standard names

17.9 Thumb rules

> 👍 Every CTA: name, group, location.
>
> 👍 Component, not copy-paste.
>
> 👍 One click, one event.

17.10 Notes to remember

> 📝 Trigger: `[data-event^="cta_"], [data-event^="cta_"] *`.
>
> 📝 Phone, chat, email have their own events.

Chapter 17 Glossary

Term	Simple meaning
CTA	Call to action (button or link asking for a next step)
Data attribute	An HTML attribute starting with `data-`
Auto-Event Variable	GTM variable reading the clicked element
`closest()`	Finds the nearest matching parent element
CTA location	Where on the page the CTA sits

Chapter 17 Quiz

1. Which event name follows the pattern?

- A) `button_click`
- B) `click1`
- C) `cta`
- D) `cta_weight_loss_get_started`

2. Which attribute holds the exact event name?

- A) `id`
- B) `data-event`
- C) `class`
- D) `style`

3. Why use `closest()`?

- A) Clicks may land on an icon inside the button
- B) To slow clicks
- C) For styling
- D) To read emails

4. Which `cta_location` values are used?

- A) hero, body, inline, sticky, nav, footer, card, modal
- B) Any text
- C) Page titles
- D) Visitor names

5. What tracks "Call Us"?

- A) `scroll_depth`
- B) `cta_call`
- C) `phone_click`
- D) `page_view`

6. What should you do to avoid double counting?

- A) Use both
- B) Choose one method per site
- C) Nothing
- D) Add more triggers

7. What's `cta_text` limited to?

- A) Unlimited text
- B) Emails
- C) Short, non-personal visible text (≤100 chars)
- D) User input

8. What extra event fires when a CTA goes to the portal?

- A) None
- B) `exit_to_portal`

- C) `purchase`
- D) `sign_up`

9. Why a CTA component?

- A) It's required by Google
- B) It hides clicks
- C) Attributes are never forgotten
- D) It's prettier

10. In the wedding-gates analogy, CTA location is…

- A) Which gate guests used
- B) The food
- C) The band
- D) The car

Sources

1. Google Analytics Help, Funnel exploration 🟢
2. Ongo Weight Loss, Book Four Chapter 39 (data attributes pattern), internal 🔵

Answer key, Chapter 17: 1-D · 2-B · 3-A · 4-A · 5-C · 6-B · 7-C · 8-B · 9-C · 10-A

LEVEL 2 · CHAPTER 18

Chapter 18: Internal Link Tracking

"How many users clicked the link from my GLP-1 article to my semaglutide page?" This chapter makes that question answerable, for every internal link, by position and funnel stage.

> 💡 **Trivia**
>
> GA4 path explorations show the **top 5 nodes** per step by default, expandable to **20**; everything else is grouped as "Others". [1] ● Official (Google)
>
> Page paths show where people went. Link events show **which link** sent them, even when it's outside the top 20.

18.1 What we capture

Parameter	Example	Why
`link_source`	`/blog/what-is-glp-1/`	Page the link is on
`link_target`	`/services/weight-loss/glp-1/semaglutide/`	Page it goes to
`link_position`	body, related, nav, footer, breadcrumb, toc, card	Which placement works
`link_text`	"our semaglutide guide" (≤100)	Which anchor text works
`source_stage` / `target_stage`	mofu → bofu	Funnel movement
`target_page_type`	medication	Movement into money pages
`link_id` (optional)	`rel-sema-01`	Track one specific editorial link

18.2 Developer: mark link positions

Most links can be classified by **where they sit** in the layout. Add one attribute on containers, not on every link:

```
<nav data-link-position="nav">…</nav>
<article data-link-position="body">…</article>
<aside data-link-position="related">…</aside>
<footer data-link-position="footer">…</footer>
<ol data-link-position="breadcrumb">…</ol>
```

Target stage and page type: add to internal links rendered by a shared `<SmartLink>` component that looks up the target in the content map:

```
// components/SmartLink.tsx
import Link from 'next/link'
import { getPageMeta } from '@/lib/content-map'

export function SmartLink({ href, children, linkId }: { href: string; children: React.ReactNode; linkId?: string }) {
  const target = getPageMeta(href)
  return (
    <Link
      href={href}
      data-target-stage={target?.funnel_stage ?? 'none'}
      data-target-type={target?.page_type ?? 'unknown'}
      data-link-id={linkId}
    >
      {children}
    </Link>
  )
}
```

18.3 GTM build

Item	Setting
Trigger	**Click – Just Links**; Some Link Clicks: `Click URL Hostname` is in our **own-domains list** (variable `RegEx – own domains`, which covers ongoweightloss.com, weightclinicsresults.com and every child site; see Appendix D); **exclude** clicks whose element has `data-event` starting `cta_` (CTAs are tracked separately) and portal links
Variables	`CJS – link position` (closest `[data-link-position]`), `CJS – click path`, `Click Text` (trim ≤100), `CJS – target stage/type`, `DLV – funnel_stage` (source stage)
Tag	GA4 Event `internal_link_click` with parameters from 18.1; `link_source` = `{{Page Path}}`
Wait for tags	Not needed (GA4 uses beacon transport)

```
// CJS – link position
function () {
  var el = {{Click Element}};
  var box = el && el.closest ? el.closest('[data-link-position]') : null;
  return box ? box.getAttribute('data-link-position') : 'other';
}
```

> **Desi analogy: mohalle ke signboard**
>
> Mohalle mein teen signboard hain "Hospital →". Kaunsa board sabse zyada log follow karte hain? Agar har board pe ek chhota counter ho, toh pata chal jaata hai. **internal_link_click har signboard ka counter hai.**

18.4 Reports you'll build

Question	Where
Top links from blog to money pages	Exploration: event = internal_link_click; source_stage in (tofu, mofu); target_stage = bofu; rows: link_source, link_target
Best link positions	Breakdown by link_position

Question	Where
Best anchor text for one target	Filter link_target; rows: link_text
Did a specific editorial link get clicks?	Filter link_id
Click rate of a link	Clicks ÷ page views of link_source (Looker calculated field)

Full analysis: Chapter 51. 🔵

18.5 Tests

Test	Expected
Click a body link on a blog to the semaglutide page	`internal_link_click` with position=body, target_stage=bofu
Click the nav "Pricing"	position=nav
Click a CTA button	**No** internal_link_click (CTA event instead)
Click a link to weightclinicsresults.com from main site	internal_link_click (cross-site internal)
Click an external source link	outbound_click_source (Chapter 19), not internal

> 🟧 **Ongo Weight Loss example**
>
> Your example question becomes one exploration filter: `link_source` = `/blog/what-is-glp-1/` and `link_target` contains `/semaglutide/` . The count, divided by that article's page views, is the link's click-through rate. 🔵

18.6 Do's and Don'ts

✅ Do	❌ Don't
Mark positions on containers	Hand-tag every link
Exclude CTAs from link tracking	Double count CTA clicks
Use path-only parameters	Send full URLs with queries
Use `link_id` for key editorial links	Rely only on link text

18.7 Pros and cons of link tracking

✅ Pros	❌ Cons
Proves which links move people	More events (volume)
Guides internal linking strategy	Requires content-map lookup
Works beyond the top 20 paths	Nav links can dominate; filter by position

18.8 Where freshers go wrong in Chapter 18 topics

Mistake	Why it happens	Avoid it by	Fix it by
Counting nav clicks as "editorial" success	No position filter	link_position	Filtering body/related
CTA counted twice	No exclusion	Trigger exception	Adding exclusion
Child-site links counted as outbound	Hostname regex incomplete	18.3	Updating regex

18.9 Thumb rules

> 👍 Count every signboard.
>
> 👍 Position tells you placement; text tells you wording.
>
> 👍 CTAs and links are tracked separately.

18.10 Notes to remember

> 📝 `internal_link_click`: source, target, position, text, stages, target type.
>
> 📝 Link CTR = link clicks ÷ source page views.

Chapter 18 Glossary

Term	Simple meaning
Internal link	A link to another page on our sites
Link position	Where the link sits (body, nav, footer...)
Anchor text	The clickable words
Link CTR	Clicks on a link ÷ views of its page
Editorial link	A link written into content by a writer

Chapter 18 Quiz

1. Which event records internal link clicks?

- A) `internal_link_click`
- B) `exit_to_portal`
- C) `scroll_depth`
- D) `page_view`

2. Where should `data-link-position` be added?

- A) On layout containers (nav, article, aside, footer)
- B) Nowhere
- C) On images only
- D) On every word

3. Should CTA clicks also fire internal_link_click?

- A) No; excluded
- B) Yes
- C) Twice
- D) Randomly

4. What does `target_stage` show?

- A) The browser
- B) The visitor's age
- C) The source medium
- D) The funnel stage of the destination page

5. How is link CTR calculated?

- A) Page views ÷ clicks
- B) Not possible
- C) Link clicks ÷ source page views
- D) Users × sessions

6. What GTM trigger type is used?

- A) History Change
- B) Click – Just Links
- C) Scroll Depth
- D) Timer

7. What does path exploration show by default per step?

- A) All nodes
- B) Top 5 nodes (up to 20)
- C) 1 node
- D) 100 nodes

8. How is a link to weightclinicsresults.com from the main site classified?

- A) Outbound
- B) Portal exit
- C) Internal (in the hostname regex)
- D) Ignored

9. What is `link_id` for?

- A) Storing emails
- B) Styling
- C) Tracking users
- D) Tracking specific editorial links

10. In the signboard analogy, internal_link_click is…

- A) The traffic police
- B) The road
- C) A counter on each signboard
- D) The hospital

Sources

1. Google Analytics Help, Path exploration 🟢
2. Ongo Weight Loss, Book Three (hub-and-spoke linking) and Book Four (internal_link_click), internal 🔵

Answer key, Chapter 18: 1-A · 2-A · 3-A · 4-D · 5-C · 6-B · 7-B · 8-C · 9-D · 10-C

LEVEL 2 · CHAPTER 19

Chapter 19: Outbound, Phone, Email and Chat Tracking

Some of the most valuable actions don't happen on our pages at all: a phone call, an email, a chat, or a click to a source. This chapter tracks each one without collecting anything personal.

> 💡 **Trivia**
>
> GA4's enhanced measurement can automatically record outbound clicks as a `click` event. [1] ● Official (Google)
>
> We keep that on as a backup, but we add our own named events so reports separate source citations from social links and partner sites.

19.1 The events

Event	Trigger	Parameters	Key event
`outbound_click_source`	Click to a citation domain (fda.gov, nih.gov, pubmed, cdc.gov, nejm.org, doi.org ...)	link_domain, link_url_path, page_type	—
`outbound_click_social`	Click to youtube.com, instagram.com, facebook.com, pinterest.com, linkedin.com	link_domain, page_type	—
`outbound_click_partner`	Click to approved partner domains	link_domain, page_type	—
`phone_click`	Click on `tel:` link	contact_location, page_type	✅
`email_click`	Click on `mailto:` link	contact_location, page_type	Optional
`chat_open`	Chat widget opened	contact_location, page_type	✅

19.2 GTM triggers

Event	Trigger
`outbound_click_*`	Click – Just Links; Click URL Hostname **does not** match our domains regex; then a **Lookup Table** variable maps hostname → context (`source` , `social` , `partner` , default `other`) and the tag name is `outbound_click_{{context}}`
`phone_click`	Click – Just Links; Click URL starts with `tel:`
`email_click`	Click – Just Links; Click URL starts with `mailto:`
`chat_open`	Custom Event from the chat widget's API (e.g., `chat_widget_opened`) or a click on the chat launcher with `data-event="chat_open"`

Privacy: never send the phone number or email address as a parameter. The link itself contains them; send only `contact_location` (header, footer, contact_page, sticky). 🔵

> 🪔 **Desi analogy: dukaan ka darwaaza**
>
> Dukaan se bahar jaate log kahan jaate hain: bank, doctor, ya ghar? Dukaandaar sirf ginti karta hai "kaunse darwaaze se, kis taraf", unke ghar ka pata nahi likhta. **Outbound tracking wahi ginti hai.**

19.3 Chat widgets and privacy

Check	Why
Does the chat vendor receive health questions?	It may need a BAA (Chapter 4)
Does the chat widget load its own trackers?	Review with legal
Do we only send `chat_open` (not messages) to GA4?	✅ Required

19.4 Tests

Test	Expected
Click an FDA source link in an article	`outbound_click_source`, link_domain=fda.gov
Click the Instagram icon in footer	`outbound_click_social`
Click phone in header	`phone_click`, contact_location=header
Open chat	`chat_open`
Parameters contain a phone number or email?	❌ Must be none

> 🟧 **Ongo Weight Loss example**
>
> Writers add source links (Book Six); `outbound_click_source` shows whether readers actually check them. High clicks on a particular citation can signal readers want more detail on that point: a content idea. 🔵

19.5 Do's and Don'ts

✅ Do	❌ Don't
Separate source, social and partner clicks	Lump all outbound clicks together
Track contact actions by location	Send phone numbers or emails
Review chat vendors for BAA needs	Send chat messages to GA4
Keep EM outbound clicks as backup	Rely on it alone for reporting

19.6 Pros and cons

✅ Pros	❌ Cons
See high-intent contact actions	Can't see call outcomes (offline)
Learn which sources readers trust	Domain lists need upkeep
Simple to build	—

19.7 Where freshers go wrong in Chapter 19 topics

Mistake	Why it happens	Avoid it by	Fix it by
Phone number in `link_url`	Using Click URL	19.2 privacy note	Removing parameter
Child sites counted as outbound	Regex gap	Shared hostname list	Updating list
Chat messages in events	Vendor integration default	19.3	Disabling

19.8 Thumb rules

> 👍 Count the door, not the address.
>
> 👍 Source, social, partner: three buckets.
>
> 👍 Chat open yes, chat content never.

19.9 Notes to remember

> 📝 Key events: `phone_click`, `chat_open`.
>
> 📝 Lookup table maps hostname → outbound context.

Chapter 19 Glossary

Term	Simple meaning
Outbound click	A click to another website
`tel:` link	A link that starts a phone call
`mailto:` link	A link that opens an email
Lookup table	GTM variable mapping inputs to outputs
Contact location	Where the contact link sits

Chapter 19 Quiz

1. Which event tracks clicks to FDA or NIH links?

- A) `page_view`
- B) `outbound_click_source`
- C) `phone_click`
- D) `internal_link_click`

2. What should never be sent with phone_click?

- A) contact_location
- B) The phone number
- C) Event name
- D) page_type

3. How are outbound contexts assigned?

- A) Randomly
- B) Lookup table by hostname
- C) By visitor name
- D) By colour

4. Which trigger catches phone links?

- A) Timer
- B) Scroll Depth
- C) Click URL starts with `tel:`
- D) History change

5. Which is a key event?

- A) `scroll_depth`
- B) `chat_open`
- C) `faq_open`
- D) `outbound_click_social`

6. What about chat messages?

- A) Sent if short
- B) Sent to Clarity
- C) Sent in full
- D) Never sent to GA4

7. Why might a chat vendor need a BAA?

- A) It uses colours
- B) It may receive health questions
- C) It's free
- D) It shows a logo

8. What's EM outbound click used for?

- A) Patient tracking
- B) Nothing

- C) Backup
- D) Main reporting

9. Which domains are "social"?

- A) YouTube, Instagram, Facebook, Pinterest, LinkedIn
- B) FDA, NIH
- C) Our child sites
- D) The portal

10. In the dukaan-darwaaza analogy, we record…

- A) Home addresses
- B) Names
- C) Nothing
- D) Which door and direction, not the home address

Sources

1. Google Analytics Help, Enhanced measurement events 🟢
2. Ongo Weight Loss, Book Six (source links) and Book Four (outbound events), internal 🔵

Answer key, Chapter 19: 1-B · 2-B · 3-B · 4-C · 5-B · 6-D · 7-B · 8-C · 9-A · 10-D

Chapter 20: Engagement Events: Scroll, Video and FAQs

Did people actually read the page? Engagement events answer that: how far they scrolled, which videos they watched, which FAQs they opened. Used carefully, they show content quality.

> 💡 **Trivia**
>
> GA4 counts a session as **engaged** if it lasts 10+ seconds, has a key event, or has 2+ page views; enhanced measurement's built-in scroll event fires only at **90%** depth. [1] 🟢 Official (Google)
>
> 90% alone hides the middle of the story. We add 25/50/75 so you can see where readers stop.

20.1 Scroll depth

Item	Setting
GTM trigger	**Scroll Depth**, Vertical, percentages `25,50,75,90`; enable on **Window Load** (or DOM Ready)
Conditions	Only Zone A content pages: `DLV – page_type` is one of blog_article, blog_hub, service, medication, pricing, how_it_works (use a RegEx table variable)
Tag	GA4 Event `scroll_depth` · `percent_scrolled` = `{{Scroll Depth Threshold}}` · `page_type`, `content_cluster`
EM scroll	Turn **off** EM scrolls to avoid two scroll events (or keep and ignore; choose one)

Next.js note: the GTM scroll trigger resets on history changes when "History Change" is used; because we fire page views from `page_meta`, add a **Custom Event** `page_meta` reset: use the scroll trigger with "Enable this trigger on: Custom Event page_meta" where supported, or push scroll events from a small client hook. 🔵

```
// hooks/useScrollDepth.ts (alternative to GTM scroll trigger)
'use client'
import { useEffect } from 'react'
import { usePathname } from 'next/navigation'
import { track } from '@/lib/analytics'

export function useScrollDepth(marks = [25, 50, 75, 90]) {
  const pathname = usePathname()
  useEffect(() => {
    const fired = new Set<number>()
    const onScroll = () => {
      const h = document.documentElement
      const pct = Math.round(((h.scrollTop + window.innerHeight) / h.scrollHeight) * 100)
      marks.forEach(m => { if (pct >= m && !fired.has(m)) { fired.add(m); track('scroll_depth', { percent_scrolled: m
}) } })
    }
    window.addEventListener('scroll', onScroll, { passive: true })
    return () => window.removeEventListener('scroll', onScroll)
  }, [pathname])
}
```

20.2 Video

Video type	Method
YouTube embeds	GTM **YouTube Video** trigger (Start, Complete, Progress 25/50/75) → GA4 `video_start`, `video_progress` (video_percent), `video_complete`, `video_title`
`@next/third-parties` YouTubeEmbed (lite)	The lite player loads the real iframe on click; ensure `enablejsapi=1` in params so GTM can listen after play
Self-hosted video	Push `video_*` events from the player's play/ended handlers

20.3 FAQs

```
<details
  data-faq-id="sema-nausea-q3"
  onToggle={(e) => (e.currentTarget as HTMLDetailsElement).open &&
    track('faq_open', { faq_id: 'sema-nausea-q3' })}
>
  <summary>Will nausea come back when my dose increases?</summary>
  …
</details>
```

`faq_id` is a short stable ID, not the question text. 🔵

> 🪔 **Desi analogy: kitaab mein bookmark**
>
> Library waale dekhte hain kitaab mein bookmark kahan tak laga hai: page 20 pe ruk gaye ya poori padhi. **Scroll depth wahi bookmark hai.**

20.4 Reading engagement correctly

Signal	Meaning	Caution
75% scroll on a guide	Read most of it	Short pages reach 90% easily
FAQ opens high on one question	Strong interest; maybe answer it higher up	—
Video 25% but rarely 75%	Hook works, middle loses people	Check Clarity
High engagement, no next step	Content good, links/CTA weak	Chapter 51–52

> 🟧 **Ongo Weight Loss example**
>
> The planned side-effect guides (Book Six) have FAQs and long sections. `scroll_depth` by `content_cluster` will show whether readers reach the "when to call your provider" section, which matters for both safety and quality. 🔵

20.5 Do's and Don'ts

✅ Do	❌ Don't
Track 25/50/75/90 on content pages	Track scroll on every tiny page
Use stable FAQ IDs	Send full question text
Pick one scroll method	Run EM scroll and custom both without plan
Reset scroll per route	Let thresholds carry across pages

20.6 Pros and cons

✅ Pros	❌ Cons
Shows reading depth	More events
Guides content edits	Page length biases scroll %
Video and FAQ insight	Embeds need setup

20.7 Where freshers go wrong in Chapter 20 topics

Mistake	Why it happens	Avoid it by	Fix it by
Scroll events only on first page	SPA navigation	20.1 note	Reset on page_meta
YouTube not tracked	JS API not enabled	20.2	Adding enablejsapi
Comparing scroll across very different page lengths	Ignoring bias	20.4	Compare similar templates

20.8 Thumb rules

👍 Bookmark at 25, 50, 75, 90.

👍 IDs, not text.

👍 Compare like with like.

20.9 Notes to remember

📝 EM scroll = 90% only.

📝 Video events need the YouTube JS API.

Chapter 20 Glossary

Term	Simple meaning
Scroll depth	How far down a page a visitor scrolled
Threshold	A percentage mark (25, 50…)
JS API	A way for scripts to talk to an embedded player
`<details>`	HTML element for expandable content
Engagement	Signs that a visitor interacted meaningfully

Chapter 20 Quiz

1. At what depth does GA4's built-in scroll event fire?

- A) 25%
- B) 50%
- C) 90%
- D) 100%

2. Which thresholds do we track?

- A) 25, 50, 75, 90
- B) 100 only
- C) None
- D) 10 only

3. What's sent with faq_open?

- A) The visitor's email
- B) The full question text
- C) A short stable faq_id
- D) Nothing

4. What does YouTube tracking need?

- A) A BAA
- B) A new property
- C) The JS API enabled
- D) Clarity

5. Why reset scroll tracking per route?

- A) SPA navigation doesn't reload pages
- B) For colours
- C) For SEO
- D) It's not needed

6. What does high engagement but few next steps suggest?

- A) Bad content
- B) Broken GA4
- C) Too many links
- D) Weak links or CTAs

7. Which pages get scroll tracking?

- A) Payment pages
- B) Zone A content pages
- C) The portal
- D) None

8. What's a caution about scroll %?

- A) It's illegal
- B) It measures names
- C) It's always exact
- D) Short pages reach 90% easily

9. Should EM scroll and custom scroll both run unplanned?

- A) Only in staging
- B) Only on mobile
- C) No; pick one method
- D) Yes always

10. In the bookmark analogy, scroll depth is…

- A) Where the reader stopped
- B) The book's price
- C) The author
- D) The library

Sources

1. Google Analytics Help, Enhanced measurement events
2. Next.js Docs, Third-party libraries (YouTubeEmbed)

Answer key, Chapter 20: 1-C · 2-A · 3-C · 4-C · 5-A · 6-D · 7-B · 8-D · 9-C · 10-A

LEVEL 2 · CHAPTER 21

Chapter 21: Tracking Tools and Calculators (Without Their Inputs)

Tools like a BMI calculator are popular and useful, but what people type into them is health information. We track that a tool was used and finished, never what was entered or the result.

> 💡 **Trivia**
>
> HHS lists **symptom-checker inputs** as an example where HIPAA can apply even on a public, unauthenticated page. [1] 🟢 Official (HHS)
>
> A calculator is close cousin to a symptom checker. Treat its inputs as untouchable.

21.1 Tool events

Event	When	Parameters	Never
`tool_start_[tool]`	First interaction with the tool	tool_name, page_type	Values
`tool_complete_[tool]`	Result shown	tool_name, page_type	Values, result, category (e.g., "obese")
`tool_cta_click`	CTA under the result	tool_name, cta_name	Result

Examples: `tool_start_bmi_calculator`, `tool_complete_bmi_calculator`. 🔵

21.2 Developer pattern

```
'use client'
import { useRef } from 'react'
import { track } from '@/lib/analytics'

export function BmiCalculator() {
  const started = useRef(false)
  const onFirstInput = () => {
    if (!started.current) { started.current = true; track('tool_start_bmi_calculator', {
tool_name: 'bmi_calculator' }) }
  }
  const onCalculate = () => {
    // compute and show result locally — DO NOT send the numbers or the category
    track('tool_complete_bmi_calculator', { tool_name: 'bmi_calculator' })
  }
  return (
    <form onSubmit={(e) => { e.preventDefault(); onCalculate() }} data-clarity-
mask="True">
      <input name="height" onChange={onFirstInput} />
      <input name="weight" onChange={onFirstInput} />
      <button type="submit">Calculate</button>
    </form>
  )
}
```

Also: the form uses `method` handled in JS (no GET submission), so values never appear in the URL; the tool area is masked for Clarity (`data-clarity-mask="True"`), and legal decides whether Clarity runs on tool pages at all. 🔵

> 🪔 **Desi analogy: dawai ki dukaan ka weighing machine**
>
> Chemist ki dukaan pe weighing machine pe log wazan dekhte hain. Dukaandaar ginta hai "aaj 50 logon ne machine use ki", lekin kisi ka wazan register mein nahi likhta. **Tool tracking wahi ginti hai.**

21.3 Privacy checklist for every tool

☐	Check
☐	No input or result values in events, URLs, page titles

☐	Check
☐	Form doesn't submit via GET
☐	Clarity masked or excluded
☐	Result category not sent (e.g., "overweight")
☐	Legal has zoned the tool page
☐	DebugView shows only tool_name and page_type

21.4 What tool data tells you

Metric	Use
Starts per page view	Is the tool visible/inviting?
Completes ÷ starts	Is it easy to finish?
Tool CTA clicks ÷ completes	Does the result lead to a next step?
Sessions with tool → exit_to_portal	Does the tool support leads?

> 🟧 **Ongo Weight Loss example**
>
> Book Four planned `tool_complete_eligibility_checker` as a key event. If the eligibility checker asks health questions, it belongs inside the BAA/portal route (Zone C), not GA4. A public BMI calculator with no data sent can stay in GA4. 🔵

21.5 Do's and Don'ts

✅ Do	❌ Don't
Track start and complete	Track inputs or results
Mask tools in Clarity	Record tool sessions unmasked
Keep values out of URLs	Submit tools via GET

✅ Do	❌ Don't
Measure completion rate	Use results to segment users

21.6 Pros and cons

✅ Pros	❌ Cons
Shows tool usefulness safely	No result-level insight
Supports lead analysis	Some teams want more data (it's not allowed)
Simple to build	—

21.7 Where freshers go wrong in Chapter 21 topics

Mistake	Why it happens	Avoid it by	Fix it by
Sending BMI category as a parameter	Wanting segments	21.3	Removing; reprocessing
Results in the URL	GET forms	JS handling	Fixing form
Clarity recording inputs	Default settings	Mask attribute	Masking/excluding

21.8 Thumb rules

> 👍 Count the use, never the numbers.
>
> 👍 No GET, no leaks.
>
> 👍 Health questions = BAA route.

21.9 Notes to remember

📝 Tool events: start, complete, tool CTA.

📝 Eligibility with health questions = Zone C.

Chapter 21 Glossary

Term	Simple meaning
Tool event	An event showing a tool was used
GET submission	A form that puts values in the URL
Completion rate	Completes ÷ starts
Masking	Hiding content in recordings
Result category	A label from a tool's output (not allowed in analytics)

Chapter 21 Quiz

1. What does HHS list as a public-page HIPAA example?

- A) Blog titles
- B) Logos
- C) Page speed
- D) Symptom-checker inputs

2. What's tracked for a BMI calculator?

- A) BMI category
- B) Height and weight
- C) BMI result
- D) Start and complete only

3. Why avoid GET form submissions?

- A) They're required
- B) They're slow
- C) They're ugly
- D) Values appear in the URL

4. What's the Clarity setting for tools?

- A) `data-clarity-mask="True"` or exclusion
- B) Unmask
- C) Relaxed mode
- D) No change

5. Where does a health-question eligibility checker belong?

- A) The BAA/portal route (Zone C)
- B) GSC
- C) Clarity
- D) GA4 key events

6. What does completes ÷ starts show?

- A) How easy the tool is to finish
- B) Patient health
- C) Revenue
- D) Search ranking

7. Can the result category be sent?

- A) Yes
- B) Only to Looker
- C) No
- D) If anonymised

8. What's `tool_cta_click` for?

- A) Typing in fields
- B) Payments
- C) Scrolls
- D) CTA clicks under the tool result

9. Who zones the tool page?

- A) Legal
- B) Google
- C) The tool itself
- D) Visitors

10. In the weighing-machine analogy, we record…

- A) Names
- B) Everyone's weight
- C) How many used it, not their weight
- D) Nothing

Sources

1. HHS OCR, Online tracking technologies bulletin
2. Microsoft Learn, Clarity masking

Answer key, Chapter 21: 1-D · 2-D · 3-D · 4-A · 5-A · 6-A · 7-C · 8-D · 9-A · 10-C

LEVEL 2 · CHAPTER 22

Chapter 22: Forms and Leads

You defined leads as eligibility completions and sign-ups. Most of those happen in or near the portal, where GA4 can't go. This chapter shows what the public site can safely measure, and how leads reach your reports anyway.

> 💡 **Trivia**
>
> HHS gives "appointment requests" and "patient portal login and registration" as examples where HIPAA can apply even when the page is unauthenticated. [1] ● Official (HHS)
>
> Sign-up and registration forms are exactly those examples.

22.1 Form types and rules

Form	Zone	GA4 events	Data flow for counts
Newsletter (email only, no health info)	A/B	`form_start_newsletter`, `form_submit_newsletter`	GA4 OK (never the email)
General contact ("How can we help?")	B	`form_submit_contact` only if legal approves; message text never	Messages go to support tool (may need BAA)
Eligibility questions	C	None	BAA vendor/portal → aggregated
Account sign-up / registration	C	None	BAA vendor/portal → aggregated
Payment	C	None	Backend → aggregated

22.2 What the public site measures as lead signals

Signal	Event	Meaning
Clicked a CTA toward eligibility	`cta_*_check_eligibility`	Intent
Left for the portal	`exit_to_portal`	Started the lead path
Contact form submitted (approved)	`form_submit_contact`	Contact lead
Called or chatted	`phone_click` , `chat_open`	Contact intent

Leads (eligibility completed, sign-ups) and **payments** come from the portal/backend as daily counts (Chapter 24). 🔵

> 🪔 **Desi analogy: shaadi ka RSVP**
>
> Card bhejne waale ko pata hai kitne cards gaye (CTA clicks) aur kitne log hall ke gate tak aaye (exit_to_portal). Andar kaun kya kha raha hai, woh caterer ka register hai (portal), aur woh sirf ginti batata hai. **Lead count caterer se aata hai, naam nahi.**

22.3 Non-health form tracking (developer)

```
<form
  data-form-name="newsletter"
  onFocus={() => startedOnce('newsletter')}
  onSubmit={async (e) => {
    e.preventDefault()
    const ok = await subscribe(new FormData(e.currentTarget)) // POST to server
    if (ok) track('form_submit_newsletter', { form_name: 'newsletter' })
  }}
>
```

Rule	Why
Fire submit **after** server success	Avoid counting failed submits
`method="post"` / JS submit	No values in URL
No field values in events	Privacy

Rule	Why
Thank-you state without query params	No `?email=`

22.4 GA4 recommended events?

Recommended event	Use here?
`generate_lead`	Only for **non-health** contact forms if you want GA4's lead reports; **not** for eligibility/sign-ups (Zone C)
`sign_up`	❌ (portal)
`purchase`	❌ in GA4 by default (Chapter 24)

> 🟧 **Ongo Weight Loss example**
>
> Your daily dashboard will show three layers side by side: GA4 intent signals (CTA clicks, `exit_to_portal`), aggregated **leads** (eligibility completions, sign-ups) and aggregated **payments**, all by date and channel. That's your full funnel, privacy-safe. 🔵

22.5 Do's and Don'ts

✅ Do	❌ Don't
Zone every form	Track every form the same way
Fire submit after server success	Fire on button click
Use aggregated lead counts	Send sign-up events from the portal to GA4
Keep emails out of URLs	Use GET forms

22.6 Pros and cons

✅ Pros	❌ Cons
Safe, defensible lead measurement	Less per-user detail
Clear separation of intent vs lead	Needs backend feed
Fewer false submits	—

22.7 Where freshers go wrong in Chapter 22 topics

Mistake	Why it happens	Avoid it by	Fix it by
`generate_lead` on eligibility	Copying ecommerce guides	22.4	Removing
Submit event on click	Easy trigger	22.3	Success-based push
Contact message text in events	Debugging	Rules	Removing

22.8 Thumb rules

> 👍 Intent in GA4, leads by count.
>
> 👍 Fire on success, not on click.
>
> 👍 Health forms live in Zone C.

22.9 Notes to remember

> 📝 Lead = eligibility completed or sign-up (aggregated).
>
> 📝 Public lead signals: CTA click, exit_to_portal, contact, phone, chat.

Chapter 22 Glossary

Term	Simple meaning
Lead	A person who completed eligibility or signed up
Lead signal	A public action showing intent
`generate_lead`	GA4 recommended event for leads
POST submission	Sending form data without putting it in the URL
Success state	Confirmation that the server accepted the form

Chapter 22 Quiz

1. Where are eligibility forms tracked?

- A) Clarity
- B) GSC
- C) GA4 with answers
- D) BAA/portal route as counts (Zone C)

2. When should form_submit fire?

- A) On page load
- B) After server success
- C) Never
- D) On button click

3. Which form can use GA4 events?

- A) Payment
- B) Newsletter (no health info)
- C) Registration
- D) Eligibility questions

4. Is `generate_lead` used for eligibility?

- A) Only on mobile
- B) Twice

- C) No
- D) Yes

5. Which is a public lead signal?

- A) Eligibility answers
- B) `exit_to_portal`
- C) Patient name
- D) Payment card

6. What do HHS list as HIPAA examples on public pages?

- A) Blog images
- B) Appointment requests and portal registration
- C) Page speed
- D) Fonts

7. Where do lead counts come from?

- A) GSC
- B) Guesswork
- C) Portal/backend aggregated daily
- D) Clarity recordings

8. What should never appear in URLs?

- A) Slugs
- B) UTM parameters
- C) Emails or form values
- D) Page paths

9. Should contact message text be sent?

- A) Yes
- B) Never
- C) If short
- D) Only to Looker

10. In the RSVP analogy, lead counts come from...

- A) Guests' diaries
- B) The invitation

- C) The caterer's register (counts only)
- D) The band

Sources

1. HHS OCR, Online tracking technologies bulletin
2. Ongo Weight Loss, lead and conversion definitions (team lead, 30 October 2026), internal

Answer key, Chapter 22: 1-D · 2-B · 3-B · 4-C · 5-B · 6-B · 7-C · 8-C · 9-B · 10-C

LEVEL 2 · CHAPTER 23

Chapter 23: Portal Exit, Child Sites and Cross-Domain Tracking

Visitors may start on weightclinicsresults.com, read on ongoweightloss.com and then leave for the patient portal. This chapter stitches the public journey together across sites, and hands the portal just enough non-personal context to count leads and payments by channel.

> 💡 **Trivia**
>
> For GA4 cross-domain measurement, "the tag on each page must use the same tag ID (i.e., the same "G-" ID) from the same web data stream." GA4 then adds a `_gl` parameter to links between your domains. [1] ● Official (Google)
>
> Same tag, configured domains, and a journey across sites becomes one session.

23.1 Two different boundaries

Boundary	Example	Approach
Public ↔ public	weightclinicsresults.com → ongoweightloss.com	**GA4 cross-domain** (same G- ID, `_gl`)
Public → portal	ongoweightloss.com → portal	**No GA4 in the portal.** Fire `exit_to_portal`; pass **non-personal context** only if legal approves

23.2 Cross-domain between public sites (setup)

Step	Action
1	All public sites load the **same GTM container** (or same G- ID)

Step	Action
2	GA4 Admin → Data streams → Web → Configure tag settings → **Configure your domains** → add `ongoweightloss.com`, `weightclinicsresults.com`, every child domain (match type "contains" or "ends with")
3	**Do not** add the portal domain
4	Make sure redirects keep the `_gl` parameter
5	Test: click from child site to main site; destination URL shows `?_gl=…`; DebugView shows the same session
6	Our `CJS – clean page_location` keeps only `utm_*` **after** gtag reads `_gl` (gtag processes `_gl` on load; reports don't need it)

23.3 `exit_to_portal` (the last public event)

Item	Setting
Developer	Portal links get `data-exit-portal="true"` (CTA component)
GTM trigger	Click – Just Links; `Click URL Hostname` equals the portal hostname **or** `data-exit-portal` = true
Tag	GA4 Event `exit_to_portal` · source_page = `{{Page Path}}`, cta_name, site_id, page_type, funnel_stage
Key event	✅ Yes (Chapter 26)

> 🪔 **Desi analogy: hospital ka visitor pass**
>
> Lobby ka guard likhta hai "ek visitor OPD ki taraf gaya, Gate 2 se, Dadar wali bus se aaya tha." Naam nahi, bas raasta. Andar OPD apna register rakhta hai. **exit_to_portal guard ki entry hai; portal OPD ka register.**

23.4 Passing non-personal context to the portal (optional, legal approval)

Goal: count leads and payments **by channel group and landing-page group** without identifiers.

Step	Detail
1	On the public site, when a session starts, store two values in **sessionStorage** (first-party, non-personal): `ch_group` (organic_search, organic_social, email, referral, direct, child_site) and `lp_group` (e.g., `blog_semaglutide`, `service_weight_loss`)
2	When the visitor clicks to the portal, append them: `https://portal…/start?ctx_ch=organic_search&ctx_lp=blog_semaglutide&ctx_site=ongo_main`
3	The portal (BAA-covered) stores these context fields with the sign-up/payment record **inside the BAA environment**
4	Backend exports **daily aggregated counts** by date × ctx_ch × ctx_lp × ctx_site (Chapter 24)
5	**Never** pass the GA4 client ID, `_gl`, emails or any identifier to the portal

```
// lib/visit-context.ts (public site only)
export function setVisitContext(meta: { page_type: string; content_cluster: string }) {
  if (sessionStorage.getItem('ctx_lp')) return            // first page of the session
only
  sessionStorage.setItem('ctx_lp', `${meta.page_type}_${meta.content_cluster}`)
  sessionStorage.setItem('ctx_ch', channelGroup(document.referrer, location.search))
}

function channelGroup(ref: string, search: string): string {
  const p = new URLSearchParams(search)
  const med = (p.get('utm_medium') || '').toLowerCase()
  if (med === 'social') return 'organic_social'
  if (med === 'email') return 'email'
  if (!ref) return 'direct'
  const host = new URL(ref).hostname
  if (/google\.|bing\.|duckduckgo\.|yahoo\./.test(host)) return 'organic_search'
  if (/weightclinicsresults\.com$/.test(host)) return 'child_site'
  if (/(facebook|instagram|youtube|pinterest|linkedin|reddit)\./.test(host)) return
'organic_social'
  return 'referral'
}

export function portalUrl(base: string) {
  const u = new URL(base)
  u.searchParams.set('ctx_ch', sessionStorage.getItem('ctx_ch') || 'unknown')
  u.searchParams.set('ctx_lp', sessionStorage.getItem('ctx_lp') || 'unknown')
  u.searchParams.set('ctx_site', process.env.NEXT_PUBLIC_SITE_ID || 'ongo_main')
  return u.toString()
}
```

Why this is safer than sending payments to GA4: Google never receives a link between a browser ID and a health payment; the portal only receives broad, non-identifying categories. Legal still approves. 🔵

23.5 Tests

Test	Expected
Child → main site	Same session in DebugView; `_gl` on landing URL
Main → portal	`exit_to_portal` fires; portal URL has ctx_ *only (if approved); no** `_gl`
Portal pages	No GTM, no GA4, no Clarity requests (Network tab)

> 🟧 **Ongo Weight Loss example**
>
> You said weightclinicsresults.com and future child sites will also link to the portal. With `ctx_site` , the daily payment table can show "payments that started on weightclinicsresults.com" vs the main site, without any patient data leaving the BAA environment. 🔵

23.6 Do's and Don'ts

✅ Do	❌ Don't
Use one G- ID across public sites	Use separate properties per child site
Exclude the portal from cross-domain	Add `_gl` to portal links
Pass broad context only, with approval	Pass client IDs or emails
Test redirects keep `_gl`	Assume they do

23.7 Pros and cons

✅ Pros	❌ Cons
Full public journeys across sites	Requires same tag everywhere
Leads/payments by channel without identifiers	Context is broad, not per user
Clear legal boundary	Portal team must store and aggregate context

23.8 Where freshers go wrong in Chapter 23 topics

Mistake	Why it happens	Avoid it by	Fix it by
Child site counted as referral	Domain not configured	23.2	Adding domain
`_gl` reaching the portal	Portal added to domain list	Step 3	Removing
ctx values from the last page, not first	Overwriting	"first page only"	Fixing logic

23.9 Thumb rules

> 👍 Public sites: one journey. Portal: one door.
>
> 👍 Context, never identity.
>
> 👍 No `_gl` beyond the lobby.

23.10 Notes to remember

> 📝 Configure your domains: all public sites, not the portal.
>
> 📝 ctx_ch, ctx_lp, ctx_site: optional, legal-approved, non-personal.

Chapter 23 Glossary

Term	Simple meaning
Cross-domain measurement	Treating visits across your domains as one journey
`_gl`	GA4's linker parameter carrying IDs between domains
Visit context	Broad, non-personal info about how a visit started
sessionStorage	Browser storage cleared when the tab closes
Channel group	A broad traffic type (organic search, social...)

Chapter 23 Quiz

1. What does GA4 cross-domain require?

- A) The same G- ID on all domains
- B) A BAA
- C) Different properties
- D) Clarity

2. Should the portal be in the cross-domain list?

- A) Only on weekends
- B) Yes
- C) Only for payments
- D) No

3. What's the last public event?

- A) `purchase`
- B) `page_view`
- C) `sign_up`
- D) `exit_to_portal`

4. What context may be passed to the portal (with approval)?

- A) GA4 client ID
- B) Email

- C) Broad channel and landing-page groups
- D) `_gl`

5. Where are context fields stored with sign-ups?

- A) In GA4
- B) In GSC
- C) In Clarity
- D) Inside the BAA environment

6. How do you test cross-domain?

- A) Look at the logo
- B) Ask users
- C) Count pages
- D) Check `_gl` on landing URL and same session in DebugView

7. Should GTM load on portal pages?

- A) Only for Clarity
- B) Yes
- C) Only in staging
- D) No

8. What does `ctx_site` identify?

- A) The doctor
- B) The browser
- C) Which public site the visit started on
- D) The patient

9. When is ctx_lp set?

- A) On every page
- B) Never
- C) In the portal
- D) On the first page of the session

10. In the hospital analogy, exit_to_portal is…

- A) The ICU
- B) The guard's entry at the lobby door

- C) The OPD register
- D) The patient file

Sources

1. Google Analytics Help, Cross-domain measurement
2. Google Analytics Help, HIPAA and Google Analytics
3. Ongo Weight Loss, portal and child-site structure (team lead, 30 October 2026), internal

Answer key, Chapter 23: 1-A · 2-D · 3-D · 4-C · 5-D · 6-D · 7-D · 8-C · 9-D · 10-B

LEVEL 2 · CHAPTER 24

Chapter 24: Counting Leads and Payments from the Backend

Payment is your real conversion, and it happens inside the portal. This chapter builds a daily, de-identified feed of leads and payments into BigQuery or Google Sheets so Looker Studio can show them next to GA4 data.

> 💡 **Trivia**
>
> Google describes the GA4 Measurement Protocol as a way to "augment automatic collection," including "offline conversions," and recommends the Data Manager API for future server-to-server integrations. [1] ● Official (Google)
>
> GA4 can receive server events, but for health payments the safer design keeps identifiers out of Google entirely.

24.1 The recommended design: daily aggregated table

```
Portal / payment system (BAA environment)
   → nightly job (SQL or script, runs inside BAA environment)
   → aggregates yesterday's records into counts (no identifiers)
   → writes rows to BigQuery table `ops.daily_conversions` (or a Google Sheet)
   → Looker Studio blends with GA4 / GSC by date (+ ctx groups)
```

24.2 The table schema

Column	Type	Example	Notes
`date`	DATE	2026-11-03	Reporting time zone = GA4's
`site_id`	STRING	ongo_main	From ctx_site (Chapter 23)
`channel_group`	STRING	organic_search	From ctx_ch

Column	Type	Example	Notes
`landing_group`	STRING	blog_semaglutide	From ctx_lp
`portal_starts`	INT64	42	Optional
`eligibility_completed`	INT64	18	**Lead**
`signups`	INT64	15	**Lead**
`payments`	INT64	9	**Conversion**
`revenue_band` or `revenue`	STRING / NUMERIC	—	Only if legal approves; totals per row, not per patient

Small-number rule: if a row has very small counts (e.g., fewer than 5), legal may require grouping rows (e.g., `landing_group = other`) to avoid re-identification. Never include dates of birth, states with tiny counts, product strength per patient, or anything individual. 🔵

24.3 Example aggregation (inside the BAA environment)

```
-- Runs in the portal/BAA database, NOT in Google
SELECT
  DATE(created_at AT TIME ZONE 'America/New_York')        AS date,
  COALESCE(ctx_site, 'unknown')                             AS site_id,
  COALESCE(ctx_ch, 'unknown')                               AS channel_group,
  COALESCE(ctx_lp, 'unknown')                               AS landing_group,
  COUNT(*) FILTER (WHERE event = 'eligibility_completed')   AS eligibility_completed,
  COUNT(*) FILTER (WHERE event = 'signup')                  AS signups,
  COUNT(*) FILTER (WHERE event = 'payment_completed')       AS payments
FROM portal_events
WHERE created_at >= CURRENT_DATE - INTERVAL '1 day'
  AND created_at <  CURRENT_DATE
GROUP BY 1,2,3,4;
```

The output rows (counts only) are then loaded to BigQuery (Chapter 31) via a service account with **write access to one dataset only**, or appended to a Google Sheet. 🔵

Counting Payments Without PHI



Totals travel; people never do.

> 🪔 **Desi analogy: mandir ki daan-peti**
>
> Mandir ki daan-peti shaam ko gini jaati hai: aaj kitna aaya, kis din. Kaun kitna daal gaya, woh nahi likha jaata. **Payment feed daan-peti ki ginti hai:** total, bina naam.

24.4 Loading options

Option	How	Best for
BigQuery table	Backend job uses BigQuery API (service account) to append rows daily	Long history, blending with GA4 export
Google Sheet	Backend appends rows via Sheets API, or ops pastes a daily CSV	Fast start; small volume
CSV upload to BigQuery	Manual weekly upload	Temporary

24.5 What about GA4 Measurement Protocol?

Approach	Verdict
Send `purchase` to GA4 with the browser's client ID	❌ Not recommended: links a browser identifier at Google to a health payment; only with explicit legal approval
Send aggregated daily totals to GA4	Not possible as totals (GA4 expects events)
Use a BAA-covered CDP that strips identifiers and forwards **de-identified** events	Possible; vendor and legal must confirm what Google receives

24.6 Reconciling the numbers

Comparison	What to expect
`exit_to_portal` (GA4) vs `portal_starts` (backend)	Similar trend; GA4 lower (blockers, consent)
Leads ÷ `exit_to_portal`	Portal completion rate
Payments ÷ leads	Lead-to-patient rate

> 🟧 **Ongo Weight Loss example**
>
> Your daily review becomes: "Yesterday: 1,240 sessions, 96 exit_to_portal, 18 eligibility completed, 15 sign-ups, 9 payments; organic search drove 5 payments, mostly from semaglutide blog landings." (Numbers illustrative.) All without a single patient identifier in Google. 🔵

24.7 Do's and Don'ts

✅ Do	❌ Don't
Aggregate inside the BAA environment	Export raw portal records to Google
Use the same time zone as GA4	Mix time zones
Group tiny counts	Publish rows that could identify someone
Limit the loader's access to one dataset	Give broad Cloud permissions

24.8 Pros and cons

✅ Pros	❌ Cons
Real conversions in reports safely	No per-user attribution
Simple, auditable	Needs a backend job
Works for all child sites	Daily, not real-time

24.9 Where freshers go wrong in Chapter 24 topics

Mistake	Why it happens	Avoid it by	Fix it by
Exporting patient IDs "for joining"	Wanting detail	24.2 schema	Removing
Different time zones	Defaults	Same zone as GA4	Converting
Treating GA4 `exit_to_portal` as leads	Similar trend	22.1 definitions	Labelling correctly

24.10 Thumb rules

👍 Count inside, report outside.

👍 Totals, not people.

👍 Same date, same time zone.

24.11 Notes to remember

📝 Table: date, site, channel, landing group, leads, payments.

📝 Measurement Protocol purchases with client IDs: not by default.

Chapter 24 Glossary

Term	Simple meaning
Offline/backend conversion	A conversion recorded outside the browser
Aggregated table	Daily totals with no individual rows
Re-identification	Figuring out who a record belongs to
Service account	A non-human account used by a program

Term	Simple meaning
Reconciliation	Comparing numbers from two systems

Chapter 24 Quiz

1. Where should payment records be aggregated?

- A) In Clarity
- B) Inside the BAA environment
- C) In GSC
- D) In GA4

2. Which column is a lead?

- A) `eligibility_completed`
- B) `site_id`
- C) `date`
- D) `payments`

3. What's recommended instead of GA4 purchase events with client IDs?

- A) Nothing
- B) Clarity recordings
- C) More pixels
- D) Daily aggregated counts

4. What's the small-number rule for?

- A) Saving storage
- B) Design
- C) Avoiding re-identification
- D) Faster queries

5. What time zone should the table use?

- A) Local browser
- B) Random
- C) Same as GA4

- D) UTC only

6. Which loader permission is right?

- A) Write to one dataset only
- B) Project owner
- C) None
- D) Organisation admin

7. What does payments ÷ leads show?

- A) Lead-to-patient rate
- B) Bounce rate
- C) Page speed
- D) Ad spend

8. Why is `exit_to_portal` lower than portal starts?

- A) Portal double counts
- B) It's always equal
- C) GA4 is broken
- D) Blockers and consent reduce GA4 data

9. What does Google recommend for future server-to-server integrations?

- A) Clarity
- B) FTP
- C) Email
- D) Data Manager API

10. In the daan-peti analogy, the feed is…

- A) A guest list
- B) A photo album
- C) Totals counted, no names
- D) A list of donors

Sources

1. Google Developers, Measurement Protocol (GA4) ●

2. Google Analytics Help, HIPAA and Google Analytics 🟢
3. Ongo Weight Loss, conversion definitions and portal setup (team lead, 30 October 2026), internal 🔵

Answer key, Chapter 24: 1-B · 2-A · 3-D · 4-C · 5-C · 6-A · 7-A · 8-D · 9-D · 10-C

LEVEL 2 · CHAPTER 25

Chapter 25: Custom Dimensions and Metrics

GA4 only shows your parameters in reports after you register them. This chapter lists exactly which ones to register, with what scope, and in what order.

> 💡 **Trivia**
>
> GA4 limits how many custom dimensions a standard property can have (event-scoped and user-scoped have separate quotas), and **new custom dimensions aren't retroactive**: data appears only from the day you create them. [1] ● Industry summary of Google help pages
>
> Register on day one, before traffic, so nothing is lost.

25.1 How to create one

GA4 → Admin → **Custom definitions** → **Create custom dimension** → name, scope, description, event parameter → Save.

25.2 The register (create all of these)

Display name	Scope	Event parameter	Used in
Site	Event	`site_id`	All reports
Page type	Event	`page_type`	Landing, content
Page group (silo)	Event	`page_group`	Silo performance
Funnel stage	Event	`funnel_stage`	Funnels
Content cluster	Event	`content_cluster`	Cluster performance
Primary entity	Event	`primary_entity`	Medication pages

Display name	Scope	Event parameter	Used in
Hub parent	Event	`hub_parent`	Hub/spoke
Template	Event	`template`	Design/tech
Content ID	Event	`content_id`	Joining with CMS
Privacy zone	Event	`privacy_zone`	Audits
Author role	Event	`author_role`	Content
Publish month	Event	`publish_month`	Content age
CTA name	Event	`cta_name`	CTA reports
CTA group	Event	`cta_group`	CTA reports
CTA location	Event	`cta_location`	CTA placement
CTA text	Event	`cta_text`	Label tests
Link source	Event	`link_source`	Internal links
Link target	Event	`link_target`	Internal links
Link position	Event	`link_position`	Internal links
Link text	Event	`link_text`	Anchor tests
Source stage	Event	`source_stage`	Funnel moves
Target stage	Event	`target_stage`	Funnel moves
Target page type	Event	`target_page_type`	Funnel moves
Link domain	Event	`link_domain`	Outbound
Contact location	Event	`contact_location`	Contact actions
Tool name	Event	`tool_name`	Tools
Form name	Event	`form_name`	Forms
FAQ ID	Event	`faq_id`	FAQs
Percent scrolled	Event	`percent_scrolled`	Engagement

Display name	Scope	Event parameter	Used in
Metric name	Event	`metric_name`	Web vitals
Metric rating	Event	`metric_rating`	Web vitals
Requested path	Event	`requested_path`	404s
Source page	Event	`source_page`	Portal exits

Custom metric: `metric_value` (Web vitals) as a **custom metric** (unit: standard; or milliseconds for LCP/INP). 🔵

> 🪔 **Desi analogy: shaadi ki seating chart**
>
> Mehmaan aa gaye, lekin agar table pe naam ka card nahi, toh koi nahi jaanta kaun kahan baithega. **Custom dimension woh name card hai:** parameter aaya, lekin card ke bina report mein jagah nahi milti.

25.3 Prioritise if you hit the limit

Priority	Dimensions
1 (must)	site_id, page_type, page_group, funnel_stage, content_cluster, cta_name, cta_location, link_position, link_target, source_stage, target_stage
2	primary_entity, hub_parent, cta_group, link_source, link_text, tool_name, form_name, metric_name, metric_rating
3	template, content_id, author_role, publish_month, faq_id, percent_scrolled, requested_path, privacy_zone

Everything is still in **BigQuery** even if not registered. 🔵

25.4 Rules

Rule	Why
Register before sending traffic	Not retroactive
Parameter name must match exactly	Case-sensitive

Rule	Why
Avoid high-cardinality dimensions in reports	e.g., link_text can create "(other)" rows; use BigQuery for detail
Never register personal data	It shouldn't exist anyway

> 🟧 **Ongo Weight Loss example**
>
> With `funnel_stage` and `target_stage` registered, a single exploration answers "How many TOFU readers clicked to BOFU pages this week?" without any URL regex. 🔵

25.5 Do's and Don'ts

✅ Do	❌ Don't
Create the register on day one	Create dimensions months later
Match names exactly	Typos ("cta_locaton")
Prioritise by reporting need	Register everything without thought
Use BigQuery for high-cardinality detail	Overload reports with unique text

25.6 Pros and cons

✅ Pros	❌ Cons
Parameters usable in reports and Looker	Quotas limit numbers
Business-friendly names	Not retroactive
Consistent across dashboards	Cardinality issues

25.7 Where freshers go wrong in Chapter 25 topics

Mistake	Why it happens	Avoid it by	Fix it by
"Dimension shows no data"	Created after traffic, or name typo	25.4	Checking names; waiting 24–48h
"(other)" rows	High cardinality	Use BigQuery	Narrowing reports
Duplicate dimensions	No register	Appendix F	Archiving one

25.8 Thumb rules

👍 No name card, no seat in reports.

👍 Register first, send second.

👍 Detail lives in BigQuery.

25.9 Notes to remember

📝 Admin → Custom definitions → Create custom dimension.

📝 Priority 1: site, page type, group, stage, cluster, CTA name/location, link position/target, stages.

Chapter 25 Glossary

Term	Simple meaning
Custom dimension	A registered parameter usable in reports
Custom metric	A registered numeric parameter
Cardinality	How many unique values a dimension has

Term	Simple meaning
(other)	GA4's grouping when too many values exist
Retroactive	Applying to past data (custom dimensions don't)

Chapter 25 Quiz

1. Are custom dimensions retroactive?

- A) Yes
- B) For 30 days
- C) Only in Looker
- D) No

2. Where are they created?

- A) GSC
- B) GTM
- C) Clarity
- D) Admin → Custom definitions

3. Which scope is `cta_location`?

- A) Item
- B) Session
- C) User
- D) Event

4. Where is unregistered parameter data still available?

- A) Clarity
- B) BigQuery
- C) Nowhere
- D) GSC

5. Which is priority 1?

- A) funnel_stage
- B) faq_id

- C) publish_month
- D) template

6. What causes "(other)" rows?

- A) High cardinality
- B) Low traffic
- C) Consent
- D) Fonts

7. Must parameter names match exactly?

- A) Roughly
- B) No
- C) Only first letter
- D) Yes, case-sensitive

8. Which should be a custom metric?

- A) `metric_value`
- B) `site_id`
- C) `page_type`
- D) `cta_name`

9. When should the register be created?

- A) Day one, before traffic
- B) After a year
- C) After complaints
- D) Never

10. In the seating-chart analogy, a custom dimension is…

- A) The name card that gives a seat in reports
- B) The music
- C) The food
- D) The hall

Sources

1. Google Analytics Help, Dimensions and metrics (limits vary; check current help)
2. Ongo Weight Loss, Event Dictionary v2 (Chapter 16), internal

Answer key, Chapter 25: 1-D · 2-D · 3-D · 4-B · 5-A · 6-A · 7-D · 8-A · 9-A · 10-A

LEVEL 2 · CHAPTER 26

Chapter 26: Key Events: What Counts as Success in GA4

GA4 calls important events "key events". Mark too many and every report looks like a success; mark too few and you can't see progress. This chapter picks the right ones for Ongo Weight Loss.

> 💡 **Trivia**
>
> GA4 funnel explorations count a user who completes a closed funnel several times **only once**, for the first sequence. [1] 🟢 Official (Google)
>
> Key events and funnels count differently. Know which one you're reading.

26.1 Our key events

Key event	Why	Counting
`exit_to_portal`	Strongest public signal of lead intent	Once per session
`phone_click`	Direct contact	Once per event
`chat_open`	Direct contact	Once per session
`form_submit_contact`	Contact lead (non-health form, legal-approved)	Once per event
(optional) `cta_*_get_started`	Use only if you want CTA-level key events; otherwise analyse as regular events	Once per session

Not key events in GA4: eligibility completed, sign-ups, payments (they come from the backend feed; Chapter 24). 🔵

26.2 How to mark them

GA4 → Admin → **Events** → toggle **Mark as key event** for events that already exist; or Admin → **Key events** → **New key event** → enter the exact name (works before the event first arrives). Set counting method to **Once per session** where repeated clicks shouldn't inflate numbers (GA4 offers "once per event" or "once per session"). 🔵

> 🪔 **Desi analogy: cricket mein run vs wicket**
>
> Har gend pe kuch hota hai, lekin scoreboard pe sirf runs aur wickets bade akshar mein. **Key events scoreboard ke bade number hain:** baaki details commentary mein.

26.3 Lead and conversion naming in reports

Label in dashboards	Source	Definition
Portal exits	GA4 `exit_to_portal`	Public lead intent
Contact actions	GA4 phone/chat/contact	Contact intent
Leads	Backend: eligibility completed + sign-ups	Your lead definition
Payments (conversions)	Backend: payments	Your conversion definition

Never label GA4 key events as "conversions" in reports; GA4 renamed conversions to key events, and our real conversion is payment. 🔵

26.4 Key event rates to watch daily

Rate	Formula
Portal exit rate	`exit_to_portal` sessions ÷ sessions
Contact rate	contact key events ÷ sessions
Lead rate	leads ÷ `exit_to_portal`
Payment rate	payments ÷ leads
Session-to-payment	payments ÷ sessions

> 🟧 **Ongo Weight Loss example**
>
> Book Four listed `portal_signup_complete` and `portal_payment_complete` as key events. Book Seven moves both to the backend feed. In GA4, `exit_to_portal` takes their place as the primary key event. 🔵

26.5 Do's and Don'ts

✅ Do	❌ Don't
Keep key events few and meaningful	Mark scroll or page_view as key events
Use once-per-session where clicks repeat	Inflate counts with repeat clicks
Label leads and payments from the backend	Call GA4 key events "conversions"
Review the list quarterly	Add key events ad hoc

26.6 Pros and cons

✅ Pros	❌ Cons
Clear success signals	Not the full funnel alone
Used in attribution reports	Must explain proxy vs real conversion
Simple daily monitoring	—

26.7 Where freshers go wrong in Chapter 26 topics

Mistake	Why it happens	Avoid it by	Fix it by
15 key events	"Everything is important"	26.1	Trimming
Key event by "once per event" for CTAs	Default	26.2	Once per session

Mistake	Why it happens	Avoid it by	Fix it by
Reporting exits as leads	Similar	26.3	Correct labels

26.8 Thumb rules

> 👍 Few key events, clearly named.
>
> 👍 GA4 sees intent; the backend sees results.
>
> 👍 Conversion = payment, always.

26.9 Notes to remember

> 📝 Primary key event: `exit_to_portal`.
>
> 📝 Leads and payments come from the backend feed.

Chapter 26 Glossary

Term	Simple meaning
Key event	An event marked as important in GA4
Counting method	Once per event or once per session
Proxy metric	A stand-in signal for something harder to measure
Lead rate	Leads ÷ portal exits
Payment rate	Payments ÷ leads

Chapter 26 Quiz

1. What's the primary key event?

- A) `exit_to_portal`
- B) `page_view`
- C) `scroll_depth`
- D) `faq_open`

2. Are payments a GA4 key event here?

- A) Only on mobile
- B) No; they come from the backend feed
- C) Twice
- D) Yes

3. Which counting method suits repeated CTA clicks?

- A) Random
- B) Never
- C) Once per session
- D) Once per event

4. What should GA4 key events not be called in reports?

- A) "Contact actions"
- B) "Portal exits"
- C) "Key events"
- D) "Conversions"

5. What's the payment rate?

- A) Clicks ÷ impressions
- B) Sessions ÷ users
- C) Leads ÷ sessions
- D) Payments ÷ leads

6. Can you create a key event before it first arrives?

- A) No
- B) Only in GTM
- C) Only in BigQuery
- D) Yes, via Admin → Key events → New key event

7. Which should never be a key event?

- A) `chat_open`
- B) `phone_click`
- C) `scroll_depth`
- D) `exit_to_portal`

8. How often should the key-event list be reviewed?

- A) Never
- B) Every 10 years
- C) Quarterly
- D) Hourly

9. How does a closed funnel count repeat completions?

- A) Twice
- B) Never
- C) Once, for the first sequence
- D) Every time

10. In the cricket analogy, key events are...

- A) Every ball's commentary
- B) Runs and wickets on the scoreboard
- C) The pitch
- D) The umpire

Sources

1. Google Analytics Help, Funnel exploration
2. Ongo Weight Loss, Book Four Appendix C and conversion definitions, internal

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Answer key, Chapter 26: 1-A · 2-B · 3-C · 4-D · 5-D · 6-D · 7-C · 8-C · 9-C · 10-B

LEVEL 2 · CHAPTER 27

Chapter 27: Minimal UTMs and Channel Groups

You asked for as few UTMs as possible and more events. This chapter sets exactly where UTMs are needed, where they're harmful, and how a custom channel group keeps reports clean.

> 💡 **Trivia**
>
> GA4 detects organic search, referrals and direct traffic on its own. UTMs are only needed when GA4 can't know the source: links you place in social bios, emails, QR codes and partner sites. 🔵 Expert
>
> Fewer UTMs, used correctly, give cleaner data than many UTMs used carelessly.

27.1 The three-situation rule

Situation	Use	Why
Google/Bing search, other sites linking to us, typed visits	**No UTM**	GA4 classifies automatically
Links **inside** our sites (banners, CTAs, related posts, child site → main site)	**Never UTMs; use events** (`internal_link_click`, `cta_*`)	UTMs on internal links start a new session source and break attribution
Links **we place outside** (social bios and posts, email, QR, partners, press kits)	**UTMs, following the convention**	GA4 can't know the context otherwise

27.2 The UTM convention (from Book Five, extended)

Parameter	Values
`utm_source`	youtube, instagram, facebook, pinterest, linkedin, reddit, threads, newsletter, partner_[name], qr_[place]
`utm_medium`	social, email, qr, partner, referral
`utm_campaign`	`pillar_yyyy_mm` or `event_yyyy`
`utm_content`	`format_topic_v#` or placement (`bio_link`)

Rules: lowercase, underscores, no personal or health data, built in the shared UTM sheet. 🔵

> 🪔 **Desi analogy: courier ka label**
>
> Bahar se aane waale parcel pe label chahiye: kahan se aaya. Lekin ghar ke andar ek kamre se doosre kamre mein chai le jaate waqt koi label nahi lagata. **UTM bahar ke parcel ke liye hai, ghar ke andar ke liye nahi.**

27.3 Internal promotions without UTMs

Need	Event instead
Which homepage banner drove clicks	`cta_*` with `cta_location = hero`
Which related-post block works	`internal_link_click` with `link_position = related`
Child site sending traffic to main site	Cross-domain (Chapter 23) + `site_id`

27.4 Custom channel group

GA4 → Admin → **Channel groups** → copy the default group → name it "OWL channels" (OWL = Ongo Weight Loss) → adjust:

Channel	Rule (examples)
Organic Search	Default rule

Channel	Rule (examples)
Organic Social	Default rule **or** medium = social
Email	medium = email
QR	medium = qr
Partner	medium = partner
Child Sites	session source matches child domains (if any journeys start there without cross-domain)
AI Assistants	source matches chatgpt.com, perplexity.ai, copilot.microsoft.com, gemini.google.com (check referrer names)
Referral	Default
Direct	Default

Why an AI channel: AI assistants increasingly send referral visits; grouping them shows the trend. 🔵

27.5 Auditing UTMs

Check (monthly)	How
Internal UTMs	Explore: session source/medium where page_referrer is our domain and utm present
Inconsistent spellings	Source/medium report: "Instagram" vs "instagram"
UTMs with personal data	Search campaign/content for "@" or digits patterns

> 🟧 **Ongo Weight Loss example**
>
> You said you want the least UTMs. With this rule, the only UTMs live in social bios/posts, newsletters, QR codes and partner links: perhaps a few dozen in a shared sheet. Everything on our own sites is measured with events. 🔵

27.6 Do's and Don'ts

✅ Do	❌ Don't
UTMs only on links we place outside	UTMs on internal links
Follow the convention	Mix capitalisation
Use events for internal promotions	Use UTMs to "track banners"
Add an AI assistants channel	Lump AI traffic into referral forever

27.7 Pros and cons of minimal UTMs

✅ Pros	❌ Cons
Clean attribution	Some off-site links may be missed if not tagged
Less admin	Requires discipline in social/email teams
Fewer errors	—

27.8 Where freshers go wrong in Chapter 27 topics

Mistake	Why it happens	Avoid it by	Fix it by
UTMs on child-site links to main site	Wanting to see the source	Cross-domain	Removing UTMs
"Instagram" and "instagram" both appear	Manual typing	UTM sheet	Lowercase rule
No UTM on newsletter links	Forgetting	Email checklist	Adding

27.9 Thumb rules

👍 Label parcels from outside, not chai inside.

👍 Internal = events; external we place = UTMs.

👍 One sheet, one spelling.

27.10 Notes to remember

📝 Custom channel group "OWL channels", including AI assistants.

📝 Monthly UTM audit.

Chapter 27 Glossary

Term	Simple meaning
UTM	Tags on a link telling analytics its source
Channel group	Rules grouping traffic into channels
Internal promotion	A banner or link promoting our own pages
Self-referral	Our own site appearing as a traffic source
AI assistant referral	Traffic from AI chat tools

Chapter 27 Quiz

1. Should internal links have UTMs?

- A) Never
- B) Sometimes
- C) Only on blogs
- D) Always

2. Where are UTMs needed?

- A) Organic search
- B) Direct visits
- C) Every internal link
- D) Links we place outside our sites

3. What replaces UTMs for homepage banners?

- A) CTA events with cta_location
- B) Clarity
- C) GSC
- D) More UTMs

4. Which medium for newsletters?

- A) social
- B) cpc
- C) direct
- D) email

5. Why add an AI assistants channel?

- A) It's required
- B) To see referral trends from AI tools
- C) For ads
- D) To block AI

6. Should UTM values be lowercase?

- A) Uppercase
- B) No
- C) Yes
- D) Mixed

7. How do child-site visits stay in the journey?

- A) Cross-domain measurement
- B) UTMs
- C) New properties
- D) Clarity

8. What's checked in the monthly UTM audit?

- A) Internal UTMs, inconsistent spelling, personal data
- B) Logo
- C) Page speed
- D) Fonts

9. Where are UTMs built?

- A) Memory
- B) The shared UTM sheet
- C) In emails by hand
- D) Random tools

10. In the courier analogy, UTMs are for...

- A) The kitchen
- B) Parcels from outside, not chai inside the house
- C) Every cup of chai
- D) Nothing

Sources

1. Ongo Weight Loss, Book Five Chapter 40 (UTM convention), internal 🔵
2. Google Analytics Help, Cross-domain measurement 🟢

Answer key, Chapter 27: 1-A · 2-D · 3-A · 4-D · 5-B · 6-C · 7-A · 8-A · 9-B · 10-B

LEVEL 2 · CHAPTER 28

Chapter 28: Microsoft Clarity: Setup and What It Gives You

GA4 tells you what happened. Clarity shows how it happened: where people clicked, how far they scrolled, where they got stuck. This chapter sets Clarity up safely for a health site.

> 💡 **Trivia**
>
> In Clarity, input boxes and dropdowns are masked in **all** modes, and you can mask any element by adding `data-clarity-mask="True"`. Masking changes "can't be applied retroactively." [1] 🟢 Official (Microsoft)
>
> Set masking right before recording, because old recordings keep whatever they captured.

28.1 What Clarity gives you

Feature	What it shows	Use
Heatmaps (click, scroll, area)	Where people click and how far they scroll	Find ignored CTAs, misleading non-clickable elements
Session recordings	Masked replays of visits	Understand confusion and drop-offs
Rage clicks	Repeated clicks in frustration	Broken buttons
Dead clicks	Clicks on non-clickable elements	Elements that look like links
Quick backs	Leaving and returning fast	Mismatched pages
Excessive scrolling	Hunting for information	Answer not near the top
Dashboard and segments	Device, page, source filters	Spot problem pages

Feature	What it shows	Use
GA4 integration	Clarity sessions linked to GA4 segments	Watch sessions from a GA4 segment

28.2 Safe setup (step by step)

Step	Action
1	clarity.microsoft.com → New project → company account; site URL
2	**Settings → Masking → Strict** (masks all text by default)
3	Unmask only safe UI chrome if needed (`data-clarity-unmask="true"` on nav/headers)
4	**Install via GTM**: Clarity's GTM template or custom HTML tag in folder 09; trigger **All Pages** with an **exception**: `privacy_zone` = B (if legal says no) or page_type in (eligibility_start, contact, tool)
5	Consent: require `analytics_storage` (or the CMP's analytics category); use Clarity's consent signal per its docs
6	**Never** add Clarity to the portal or any Zone C page
7	Settings → **IP blocking** for internal traffic
8	Settings → Setup → **Google Analytics integration** → connect GA4
9	Test with your own visit: open a recording and confirm text is masked
10	Record in the change log

```
<!-- Mask any area that could show personal info -->
<section data-clarity-mask="True"> … </section>
```

> **Desi analogy: CCTV with dhundla shisha**
>
> Dukaan mein CCTV lagta hai taaki pata chale log kahan atakte hain, lekin cash counter aur trial room par dhundla shisha hota hai. **Clarity CCTV hai; Strict masking dhundla shisha.**

28.3 What Clarity is NOT for

❌	Why
Portal, intake, eligibility questions, payment	Health data; no BAA
Identifying individuals	Recordings must stay anonymous
Exact counts	Use GA4/BigQuery for numbers

28.4 How to use Clarity weekly

Task	Time
Check rage/dead clicks on top 10 pages	10 min
Watch 5 recordings from sessions that clicked a CTA but didn't exit to portal	15 min
Scroll heatmap on new articles	10 min
Note issues → tickets for design/dev	10 min

> 🟧 **Ongo Weight Loss example**
>
> Book Four's audit noted Clarity on the site and flagged masking and page scope as the items closest to the risk line. Week 1 action: set Strict masking, exclude forms/tools/eligibility pages, confirm it's absent from the portal, and log the change. 🔵

28.5 Do's and Don'ts

✅ Do	❌ Don't
Use Strict masking	Leave Balanced default
Exclude Zone B forms and all Zone C	Install site-wide without exceptions
Link to GA4	Use Clarity numbers as official counts
Turn findings into tickets	Watch recordings without action

28.6 Pros and cons

✅ Pros	❌ Cons
Free, powerful UX insight	No BAA
Explains GA4 drop-offs	Can't be used on sensitive pages
Easy setup	Recordings need careful masking

28.7 Where freshers go wrong in Chapter 28 topics

Mistake	Why it happens	Avoid it by	Fix it by
Balanced masking left on	Default	Step 2	Switching to Strict
Clarity on contact form	Site-wide tag	Step 4 exception	Excluding
Thinking old recordings get masked	Assumption	Trivia	Deleting risky recordings per legal

28.8 Thumb rules

> 👍 CCTV yes, dhundla shisha always.
>
> 👍 Recordings explain, GA4 counts.
>
> 👍 Never in the portal.

28.9 Notes to remember

> 📝 Strict masking; `data-clarity-mask="True"`.
>
> 📝 Masking isn't retroactive.

Chapter 28 Glossary

Term	Simple meaning
Heatmap	Visual of clicks or scrolls on a page
Session recording	Masked replay of a visit
Rage click	Repeated frustrated clicks
Dead click	Click on something not clickable
Quick back	Leaving a page and quickly returning

Chapter 28 Quiz

1. Which masking mode should we use?

- A) Relaxed
- B) Strict
- C) None
- D) Balanced

2. Are masking changes retroactive?

- A) No
- B) After a day
- C) Yes
- D) Only on mobile

3. Where must Clarity never run?

- A) Blog articles
- B) The portal and Zone C pages
- C) About page
- D) The homepage

4. What's a dead click?

- A) A click on a non-clickable element
- B) A video play
- C) A scroll
- D) A purchase

5. Which attribute masks an element?

- A) `class="mask"`
- B) `aria-hidden`
- C) `data-clarity-mask="True"`
- D) `data-hide`

6. What's Clarity's GA4 integration for?

- A) Ads
- B) Replacing GA4
- C) Linking Clarity sessions to GA4 segments
- D) Sending PHI

7. Should Clarity numbers be official counts?

- A) Only on Fridays
- B) Only for payments
- C) Yes
- D) No; use GA4/BigQuery

8. How is Clarity installed?

- A) Through GSC
- B) Not at all
- C) Hard-coded everywhere including portal
- D) Via GTM with exceptions and consent

9. What's a weekly Clarity task?

- A) Change prices
- B) Watch sessions that clicked a CTA but didn't exit to portal
- C) Delete GA4
- D) Identify patients

10. In the CCTV analogy, Strict masking is...

- A) Frosted glass over sensitive areas
- B) A brighter camera
- C) A cashier
- D) No camera

Sources

1. Microsoft Learn, Clarity masking 🟢
2. Ongo Weight Loss, Book Four Chapter 44 (Clarity risk note), internal 🔵

Answer key, Chapter 28: 1-B · 2-A · 3-B · 4-A · 5-C · 6-C · 7-D · 8-D · 9-B · 10-A

LEVEL 2 · CHAPTER 29

Chapter 29: Consent Implementation in GTM (Step by Step)

Chapter 5 chose the approach. This chapter builds it: the CMP template, default and update states, consent checks on every tag, Global Privacy Control, and how to prove it works.

> 💡 **Trivia**
>
> In Consent Mode's **advanced** mode, Google tags load with consent denied and send measurements **without cookies**; only when consent is granted do they send full data. In **basic** mode, tags don't load until the visitor interacts with the banner. [1] ● Official (Google)
>
> Basic is simpler and stricter; it's our default unless legal chooses advanced.

29.1 Build sequence

#	Where	Action
1	CMP dashboard	Add all public domains; set banner text (legal-approved); categories: Necessary, Analytics, Functional, Marketing (Marketing unused for now)
2	CMP dashboard	Enable **Google Consent Mode v2** support; enable **GPC** honouring for applicable states
3	GTM → Templates → Search Gallery	Add the **CMP's official template** (CookieYes or Cookiebot)
4	GTM → Tags → New	Tag from template; enter CMP ID; **Default consent**: per legal (e.g., analytics denied until choice); region rules if legal wants state-based defaults
5	Trigger	**Consent Initialization – All Pages**

#	Where	Action
6	GTM → Admin → Container settings	Tick **Enable consent overview**
7	Each GA4 tag	Advanced settings → Consent settings → **Require additional consent**: `analytics_storage`
8	Clarity tag	Require `analytics_storage` (or CMP analytics category)
9	Optional	Push `consent_update` event for reporting consent rates (no personal data)
10	Remove	Any inline consent default in Next.js if the template sets defaults (avoid conflicts; Chapter 13)

29.2 Region-specific defaults (if legal wants)

Region	Default `analytics_storage`	Notes
States legal flags as opt-in for health data	denied	Banner asks first
Other U.S. states	granted with opt-out (if legal approves)	Honour GPC → denied
Outside U.S.	denied	Rare traffic; safe default

Templates usually support region codes (e.g., `US-WA`). 🔵

> 🪔 **Desi analogy: ghar ke darwaaze pe pehle poochna**
>
> Achhe mehmaan ghar mein ghusne se pehle poochte hain "Andar aa sakte hain?" Kuch gharon mein seedha aa sakte ho, kuch mein pehle jootey bahar. **Region defaults har ghar ka niyam hain.**

29.3 Testing consent

Test	Expected
Fresh incognito visit, no choice yet (opt-in region)	GA4 and Clarity **not** sending (basic)

Test	Expected
Accept analytics	GA4 `page_view` sent; Clarity loads
Reject analytics	No GA4/Clarity requests
Browser with GPC on (opt-out region)	Treated as opt-out
Change choice later	Tags stop/start accordingly
GTM Preview → Consent tab	Shows default and update states in order

29.4 Measuring consent impact

Metric	How
Consent rate	(if `consent_update` events are sent after consent) not measurable for refusals in basic mode; use CMP's own dashboard
Data gap	Compare backend portal starts vs GA4 `exit_to_portal` trend

> 🟧 **Ongo Weight Loss example**
>
> Main site, weightclinicsresults.com and every child site share one CMP account and one GTM container, so one banner configuration and one set of consent checks cover all public sites at once. 🔵

29.5 Do's and Don'ts

✅ Do	❌ Don't
Use the CMP's official GTM template	Hand-code consent logic
Add consent checks to every tag	Forget Clarity
Test reject paths, not just accept	Test only "Accept all"
Keep banner text legal-approved	Write your own legal text

29.6 Pros and cons of basic mode

✅ Pros	❌ Cons
Strictest: nothing before choice	No modelling of declined users
Simple to explain	Lower measured traffic
Lower legal risk	—

29.7 Where freshers go wrong in Chapter 29 topics

Mistake	Why it happens	Avoid it by	Fix it by
GA4 fires before choice	Missing consent checks	Step 7	Adding checks
Two defaults conflict	Inline + template	Step 10	Removing one
Not testing GPC	Unknown feature	29.3	Testing

29.8 Thumb rules

👍 Ask, then measure.

👍 Every tag checks consent.

👍 Test reject as hard as accept.

29.9 Notes to remember

📝 CMP template on Consent Initialization; consent overview on.

📝 Basic mode by default; advanced only with legal approval.

Chapter 29 Glossary

Term	Simple meaning
Default consent	The state before a visitor chooses
Consent update	The state after a choice
Consent overview	GTM screen showing each tag's consent settings
Region default	Different defaults by location
Template	A ready-made GTM tag from the gallery

Chapter 29 Quiz

1. Which trigger does the CMP tag use?

- A) All Pages
- B) Click
- C) Consent Initialization – All Pages
- D) Timer

2. What's our default consent mode?

- A) Basic (unless legal chooses advanced)
- B) Random
- C) None
- D) Advanced always

3. What must every GA4 tag require?

- A) `ad_storage`
- B) `analytics_storage` consent
- C) Nothing
- D) `security_storage`

4. Where do you see consent states in testing?

- A) Looker Studio
- B) Bing

- C) GSC
- D) GTM Preview → Consent tab

5. What should happen when analytics is rejected?

- A) Only page views
- B) Only Clarity
- C) No GA4/Clarity requests
- D) Full data sent

6. What's GPC?

- A) A GA4 report
- B) A CMS
- C) A cookie banner design
- D) A browser opt-out signal to honour

7. What causes conflicting defaults?

- A) Images
- B) Inline defaults plus template defaults
- C) Too many sites
- D) Fonts

8. Which tag also needs consent checks?

- A) None
- B) Only GA4
- C) Only CMP
- D) Clarity

9. How many CMP setups for all public sites?

- A) None
- B) One per visitor
- C) One shared account and container
- D) One per page

10. In the ghar analogy, region defaults are…

- A) Each house's rule for entering
- B) Shoes

- C) Guests
- D) Doors

Sources

1. Google, Consent mode overview
2. Google Tag Manager Help, Consent mode and CMPs

Answer key, Chapter 29: 1-C · 2-A · 3-B · 4-D · 5-C · 6-D · 7-B · 8-D · 9-C · 10-A

LEVEL 2 · CHAPTER 30

Chapter 30: QA, Debugging and the Release Checklist

Tracking breaks quietly: a renamed button, a new template, a route that skips `PageMeta`. This chapter gives you the tools and a repeatable checklist to catch problems before they reach your reports.

> 💡 **Trivia**
>
> GA4's user explorer lets Editors request deletion of a user's data; it stops appearing within 24 hours and is permanently removed within 63 days. [1] ● Official (Google)
>
> Deletion exists for emergencies. The real goal is never collecting the wrong data in the first place.

30.1 QA tools

Tool	Use
GTM Preview (Tag Assistant)	See which tags fired on each event, with variable values and consent state
GA4 DebugView	See events arriving in GA4 in real time (debug mode on)
Browser DevTools → Network	Filter `collect?v=2` (GA4), `clarity.ms`; inspect parameters
Omnibug (extension)	Decodes analytics requests in a readable list
dataLayer inspector (extension or console `dataLayer`)	See pushes in order
GA4 Realtime	Rough check after publish

30.2 The QA script (run for every release)

#	Step	Pass
1	Open Preview on staging	Container connects
2	Load home	1 page_view, correct metadata
3	Navigate to a blog article (client-side)	1 page_view; funnel_stage correct
4	Back / forward / refresh	1 page_view each
5	Click a body internal link to a money page	internal_link_click with correct params
6	Click a hero CTA	cta_*; no internal_link_click
7	Click a portal CTA	cta_* + exit_to_portal; no `_gl` to portal
8	Phone / chat / email	Correct events, no numbers/emails in params
9	Scroll an article	scroll_depth 25/50/75/90 once each
10	Open an FAQ, play a video	faq_open, video_*
11	Use a tool	tool_start/complete; **no values**
12	Visit a 404	not_found page_view + page_not_found (path only)
13	Child site → main site	Same session; `_gl` present
14	Consent: reject	No GA4/Clarity requests
15	Personal-data scan	Search all hits for `@`, phone patterns, `email=`, `name=` → none
16	Portal pages	No GTM/GA4/Clarity requests

> 🪔 **Desi analogy: shaadi se pehle rehearsal**
>
> Shaadi se ek din pehle stage, mic, light, band sab check hota hai. Asli din kuch nahi bigadta. **QA script rehearsal hai.**

30.3 Debugging common problems

Symptom	Likely cause	Fix
Two page_views per navigation	History-change page views on	GA4 setting off (Chapter 15)
No page_view after navigation	`PageMeta` missing on template	Add component; build test
CTA event missing on icon clicks	Click Element is child	`closest()` variable
Parameters "(not set)" in reports	Not registered / typo	Custom definitions (Chapter 25)
Child site appears as referral	Domain not configured	Configure your domains
Data appears hours later	Normal processing time	Use DebugView/Realtime for tests
Values with personal data	Raw URL/Click URL used	Clean variables; redact; legal review

30.4 Monitoring after release

Check	When
GA4 Realtime: page_view and key events present	Right after publish
Daily dashboard anomaly (Chapter 56)	Next morning
Event count comparison (BigQuery) vs previous week	Weekly
Personal-data scan in BigQuery	Weekly

30.5 If personal data was collected

Step	Action
1	Stop the source (pause tag / fix code) immediately
2	Tell team lead and legal the same day
3	Use GA4 **data deletion requests** (Admin) for affected parameters/fields as advised
4	Delete affected BigQuery rows/tables per legal

Step	Action
5	Log the incident and the fix

> 🟧 **Ongo Weight Loss example**
>
> Make steps 15 and 16 (personal-data scan and portal check) mandatory for **every** website release, not just tracking changes. A new form or template can leak data even when nobody touched GTM. 🔵

30.6 Do's and Don'ts

✅ Do	❌ Don't
Run the full script each release	Test only the new event
Test on staging first	Debug on live with real users
Scan for personal data every time	Assume it's fine
Log results	Rely on memory

30.7 Pros and cons of formal QA

✅ Pros	❌ Cons
Catches silent breaks	Takes 30–45 minutes
Protects privacy	Needs staging environment
Builds trust in data	—

30.8 Where freshers go wrong in Chapter 30 topics

Mistake	Why it happens	Avoid it by	Fix it by
Testing only first load	Forgetting SPA	Steps 3–4	Adding navigation tests
Publishing without Preview	Hurry	Change control	Rolling back version
Not checking the portal	Out of sight	Step 16	Adding check

30.9 Thumb rules

👍 Rehearse before the wedding.

👍 Every release, every check.

👍 Personal-data scan is never optional.

30.10 Notes to remember

📝 16-step QA script (Appendix K).

📝 GA4 deletion: hidden within 24 hours, removed within 63 days.

Chapter 30 Glossary

Term	Simple meaning
Preview mode	GTM's testing mode
DebugView	GA4 real-time debug screen
Hit	A single request sent to GA4
Regression	Something that used to work breaking
Data deletion request	GA4 feature to remove collected data

Chapter 30 Quiz

1. Which tool shows tags fired per event?

- A) GTM Preview
- B) Bing
- C) GSC
- D) Looker Studio

2. What's the GA4 request filter in Network?

- A) `gtm.js` only
- B) `robots.txt`
- C) `collect?v=2`
- D) `favicon`

3. What causes two page_views per navigation?

- A) History-change page views still on
- B) Consent
- C) Fonts
- D) Too many links

4. What must be checked on portal pages?

- A) No GTM/GA4/Clarity requests
- B) Full tracking
- C) Nothing
- D) Clarity only

5. When should the personal-data scan run?

- A) On complaints
- B) Never
- C) Every release
- D) Once a year

6. What's step 1 if personal data was collected?

- A) Publish more
- B) Wait a month

- C) Ignore
- D) Stop the source immediately

7. How long until deleted GA4 data stops appearing?

- A) Within 24 hours
- B) 5 minutes
- C) Never
- D) 1 year

8. Where should testing happen first?

- A) Portal
- B) Live
- C) GSC
- D) Staging

9. What does Omnibug do?

- A) Sends emails
- B) Writes code
- C) Decodes analytics requests
- D) Builds dashboards

10. In the rehearsal analogy, the QA script is…

- A) The wedding
- B) The food
- C) The rehearsal before the wedding
- D) The photos

Sources

1. Google Analytics Help, User explorer (data deletion) 🟢
2. Ongo Weight Loss, release process, internal 🔵

Answer key, Chapter 30: 1-A · 2-C · 3-A · 4-A · 5-C · 6-D · 7-A · 8-D · 9-C · 10-C

03

LEVEL 3

Data Platforms

BigQuery from zero, the GA4 export, SQL and the query library, Search Console and bulk export, Bing, Core Web Vitals, social data and benchmarking.

31 · BigQuery 101: What It Is and Why You Need It

32 · Setting Up GA4 → BigQuery Export (Click by Click)

33 · Understanding the GA4 BigQuery Tables

34 · SQL Basics for Marketers (From Zero)

35 · The Ongo Weight Loss Query Library (Paths, Funnels, Attribution)

36 · Google Search Console: Setup and Every Report Explained

37 · Search Console Bulk Export to BigQuery

38 · Joining Search Console and GA4 (Before and After the Click)

39 · Bing Webmaster Tools, Step by Step

40 · Core Web Vitals and CrUX (What Replaced the CrUX Connector)

41 · Social Platform Data in Your Reporting

42 · Competitor Benchmarking

LEVEL 3 · CHAPTER 31

Chapter 31: BigQuery 101: What It Is and Why You Need It

BigQuery is Google's data warehouse. GA4 and Search Console can both copy their raw data into it every day. Once there, you can see every visitor's full path, keep data forever, and power fast dashboards.

> 💡 **Trivia**
>
> The BigQuery sandbox needs no credit card. It gives **10 GiB** of storage (a lifetime limit not refunded when you delete data) and **1 TiB** of query processing per month, but tables expire after **60 days**. [1] ● Official (Google Cloud)
>
> The sandbox is perfect for learning; the 60-day expiry is why production needs billing turned on.

31.1 What BigQuery is, simply

Concept	Plain meaning
Project	Your Google Cloud workspace (billing, permissions)
Dataset	A folder of tables (e.g., `analytics_123456789` for GA4)
Table	Rows and columns (e.g., `events_20261103`)
Query	A SQL question you ask the data
Bytes processed	What you pay for when querying
Storage	What you pay for keeping data

31.2 What BigQuery lets you do that GA4 can't

Need	GA4 UI	BigQuery
Every user's full path, entry → exit	Top paths only (≤20 nodes per step)	✅ Every session
Data older than 14 months for explorations	❌	✅ Forever (with billing)
Unsampled, no thresholds	Sometimes limited	✅ Raw rows
First landing page → leads/payments by channel (with backend table)	❌	✅ Join tables
Search Console data beyond 16 months, more query rows	❌	✅ (bulk export)
Fast Looker dashboards without GA4 quota errors	Sometimes slow	✅

31.3 Costs (plain English, check current pricing)

Item	Free amount	Beyond free
Query processing	1 TiB per month	Pay per TiB scanned (on-demand)
Storage	10 GiB (sandbox lifetime; billing-enabled projects also have a free tier)	Pay per GiB per month
GA4 daily export	Free to export	Storage/query costs apply
GA4 streaming export	—	Extra charge ($0.05 per GB, ~600,000 events per GB) [2] 🟢

At our expected size, costs should stay very low; set **budgets and alerts** anyway (Chapter 32). 🔵

> 🪔 **Desi analogy: godown aur dukaan**
>
> Dukaan (GA4) mein shelf pe sirf zaroori maal dikhta hai. Peeche godown (BigQuery) mein har boru, har packet rakha hai. Jab poora hisaab chahiye, godown jaate ho. **BigQuery godown hai.**

31.4 Sandbox vs billing-enabled

	Sandbox	Billing enabled
Card needed	No	Yes
Table expiry	60 days	None (you control)
Streaming export	❌	✅
Good for	Learning, first tests	Real history and dashboards

Recommendation: start in sandbox for a week to learn, then enable billing with a **budget alert** (e.g., $25/month) before relying on it. 🔵

31.5 Datasets we'll create

Dataset	Source	Chapter
`analytics_<property_id>`	GA4 export (auto-created)	32
`searchconsole_ongo`	GSC bulk export	37
`ops`	Daily leads/payments (aggregated, Chapter 24)	24
`reporting`	Our saved views/tables for Looker	35, 58
`crux` (optional)	Public CrUX dataset queries	40

> 🟧 **Ongo Weight Loss example**
>
> Your wish to "track each user from entry to exit, page by page" is fully possible only in BigQuery. GA4's screens show samples of journeys; BigQuery holds every one, ready for the path and funnel queries in Chapter 35. 🔵

31.6 Do's and Don'ts

✅ Do	❌ Don't
Learn in the sandbox	Rely on sandbox tables (60-day expiry)
Enable billing with a budget alert	Enable billing without alerts
Keep datasets organised	Dump everything in one dataset
Use the same region for all datasets	Mix regions (joins fail)

31.7 Pros and cons of BigQuery

✅ Pros	❌ Cons
Every event, forever	Requires SQL
Joins GA4, GSC and backend tables	Small costs
Fast dashboards	Setup steps

31.8 Where freshers go wrong in Chapter 31 topics

Mistake	Why it happens	Avoid it by	Fix it by
Losing data after 60 days	Staying in sandbox	Enable billing	Re-export not possible; act early
Datasets in different regions	Defaults	Choose one region	Recreate datasets
Huge query bills	`SELECT *` on all tables	Chapter 34 habits	Partition filters

31.9 Thumb rules

👍 GA4 is the shop; BigQuery is the godown.

👍 Sandbox to learn, billing to keep.

👍 One region for everything.

31.10 Notes to remember

📝 Sandbox: 10 GiB storage, 1 TiB queries/month, 60-day expiry.

📝 Datasets: analytics_, searchconsole_, ops, reporting.

Chapter 31 Glossary

Term	Simple meaning
BigQuery	Google's cloud data warehouse
Dataset	A folder of tables
Sandbox	Free BigQuery mode with limits
Bytes processed	Amount of data a query reads (cost driver)
Region	Where data is stored geographically

Chapter 31 Quiz

1. When do sandbox tables expire?

- A) After 1 day
- B) After 10 years
- C) After 60 days
- D) Never

2. How much query processing is free per month?

- A) 100 TiB
- B) 1 GB
- C) 1 TiB
- D) None

3. What can BigQuery show that GA4 UI can't?

- A) Every session's full path
- B) Fonts
- C) Logos
- D) Ads

4. What drives query cost?

- A) Users
- B) Colours
- C) Number of charts
- D) Bytes processed

5. What's the recommendation for billing?

- A) Use personal card
- B) Enable with a budget alert after learning
- C) Enable without alerts
- D) Never enable

6. Why keep datasets in the same region?

- A) It's prettier
- B) No reason
- C) For SEO
- D) Joins across regions fail

7. Which dataset holds daily leads/payments?

- A) `crux`
- B) `ops`
- C) `analytics_…`
- D) None

8. What's the streaming export extra cost?

- A) $0.05 per GB
- B) $100 per event
- C) Unknown
- D) Free

9. Is sandbox storage refunded when you delete data?

- A) Yes
- B) No
- C) Monthly
- D) Partly

10. In the godown analogy, BigQuery is…

- A) The signboard
- B) The shop shelf
- C) The warehouse with every item
- D) The cash counter

Sources

1. Google Cloud, BigQuery sandbox ●
2. Google Analytics Help, BigQuery Export ●

Answer key, Chapter 31: 1-C · 2-C · 3-A · 4-D · 5-B · 6-D · 7-B · 8-A · 9-B · 10-C

LEVEL 3 · CHAPTER 32

Chapter 32: Setting Up GA4 → BigQuery Export (Click by Click)

This is the tutorial you asked for: from an empty Google Cloud account to GA4 data arriving daily in BigQuery, with costs under control.

> 💡 **Trivia**
>
> GA4's daily export "Exports all the raw, unsampled event data once per day from the previous day," typically arriving by mid-afternoon in the property's time zone. Standard properties can export up to **1 million events per day**. [1] 🟢 Official (Google)
>
> Once exported, data can't be re-exported, so set it up early.

32.1 Before you start

Need	Detail
Google account	Company account (not personal)
GA4 role	**Editor** or above on the property
Cloud role	**Owner** on the Google Cloud project
Decision	Data **location** (e.g., `US` multi-region); can't easily change later

32.2 Step 1: create the Google Cloud project

#	Click
1	Go to console.cloud.google.com → sign in with the company account
2	Top bar project picker → **New project** → name `ongo-analytics` → Create

#	Click
3	Note the **Project ID** (e.g., `ongo-analytics-412345`)
4	Menu → **APIs & Services → Library** → search "BigQuery API" → **Enable**
5	(For GSC export later) also enable **BigQuery Storage API**

32.3 Step 2: billing and budget (recommended)

#	Click
1	Menu → **Billing** → link a company billing account (or stay in sandbox for testing)
2	Billing → **Budgets & alerts** → Create budget → scope: project `ongo-analytics` → amount e.g. $25/month → alerts at 50%, 90%, 100% → email team lead
3	BigQuery → **Admin → Quotas** (optional): set a custom daily query quota per user to prevent accidental huge queries

> 🍛 **Desi analogy: bijli ka meter**
>
> Naye ghar mein bijli lagwate hi meter aur MCB lagate ho, taaki koi galti se poora bill na badha de. **Budget alert aur quota wahi MCB hain.**

32.4 Step 3: link GA4 to BigQuery

#	Click
1	GA4 → **Admin** → Product links → **BigQuery links** → **Link**
2	**Choose a BigQuery project** → select `ongo-analytics` → Confirm
3	**Data location** → choose (same as other datasets, e.g., US)
4	**Configure data streams and events** → select the Public sites web stream; optionally **exclude** noisy events (e.g., `scroll_depth` if volume becomes an issue)
5	Frequency: **Daily** ✅; **Streaming** only if billing is on and you need same-day data (note: streaming lacks user-attribution fields for new users)

#	Click
6	Review → **Submit**
7	Open the link details and check the built-in identity has the **BigQuery User** role (older links used `firebase-measurement@system.gserviceaccount.com`)

32.5 Step 4: confirm data arrives

When	Check
Within ~24 hours	BigQuery → project → dataset `analytics_<property_id>` appears
Next day	Table `events_YYYYMMDD` for yesterday
If streaming	`events_intraday_YYYYMMDD` during the day

Test query (cheap: one day):

```
SELECT event_name, COUNT(*) AS events
FROM `ongo-analytics-412345.analytics_123456789.events_20261103`
GROUP BY event_name
ORDER BY events DESC;
```

32.6 Step 5: access for the team

Person	Role on the project
Team lead	Owner
On-page SEO / analytics	BigQuery Data Editor + BigQuery Job User
Viewers (Looker)	BigQuery Data Viewer + BigQuery Job User
Backend loader (Chapter 24)	Service account with Data Editor on `ops` dataset only

32.7 Step 6: protect against mistakes

Protection	How
Budget alert	Done in 32.3
Partition/date filters in every query	Use `_TABLE_SUFFIX BETWEEN …` (Chapter 34)
Saved views for Looker	Looker reads small views, not raw tables (Chapter 58)
Personal-data scan query	Weekly (Chapter 35)

> 🟧 **Ongo Weight Loss example**
>
> Because no tracking is live yet, link BigQuery **in the same week** you publish the new GTM setup. Day one of clean tracking becomes day one of your BigQuery history: no gaps, no messy early data. 🔵

32.8 Do's and Don'ts

✅ Do	❌ Don't
Use a company project and account	Link to a personal project
Set data location carefully	Change location later (needs relink)
Start with daily export	Turn on streaming without need
Set budgets and quotas	Leave costs unwatched

32.9 Pros and cons of daily vs streaming

	Daily	Streaming
✅	Complete, includes attribution fields; free to export	Near real-time

	Daily	Streaming
❌	Next-day data	Extra cost; missing user-attribution fields for new users

32.10 Where freshers go wrong in Chapter 32 topics

Mistake	Why it happens	Avoid it by	Fix it by
"Link" button greyed out	Missing Owner on project	32.1 roles	Granting Owner
No tables after a day	Link to wrong project	Check link details	Relink
Query scans months of data	No date filter	32.7	Add `_TABLE_SUFFIX`

32.11 Thumb rules

👍 Link early; exports can't go back in time.

👍 Meter and MCB before switching on.

👍 One day first, then wider.

32.12 Notes to remember

📝 GA4 Admin → Product links → BigQuery links → Link.

📝 Daily export: previous day, mid-afternoon; 1M events/day limit.

Chapter 32 Glossary

Term	Simple meaning
Project ID	Unique name of a Cloud project

Term	Simple meaning
Daily export	Once-a-day full table of yesterday's events
Streaming export	Near-real-time intraday table
Budget alert	Email when spending reaches a threshold
Service account	Non-human account for programs

Chapter 32 Quiz

1. Which GA4 role is needed to link BigQuery?

- A) Editor or above
- B) Analyst only
- C) None
- D) Viewer

2. Which Cloud role is needed?

- A) Billing only
- B) Owner on the project
- C) None
- D) Viewer

3. Where is the link created in GA4?

- A) GTM
- B) Admin → Product links → BigQuery links
- C) Explore
- D) Reports → Engagement

4. What's the daily table name pattern?

- A) `sessions_*`
- B) `pages_daily`
- C) `ga4_table`
- D) `events_YYYYMMDD`

5. What does streaming lack for new users?

- A) Timestamps
- B) Page paths
- C) Event names
- D) User-attribution fields

6. Why set a budget alert?

- A) It's decorative
- B) Required by GA4
- C) To catch unexpected costs
- D) For SEO

7. Can exported data be re-exported?

- A) Yes, anytime
- B) No
- C) Monthly
- D) On request

8. Which API must be enabled?

- A) Maps API
- B) BigQuery API
- C) Gmail API
- D) YouTube API

9. When should BigQuery be linked?

- A) After launch problems
- B) The same week new tracking goes live
- C) After a year
- D) Never

10. In the bijli analogy, budgets and quotas are…

- A) The meter and MCB
- B) The bulbs
- C) The switches
- D) The wires

Sources

1. Google Analytics Help, BigQuery Export
2. Google Analytics Help, Set up BigQuery Export

Answer key, Chapter 32: 1-A · 2-B · 3-B · 4-D · 5-D · 6-C · 7-B · 8-B · 9-B · 10-A

LEVEL 3 · CHAPTER 33

Chapter 33: Understanding the GA4 BigQuery Tables

GA4's export looks strange at first: one row per event, with parameters packed inside nested lists. Once you see the shape, every query becomes a pattern you can reuse.

> 💡 **Trivia**
>
> GA4 streaming export writes to `events_intraday_YYYYMMDD`, and Google recommends querying the daily `events_YYYYMMDD` tables for a stable dataset. [1] ● Official (Google)
>
> Intraday is a draft; daily is the final copy.

33.1 One row = one event

Column	Type	Example	Meaning
`event_date`	STRING	`20261103`	Date in property time zone
`event_timestamp`	INTEGER	microseconds since 1970	Exact time
`event_name`	STRING	`page_view`	The event
`event_params`	REPEATED RECORD	key/value list	All parameters
`user_pseudo_id`	STRING	`123.456`	Client ID (browser)
`user_properties`	REPEATED RECORD	—	User-scoped values (if any)
`traffic_source`	RECORD	source, medium, name	**First-user** acquisition

Column	Type	Example	Meaning
`collected_traffic_source`	RECORD	manual_source, manual_medium, manual_campaign_name...	UTMs/collected on the event
`session_traffic_source_last_click`	RECORD	—	Session attribution (newer field; check availability)
`device`	RECORD	category, browser, os	Device info
`geo`	RECORD	country, region, city	Location (legal may restrict use)
`stream_id` , `platform`	STRING	—	Stream info

33.2 Reading `event_params`

`event_params` is a list like:

```
[ {key: 'page_location', value: {string_value: 'https://…/blog/x/'}},
  {key: 'ga_session_id', value: {int_value: 1730650000}},
  {key: 'page_type',     value: {string_value: 'blog_article'}} ]
```

To read one parameter, use a subquery:

```
(SELECT value.string_value FROM UNNEST(event_params) WHERE key = 'page_type') AS
page_type,
(SELECT value.int_value    FROM UNNEST(event_params) WHERE key = 'ga_session_id') AS
ga_session_id
```

Tip: numbers may arrive as `int_value` , `float_value` or `double_value` ; use `COALESCE` when unsure. 🔵

> 🍛 **Desi analogy: tiffin ka dabba andar dabba**
>
> Ek bade dabbe mein chhote-chhote dabbe: dal, sabzi, roti. Dal chahiye toh bada dabba khol ke dal waala nikaalo. **event_params bada dabba hai; UNNEST se chhota dabba nikaalte ho.**

33.3 Building a session key

GA4 sessions are identified by **user_pseudo_id + ga_session_id**:

```
CONCAT(user_pseudo_id, '.',
  CAST((SELECT value.int_value FROM UNNEST(event_params) WHERE key='ga_session_id') AS
STRING)
) AS session_key
```

33.4 Useful built-in parameters

Parameter	On which events	Use
`page_location`, `page_title`, `page_referrer`	page_view and most events	Page info (our cleaned versions)
`ga_session_id`, `ga_session_number`	All	Session key, new vs returning
`engagement_time_msec`	user_engagement and others	Time on page
`entrances`	page_view (1 on the landing page view)	Landing pages
`session_engaged`	Various	Engaged sessions

Plus our custom parameters: `page_type`, `funnel_stage`, `cta_name`, `link_target`, etc. (Chapter 16).

33.5 Date filtering with wildcard tables

```
FROM `ongo-analytics-412345.analytics_123456789.events_*`
WHERE _TABLE_SUFFIX BETWEEN '20261101' AND '20261107'
```

Always filter `_TABLE_SUFFIX`; without it the query scans **every day** and costs more. 🔵

> 🟧 **Ongo Weight Loss example**
>
> Because every `page_view` carries `funnel_stage` and `content_cluster`, you can answer "entry pages by funnel stage" with one simple query: filter `event_name='page_view'` and `entrances=1`, group by `funnel_stage`. No URL guessing. 🔵

33.6 Do's and Don'ts

✅ Do	❌ Don't
Query daily tables for final numbers	Use intraday for reports
Build session keys correctly	Count sessions by user only
Filter `_TABLE_SUFFIX`	Scan all dates
Save common subqueries as views	Rewrite them every time

33.7 Pros and cons of event-level tables

✅ Pros	❌ Cons
Complete detail	Nested fields need UNNEST
Flexible analysis	Easy to write costly queries
Same model as GA4	Learning curve

33.8 Where freshers go wrong in Chapter 33 topics

Mistake	Why it happens	Avoid it by	Fix it by
Sessions counted by user_pseudo_id	Ignoring ga_session_id	33.3	Session key

Mistake	Why it happens	Avoid it by	Fix it by
Missing values (null)	Wrong value type	33.2 tip	COALESCE types
Different numbers vs GA4 UI	GA4 uses estimates, thresholds, modelling	Explain in reports	Accept small gaps

33.9 Thumb rules

👍 One row, one event.

👍 UNNEST to open the dabba.

👍 Session = user + session ID.

33.10 Notes to remember

📝 Landing page = page_view with `entrances = 1`.

📝 Always `_TABLE_SUFFIX BETWEEN …`.

Chapter 33 Glossary

Term	Simple meaning
Schema	The structure of a table
Repeated record	A list of structured items in one cell
UNNEST	SQL function to open a list into rows
Wildcard table	Querying many daily tables with `events_*`
`_TABLE_SUFFIX`	The date part of a wildcard table name

Chapter 33 Quiz

1. What is one row in the GA4 export?

- A) One event
- B) One user
- C) One day
- D) One page

2. Which tables should reports use?

- A) Daily `events_YYYYMMDD`
- B) None
- C) Clarity tables
- D) Intraday only

3. How do you read a parameter?

- A) From GTM
- B) Directly as a column
- C) Subquery with UNNEST(event_params)
- D) From GSC

4. What forms a session key?

- A) user_pseudo_id + ga_session_id
- B) device category
- C) event_name
- D) page_title

5. Which parameter marks the landing page view?

- A) `cta_name`
- B) `scroll = 90`
- C) `entrances = 1`
- D) `faq_id`

6. What does `traffic_source` describe?

- A) Consent
- B) Device
- C) First-user acquisition

- D) Each event's page

7. Why filter `_TABLE_SUFFIX`?

- A) To add users
- B) It's optional decoration
- C) For colours
- D) To limit dates and cost

8. Why might BigQuery differ slightly from GA4 UI?

- A) GA4 UI uses estimates, thresholds and modelling
- B) Fonts
- C) BigQuery is wrong
- D) Consent banners

9. What does `user_pseudo_id` represent?

- A) The browser's client ID
- B) A page
- C) An email
- D) A patient ID

10. In the tiffin analogy, UNNEST…

- A) Opens the inner boxes
- B) Washes the dabba
- C) Cooks the food
- D) Throws the tiffin away

Sources

1. Google Analytics Help, BigQuery Export 🟢
2. Ongo Weight Loss, Event Dictionary v2 (Chapter 16), internal 🔵

Answer key, Chapter 33: 1-A · 2-A · 3-C · 4-A · 5-C · 6-C · 7-D · 8-A · 9-A · 10-A

LEVEL 3 · CHAPTER 34

Chapter 34: SQL Basics for Marketers (From Zero)

SQL is how you ask BigQuery questions. You need about ten words to answer most analytics questions. This chapter teaches them with GA4 examples you can copy.

> 💡 **Trivia**
>
> BigQuery charges on-demand queries by the bytes they read, and it is a columnar store, so selecting only the columns you need reads less data. Google's own best-practice guide says to avoid `SELECT *`. [1] ● Official (Google Cloud)
>
> Fewer columns, fewer bytes, smaller bill.

34.1 The ten words you need

Word	What it does	Example
`SELECT`	Which columns to show	`SELECT event_name`
`FROM`	Which table	`FROM ...events_*`
`WHERE`	Filter rows	`WHERE event_name = 'page_view'`
`GROUP BY`	Group rows to count	`GROUP BY event_name`
`COUNT()`	Count rows	`COUNT(*)`
`COUNT(DISTINCT …)`	Count unique values	`COUNT(DISTINCT user_pseudo_id)`
`ORDER BY`	Sort	`ORDER BY events DESC`
`LIMIT`	Show only N rows (doesn't reduce cost!)	`LIMIT 20`

Word	What it does	Example
`WITH`	Name a step (CTE)	`WITH pv AS (…)`
`JOIN`	Combine tables	`JOIN ops.daily_conversions USING (date)`

34.2 Your first five queries

1. Events by name (one week)

```
SELECT event_name, COUNT(*) AS events
FROM `PROJECT.analytics_PROPERTY.events_*`
WHERE _TABLE_SUFFIX BETWEEN '20261101' AND '20261107'
GROUP BY event_name
ORDER BY events DESC;
```

2. Users and sessions per day

```
SELECT event_date,
  COUNT(DISTINCT user_pseudo_id) AS users,
  COUNT(DISTINCT CONCAT(user_pseudo_id, '.', CAST(
    (SELECT value.int_value FROM UNNEST(event_params) WHERE key='ga_session_id') AS
STRING))) AS sessions
FROM `PROJECT.analytics_PROPERTY.events_*`
WHERE _TABLE_SUFFIX BETWEEN '20261101' AND '20261107'
GROUP BY event_date
ORDER BY event_date;
```

3. Top pages by page views

```
SELECT
  (SELECT value.string_value FROM UNNEST(event_params) WHERE key='page_location') AS
page,
  COUNT(*) AS views
FROM `PROJECT.analytics_PROPERTY.events_*`
WHERE _TABLE_SUFFIX BETWEEN '20261101' AND '20261107'
  AND event_name = 'page_view'
GROUP BY page
ORDER BY views DESC
LIMIT 25;
```

4. Landing pages (entrances)

```
SELECT
  (SELECT value.string_value FROM UNNEST(event_params) WHERE key='page_location') AS
landing_page,
  COUNT(*) AS entrances
FROM `PROJECT.analytics_PROPERTY.events_*`
WHERE _TABLE_SUFFIX BETWEEN '20261101' AND '20261107'
  AND event_name = 'page_view'
  AND (SELECT value.int_value FROM UNNEST(event_params) WHERE key='entrances') = 1
GROUP BY landing_page
ORDER BY entrances DESC;
```

5. CTA clicks by name and location

```
SELECT
  (SELECT value.string_value FROM UNNEST(event_params) WHERE key='cta_name') AS cta_name,
  (SELECT value.string_value FROM UNNEST(event_params) WHERE key='cta_location') AS
cta_location,
  COUNT(*) AS clicks
FROM `PROJECT.analytics_PROPERTY.events_*`
WHERE _TABLE_SUFFIX BETWEEN '20261101' AND '20261107'
  AND STARTS_WITH(event_name, 'cta_')
GROUP BY cta_name, cta_location
ORDER BY clicks DESC;
```

> 🍛 **Desi analogy: kirana list**
>
> SELECT matlab "kya chahiye" (aata, chawal), FROM matlab "kis dukaan se", WHERE matlab "sirf brand X", GROUP BY matlab "har item ka total". **SQL ek kirana list hai jo machine samajhti hai.**

34.3 Cost habits

Habit	Why
Always filter `_TABLE_SUFFIX`	Reads fewer days
Select only needed columns	Reads fewer columns
Check the **"This query will process X"** estimate before Run	Avoid surprises

Habit	Why
`LIMIT` does **not** reduce bytes	Don't rely on it for cost
Save repeated logic as a **view** or a scheduled table	Reuse, cheaper dashboards
Use **Preview** tab to look at a table	Free; queries cost

34.4 Saving and scheduling

Action	Where
Save query	Editor → **Save → Save query** (project-shared)
Save as view	Editor → **Save → Save view** → dataset `reporting`
Schedule a query	Editor → **Schedule** → daily, destination table in `reporting`, write mode **Overwrite** or **Append**

> 🟧 **Ongo Weight Loss example**
>
> Schedule query 4 (landing pages) to run daily at 9 am into `reporting.landing_pages_daily`, adding `event_date`. Looker then reads a tiny table instead of raw events, so the daily dashboard loads in seconds. 🔵

34.5 Do's and Don'ts

✅ Do	❌ Don't
Check bytes before running	Run blind
Comment your SQL with `--`	Leave mystery queries
Name results clearly (`AS clicks`)	Leave default names `f0_`
Save useful queries	Retype from memory

34.6 Pros and cons of learning SQL

✅ Pros	❌ Cons
Answers any question	Takes practice
Reusable	Mistakes can be costly without habits
Career skill	—

34.7 Where freshers go wrong in Chapter 34 topics

Mistake	Why it happens	Avoid it by	Fix it by
Thinking LIMIT saves money	Common myth	34.3	Date filters
Counting users with COUNT(*)	Rows ≠ users	COUNT(DISTINCT)	Rewrite
Forgetting GROUP BY columns	Syntax error	Group every non-counted column	Add to GROUP BY

34.8 Thumb rules

> 👍 Filter dates first, always.
>
> 👍 Distinct for people, plain count for events.
>
> 👍 Save it if you'll ask it twice.

34.9 Notes to remember

> 📝 Ten words: SELECT, FROM, WHERE, GROUP BY, COUNT, DISTINCT, ORDER BY, LIMIT, WITH, JOIN.
>
> 📝 LIMIT doesn't cut cost.

Chapter 34 Glossary

Term	Simple meaning
SQL	Language for asking databases questions
CTE	A named step created with `WITH`
View	A saved query that acts like a table
Scheduled query	A query BigQuery runs automatically
Columnar storage	Data stored by column, so fewer columns cost less

Chapter 34 Quiz

1. Which word filters rows?

- A) ORDER BY
- B) LIMIT
- C) WHERE
- D) SELECT

2. Does LIMIT reduce bytes processed?

- A) Yes
- B) Sometimes doubles it
- C) No
- D) Only on Mondays

3. How do you count unique users?

- A) COUNT(DISTINCT user_pseudo_id)
- B) COUNT(*)
- C) SUM(users)
- D) MAX(user)

4. Why avoid SELECT *?

- A) Reads more columns and costs more
- B) No reason
- C) It's slower to type only
- D) It's illegal

5. Where do you see the cost estimate?

- A) In GSC
- B) After a month
- C) "This query will process X" before Run
- D) In GTM

6. How do you make Looker load faster?

- A) Bigger fonts
- B) More raw tables
- C) Scheduled small tables or views
- D) Fewer users

7. What does WITH create?

- A) A new dataset
- B) A named step (CTE)
- C) A chart
- D) A user

8. Which is free for looking at a table?

- A) JOIN
- B) Scheduled query
- C) Preview tab
- D) SELECT *

9. What does `--` do in SQL?

- A) Starts a comment
- B) Subtracts tables
- C) Ends the query
- D) Deletes data

10. In the kirana analogy, FROM is…

- A) Which shop
- B) Which brand
- C) The total
- D) Which item

Sources

1. Google Cloud, Optimize query computation (avoid SELECT *) 🟢
2. Google Cloud, Scheduling queries 🟢

Answer key, Chapter 34: 1-C · 2-C · 3-A · 4-A · 5-C · 6-C · 7-B · 8-C · 9-A · 10-A

LEVEL 3 · CHAPTER 35

Chapter 35: The Ongo Weight Loss Query Library (Paths, Funnels, Attribution)

These are the queries that deliver your "entry to exit, page by page" vision: full session paths, entry and exit pages, the TOFU → BOFU → portal funnel, first-touch content credit, and internal-link flow.

> 💡 **Trivia**
>
> GA4's path exploration shows the top 5 nodes per step by default and up to 20 when expanded. [1] ● Official (Google)
>
> BigQuery has no such ceiling: every session's full path is a row you can read.

35.1 Base view: clean page views (build once)

Save as view `reporting.v_page_views`:

```
SELECT
  PARSE_DATE('%Y%m%d', event_date) AS date,
  event_timestamp,
  user_pseudo_id,
  CONCAT(user_pseudo_id, '.', CAST((SELECT value.int_value FROM UNNEST(event_params)
    WHERE key='ga_session_id') AS STRING)) AS session_key,
  REGEXP_REPLACE((SELECT value.string_value FROM UNNEST(event_params)
    WHERE key='page_location'), r'\?.*$', '') AS page,
  (SELECT value.string_value FROM UNNEST(event_params) WHERE key='page_type') AS
page_type,
  (SELECT value.string_value FROM UNNEST(event_params) WHERE key='funnel_stage') AS
funnel_stage,
  (SELECT value.string_value FROM UNNEST(event_params) WHERE key='content_cluster') AS
cluster,
  (SELECT value.string_value FROM UNNEST(event_params) WHERE key='site_id') AS site_id,
  (SELECT value.int_value FROM UNNEST(event_params) WHERE key='entrances') AS entrances,
  collected_traffic_source.manual_source AS utm_source,
  collected_traffic_source.manual_medium AS utm_medium
FROM `PROJECT.analytics_PROPERTY.events_*`
WHERE event_name = 'page_view'
  AND _TABLE_SUFFIX >= FORMAT_DATE('%Y%m%d', DATE_SUB(CURRENT_DATE(), INTERVAL 90 DAY))
```

Change the 90-day window when needed. 🔵

35.2 Query A: every session's path, entry → exit

```
WITH ordered AS (
  SELECT session_key, page, event_timestamp,
    ROW_NUMBER() OVER (PARTITION BY session_key ORDER BY event_timestamp) AS step,
    COUNT(*) OVER (PARTITION BY session_key) AS steps
  FROM `PROJECT.reporting.v_page_views`
  WHERE date BETWEEN '2026-11-01' AND '2026-11-07'
)
SELECT session_key,
  STRING_AGG(page, ' > ' ORDER BY step) AS path,
  MAX(steps) AS pages_in_session
FROM ordered
GROUP BY session_key
ORDER BY pages_in_session DESC
LIMIT 100;
```

Most common paths: wrap the above and `GROUP BY path` with `COUNT(*)`.

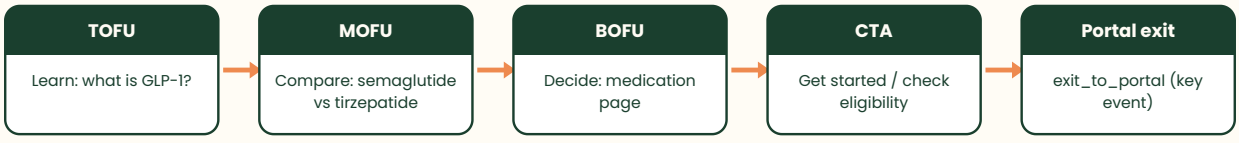
35.3 Query B: entry page and exit page per session

```
WITH s AS (
  SELECT session_key,
    ARRAY_AGG(page ORDER BY event_timestamp LIMIT 1)[OFFSET(0)] AS entry_page,
    ARRAY_AGG(page ORDER BY event_timestamp DESC LIMIT 1)[OFFSET(0)] AS exit_page,
    COUNT(*) AS pages
  FROM `PROJECT.reporting.v_page_views`
  WHERE date BETWEEN '2026-11-01' AND '2026-11-07'
  GROUP BY session_key
)
SELECT entry_page, exit_page, COUNT(*) AS sessions, ROUND(AVG(pages),1) AS avg_pages
FROM s
GROUP BY entry_page, exit_page
ORDER BY sessions DESC
LIMIT 50;
```

35.4 Query C: next page after a page (page-to-page flow)

```
WITH ordered AS (
  SELECT session_key, page,
    LEAD(page) OVER (PARTITION BY session_key ORDER BY event_timestamp) AS next_page
  FROM `PROJECT.reporting.v_page_views`
  WHERE date BETWEEN '2026-11-01' AND '2026-11-07'
)
SELECT page, IFNULL(next_page, '(exit)') AS next_page, COUNT(*) AS moves
FROM ordered
WHERE page LIKE '%/blog/%'
GROUP BY page, next_page
ORDER BY moves DESC
LIMIT 100;
```

The Ongo Weight Loss Content Funnel



Measure every step, judge every page by its job.

> 🪔 **Desi analogy: railway ka route chart**
>
> Har yatri ka ticket batata hai: kis station se chadha, kahan-kahan ruka, kahan utra. Query A poora route hai, Query B chadhne-utarne ka station, Query C "agla station". **Teeno milke poora railway chart.**

35.5 Query D: funnel TOFU → MOFU → BOFU → portal

```
WITH sess AS (
  SELECT session_key,
    MAX(IF(funnel_stage='tofu',1,0)) AS saw_tofu,
    MAX(IF(funnel_stage='mofu',1,0)) AS saw_mofu,
    MAX(IF(funnel_stage='bofu',1,0)) AS saw_bofu
  FROM `PROJECT.reporting.v_page_views`
  WHERE date BETWEEN '2026-11-01' AND '2026-11-30'
  GROUP BY session_key
),
portal AS (
  SELECT DISTINCT CONCAT(user_pseudo_id, '.', CAST((SELECT value.int_value FROM
UNNEST(event_params)
    WHERE key='ga_session_id') AS STRING)) AS session_key
  FROM `PROJECT.analytics_PROPERTY.events_*`
  WHERE _TABLE_SUFFIX BETWEEN '20261101' AND '20261130'
    AND event_name = 'exit_to_portal'
)
SELECT
  COUNT(*) AS sessions,
  SUM(saw_tofu) AS tofu,
  SUM(IF(saw_tofu=1 AND saw_mofu=1,1,0)) AS tofu_to_mofu,
  SUM(IF(saw_tofu=1 AND saw_bofu=1,1,0)) AS tofu_to_bofu,
  SUM(IF(saw_bofu=1 AND p.session_key IS NOT NULL,1,0)) AS bofu_to_portal
FROM sess s
LEFT JOIN portal p USING (session_key);
```

This is an **"any order"** funnel; Chapter 45 shows the strict-order version in GA4.

35.6 Query E: first-touch content credit for portal exits

Which **landing page** started sessions that ended in `exit_to_portal`?

```
WITH landing AS (
  SELECT session_key, page AS landing_page, cluster, utm_source, utm_medium
  FROM `PROJECT.reporting.v_page_views`
  WHERE entrances = 1 AND date BETWEEN '2026-11-01' AND '2026-11-30'
),
portal AS (
  SELECT DISTINCT CONCAT(user_pseudo_id, '.', CAST((SELECT value.int_value FROM
UNNEST(event_params)
    WHERE key='ga_session_id') AS STRING)) AS session_key
  FROM `PROJECT.analytics_PROPERTY.events_*`
  WHERE _TABLE_SUFFIX BETWEEN '20261101' AND '20261130' AND event_name='exit_to_portal'
)
SELECT l.landing_page, l.cluster,
  COUNT(*) AS sessions,
  COUNTIF(p.session_key IS NOT NULL) AS portal_exits,
  ROUND(SAFE_DIVIDE(COUNTIF(p.session_key IS NOT NULL), COUNT(*))*100, 2) AS
portal_rate_pct
FROM landing l LEFT JOIN portal p USING (session_key)
GROUP BY l.landing_page, l.cluster
HAVING sessions >= 30
ORDER BY portal_exits DESC;
```

User-level first touch: replace session landing with each user's **first ever** landing page (`ROW_NUMBER() OVER (PARTITION BY user_pseudo_id ORDER BY event_timestamp) = 1`) and count users who **ever** reached the portal. 🔵

35.7 Query F: internal-link flow

```
SELECT
  REGEXP_REPLACE((SELECT value.string_value FROM UNNEST(event_params) WHERE
key='page_location'), r'\?.*$','') AS from_page,
  (SELECT value.string_value FROM UNNEST(event_params) WHERE key='link_target') AS
to_page,
  (SELECT value.string_value FROM UNNEST(event_params) WHERE key='link_position') AS
position,
  COUNT(*) AS clicks
FROM `PROJECT.analytics_PROPERTY.events_*`
WHERE _TABLE_SUFFIX BETWEEN '20261101' AND '20261130'
  AND event_name = 'internal_link_click'
GROUP BY from_page, to_page, position
ORDER BY clicks DESC;
```

Pair with page views of `from_page` to get **link CTR** (Chapter 51).

35.8 Query G: linking GA4 to backend totals (aggregated)

```
WITH ga AS (
  SELECT PARSE_DATE('%Y%m%d', event_date) AS date, COUNT(*) AS portal_exits
  FROM `PROJECT.analytics_PROPERTY.events_*`
  WHERE _TABLE_SUFFIX BETWEEN '20261101' AND '20261130' AND event_name='exit_to_portal'
  GROUP BY date
)
SELECT g.date, g.portal_exits, d.channel_group,
  d.signups + d.eligibility_completed AS leads, d.payments
FROM ga g
LEFT JOIN `PROJECT.ops.daily_conversions` d USING (date)
ORDER BY g.date;
```

Totals only, no individuals (Chapter 24). 🔵

35.9 Query H: weekly personal-data scan

```
SELECT event_name, ep.key, ep.value.string_value
FROM `PROJECT.analytics_PROPERTY.events_*`, UNNEST(event_params) ep
WHERE _TABLE_SUFFIX BETWEEN FORMAT_DATE('%Y%m%d', DATE_SUB(CURRENT_DATE(), INTERVAL 7
DAY))
                       AND FORMAT_DATE('%Y%m%d', CURRENT_DATE())
  AND (REGEXP_CONTAINS(ep.value.string_value, r'@')
    OR REGEXP_CONTAINS(ep.value.string_value, r'\d{3}[-. ]?\d{3}[-. ]?\d{4}')
    OR REGEXP_CONTAINS(LOWER(ep.value.string_value), r'(email|phone|name|dob)='))
LIMIT 100;
```

Expected result: zero rows. Any row is an incident (Chapter 30). 🔵

> 🟧 **Ongo Weight Loss example**
>
> Query E answers your core question: "Which blog articles start journeys that reach the portal?" Combine it with Chapter 24's aggregated payments by channel and you can report "Organic GLP-1 cluster articles started 41% of portal exits this month" without any patient-level data. 🔵

35.10 Do's and Don'ts

✅ Do	❌ Don't
Build the base view first	Repeat UNNEST code everywhere
Set minimum session thresholds (HAVING)	Rank pages with 3 sessions
Run the personal-data scan weekly	Skip when busy
Keep backend joins aggregated	Join on user identifiers

35.11 Pros and cons of BigQuery paths vs GA4 path exploration

	BigQuery	GA4 exploration
✅	Every session, any length	No SQL, visual
❌	Requires SQL	Top nodes only, thresholds

35.12 Where freshers go wrong in Chapter 35 topics

Mistake	Why it happens	Avoid it by	Fix it by
Paths full of query strings	Not cleaning URLs	Base view regex	Fix view
Funnel numbers too high	"Any order" misunderstood	Label funnel type	Strict query
Small pages ranked top	No threshold	HAVING	Add threshold

35.13 Thumb rules

👍 One base view, many questions.

👍 Every ticket shows the full route.

👍 Scan for personal data, expect zero.

35.14 Notes to remember

📝 Queries A–H: path, entry/exit, next page, funnel, first touch, internal links, backend join, PII scan.

📝 Full library in Appendix H.

Chapter 35 Glossary

Term	Simple meaning
Window function	Calculation over related rows (e.g., ROW_NUMBER)
LEAD()	Gets the next row's value
STRING_AGG	Joins values into one text
First touch	Credit to the first landing page
HAVING	Filter after grouping

Chapter 35 Quiz

1. Which function gets the next page?

- A) WHERE
- B) LIMIT
- C) COUNT()

- D) LEAD()

2. What's the base view for?

- A) Storing images
- B) Reusing clean page-view rows
- C) Consent
- D) Sending emails

3. How is the landing page identified?

- A) Random
- B) `entrances = 1`
- C) Last page
- D) Longest page

4. What's Query D's funnel type?

- A) Any order
- B) None
- C) Strict order
- D) Timed

5. What's the expected result of the PII scan?

- A) All emails
- B) Exactly 10
- C) Zero rows
- D) Many rows

6. Why use HAVING sessions >= 30?

- A) For colours
- B) To slow queries
- C) Required by GA4
- D) Avoid ranking tiny samples

7. How is backend data joined?

- A) By patient ID
- B) By phone
- C) By email

- D) By date and channel totals only

8. What does STRING_AGG make?

- A) A user
- B) A table
- C) A chart
- D) A path text from pages

9. What does GA4 path exploration show by default?

- A) Nothing
- B) 1,000 nodes
- C) Every session
- D) Top 5 nodes per step

10. In the railway analogy, Query B is...

- A) The ticket price
- B) The full route
- C) The train name
- D) Boarding and getting-off stations

Sources

1. Google Analytics Help, Path exploration
2. Google Cloud, Window function calls

Answer key, Chapter 35: 1-D · 2-B · 3-B · 4-A · 5-C · 6-D · 7-D · 8-D · 9-D · 10-D

Chapter 36: Google Search Console: Setup and Every Report Explained

Search Console shows what happens before the click: which searches show our pages, how often, at what position, and whether Google can crawl and index them. GA4 can't see any of that.

> 💡 **Trivia**
>
> Search Console's Performance report keeps **16 months** of data. [1] 🟢 Official (Google)
>
> That's why Chapter 37's bulk export matters: after 16 months, the UI forgets.

36.1 Property types

Type	Covers	Verify by
Domain property (recommended)	All subdomains and http/https	DNS TXT record
URL-prefix property	One exact prefix	HTML tag, file, GA, GTM

Setup: one **domain property per public domain** (main site, weightclinicsresults.com, each child site). Never add the portal. 🔵

36.2 Verification steps (domain property)

#	Step
1	search.google.com/search-console → Add property → **Domain** → enter `ongoweightloss.com`
2	Copy the TXT record

#	Step
3	Developer/DNS admin adds it at the DNS provider
4	Click **Verify** (can take minutes to hours)
5	Settings → **Users and permissions** → add team (Full or Restricted)
6	Submit sitemap: Indexing → **Sitemaps** → `https://…/sitemap.xml`

36.3 Reports tour

Report	What it tells you	Look at it
Performance → Search results	Clicks, impressions, CTR, average position by query, page, country, device, date, search appearance	Daily (trend), weekly (detail)
Performance → Discover / News	If we appear there	Monthly
URL Inspection	One URL's index status, canonical chosen, last crawl, live test	When publishing / fixing
Indexing → Pages	Indexed vs not indexed, with reasons	Weekly
Indexing → Sitemaps	Submitted vs discovered	After changes
Indexing → Removals	Temporary removals	Rarely
Experience → Core Web Vitals	Good/needs improvement/poor URL groups (field data)	Weekly
Experience → HTTPS	HTTPS issues	Monthly
Enhancements (e.g., Breadcrumbs, FAQ if present)	Structured data validity	After releases
Security & Manual actions	Penalties or hacking	Weekly glance
Links	Top linked pages, linking sites, anchor text	Monthly
Settings → Crawl stats	Googlebot requests, response codes, host status	Monthly / after migrations

> 🪔 **Desi analogy: dukaan ke bahar ka CCTV**
>
> GA4 dukaan ke andar ka CCTV hai. Search Console bahar sadak ka CCTV: kitne log guzre (impressions), kitne ne board dekha, kitne andar aaye (clicks). **Andar aane se pehle ki kahaani GSC batata hai.**

36.4 Performance report filters you'll use daily

Filter	Use
Query contains "semaglutide"	Entity performance
Page contains `/blog/`	Blog vs service
Compare last 28 days vs previous	Trend
Device	Mobile vs desktop
Search appearance	Rich results
Regex (Custom)	Groups of queries, e.g., question queries

Example regex for question queries: `^(what|how|why|can|does|is|when)\b` (no pipes in our tables; type it in GSC). 🔵

36.5 Key metrics, defined

Metric	Meaning	Watch out
Impressions	Times a link to our site appeared in results	Counted even if not scrolled into view in some layouts
Clicks	Clicks from results to our site	Not equal to GA4 sessions
CTR	Clicks ÷ impressions	Depends on position and SERP features
Average position	Average of the topmost position	Average hides spread

36.6 Health-specific checks

Check	Why
Queries with brand drug names landing on commercial pages	Confirm content is educational and compliant (Book Six)
Pages with personal-sounding URLs	Should never exist
Indexed pages that should be `noindex` (thank-you pages)	Fix `index_state`
YMYL pages losing impressions after core updates	Review E-E-A-T (Book Three)

> 🟧 **Ongo Weight Loss example**
>
> Weekly ritual: Performance → Pages → filter `/services/weight-loss/` → compare 28 days. If a medication page's impressions drop >20% while position holds, check SERP features (new AI Overview?) before rewriting. 🔵

36.7 Do's and Don'ts

✅ Do	❌ Don't
Use domain properties	Rely on one URL-prefix property
Submit sitemaps per site	Forget child sites
Give team Restricted access where possible	Give everyone Owner
Compare periods	Judge single days

36.8 Pros and cons of Search Console

✅ Pros	❌ Cons
Only source of Google query data	16-month limit in UI

✅ Pros	❌ Cons
Free, official	Some queries anonymised
Index and crawl diagnostics	1,000-row UI export limit

36.9 Where freshers go wrong in Chapter 36 topics

Mistake	Why it happens	Avoid it by	Fix it by
Expecting GSC clicks = GA4 sessions	Different measurements	36.5	Explain gap
Missing child sites	One property only	36.1	Add each
Panicking over one-day drop	Noise	Compare 28 days	Wait and compare

36.10 Thumb rules

👍 GSC is the street camera; GA4 is the shop camera.

👍 One domain property per public domain.

👍 28 days vs 28 days, not today vs yesterday.

36.11 Notes to remember

📝 16 months of data in the UI.

📝 Never add the portal.

Chapter 36 Glossary

Term	Simple meaning
Domain property	GSC property covering a whole domain
Impression	Our link appeared in results
CTR	Clicks divided by impressions
URL Inspection	Tool to check one URL's index status
Crawl stats	Report on Googlebot's requests

Chapter 36 Quiz

1. How long does the Performance report keep data?

- A) 16 months
- B) Forever
- C) 5 years
- D) 30 days

2. Which property type is recommended?

- A) Domain property
- B) URL-prefix only
- C) Portal property
- D) None

3. How is a domain property verified?

- A) Phone
- B) DNS TXT record
- C) GA4 event
- D) Email

4. What does CTR mean?

- A) Clicks ÷ impressions
- B) Pages ÷ users

- C) Time ÷ page
- D) Users ÷ sessions

5. Which report shows one URL's index status?

- A) Performance
- B) Links
- C) Removals
- D) URL Inspection

6. Should the portal be added to GSC?

- A) Yes
- B) Only on Fridays
- C) No
- D) As URL-prefix

7. Are GSC clicks equal to GA4 sessions?

- A) No, they measure different things
- B) Always equal
- C) GSC is double
- D) GA4 is zero

8. What's the UI export limit?

- A) 1 million rows
- B) Unlimited
- C) 1,000 rows
- D) 10 rows

9. Best comparison period?

- A) 1 hour vs 1 hour
- B) None
- C) Today vs yesterday
- D) 28 days vs previous 28 days

10. In the CCTV analogy, GSC is...

- A) The camera inside
- B) The cash counter

- C) The door lock
- D) The street camera outside the shop

Sources

1. Google Search Console Help, Performance report
2. Google Search Console Help, Add a website property

Answer key, Chapter 36: 1-A · 2-A · 3-B · 4-A · 5-D · 6-C · 7-A · 8-C · 9-D · 10-D

LEVEL 3 · CHAPTER 37

Chapter 37: Search Console Bulk Export to BigQuery

The Search Console screen shows at most 1,000 rows and forgets after 16 months. Bulk export sends every day's search data into BigQuery, where it stays as long as you keep it and joins with GA4.

> 💡 **Trivia**
>
> Search Console's bulk export starts from the day you set it up and **does not backfill** earlier data. The first export can take up to **48 hours** to arrive. [1] ● Official (Google)
>
> Same lesson as GA4: switch it on early.

37.1 What you get

Table	Grain	Key columns
`searchdata_site_impression`	Property-level, per day × query × country × device × search type	`data_date`, `query`, `is_anonymized_query`, `country`, `device`, `search_type`, `impressions`, `clicks`, `sum_top_position`
`searchdata_url_impression`	URL-level, per day × URL × query × …	Adds `url`, `sum_position`, search-appearance flags (e.g., `is_amp_story` …)
`ExportLog`	One row per export	Which dates/tables exported

Anonymised queries: rare queries are hidden for privacy; they appear with `is_anonymized_query = true` and an empty query. [2] ●

37.2 Setup steps

#	Where	Step
1	Cloud console	Use the **billing-enabled** project `ongo-analytics` (sandbox not supported for this)
2	APIs & Services	Enable **BigQuery API** and **BigQuery Storage API**
3	IAM & Admin → IAM → **Grant access**	Principal: `search-console-data-export@system.gserviceaccount.com`; roles: **BigQuery Job User** and **BigQuery Data Editor**
4	Search Console (domain property)	**Settings → Bulk data export**
5	Form	Project ID; dataset name (default `searchconsole`; we use `searchconsole_main` for the main site); **location** = same as GA4 dataset
6	Click	**Continue** → wait up to 48 hours
7	Repeat	For each child-site property, using a different dataset name (e.g., `searchconsole_wcr`)
8	Optional	Set partition expiration (must be at least 14 days) if you want auto-cleanup; we recommend none

> 🪔 **Desi analogy: roz ka akhbaar band karwaana**
>
> Akhbaar waala sirf aaj se dena shuru karega, pichhle saal ke purane akhbaar nahi laayega. Isliye jitni jaldi bandh karwao (subscribe), utna lamba record. **Bulk export bhi aaj se shuru hota hai.**

37.3 Core queries

Clicks, impressions, CTR, position by day (property)

```
SELECT data_date,
  SUM(clicks) AS clicks,
  SUM(impressions) AS impressions,
  SAFE_DIVIDE(SUM(clicks), SUM(impressions)) AS ctr,
  SAFE_DIVIDE(SUM(sum_top_position), SUM(impressions)) + 1 AS avg_position
FROM `PROJECT.searchconsole_main.searchdata_site_impression`
WHERE data_date BETWEEN '2026-11-01' AND '2026-11-30'
  AND search_type = 'WEB'
GROUP BY data_date
ORDER BY data_date;
```

Top queries per URL (page-level)

```
SELECT url, query,
  SUM(clicks) AS clicks, SUM(impressions) AS impressions,
  SAFE_DIVIDE(SUM(sum_position), SUM(impressions)) + 1 AS avg_position
FROM `PROJECT.searchconsole_main.searchdata_url_impression`
WHERE data_date BETWEEN '2026-11-01' AND '2026-11-30'
  AND search_type = 'WEB' AND NOT is_anonymized_query
GROUP BY url, query
HAVING impressions >= 50
ORDER BY clicks DESC
LIMIT 500;
```

Striking-distance queries (positions 8–20, many impressions)

```
SELECT query, url,
  SUM(impressions) AS impressions, SUM(clicks) AS clicks,
  SAFE_DIVIDE(SUM(sum_position), SUM(impressions)) + 1 AS avg_position
FROM `PROJECT.searchconsole_main.searchdata_url_impression`
WHERE data_date >= DATE_SUB(CURRENT_DATE(), INTERVAL 28 DAY)
  AND search_type = 'WEB' AND NOT is_anonymized_query
GROUP BY query, url
HAVING avg_position BETWEEN 8 AND 20 AND impressions >= 200
ORDER BY impressions DESC;
```

Entity groups (use `CASE WHEN REGEXP_CONTAINS(query, 'semaglutide') THEN 'semaglutide' …`) to report Semaglutide, Tirzepatide, GLP-1 and weight-loss topic families. 🔵

Position formula: Google's documented method adds 1 because positions in the export are zero-based sums. [2] 🟢

37.4 Cost and size

Factor	Note
Table size	Grows daily; URL table is the largest
Query cost	Always filter `data_date` (tables are partitioned by it)
Dashboards	Build small scheduled tables (e.g., `reporting.gsc_page_daily`)

> 🟧 **Ongo Weight Loss example**
>
> The striking-distance query becomes the on-page SEO team's weekly worklist: pages ranking 8–20 for valuable queries with high impressions are the fastest wins. Pair each with its GA4 engagement (Chapter 38) before choosing which to refresh. 🔵

37.5 Do's and Don'ts

✅ Do	❌ Don't
Set up on day one	Wait for "later"
Use separate datasets per property	Mix properties in one dataset
Filter `data_date`	Scan the whole table
Exclude anonymised queries when grouping by query	Treat blank queries as one keyword

37.6 Pros and cons of bulk export

✅ Pros	❌ Cons
All rows, no 1,000 limit	No backfill
Keeps data beyond 16 months	Billing required
Joins with GA4 and CrUX	SQL needed

37.7 Where freshers go wrong in Chapter 37 topics

Mistake	Why it happens	Avoid it by	Fix it by
Export fails	Service account roles missing	Step 3	Grant both roles
Wrong average position	Forgetting `+ 1` and weighting	37.3 formula	Rewrite
Location mismatch with GA4	Default chosen	Step 5	Recreate (hard)

37.8 Thumb rules

👍 Subscribe today; there's no old newspaper.

👍 Weighted position, plus one.

👍 Striking distance is the weekly worklist.

37.9 Notes to remember

📝 Service account: search-console-data-export@system.gserviceaccount.com (Job User + Data Editor).

📝 Tables: searchdata_site_impression, searchdata_url_impression, ExportLog.

Chapter 37 Glossary

Term	Simple meaning
Bulk export	Daily copy of GSC data into BigQuery
Anonymised query	Rare query hidden for privacy
Striking distance	Ranking just off page one (8–20)
Partition	A slice of a table by date

Term	Simple meaning
ExportLog	Table recording each export

Chapter 37 Quiz

1. Does bulk export backfill old data?

- A) No
- B) Yes, 16 months
- C) Only for clicks
- D) Yes, 5 years

2. How long can the first export take?

- A) 5 minutes
- B) Never arrives
- C) Up to 48 hours
- D) A month

3. Which service account needs access?

- A) None
- B) Any Gmail
- C) firebase-measurement@...
- D) search-console-data-export@system.gserviceaccount.com

4. Which roles does it need?

- A) Viewer only
- B) BigQuery Job User and Data Editor
- C) Owner and Billing
- D) None

5. How is average position calculated?

- A) Random
- B) Sum of positions ÷ impressions, plus 1
- C) Max position

- D) Clicks ÷ impressions

6. What's a striking-distance query?

- A) Ranking 8–20 with good impressions
- B) No impressions
- C) Ranking 1
- D) Paid query

7. Does bulk export work in the sandbox?

- A) Only for URLs
- B) Only on weekends
- C) No, billing is required
- D) Yes

8. Which column marks hidden rare queries?

- A) `country`
- B) `secret`
- C) `is_hidden`
- D) `is_anonymized_query`

9. What should every query filter?

- A) Nothing
- B) `font`
- C) `data_date`
- D) `user_id`

10. In the akhbaar analogy, bulk export...

- A) Brings last year's papers
- B) Comes weekly
- C) Is free forever
- D) Starts from the day you subscribe

Sources

1. Google Search Console Help, Start a new bulk data export

2. Google Search Console Help, About bulk data export / table guidelines

Answer key, Chapter 37: 1-A · 2-C · 3-D · 4-B · 5-B · 6-A · 7-C · 8-D · 9-C · 10-D

LEVEL 3 · CHAPTER 38

Chapter 38: Joining Search Console and GA4 (Before and After the Click)

Search Console knows the search; GA4 knows what happened next. Joined by page, they tell one story: "this query brought these clicks, and those visitors did this."

> 💡 **Trivia**
>
> GA4 can link to Search Console so that "Google organic search queries" and "Google organic search traffic" reports appear in GA4, but the Search Console data in those reports isn't joined to GA4 events at the user level. [1] 🟢 Official (Google)
>
> The join is always at **page level**, never person level.

38.1 Three ways to combine

Method	Effort	Best for
GA4 ↔ GSC link (Admin → Product links → Search Console links)	5 minutes	Quick look inside GA4
Looker Studio blend (GSC connector + GA4 connector on landing page)	1 hour	Dashboards
BigQuery join (bulk export + GA4 export on URL)	Half a day	Full control, history

38.2 Linking GA4 and GSC

#	Step
1	GA4 → Admin → Product links → **Search Console links** → Link
2	Choose the GSC property (you must be a verified owner in GSC and Editor in GA4)
3	Choose the web stream → Submit
4	Reports → Library → **publish** the Search Console collection so the reports appear in the left menu
5	Repeat for each child-site property (one link per GSC property)

38.3 URL matching rules

Problem	Example	Fix
Protocol/host differences	GSC: `https://www.…/a/` GA4: `/a/`	Strip to path in both
Trailing slash	`/a` vs `/a/`	Normalise to trailing slash
Query strings	GA4 has `?utm…`	Our page_location is cleaned; also regex-strip
Case	`/A/` vs `/a/`	LOWER() both

38.4 The BigQuery join

```
WITH gsc AS (
  SELECT LOWER(REGEXP_REPLACE(url, r'^https?://[^/]+', '')) AS path,
    SUM(clicks) AS gsc_clicks, SUM(impressions) AS impressions
  FROM `PROJECT.searchconsole_main.searchdata_url_impression`
  WHERE data_date BETWEEN '2026-11-01' AND '2026-11-30' AND search_type='WEB'
  GROUP BY path
),
ga AS (
  SELECT LOWER(REGEXP_REPLACE(page, r'^https?://[^/]+', '')) AS path,
    COUNT(*) AS organic_entrances
  FROM `PROJECT.reporting.v_page_views`
  WHERE entrances = 1 AND date BETWEEN '2026-11-01' AND '2026-11-30'
  GROUP BY path
)
SELECT gsc.path, impressions, gsc_clicks, organic_entrances
FROM gsc LEFT JOIN ga USING (path)
ORDER BY gsc_clicks DESC;
```

To restrict GA4 to organic, add the session source to the base view (or use `collected_traffic_source`/session attribution fields) and filter `google / organic`. 🔵

> 🪔 **Desi analogy: shaadi ka card aur mehmaan register**
>
> Card kitne logon ko bheje (impressions), kitne aaye (clicks) — yeh card waale ke paas. Aakar kya khaaya, kitni der ruke — yeh register mein. **Dono milao toh poori shaadi ka hisaab.**

38.5 Questions the join answers

Question	Columns
Pages with many impressions but low CTR	impressions, ctr
Pages with clicks but poor engagement	gsc_clicks, engagement rate
Pages whose organic visitors reach the portal	gsc_clicks, portal exits (Query E)
Queries → page → CTA	GSC query per URL + GA4 CTA per page

38.6 Other connectors you may use

Connector	Use	Note
Looker Studio **Search Console** connector	Site or URL impression tables	Choose "URL impression" for page level
Looker Studio **BigQuery** connector	Our joined tables	Fastest, most flexible
Google Sheets + GSC add-ons (e.g., Search Analytics for Sheets)	Quick pulls	Third-party; check permissions 🟠
GSC API	Custom scripts, backfill up to 16 months	Developer task

> 🟧 **Ongo Weight Loss example**
>
> The day bulk export starts, also run a one-time **GSC API backfill** of the last 16 months into `searchconsole_main.backfill_pages` (developer script, page + query + date). That closes the "no backfill" gap from Chapter 37. 🔵

38.7 Do's and Don'ts

✅ Do	❌ Don't
Normalise URLs before joining	Join raw URLs
Join on page and date	Try to join on users
Explain metric differences in reports	Expect clicks = sessions
Link every GSC property to GA4	Link only the main site

38.8 Pros and cons of each method

Method	✅	❌
GA4 link	Easy	Limited, no blending with our dimensions

Method	✅	❌
Looker blend	Visual	Slow; blend limits
BigQuery	Complete	Needs SQL

38.9 Where freshers go wrong in Chapter 38 topics

Mistake	Why it happens	Avoid it by	Fix it by
Join returns few matches	URL formats differ	38.3	Normalise
GSC reports missing in GA4	Collection not published	Step 4	Publish
Comparing totals across tools	Different definitions	Page-level trends	Explain

38.10 Thumb rules

> 👍 Card list plus guest register equals the full wedding.
>
> 👍 Join on page, never person.
>
> 👍 Normalise first, join second.

38.11 Notes to remember

> 📝 GA4 Admin → Product links → Search Console links.
>
> 📝 Backfill 16 months via API once.

Chapter 38 Glossary

Term	Simple meaning
Join	Combining two tables on a shared column
Normalise	Make values the same format
Blend	Looker Studio's join of two sources
Backfill	Loading historical data after setup
Collection	A group of reports in GA4's library

Chapter 38 Quiz

1. At what level are GSC and GA4 joined?

- A) Person level
- B) Email level
- C) Page level
- D) Device ID

2. Where do you link GSC in GA4?

- A) Reports snapshot
- B) Explore
- C) GTM
- D) Admin → Product links → Search Console links

3. What must be published to see GSC reports in GA4?

- A) A new property
- B) A sitemap
- C) A blog post
- D) The Search Console collection

4. Why normalise URLs?

- A) For colours
- B) No reason

- C) So matching pages join correctly
- D) To change rankings

5. What fills the "no backfill" gap?

- A) GA4 exports
- B) Nothing
- C) Clarity
- D) A one-time GSC API pull of 16 months

6. Which method gives full control and history?

- A) BigQuery join
- B) GA4 link
- C) Email
- D) Screenshots

7. Which Looker connector table gives page-level data?

- A) Crawl stats
- B) Site impression
- C) Sitemaps
- D) URL impression

8. What should reports explain?

- A) Server costs
- B) Nothing
- C) Font choices
- D) Why GSC clicks differ from GA4 sessions

9. How many GSC links for child sites?

- A) None
- B) One per GSC property
- C) One per page
- D) One total

10. In the shaadi analogy, GA4 is...

- A) The guest register
- B) The caterer

- C) The band
- D) The invitation cards

Sources

1. Google Analytics Help, Search Console integration
2. Google Search Console, Search Analytics API

Answer key, Chapter 38: 1-C · 2-D · 3-D · 4-C · 5-D · 6-A · 7-D · 8-D · 9-B · 10-A

LEVEL 3 · CHAPTER 39

Chapter 39: Bing Webmaster Tools, Step by Step

Bing powers Bing search, parts of DuckDuckGo and Yahoo results, and Microsoft Copilot answers. Bing Webmaster Tools is free, imports your Search Console setup in minutes, and now shows how often Copilot cites your pages.

> 💡 **Trivia**
>
> In February 2026, Bing Webmaster Tools put an **AI Performance** report into public preview, showing total citations in AI answers, cited pages and "grounding queries", but no click data. [1] 🟡 Industry (reporting on Microsoft announcement)
>
> It's the first official window into how an AI assistant uses your content.

39.1 Setup (fastest way)

#	Step
1	bing.com/webmasters → sign in with a company Microsoft account (or Google)
2	Choose **Import from Google Search Console** → authorise → select all public properties → Import
3	Sites and sitemaps are imported and verified automatically [2] 🟢
4	Settings → **User management** → add team members (Administrator / Read-write / Read-only)
5	Never add the portal domain

Manual alternative: Add site → verify by XML file, meta tag or DNS CNAME. 🟢

39.2 Reports tour

Report	What it shows	Use
Search Performance	Clicks, impressions, CTR, average position by query/page/country/device	Weekly; compare with GSC
AI Performance (preview)	Citations in Copilot/Bing AI answers, cited pages, grounding queries	Monthly; AI visibility KPI
URL Inspection	Index status, live test, markup	Fixing pages
Site Explorer	Crawled URLs, status, links by folder	Technical review
Sitemaps	Submitted sitemaps and status	After releases
IndexNow	URLs submitted instantly	After publishing
Backlinks	Linking sites and anchors; compare with competitors	Monthly (Book Five/off-page)
Keyword Research	Bing query volumes	Topic planning
SEO Reports / Site Scan	On-page issues crawl	Monthly
Recommendations	Prioritised fixes	Monthly
Robots.txt tester	Test rules	Before changes

> **Desi analogy: doosra mohalla**
>
> Tumhari dukaan ek mohalle (Google) mein mashhoor hai, par doosre mohalle (Bing, Copilot) mein bhi graahak rehte hain. Wahan ka bhi signboard lagao. **Bing Webmaster Tools doosre mohalle ka signboard hai.**

39.3 IndexNow (tell Bing instantly)

Item	Detail
What	A protocol to notify participating engines when a URL is added, updated or deleted

Item	Detail
Who uses it	Bing, Yandex and others; Google doesn't use it 🟡
How (Next.js)	Generate a key file at `/<key>.txt`; on publish/update, POST the URL list to `https://api.indexnow.org/indexnow` with `host`, `key`, `urlList`
Where to hook	CMS publish webhook or deploy step
Check	Bing Webmaster Tools → IndexNow → submitted URLs

```
{ "host": "www.ongoweightloss.com",
  "key": "YOUR_KEY",
  "keyLocation": "https://www.ongoweightloss.com/YOUR_KEY.txt",
  "urlList": ["https://www.ongoweightloss.com/blog/glp-1-side-effects/"] }
```

39.4 What to report from Bing

Metric	Frequency	Why
Bing clicks and impressions	Weekly	Second search engine share
AI citations and cited pages	Monthly	AI visibility trend
Top grounding queries	Monthly	Which topics AI associates with us
Crawl errors	Weekly	Technical health
IndexNow submissions	After publishes	Freshness

Bing data in Looker: there's no official Google connector; export CSVs to Sheets or BigQuery (or use the Bing Webmaster API via a developer script) on a weekly schedule. 🔵

> 🟧 **Ongo Weight Loss example**
>
> Add a small "AI visibility" tile on the monthly report: Bing AI citations, number of cited pages and top three grounding queries. Combine with the GA4 **AI Assistants** channel (Chapter 27) to show "AI cites us" next to "AI sends us visitors." 🔵

39.5 Do's and Don'ts

✅ Do	❌ Don't
Import from GSC	Verify each site manually without need
Hook IndexNow into publishing	Spam IndexNow with unchanged URLs
Track AI citations monthly	Treat citations as clicks
Compare Bing and GSC trends	Ignore Bing because it's smaller

39.6 Pros and cons of Bing Webmaster Tools

✅ Pros	❌ Cons
Free; quick import	Smaller search share
AI citation data	AI report in preview
Backlink and keyword tools included	No native Looker connector

39.7 Where freshers go wrong in Chapter 39 topics

Mistake	Why it happens	Avoid it by	Fix it by
Skipping Bing	"Only Google matters"	39.2 Copilot reach	Set up
Expecting AI citations to bring clicks	Misreading the report	Know its limits	Use channel data for visits
Submitting every URL daily to IndexNow	Misunderstanding	Changed URLs only	Fix hook

39.8 Thumb rules

👍 Put a signboard in the second mohalla too.

👍 Citations show visibility, not visits.

👍 IndexNow on publish, not on a timer.

39.9 Notes to remember

📝 Import from Google Search Console for one-click setup.

📝 AI Performance: citations, cited pages, grounding queries; no clicks.

Chapter 39 Glossary

Term	Simple meaning
Bing Webmaster Tools	Microsoft's free site-owner console
IndexNow	Protocol to notify search engines of URL changes
AI citation	Our page referenced in an AI answer
Grounding query	Query the AI used to find our content
Site Explorer	Bing's crawled-URL browser

Chapter 39 Quiz

1. What's the fastest setup?

- A) Import from Google Search Console
- B) Use GA4
- C) Email Bing
- D) Manual for every page

2. What does the AI Performance report show?

- A) Payments
- B) Heatmaps
- C) Citations, cited pages, grounding queries
- D) Clicks and revenue

3. Does it show clicks?

- A) Only for ads
- B) No
- C) Yes
- D) Only on mobile

4. What is IndexNow?

- A) A sitemap format only
- B) A protocol to notify engines of URL changes
- C) A Clarity feature
- D) A GA4 report

5. When should IndexNow fire?

- A) When URLs are added, changed or deleted
- B) Never
- C) Every minute for all URLs
- D) Once a year

6. Should the portal be added?

- A) Only Bing
- B) Sometimes
- C) No
- D) Yes

7. How does Bing data reach Looker?

- A) CSV/API export to Sheets or BigQuery
- B) Through GTM
- C) It can't, ever
- D) Native Google connector

8. Which report compares backlinks with competitors?

- A) Sitemaps
- B) Backlinks
- C) IndexNow
- D) URL Inspection

9. Does Google use IndexNow?

- A) Yes, exclusively
- B) No
- C) Only for news
- D) Only in the UK

10. In the mohalla analogy, Bing Webmaster Tools is…

- A) The cash box
- B) The shop itself
- C) A signboard in the second neighbourhood
- D) The godown

Sources

1. Koozai, Bing Webmaster Tools launches AI Performance reporting 🟡
2. Bing Webmaster Blog, Import sites from Search Console to Bing Webmaster Tools 🟢
3. IndexNow, Documentation 🟢

Answer key, Chapter 39: 1-A · 2-C · 3-B · 4-B · 5-A · 6-C · 7-A · 8-B · 9-B · 10-C

LEVEL 3 · CHAPTER 40

Chapter 40: Core Web Vitals and CrUX (What Replaced the CrUX Connector)

You asked how to use the CrUX connector. Here's the honest answer: Google switched it off at the end of November 2025. This chapter teaches the four tools that replace it and what each tells you about real users' speed.

> 💡 **Trivia**
>
> In September 2025 Google announced that the CrUX Dashboard and its Looker Studio connector would stop working after the end of November 2025, pointing users to **CrUX Vis** and to BigQuery directly. [1] 🟢 Official (Chrome for Developers)
>
> Old guides that still show the connector are out of date.

40.1 Core Web Vitals in one table

Metric	Measures	"Good" at the 75th percentile
LCP (Largest Contentful Paint)	Loading: when the main content appears	≤ 2.5 s
INP (Interaction to Next Paint)	Responsiveness to taps/clicks	≤ 200 ms
CLS (Cumulative Layout Shift)	Visual stability	≤ 0.1

Also useful: **TTFB** (server response) and **FCP** (first paint). [2] 🟢

40.2 Field data vs lab data

	Field (real users)	Lab (simulated)
Sources	CrUX, GSC CWV report, our web-vitals → GA4	Lighthouse, PageSpeed Insights lab section
Tells you	What visitors actually experienced	Why a page is slow (diagnostics)
Use for	KPIs and reporting	Fixing

> 🍛 **Desi analogy: thermometer aur doctor ka checkup**
>
> Ghar pe roz thermometer (field data) se pata chalta hai bukhar hai ya nahi. Doctor ka checkup (lab) batata hai kyun hai. **Report mein thermometer, fix ke liye doctor.**

40.3 The four replacements

Tool	Level	Freshness	Best for
CrUX Vis (cruxvis.withgoogle.com)	Origin and URL (if enough traffic)	Weekly/monthly history	Quick visual trends, no setup
PageSpeed Insights / CrUX API	URL and origin	28-day rolling, updated daily	Checking one page; scripts
CrUX on BigQuery (`chrome-ux-report`)	Origin only	Monthly release	Competitor comparison, Looker via BigQuery
GSC Core Web Vitals report	URL groups	Rolling 28 days	Which templates fail
Our own: web-vitals → GA4	Every page, every visit (consented)	Real time	Page-type detail, our own KPI

40.4 Using CrUX Vis

#	Step
1	Open cruxvis.withgoogle.com
2	Enter `https://www.ongoweightloss.com` (origin) or a URL

#	Step
3	Choose form factor (phone/desktop)
4	Read the trend for LCP, INP, CLS (good/needs improvement/poor share)
5	Add competitor origins one at a time and screenshot for the monthly report

40.5 CrUX in BigQuery (origin-level, competitors too)

```
SELECT date, origin, device,
  p75_lcp, p75_inp, p75_cls,
  fast_lcp, fast_inp, small_cls
FROM `chrome-ux-report.materialized.device_summary`
WHERE origin IN ('https://www.ongoweightloss.com', 'https://ro.co',
'https://www.hims.com')
  AND date >= '2026-01-01'
ORDER BY date, origin, device;
```

Notes: the public dataset is free to query within your 1 TiB monthly allowance; column names can change between releases, so check the table schema first. Origins with too little traffic won't appear. 🔵

Looker: connect the **BigQuery** connector to a saved query or scheduled table in `crux` dataset (your own credentials). That's the direct replacement for the old connector. 🟢

40.6 Our own field data: web-vitals → GA4

```
// app/components/WebVitals.tsx
'use client';
import { useEffect } from 'react';
import { onLCP, onINP, onCLS } from 'web-vitals';
import { track } from '@/lib/analytics';

export function WebVitals() {
  useEffect(() => {
    const send = (m: { name: string; value: number; rating: string; id: string }) =>
      track('web_vitals', {
        metric_name: m.name,
        metric_value: Math.round(m.name === 'CLS' ? m.value * 1000 : m.value),
        metric_rating: m.rating,   // good / needs-improvement / poor
        metric_id: m.id,
      });
    onLCP(send); onINP(send); onCLS(send);
  }, []);
  return null;
}
```

GTM / GA4 step	Detail
Trigger	Custom event `web_vitals`
GA4 event tag	Name `web_vitals`; params metric_name, metric_value, metric_rating
Custom definitions	metric_name, metric_rating (event dims); metric_value (event metric)
Report	Exploration: page_type × metric_name × metric_rating, or BigQuery p75

p75 in BigQuery:

```
SELECT
  (SELECT value.string_value FROM UNNEST(event_params) WHERE key='metric_name') AS
metric,
  (SELECT value.string_value FROM UNNEST(event_params) WHERE key='page_type') AS
page_type,
  APPROX_QUANTILES((SELECT value.int_value FROM UNNEST(event_params) WHERE
key='metric_value'), 100)[OFFSET(75)] AS p75
FROM `PROJECT.analytics_PROPERTY.events_*`
WHERE _TABLE_SUFFIX BETWEEN '20261101' AND '20261130' AND event_name='web_vitals'
GROUP BY metric, page_type;
```

> 🟧 **Ongo Weight Loss example**
>
> CrUX tells you the whole origin is "good"; your own web-vitals data tells you the **medication pages** have slow LCP on phones because of the hero image. That's the difference between a KPI and a fix list. 🔵

40.7 Do's and Don'ts

✅ Do	❌ Don't
Report field data (p75)	Report one Lighthouse score as the KPI
Use CrUX Vis for quick checks	Search for the retired connector
Send web-vitals to GA4	Send raw decimals for CLS without scaling
Compare competitors monthly	Compare lab scores across sites

40.8 Pros and cons of the replacements

Tool	✅	❌
CrUX Vis	No setup	Not embeddable in Looker
BigQuery CrUX	Competitors, history	Origin-only, monthly
web-vitals → GA4	Page-type detail	Consent reduces sample

40.9 Where freshers go wrong in Chapter 40 topics

Mistake	Why it happens	Avoid it by	Fix it by
Using the old CrUX connector guide	Outdated blogs	40.3	Switch to BigQuery
Chasing a 100 Lighthouse score	Lab ≠ field	40.2	Focus on p75 field

Mistake	Why it happens	Avoid it by	Fix it by
Averaging CWV values	Wrong statistic	p75	Use quantiles

40.10 Thumb rules

👍 Thermometer for reports, doctor for fixes.

👍 p75, not average.

👍 Origin from CrUX, pages from our own data.

40.11 Notes to remember

📝 Good: LCP ≤ 2.5 s, INP ≤ 200 ms, CLS ≤ 0.1.

📝 CrUX connector retired after Nov 2025; use CrUX Vis, BigQuery, GSC, web-vitals.

Chapter 40 Glossary

Term	Simple meaning
CrUX	Chrome UX Report: real-user speed data from Chrome
p75	Value that 75% of visits are at or better than
Origin	Protocol + host (e.g., https://www.site.com)
Field data	Measurements from real visitors
Lab data	Measurements from a simulated test

Chapter 40 Quiz

1. What happened to the CrUX Looker connector?

- A) Nothing
- B) Upgraded
- C) Retired after end of November 2025
- D) Made paid

2. What's "good" LCP?

- A) ≤ 200 s
- B) ≤ 10 s
- C) ≤ 2.5 s at p75
- D) ≤ 0.1

3. What's "good" INP?

- A) ≤ 2 s
- B) ≤ 10 s
- C) ≤ 200 ms
- D) ≤ 1 ms

4. What level is CrUX in BigQuery?

- A) Session level
- B) Every URL
- C) User level
- D) Origin only

5. What replaces the connector inside Looker?

- A) Nothing
- B) Clarity connector
- C) BigQuery connector with own credentials
- D) GA4 connector

6. Which data is for KPIs?

- A) Lab data
- B) Field data
- C) Screenshots
- D) Guesswork

7. What library sends vitals to GA4?

- A) Moment
- B) web-vitals
- C) Lodash
- D) jQuery

8. Why scale CLS by 1000?

- A) GA4 rejects decimals always
- B) To store it as an integer
- C) To make it look bigger
- D) No reason

9. Which statistic should be reported?

- A) Maximum
- B) Minimum
- C) p75
- D) Average

10. In the thermometer analogy, lab data is...

- A) The medicine
- B) The daily thermometer
- C) The doctor's checkup
- D) The bill

Sources

1. Chrome for Developers, CrUX Dashboard deprecation
2. web.dev, Web Vitals
3. CrUX Vis, cruxvis.withgoogle.com
4. GitHub, GoogleChrome/web-vitals

Answer key, Chapter 40: 1-C · 2-C · 3-C · 4-D · 5-C · 6-B · 7-B · 8-B · 9-C · 10-C

LEVEL 3 · CHAPTER 41

Chapter 41: Social Platform Data in Your Reporting

Social platforms measure what happens on their apps; GA4 measures what happens when people come to our site. You need both, joined by date and UTM, to see whether organic social (Book Five) builds the business.

> 💡 **Trivia**
>
> Looker Studio has no free, Google-built connectors for Instagram, Facebook, TikTok, LinkedIn or YouTube Studio analytics beyond YouTube's own connector; most teams use partner connectors (paid) or scheduled CSV/Sheets exports. [1] 🔵 Practice (Looker Studio connector gallery)
>
> Choose the method by budget, not by habit.

41.1 Two kinds of social numbers

Kind	Source	Examples
On-platform	Meta Business Suite, TikTok Studio, LinkedIn Page analytics, YouTube Studio, Pinterest Analytics	Reach, impressions, followers, engagement, saves, shares, video views, watch time
On-site	GA4 (UTMs from Book Five convention)	Sessions, engaged sessions, internal clicks, CTA clicks, portal exits

41.2 Getting on-platform data into Looker

Method	Cost	Effort	Freshness
Manual weekly CSV → Google Sheet (one tab per platform, fixed columns)	Free	20 min/week	Weekly

Method	Cost	Effort	Freshness
Partner connector (e.g., Supermetrics, Porter Metrics, Windsor.ai)	Paid	1 hour setup	Daily
Platform APIs → BigQuery (developer script)	Free (dev time)	Days	Daily
YouTube Analytics connector (Google)	Free	10 min	Daily

Recommendation for now: a standard Google Sheet per platform with these columns: `date, platform, account, followers, posts, reach, impressions, engagements, link_clicks, video_views`. Load via the Sheets connector. Upgrade to a partner connector when time cost exceeds price. 🔵

> 🍛 **Desi analogy: mele ka stall**
>
> Mele mein stall pe kitne log ruke, kitno ne dekha (on-platform). Phir kitne log tumhari asli dukaan tak aaye (on-site). **Dono ginti chahiye, warna pata nahi mela kaam aaya ya nahi.**

41.3 On-site social measurement (GA4)

Item	How
Channel	"Organic Social" in our custom channel group (Chapter 27)
Platform	`utm_source` (instagram, facebook, tiktok, youtube, linkedin, pinterest, x, reddit)
Content	`utm_campaign` pillar_yyyy_mm, `utm_content` format_topic_v# (Book Five)
Bio links	Always UTM'd link-in-bio page or direct UTM link
Untagged social	GA4 still classifies known referrers (e.g., l.instagram.com) as Organic Social

41.4 Social KPIs (what we report)

KPI	Formula	Source
Reach growth	Reach this month ÷ last month	Platform

KPI	Formula	Source
Engagement rate	Engagements ÷ reach	Platform
Link CTR	Link clicks ÷ reach	Platform
Social sessions	Sessions with Organic Social	GA4
Social engaged rate	Engaged sessions ÷ sessions	GA4
Social → BOFU rate	Social sessions reaching a BOFU page ÷ social sessions	GA4/BigQuery
Social portal exits	exit_to_portal from Organic Social	GA4
Social share of leads/payments	From `ops.daily_conversions` channel_group	Backend aggregate

41.5 The join

Join key	Example
Date + platform	Sheet row (2026-11-03, instagram) ↔ GA4 sessions where source = instagram
Campaign	Post's UTM campaign ↔ GA4 `utm_campaign`

In Looker, **blend** the Sheet (date, platform) with GA4 (date, session source) on date + platform (Chapter 59). 🔵

> 🟧 **Ongo Weight Loss example**
>
> A reel about "GLP-1 and protein" reaches 40,000 people (platform) but brings 30 sessions (GA4); a carousel reaching 8,000 brings 120 sessions with 9 portal exits. The monthly social report ranks content by **portal exits per 1,000 reach**, not by likes. 🔵

41.6 Do's and Don'ts

✅ Do	❌ Don't
Use the Book Five UTM convention	Invent new UTM names per post
Keep one fixed Sheet layout	Change columns monthly
Report on-site outcomes per platform	Report likes alone
Respect platform privacy (aggregate only)	Export follower personal data

41.7 Pros and cons of the Sheet method

✅ Pros	❌ Cons
Free	Manual time
Full control of columns	Human error
Works for every platform	Weekly freshness

41.8 Where freshers go wrong in Chapter 41 topics

Mistake	Why it happens	Avoid it by	Fix it by
Social traffic shows as Referral	Missing UTMs	UTM every link	Tag links
Comparing reach across platforms directly	Different definitions	Trend per platform	Explain
Mixed case sources (Instagram/instagram)	No convention	Lowercase rule	Case-normalise in Looker

41.9 Thumb rules

👍 Count the mela crowd and the shop visitors.

👍 Portal exits per 1,000 reach beats likes.

👍 Lowercase UTMs, always.

41.10 Notes to remember

📝 On-platform via Sheet/partner connector; on-site via GA4 UTMs.

📝 Join on date + platform.

Chapter 41 Glossary

Term	Simple meaning
On-platform metric	Measured by the social app
On-site metric	Measured by GA4 on our website
Partner connector	Third-party Looker connector, usually paid
Link-in-bio	Single link in a social profile
Engagement rate	Engagements divided by reach

Chapter 41 Quiz

1. Which tool measures on-site social visits?

- A) GA4
- B) TikTok Studio
- C) Instagram Insights
- D) Bing

2. What's the free method recommended now?

- A) Nothing
- B) Screenshots
- C) Paid connector only
- D) Weekly CSV into a fixed Google Sheet

3. What identifies the platform in GA4?

- A) `cta_name`
- B) `page_type`
- C) `utm_source`
- D) `metric_name`

4. What's the best social ranking metric here?

- A) Likes
- B) Followers only
- C) Portal exits per 1,000 reach
- D) Hashtags

5. What's the join key?

- A) Date + platform
- B) Email
- C) Patient ID
- D) Phone

6. Which platform has a free Google connector?

- A) Instagram
- B) LinkedIn
- C) YouTube
- D) TikTok

7. Why lowercase UTMs?

- A) To avoid split rows
- B) For SEO
- C) No reason
- D) Platforms require it

8. What does Engagement rate equal?

- A) Clicks ÷ sessions
- B) Posts ÷ days
- C) Engagements ÷ reach
- D) Reach ÷ followers

9. When should you upgrade to a partner connector?

- A) When followers drop
- B) Never
- C) When manual time costs more than the price
- D) Day one always

10. In the mela analogy, on-site data is…

- A) Visitors reaching the real shop
- B) Crowd at the stall
- C) The ticket seller
- D) The stall decoration

Sources

1. Looker Studio, Connector gallery 🔵
2. Ongo Weight Loss, Book Five UTM convention, internal 🔵

Answer key, Chapter 41: 1-A · 2-D · 3-C · 4-C · 5-A · 6-C · 7-A · 8-C · 9-C · 10-A

LEVEL 3 · CHAPTER 42

Chapter 42: Competitor Benchmarking

Your numbers mean more when you know where you stand. This chapter shows how to benchmark Ongo Weight Loss against Ro, Hims & Hers, Henry Meds, Mochi Health and others using estimates, public data and a monthly ritual.

> 💡 **Trivia**
>
> Third-party traffic tools such as Semrush and Similarweb **estimate** competitors' traffic from clickstream panels and ranking models; their own documentation describes the figures as estimates. [1] 🟡 Industry (vendor docs)
>
> Use them for direction and ratios, never as exact numbers.

42.1 What you can benchmark (and where from)

Area	Metric	Source	Accuracy
Organic visibility	Estimated organic traffic, keyword count, top-3 keywords	Semrush / Ahrefs	🟡 estimate
Share of voice	Our visibility % vs competitors on a fixed keyword set	Semrush Position Tracking	🟡 good for trends
Total traffic and channels	Visits, channel mix, top pages	Similarweb	🟡 estimate
Page speed	CrUX p75 LCP/INP/CLS	BigQuery CrUX (Chapter 40)	🟢 real data
Backlinks	Referring domains	Ahrefs / Semrush / Bing	🟡
AI visibility	Mentions in AI answers	Manual prompt checks; AI visibility tools	🟠

Area	Metric	Source	Accuracy
Social	Followers, posting rate, engagement	Public profiles	🟢 counts / 🟡 rates
Content	Pages per cluster, freshness	Site crawl (Screaming Frog)	🟢

42.2 Competitor set

Tier	Competitors (from project list)
Direct telehealth GLP-1	Henry Meds, Mochi Health, ShedRx, FuturHealth, IVIM Health, Fridays
Large brands	Ro, Hims & Hers, LifeMD, WeightWatchers
Clinical programs	Found, Calibrate, Form Health

Pick **5 core** for monthly tracking (mix of direct and large) and review the rest quarterly. 🔵

> 🍛 **Desi analogy: class ka report card**
>
> Apne 70 marks achhe hain ya bure, tab pata chalta hai jab class ka average dekho. **Benchmark class ka average hai.**

42.3 Share of voice setup (Semrush example)

#	Step
1	Projects → Position Tracking → domain `ongoweightloss.com`
2	Location: United States; device: mobile and desktop (two campaigns)
3	Keywords: 100–200 from the Book Three content map, tagged by cluster (weight-loss, glp-1, semaglutide, tirzepatide, cost, eligibility)
4	Competitors: 5 core domains
5	Weekly email; monthly export to Sheets → Looker

42.4 Monthly benchmark table (template)

Metric	Ongo Weight Loss	Comp 1	Comp 2	Comp 3	Comp 4	Comp 5
Share of voice (tracked set)						
Est. organic traffic (trend arrow)						
Keywords in top 3						
Referring domains						
p75 LCP mobile (CrUX)						
p75 INP mobile (CrUX)						
New/updated pages in GLP-1 cluster						
Social posts / month (main platform)						

42.5 Benchmarking rules

Rule	Why
Compare **trends and ratios**, not absolute estimates	Estimates vary by tool
Use the **same tool** every month	Tool switches break trends
Note big competitor events (rebrand, lawsuit, product change)	Explains jumps
Never copy content or claims	Book Six rules; compliance
View-only research	Don't interact with competitor accounts

42.6 Our own KPIs vs industry benchmarks

KPI	Where benchmark comes from	Use
Engagement rate	Industry studies (vary widely) 🟠	Sanity check only
Organic CTR by position	Public CTR studies 🟡	Spot low-CTR pages
Conversion rates	Rarely public for telehealth 🟠	Use our own history as the benchmark

Rule: after 90 days, **your own history** becomes the best benchmark (Chapter 66). 🔵

> 🟧 **Ongo Weight Loss example**
>
> If Ongo Weight Loss's share of voice in the "semaglutide cost" cluster is 4% while Henry Meds holds 18%, the content team's next sprint targets that cluster; the benchmark tells you **where** to compete, your GA4 tells you **whether it worked**. 🔵

42.7 Do's and Don'ts

✅ Do	❌ Don't
Fix a keyword set and tool	Change both monthly
Report trends and share	Quote estimates as facts
Include CrUX (real data)	Compare Lighthouse scores
Stay view-only	Engage competitor accounts

42.8 Pros and cons of benchmarking tools

Tool	✅	❌
Semrush/Ahrefs	Keywords, SoV, backlinks	Paid; estimates
Similarweb	Channel mix	Paid tiers; weak for small sites

Tool	✅	❌
CrUX	Real, free	Origin-level only

42.9 Where freshers go wrong in Chapter 42 topics

Mistake	Why it happens	Avoid it by	Fix it by
"Competitor gets 2M visits" as fact	Trusting estimates	🟡 label	Rephrase as estimate
Benchmarking against the wrong set	Only big brands	42.2 tiers	Rebalance
No action from benchmark	Report-only	🟧 example	Add "so what"

42.10 Thumb rules

👍 Know the class average before celebrating.

👍 Same tool, same keywords, every month.

👍 Your own history is the best benchmark after 90 days.

42.11 Notes to remember

📝 Estimates for direction; CrUX for real speed comparison.

📝 5 core competitors monthly, rest quarterly.

Chapter 42 Glossary

Term	Simple meaning
Benchmark	A reference point to compare against

Term	Simple meaning
Share of voice	Our share of visibility on a keyword set
Clickstream	Panel data on browsing used for estimates
Position tracking	Daily ranking checks for chosen keywords
Referring domain	A unique website linking to us

Chapter 42 Quiz

1. How accurate are Semrush/Similarweb traffic numbers?

- A) Official Google data
- B) Exact
- C) Audited
- D) Estimates

2. Which benchmark uses real user data?

- A) Guessing
- B) Semrush traffic
- C) CrUX
- D) Similarweb

3. What should be compared?

- A) Trends and ratios
- B) Single estimates as facts
- C) Prices only
- D) Logos

4. How many core competitors monthly?

- A) 0
- B) 5
- C) 50
- D) 1

5. Best benchmark after 90 days?

- A) None
- B) Competitor's claims
- C) Random blog
- D) Your own history

6. What does share of voice measure?

- A) Followers
- B) Ad spend
- C) Page speed
- D) Our visibility share on a keyword set

7. Should we interact with competitor accounts?

- A) Yes, follow
- B) Yes, comment
- C) Yes, like
- D) No, view-only

8. Why use the same tool every month?

- A) Switching tools breaks trends
- B) It's cheaper
- C) No reason
- D) Google requires it

9. What turns a benchmark into action?

- A) A "so what" deciding where to compete
- B) Nothing
- C) More competitors
- D) A prettier chart

10. In the report card analogy, the benchmark is...

- A) Your marks
- B) The class average
- C) The teacher
- D) The exam hall

Sources

1. Semrush, Traffic Analytics data accuracy
2. Similarweb, Data methodology
3. Chrome UX Report, BigQuery

Answer key, Chapter 42: 1-D · 2-C · 3-A · 4-B · 5-D · 6-D · 7-D · 8-A · 9-A · 10-B

04

LEVEL 4

Analysis

Reports, explorations, funnels, paths, user explorer, segments, attribution, content, links, CTAs, search, Clarity, data quality and alerts.

43 · A Tour of GA4's Standard Reports

44 · Explorations: GA4's Workbench

45 · Building Funnels (TOFU → MOFU → BOFU → Portal)

46 · Paths, Entry Pages and Exit Pages

47 · User Explorer: One Visitor's Journey, Responsibly

48 · Segments, Audiences, Cohorts and Lifetime

49 · Attribution: Who Gets Credit?

50 · Landing Page and Content Performance

51 · Internal Link Performance

52 · CTA Performance

53 · Search Performance Analysis

54 · Analysing with Microsoft Clarity

55 · Data Quality: Can We Trust the Numbers?

56 · Alerts and Anomaly Detection

LEVEL 4 · CHAPTER 43

Chapter 43: A Tour of GA4's Standard Reports

Standard reports are GA4's ready-made screens. They're quick for daily checks, and once we customise the library they show Ongo Weight Loss's own dimensions: page type, funnel stage and CTA name.

> 💡 **Trivia**
>
> GA4's data retention setting (2 or 14 months) affects explorations and funnel reports only; standard reports use aggregated data and aren't limited by it. [1] ● Official (Google)
>
> That's why yearly comparisons belong in standard reports, Looker or BigQuery, not explorations.

43.1 The left menu

Section	Reports	Daily use
Home	Overview cards, insights	Glance
Reports snapshot	Summary cards	Morning check
Realtime	Last 30 minutes	After releases
Acquisition	User acquisition (first touch), Traffic acquisition (session), (Search Console if published)	Channel trend
Engagement	Events, Key events, Pages and screens, Landing page	Content and CTAs
Retention (Lifecycle)	New vs returning, cohort retention	Weekly
User (Demographics, Tech)	Country/region, device, browser	Weekly (legal limits on use)

Section	Reports	Daily use
Advertising	Attribution paths, model comparison	Chapter 49

43.2 User acquisition vs Traffic acquisition

	User acquisition	Traffic acquisition
Question	How did **new users** first find us?	Where did **each session** come from?
Dimension prefix	"First user ..."	"Session ..."
Use	Which channels bring new people	Which channels bring visits now

> 🪔 **Desi analogy: pehli mulaqat aur roz ka aana-jaana**
>
> Dost se pehli baar kahan mile (user acquisition), aur ab woh roz kis raaste se aata hai (traffic acquisition). **Dono alag sawaal hain.**

43.3 Customising reports for Ongo Weight Loss

#	Change	How
1	Use **OWL channels** as default channel group	Admin → Channel groups; in reports, choose "Session OWL channels" dimension
2	Pages report: add secondary dimension `page_type`	Click **+** next to dimension
3	Create a **"Pages by funnel stage"** detail report	Library → Create new report → Detail → dimension `funnel_stage`, metrics Views, Engaged sessions, Key events
4	Create a **"CTA performance"** report	Detail → dimension Event name filtered to `cta_` + `cta_location`
5	Create a **"Portal exits"** report	Detail → dimension Landing page + query string → filter Event name = exit_to_portal
6	Add them to a new **Ongo Weight Loss** collection	Library → Create collection → publish

#	Change	How
7	Set **key events** as the primary metric in the collection	Report customisation

43.4 Metrics you must understand

Metric	Meaning
Users (Active users)	Users with an engaged session or first visit
Sessions	Visits (30-minute inactivity ends one)
Engaged sessions	>10 seconds, or key event, or 2+ page views
Engagement rate	Engaged sessions ÷ sessions
Average engagement time	Time the page was in focus
Views	Page views
Event count	Times an event fired
Key events	Count of key event occurrences
Session key event rate	Sessions with a key event ÷ sessions

43.5 Comparisons and filters

Feature	Use
Add comparison (top of report)	e.g., Organic Social vs Organic Search
Filter	e.g., `site_id` = wcr
Date compare	Previous period / same period last year

🟧 **Ongo Weight Loss example**

Morning check in GA4 (3 minutes): Reports snapshot → Traffic acquisition (OWL channels, yesterday vs same day last week) → Ongo Weight Loss collection → Portal exits. Anything odd → open the daily Looker dashboard (Chapter 60). 🔵

43.6 Do's and Don'ts

✅ Do	❌ Don't
Build an Ongo Weight Loss collection	Use default reports only
Compare like-for-like days	Compare Monday to Sunday
Use session dimensions for traffic	Mix first-user and session dimensions
Know metric definitions	Guess what "users" means

43.7 Pros and cons of standard reports

✅ Pros	❌ Cons
Fast, no setup	Limited dimensions per view
Not limited by retention	Thresholding can hide rows
Customisable library	Fewer visual options than Looker

43.8 Where freshers go wrong in Chapter 43 topics

Mistake	Why it happens	Avoid it by	Fix it by
Comparing user and traffic acquisition	Different scopes	43.2	Pick one per question

Mistake	Why it happens	Avoid it by	Fix it by
"(not set)" landing pages	Sessions without page_view	Check consent/page_view	Investigate
Using Realtime for trends	Fun to watch	Realtime = QA only	Use reports

43.9 Thumb rules

👍 First meeting vs daily route: know which question you're asking.

👍 Collection first, default menu second.

👍 Realtime for testing, reports for decisions.

43.10 Notes to remember

📝 Engaged session = >10 s, or key event, or 2+ page views.

📝 Retention setting affects explorations, not standard reports.

Chapter 43 Glossary

Term	Simple meaning
Collection	A custom group of reports in GA4's menu
Detail report	Table report with chosen dimensions/metrics
First user dimension	How a user first arrived
Session dimension	How a session arrived
Thresholding	GA4 hiding small rows for privacy

Chapter 43 Quiz

1. Which report shows how new users first arrived?

- A) Traffic acquisition
- B) Realtime
- C) User acquisition
- D) Pages

2. What's an engaged session?

- A) Any session
- B) Only desktop
- C) Only purchases
- D) >10 s, or key event, or 2+ page views

3. Does retention setting affect standard reports?

- A) Yes
- B) Only Realtime
- C) Only on Mondays
- D) No

4. Where do you create a collection?

- A) Clarity
- B) Library
- C) GTM
- D) GSC

5. What is Realtime for?

- A) Attribution
- B) Monthly trends
- C) QA and quick checks
- D) Billing

6. What's engagement rate?

- A) Clicks ÷ impressions
- B) Views ÷ users
- C) Users ÷ events

- D) Engaged sessions ÷ sessions

7. Which dimension prefix means per-visit?

- A) Ad ...
- B) Item ...
- C) First user ...
- D) Session ...

8. Which channel group should reports use?

- A) None
- B) Paid only
- C) OWL channels
- D) Random

9. What fair comparison avoids weekday bias?

- A) Monday vs Sunday
- B) Same weekday last week
- C) Today vs last year only
- D) Any two days

10. In the dost analogy, traffic acquisition is...

- A) Where you first met
- B) His name
- C) His phone
- D) The daily route he takes now

Sources

1. Google Analytics Help, Data retention
2. Google Analytics Help, Customize reports and the library

Answer key, Chapter 43: 1-C · 2-D · 3-D · 4-B · 5-C · 6-D · 7-D · 8-C · 9-B · 10-D

LEVEL 4 · CHAPTER 44

Chapter 44: Explorations: GA4's Workbench

Explorations let you drag dimensions and metrics into custom tables, funnels, paths and segments. They're where questions turn into answers without SQL.

> 💡 **Trivia**
>
> Explorations are limited by your data retention setting. With 14 months selected, you can explore back 14 months; with the default 2 months, only 2. [1] ● Official (Google)
>
> Set retention to 14 months on day one (Chapter 11).

44.1 Exploration techniques

Technique	Answers	Chapter
Free form	Custom tables and charts	44
Funnel exploration	Step-by-step drop-off	45
Path exploration	What users did before/after	46
Segment overlap	How segments overlap	48
User explorer	One user's timeline	47
Cohort exploration	Return behaviour by start week	48
User lifetime	Lifetime behaviour by first source	48

44.2 Anatomy of an exploration

Panel	What it holds
Variables (left)	Date range, segments, dimensions, metrics you've imported
Tab settings (middle)	Technique, rows, columns, values, filters, visualisation
Canvas (right)	The result

44.3 Building a free-form table (example)

Question: Which blog clusters send people to BOFU pages?

#	Step
1	Explore → **Blank** → name "Cluster → BOFU"
2	Variables → Dimensions **+** → import `content_cluster`, `page_type`, `Landing page + query string`
3	Metrics **+** → Sessions, Engaged sessions, Key events, Event count
4	Rows: `content_cluster`; Values: Sessions, Engaged sessions
5	Filter: Event name = `internal_link_click` and `link_target` contains `/services/` (event count)
6	Visualisation: table → switch to bar if presenting
7	**Share** with the property (others can view, not edit)

> 🍛 **Desi analogy: rasoi ka workbench**
>
> Reports thaali mein parosa hua khaana hain. Exploration rasoi ka workbench hai, jahan tum khud sabzi kaatte ho, masala milaate ho. **Naya sawaal = workbench.**

44.4 Sampling and thresholds

Issue	Sign	What to do
Sampling	Green/yellow shield icon "based on X% of data"	Shorten date range, or use BigQuery
Thresholding	Rows missing, message about thresholds	Use device-based identity, wider ranges, or BigQuery
(other) row	High-cardinality dimension	Fewer unique values (we cleaned URLs)

44.5 Our standard explorations (build once, share)

Name	Technique	Rows/steps
Content → BOFU	Free form	content_cluster × internal_link_click to services
CTA by location	Free form	cta_name × cta_location × page_type
Funnel TOFU → portal	Funnel	Chapter 45
After landing on blog	Path	Chapter 46
Portal exit users sample	User explorer	Chapter 47
Returning visitors by cluster	Cohort	Chapter 48

> 🟧 **Ongo Weight Loss example**
>
> Name every shared exploration with a prefix "OWL –" and the owner's initials, e.g., "OWL – CTA by location – AO". The team can find them, and nobody edits someone else's work by mistake. 🔵

44.6 Do's and Don'ts

✅ Do	❌ Don't
Set retention to 14 months	Leave 2 months

✅ Do	❌ Don't
Name and share explorations	Keep private untitled ones
Watch the sampling shield	Present sampled numbers as exact
Use BigQuery when limits bite	Fight sampling forever

44.7 Pros and cons of explorations

✅ Pros	❌ Cons
No SQL	Retention limit
Funnels, paths, segments	Sampling/thresholds
Quick to share	Not great for recurring dashboards

44.8 Where freshers go wrong in Chapter 44 topics

Mistake	Why it happens	Avoid it by	Fix it by
Custom dimension missing	Not registered	Chapter 25	Register (not retroactive)
Results "(other)"	Too many values	Clean URLs	Narrow dimension
Can't go back far	Retention 2 months	44.1 trivia	Change to 14 (future only)

44.9 Thumb rules

> 👍 Reports are the thaali; explorations are the workbench.
>
> 👍 Name it, share it, reuse it.
>
> 👍 Shield icon = check before presenting.

44.10 Notes to remember

📝 Seven techniques: free form, funnel, path, segment overlap, user explorer, cohort, lifetime.

📝 Prefix shared explorations "OWL –".

Chapter 44 Glossary

Term	Simple meaning
Exploration	Custom analysis workspace in GA4
Sampling	Using part of the data to estimate results
Cardinality	Number of unique values in a dimension
Technique	Type of exploration (free form, funnel…)
Canvas	Area where results appear

Chapter 44 Quiz

1. What limits how far back explorations go?

- A) Number of users
- B) Fonts
- C) Data retention setting
- D) GSC

2. Which technique shows step drop-off?

- A) Free form
- B) Segment overlap
- C) Funnel exploration
- D) Cohort

3. What does the shield icon warn about?

- A) Errors in GTM
- B) Consent
- C) Sampling
- D) Fonts

4. What causes "(other)" rows?

- A) High-cardinality dimensions
- B) Consent
- C) Too few users
- D) GSC

5. Where do you import dimensions?

- A) GTM
- B) Variables panel
- C) Canvas
- D) Admin

6. What prefix should shared explorations use?

- A) None
- B) "OWL –"
- C) "Test"
- D) Random

7. When should you switch to BigQuery?

- A) Only for images
- B) When sampling/thresholds limit you
- C) For every question
- D) Never

8. Is registering a custom dimension retroactive?

- A) Only on weekends
- B) Yes
- C) Sometimes
- D) No

9. Which technique shows one user's timeline?

- A) User explorer
- B) Cohort
- C) Funnel
- D) Free form

10. In the rasoi analogy, explorations are…

- A) The dining table
- B) The served thaali
- C) The kitchen workbench
- D) The fridge

Sources

1. Google Analytics Help, Explorations overview
2. Google Analytics Help, About data sampling

Answer key, Chapter 44: 1-C · 2-C · 3-C · 4-A · 5-B · 6-B · 7-B · 8-D · 9-A · 10-C

LEVEL 4 · CHAPTER 45

Chapter 45: Building Funnels (TOFU → MOFU → BOFU → Portal)

A funnel shows how many people move from one step to the next and where they drop. This chapter builds the Ongo Weight Loss funnels in GA4, step by step, and explains open vs closed and direct vs indirect.

> 💡 **Trivia**
>
> GA4 funnel explorations support up to **10 steps**, open or closed funnels, "directly followed by" or "indirectly followed by" between steps, elapsed time, and a "next action" view of the top 5 events after a step. [1] 🟢 Official (Google)
>
> Ten steps is plenty; good funnels use four to six.

45.1 Funnel settings explained

Setting	Meaning	Our default
Closed	Users must enter at step 1	Closed for the content funnel
Open	Users can enter at any step	Open for "BOFU → portal"
Indirectly followed by	Other events may happen between steps	Default
Directly followed by	Next event must be the step	Rare (CTA → portal)
Within (time)	Step must happen within X minutes/hours	Optional
Trended	Shows step completion over time	Weekly review
Elapsed time	Average time between steps	Useful

Setting	Meaning	Our default
Make user funnel / session funnel	Users across sessions vs one session	Session for content flow; user for multi-visit journeys

> 🍛 **Desi analogy: chhanni (sieve)**
>
> Aata chhanni se guzarta hai; har parat pe kuch ruk jaata hai. Funnel dikhata hai kis parat pe kitna ruka. **Jahan zyada rukta hai, wahan chhanni saaf karo.**

45.2 Funnel 1: Content to portal (session, closed)

Step	Condition
1 Content visit	Event `page_view` where `funnel_stage` = tofu
2 Comparison/explainer	`page_view` where `funnel_stage` = mofu
3 Service/medication page	`page_view` where `funnel_stage` = bofu
4 CTA click	Event name starts with `cta_`
5 Portal exit	`exit_to_portal`

45.3 Funnel 2: Money page to portal (session, open)

Step	Condition
1 Medication page view	`page_view` where `page_type` = medication
2 Engaged	`scroll_depth` where `percent_scrolled` ≥ 50, or `faq_open`
3 Pricing/eligibility section	`cta_*` with `cta_location` in pricing/eligibility, or internal link to pricing
4 Portal exit	`exit_to_portal`

45.4 Funnel 3: Tool to portal (user, closed)

Step	Condition
1 Tool start	`tool_start_*`
2 Tool complete	`tool_complete_*`
3 Any BOFU view	`page_view` funnel_stage = bofu
4 Portal exit	`exit_to_portal`

45.5 Building it (clicks)

#	Step
1	Explore → **Funnel exploration** template
2	Tab settings → Steps → pencil ✏️ → **Add step**
3	Each step: name it; condition "Event" → choose event → **Add parameter** (e.g., funnel_stage = tofu)
4	Choose Closed/Open; Make session funnel or user funnel
5	Breakdown: `Session OWL channels` or `site_id`
6	Toggle **Show elapsed time** and **Next action**
7	Save; name "OWL – Funnel 1 Content to portal"

45.6 Reading a funnel

Look at	Question
Biggest abandonment step	Where to improve first
Breakdown by channel	Which source brings qualified visitors
Next action after drop	What they did instead (e.g., clicked phone)
Trended view	Did a change improve the step?

Look at	Question
Segment → **Create segment from step**	Study the droppers (right-click step)

45.7 GA4 funnel vs BigQuery funnel

	GA4 funnel	BigQuery (Chapter 35)
Order	Strict (step order)	Any order unless you code order
Limits	10 steps, retention, thresholds	None
Best use	Visual, quick	History, custom logic

> 🟧 **Ongo Weight Loss example**
>
> If Funnel 2 shows 60% of medication-page visitors scroll to 50% but only 8% reach the pricing CTA, the content team tests moving the cost summary above the fold (Chapter 67), and the trended funnel shows whether step 3 rises. 🔵

45.8 Do's and Don'ts

✅ Do	❌ Don't
Keep funnels to 4–6 steps	Use 10 steps for vanity
Break down by channel	Look at totals only
Label open/closed in the name	Mix them silently
Create segments from drop steps	Guess why people dropped

45.9 Pros and cons of GA4 funnels

✅ Pros	❌ Cons
Visual and fast	Retention and threshold limits

✅ Pros	❌ Cons
Elapsed time and next action	Up to 10 steps
Segments from steps	Not ideal for scheduled reporting

45.10 Where freshers go wrong in Chapter 45 topics

Mistake	Why it happens	Avoid it by	Fix it by
Funnel shows almost nobody at step 1	Closed funnel with rare first step	Choose open or broader step 1	Edit
Parameter step returns zero	Parameter not registered or wrong value case	Chapter 25; lowercase	Fix values
Comparing user and session funnels	Different scopes	Name clearly	Rebuild

45.11 Thumb rules

👍 Clean the sieve where most flour stops.

👍 Four to six steps tell the story.

👍 Right-click the drop, study the droppers.

45.12 Notes to remember

📝 Closed = start at step 1; open = start anywhere.

📝 Up to 10 steps; elapsed time; next action top 5.

Chapter 45 Glossary

Term	Simple meaning
Funnel	Ordered steps showing drop-off
Closed funnel	Must enter at step 1
Open funnel	Can enter at any step
Abandonment	Users who didn't reach the next step
Trended funnel	Funnel completion over time

Chapter 45 Quiz

1. How many steps can a GA4 funnel have?

- A) Up to 10
- B) 100
- C) Unlimited
- D) 3

2. What's a closed funnel?

- A) Users must enter at step 1
- B) No users
- C) Private funnel
- D) Users enter anywhere

3. What does "next action" show?

- A) Top events after a step
- B) GTM tags
- C) Next month's data
- D) Search queries

4. How do you study droppers?

- A) Delete the step
- B) Email them

- C) Create segment from the step
- D) Ignore them

5. Which funnel type suits multi-visit journeys?

- A) Closed only
- B) User funnel
- C) Session funnel
- D) None

6. How many steps are ideal?

- A) 1
- B) 4–6
- C) 20
- D) 10 always

7. Why might a parameter step return zero?

- A) Font issues
- B) Too many users
- C) GSC delay
- D) Not registered or wrong value

8. What does "directly followed by" mean?

- A) The next event must be the step
- B) Any order
- C) Same user only
- D) Within a year

9. What breakdown shows qualified sources?

- A) Font
- B) Colour
- C) Browser version only
- D) Channel

10. In the sieve analogy, the funnel shows…

- A) The flour brand
- B) The flour price

- C) Where most flour stops
- D) The chakki

Sources

1. Google Analytics Help, Funnel exploration

Answer key, Chapter 45: 1-A · 2-A · 3-A · 4-C · 5-B · 6-B · 7-D · 8-A · 9-D · 10-C

LEVEL 4 · CHAPTER 46

Chapter 46: Paths, Entry Pages and Exit Pages

Where do people land, where do they go next, and where do they leave? Path exploration shows the common routes visually; landing and exit analysis tells you which pages open and close journeys.

> 💡 **Trivia**
>
> In GA4 path exploration you can choose a starting point **or** an ending point, not both, and the tree shows the top 5 nodes per step by default, expandable to 20. A new session starts after 30 minutes of inactivity. [1] ● Official (Google)
>
> Backward paths ("what led to the portal exit?") are often more useful than forward ones.

46.1 Forward and backward paths

Type	Starting/ending point	Question
Forward	Start: page or event (e.g., landing on `/blog/glp-1-side-effects/`)	What do people do after this page?
Backward	End: event (e.g., `exit_to_portal`)	What did people do before reaching the portal?

46.2 Building a path (clicks)

#	Step
1	Explore → **Path exploration** → **Start over** (top right)
2	Choose **Starting point** (forward) or **Ending point** (backward)

#	Step
3	Node type: **Page path and screen class** (or our `page_type` if imported; or Event name)
4	Pick the node (e.g., `exit_to_portal` for backward)
5	Click nodes to expand step by step
6	Node type per step can change: e.g., step +1 by Event name, step +2 by Page
7	Toggle **View unique nodes only** to collapse repeats (page reloads)
8	Save "OWL – Before portal exit (backward)"

> 🪔 **Desi analogy: gali ke nishaan**
>
> Mitti mein pairon ke nishaan dekh ke pata chalta hai log kis gali se aaye aur kahan mude. **Path exploration wahi nishaan hai.**

46.3 Landing pages

Where	How
GA4 Reports	Engagement → **Landing page** report (add `Session OWL channels` comparison)
Exploration	Dimension "Landing page + query string"
BigQuery	`entrances = 1` (Chapter 35)

Landing page scorecard columns: sessions, engagement rate, % reaching BOFU, internal link clicks per session, portal exits, portal exit rate. 🔵

46.4 Exit pages

GA4 doesn't have a built-in "exit rate" metric in standard reports; use:

Method	How
Exploration	Free form, dimension Page path, metric **Exits** (available in explorations)
BigQuery	Last page_view per session (Query B, Chapter 35)

Method	How
Special exits	`exit_to_portal` (good exit), outbound_click (neutral), plain exit (lost)

46.5 Good exit vs bad exit

Exit type	Example	Meaning
Good	`exit_to_portal`, `phone_click`, `chat_open`	Journey moved forward
Neutral	Outbound to a cited study	Research behaviour
Bad	Leaves from pricing page with no action	Friction or mismatch

Exit rate that matters: plain exits from BOFU pages ÷ BOFU page views. 🔵

46.6 Entry → exit matrix

From BigQuery Query B, build a matrix of entry cluster × exit type:

Entry cluster	% portal	% phone/chat	% plain exit
glp-1			
semaglutide			
tirzepatide			
weight-loss basics			
tools			

> **🟧 Ongo Weight Loss example**
>
> A backward path from `exit_to_portal` might show that 30% of portal exits came through the "How much does compounded semaglutide cost" page. That page then gets priority for freshness, speed and internal links from TOFU articles. 🔵

46.7 Do's and Don'ts

✅ Do	❌ Don't
Use backward paths from key events	Only look forward from home
Separate good and bad exits	Treat all exits as bad
Collapse repeat nodes	Read reload noise as journeys
Use BigQuery for every path	Expect GA4 to show all

46.8 Pros and cons of path exploration

✅ Pros	❌ Cons
Visual, quick	Top nodes only
Backward analysis	Start or end, not both
Mix node types	Thresholds on small data

46.9 Where freshers go wrong in Chapter 46 topics

Mistake	Why it happens	Avoid it by	Fix it by
Calling every exit a failure	Old bounce thinking	46.5	Classify exits
Paths cluttered with same page	Reloads/SPA	Unique nodes	Toggle
Ignoring "(other)"	Hidden nodes	Expand to 20	BigQuery

46.10 Thumb rules

👍 Follow the footprints backward from the goal.

👍 Some exits are wins.

👍 GA4 for the picture, BigQuery for every path.

46.11 Notes to remember

📝 Start OR end point; top 5 → 20 nodes.

📝 Exits metric lives in explorations.

Chapter 46 Glossary

Term	Simple meaning
Path	Sequence of pages/events
Node	One step in a path
Landing page	First page of a session
Exit page	Last page of a session
Backward path	Path ending at a chosen event

Chapter 46 Quiz

1. Can a GA4 path have both start and end points?

- A) Only for events
- B) Only on mobile
- C) Yes
- D) No, one or the other

2. What's a backward path for?

- A) What happens next year
- B) GTM debugging
- C) GSC queries
- D) What happened before a goal

3. Where is the Exits metric?

- A) Bing
- B) GSC
- C) Clarity only
- D) Explorations

4. Which is a good exit?

- A) exit_to_portal
- B) Closing the tab instantly
- C) Leaving pricing with no action
- D) A 404

5. What does "unique nodes only" remove?

- A) Key events
- B) Repeated consecutive nodes
- C) Mobile data
- D) All users

6. What ends a session after inactivity?

- A) 30 minutes
- B) 5 seconds
- C) 1 week
- D) 24 hours

7. Which BigQuery query gives entry and exit?

- A) Query G
- B) None
- C) Query H
- D) Query B

8. How many nodes per step by default?

- A) 1
- B) 50
- C) 5
- D) 100

9. What exit rate matters most?

- A) None
- B) All exits
- C) Plain exits from BOFU pages
- D) Exits from home

10. In the footprints analogy, paths show…

- A) Shop prices
- B) Weather
- C) Which lanes people took
- D) Shoe sizes

Sources

1. Google Analytics Help, Path exploration

Answer key, Chapter 46: 1-D · 2-D · 3-D · 4-A · 5-B · 6-A · 7-D · 8-C · 9-C · 10-C

LEVEL 4 · CHAPTER 47

Chapter 47: User Explorer: One Visitor's Journey, Responsibly

User explorer shows a single (anonymous) visitor's timeline: every page and event in order. It's the closest GA4 gets to "watching one person's journey" — and it must be used carefully on a health site.

> 💡 **Trivia**
>
> GA4's user explorer identifies users by "Effective user ID", which is the User ID if you send one, otherwise the device/client ID. By default it lists the top 10 users. [1] 🟢 Official (Google)
>
> We don't send User IDs, so every "user" is an anonymous browser.

47.1 What you see

Element	Detail
User list	Anonymous client IDs with sessions, event count
User snapshot	First seen, platform, location, device
Timeline	Every event and page, by session, with timestamps
Event detail	Click an event to see its parameters

47.2 Opening it

Route	Steps
Template	Explore → **User explorer** template
From a segment	In a funnel/path, right-click a step → **View users**

Route	Steps
From free form	Right-click a row → View users

47.3 Responsible-use rules

Rule	Why
Use for **pattern learning**, not watching individuals for curiosity	Respect privacy
Never try to identify a person	Re-identification risk
Never export timelines	Data minimisation
Only Zone A/B pages exist in GA4	Zone C excluded by design
Analyst role or above only	Access control (Chapter 10)

> 🪔 **Desi analogy: CCTV ki recording**
>
> Dukaan ki CCTV recording dekhte ho yeh samajhne ke liye ki log kaunsa shelf miss karte hain, kisi ka peecha karne ke liye nahi. **User explorer seekhne ke liye, jaasoosi ke liye nahi.**

47.4 Good questions for user explorer

Question	How
What do portal exiters do before exiting?	Backward path → View users on last step → review 10 timelines
Why do people abandon at the pricing step?	Funnel drop step → View users
Do tool users come back?	Cohort of tool_complete → View users
Is tracking firing correctly in real life?	Recent users → check events in order (QA)

47.5 Reading a timeline (example, anonymised)

Time	Event	Notes
10:02	page_view `/blog/what-is-glp-1/` (entrance, organic)	TOFU
10:05	scroll_depth 75	Engaged
10:06	internal_link_click → `/blog/semaglutide-vs-tirzepatide/`	Body link
10:11	faq_open "cost"	Price concern
10:12	internal_link_click → `/services/weight-loss/glp-1-medications/semaglutide/`	To BOFU
10:15	cta_semaglutide_get_started (hero)	CTA
10:15	exit_to_portal	Goal

Lesson: the cost FAQ was the turning point; make cost answers easy to find in TOFU/MOFU articles. 🔵

> 🟧 **Ongo Weight Loss example**
>
> Weekly, the on-page SEO lead reviews 10 portal-exit timelines and 10 BOFU-abandon timelines and writes three observations in the weekly report (Chapter 65). It's qualitative research, like Clarity, but with event detail. 🔵

47.6 Do's and Don'ts

✅ Do	❌ Don't
Review samples to learn patterns	Hunt for individuals
Start from a segment or step	Browse random users
Note patterns, not people	Export or screenshot timelines widely
Use it for live QA	Rely on it for counts

47.7 Pros and cons of user explorer

✅ Pros	❌ Cons
Full event detail per visitor	Anecdotal, small sample
Great for QA	Privacy care needed
Links from funnels/paths	Retention-limited

47.8 Where freshers go wrong in Chapter 47 topics

Mistake	Why it happens	Avoid it by	Fix it by
Generalising from one user	Vivid stories	Review 10+	Confirm with numbers
Treating client ID as a person	Same person, many devices	Know limits	Phrase as "browser"
Sharing timelines in chat	Convenience	47.3 rules	Summarise patterns

47.9 Thumb rules

👍 Watch to learn, not to spy.

👍 Ten timelines, then a pattern.

👍 A client ID is a browser, not a person.

47.10 Notes to remember

📝 Effective user ID = User ID or client ID (we use client ID only).

📝 Open via template or right-click "View users".

Chapter 47 Glossary

Term	Simple meaning
User explorer	GA4 tool showing one user's event timeline
Effective user ID	ID GA4 uses for a user
Client ID	Random ID stored in the browser cookie
Timeline	Events in time order
Qualitative	Insight from examples, not counts

Chapter 47 Quiz

1. What's the effective user ID for us?

- A) Phone
- B) Patient ID
- C) Client ID (no User ID sent)
- D) Email

2. How many users does it list by default?

- A) 1
- B) 1,000
- C) All
- D) Top 10

3. What's the right use?

- A) Exporting timelines
- B) Identifying people
- C) Learning patterns
- D) Spying

4. How do you open users from a funnel step?

- A) Right-click → View users
- B) GSC

- C) BigQuery only
- D) GTM Preview

5. How many timelines before a conclusion?

- A) 1
- B) 0
- C) 1,000 always
- D) 10+

6. Does user explorer include Zone C pages?

- A) Sometimes
- B) Only payments
- C) Yes
- D) No

7. Who can access it?

- A) Analyst role or above
- B) Patients
- C) Everyone public
- D) Only competitors

8. What can user explorer help QA?

- A) DNS records
- B) Events firing in order in real life
- C) Sitemaps
- D) Server costs

9. Is a client ID a person?

- A) Sometimes a doctor
- B) No, it's a browser
- C) Always unique human
- D) Yes

10. In the CCTV analogy, user explorer is for...

- A) Billing
- B) Fixing lights

- C) Following someone
- D) Learning which shelf people miss

Sources

1. Google Analytics Help, User explorer ●

Answer key, Chapter 47: 1-C · 2-D · 3-C · 4-A · 5-D · 6-D · 7-A · 8-B · 9-B · 10-D

Chapter 48: Segments, Audiences, Cohorts and Lifetime

Averages hide stories. Segments split visitors into meaningful groups (organic vs social, blog readers vs tool users); cohorts show whether people come back; lifetime shows long-run value by first source.

> 💡 **Trivia**
>
> GA4 explorations support three segment types: **user**, **session** and **event** segments, and you can compare up to four segments side by side in most techniques. [1] ● Official (Google)
>
> Pick the scope that matches the question.

48.1 Segment scopes

Scope	Includes	Example
User	All activity of users who ever matched	Users who ever completed a tool
Session	Sessions that matched	Sessions from Organic Social
Event	Only matching events	Only CTA events on medication pages

48.2 Our standard segments

Segment	Scope	Condition
Organic search sessions	Session	Session OWL channels = Organic Search
Social sessions	Session	= Organic Social
AI assistant sessions	Session	= AI Assistants

Segment	Scope	Condition
Blog entrants	Session	Landing page contains `/blog/`
BOFU viewers	Session	page_view with funnel_stage = bofu
Tool completers	User	tool_complete_*
Portal exiters	User	exit_to_portal
Engaged non-converters	Session	Engaged session = true AND no exit_to_portal
Child-site visitors	Session	site_id ≠ ongo_main

> 🪔 **Desi analogy: shaadi ke mehmaan**
>
> Mehmaan ek jaise nahi: dulhe waale, dulhan waale, office waale, padosi. Sabka average lagaoge toh kuch samajh nahi aayega. **Segment mehmaanon ki alag tables hain.**

48.3 Audiences vs segments

	Segment	Audience
Where	Explorations	Admin → Audiences
Retroactive	Yes	No (from creation)
Use	Analysis	Reports, comparisons, (ads later)
Health caution	Fine for analysis	**Don't build health-condition audiences for advertising**; paid media book will cover

48.4 Cohort exploration

Setting	Our choice
Cohort inclusion	First touch (or tool_complete)
Return criteria	Any event / page_view
Granularity	Weekly

Setting	Our choice
Metric	Active users (or retention %)
Breakdown	First user OWL channels

Question answered: of people who first came in week X, what % came back in weeks 1–4? 🔵

48.5 User lifetime

Use	Example
Lifetime engagement by first source	Do Organic Search first-timers return more than Social?
Lifetime key events by first landing page	Which entry content creates users who later exit to portal

48.6 Segment overlap

Question: How many users are both blog readers and tool users, and how many of those reached the portal? Use **Segment overlap** with three segments. 🔵

> 🟧 **Ongo Weight Loss example**
>
> If "Tool completers" have a portal exit rate five times higher than "Blog entrants", the BMI and dose-explainer tools deserve more internal links from articles, and the TOFU articles deserve tool CTAs. Segments turn a hunch into a decision. 🔵

48.7 Do's and Don'ts

✅ Do	❌ Don't
Choose scope deliberately	Use user scope for session questions
Save standard segments	Rebuild each time
Compare 2–4 segments	Compare 10 at once
Keep audiences non-sensitive	Build condition-based ad audiences

48.8 Pros and cons of segments

✅ Pros	❌ Cons
Reveal hidden differences	Thresholds on small segments
Retroactive in explorations	Scope confusion
Quick to compare	Not shared with reports

48.9 Where freshers go wrong in Chapter 48 topics

Mistake	Why it happens	Avoid it by	Fix it by
User segment inflates sessions	Includes all sessions of matching users	Session scope	Rebuild
Tiny segments	Too narrow	Wider dates	Combine conditions
Expecting audiences to fill backwards	Not retroactive	Create early	Wait

48.10 Thumb rules

> 👍 Seat guests at separate tables.
>
> 👍 Scope matches the question.
>
> 👍 Cohorts answer "do they come back?"

48.11 Notes to remember

📝 User, session, event segments; up to 4 compared.

📝 Audiences aren't retroactive.

Chapter 48 Glossary

Term	Simple meaning
Segment	Subset of data in explorations
Audience	Saved group of users in Admin
Cohort	Group sharing a start date
Retention	% returning later
Segment overlap	Venn view of segments

Chapter 48 Quiz

1. Which scope includes all activity of matching users?

- A) Page
- B) Session
- C) User
- D) Event

2. Are segments retroactive in explorations?

- A) Only audiences
- B) Yes
- C) Only weekdays
- D) No

3. Are audiences retroactive?

- A) No
- B) Always
- C) Yes
- D) Only for social

4. What does a cohort answer?

- A) What's the DNS?
- B) Which font?
- C) Do people come back?
- D) What is our CTR?

5. How many segments can be compared?

- A) 50
- B) Up to 4
- C) None
- D) 1

6. What should we not build?

- A) Organic segments
- B) Health-condition audiences for ads
- C) Tool segments
- D) Cohorts

7. Which technique shows overlap?

- A) Segment overlap
- B) Funnel
- C) Free form only
- D) Path

8. Which segment finds sessions from AI tools?

- A) Browser
- B) Session OWL channels = AI Assistants
- C) Device = mobile
- D) Country

9. What's the risk with tiny segments?

- A) Faster reports
- B) More users
- C) Thresholds hide data
- D) None

10. In the shaadi analogy, segments are…

- A) Separate tables for different guests
- B) The food
- C) The stage
- D) The band

Sources

1. Google Analytics Help, About segments
2. Google Analytics Help, Cohort exploration

Answer key, Chapter 48: 1-C · 2-B · 3-A · 4-C · 5-B · 6-B · 7-A · 8-B · 9-C · 10-A

LEVEL 4 · CHAPTER 49

Chapter 49: Attribution: Who Gets Credit?

A visitor reads an article from Google, comes back from Instagram, then exits to the portal. Which channel "caused" it? Attribution is the rulebook for sharing credit, and for Ongo Weight Loss, content-level first touch matters as much as channel credit.

> 💡 **Trivia**
>
> Since November 2023, GA4 offers data-driven attribution and last-click models only; first click, linear, time decay and position-based were removed from GA4. [1] ● Official (Google)
>
> That's one reason we compute first-touch content credit ourselves in BigQuery.

49.1 Attribution in GA4

Setting	Where	Our choice
Reporting attribution model	Admin → Attribution settings	Data-driven (default)
Key event lookback (acquisition events)	Same	30 days
Other key events lookback	Same	90 days
Channels that can receive credit	Same	Paid and organic

49.2 Three levels of "credit"

Level	Question	Tool
First touch (user)	Which channel/page first brought this person?	GA4 User acquisition; BigQuery first landing page

Level	Question	Tool
Session (last non-direct)	Which channel brought the session that converted?	GA4 Traffic acquisition
Multi-touch (data-driven)	How much did each touchpoint contribute?	GA4 Advertising → Attribution paths, Model comparison

> 🍛 **Desi analogy: cricket match ka hero**
>
> Match jeeta. Credit kise? Opener jisne shuruaat di (first touch), finisher jisne chhakka maara (last click), ya poori team (data-driven)? **Teeno sahi, sawaal alag.**

49.3 Ongo Weight Loss reporting rule

Report	Model	Why
Daily dashboard	Session channel (last non-direct)	Simple, stable
Monthly content value	**First-touch landing page** (BigQuery Query E, user-level)	Content team's question
Quarterly strategy	GA4 attribution paths + model comparison	See assist roles
Backend leads/payments	Channel from portal context param `ctx_ch` (if legal approves) else unattributed totals	No personal data

49.4 Attribution paths report

#	Step
1	Advertising → **Attribution → Attribution paths**
2	Key event: `exit_to_portal`
3	See early / mid / late touchpoints by channel
4	**Model comparison**: data-driven vs last click; channels with credit gain under data-driven are "assisters"

49.5 Why numbers don't match across tools

Source	Credit rule
GA4 reports	Data-driven (key event credit) or session last non-direct
GA4 BigQuery	Raw; you choose
Backend `ctx_ch`	Channel of the session that clicked to the portal
Social platforms	Their own view-through/click rules

Rule: always label which model a number uses. 🔵

> 🟧 **Ongo Weight Loss example**
>
> Monthly report line: "Organic search started 58% of portal-exit journeys (first touch); organic social finished 12% but assisted 21% (data-driven)." Two numbers, two models, both clearly labelled, and the team understands social's real role. 🔵

49.6 Do's and Don'ts

✅ Do	❌ Don't
Label the model on every number	Mix models in one table
Use first-touch for content value	Judge TOFU by last click
Keep lookbacks stable	Change settings monthly
Use aggregated backend channel totals	Match backend records to users

49.7 Pros and cons of models

Model	✅	❌
First touch	Shows discovery value	Ignores closers

Model	✅	❌
Last click	Simple	Undervalues content
Data-driven	Balanced	Black box; needs volume

49.8 Where freshers go wrong in Chapter 49 topics

Mistake	Why it happens	Avoid it by	Fix it by
"Blog drives no conversions"	Last-click view	First touch	Query E
Expecting first-click in GA4	Removed in 2023	49.1 trivia	BigQuery
Unlabelled numbers	Rushing	Labels	Add model name

49.9 Thumb rules

> 👍 Opener, finisher, team: say which hero you mean.
>
> 👍 Content is judged on first touch.
>
> 👍 Label the model, always.

49.10 Notes to remember

> 📝 GA4 models: data-driven, last click variants only.
>
> 📝 Lookbacks: 30 days acquisition, 90 days other.

Chapter 49 Glossary

Term	Simple meaning
Attribution	Rules for giving credit to touchpoints
First touch	Credit to the first interaction
Last non-direct click	Credit to last non-direct channel
Data-driven attribution	Credit by machine-learned contribution
Lookback window	How far back touchpoints count

Chapter 49 Quiz

1. Which models remain in GA4?

- A) None
- B) Data-driven and last click
- C) First click only
- D) Linear and time decay

2. Where do we compute first-touch content credit?

- A) Clarity
- B) GTM
- C) Bing
- D) BigQuery

3. Which model for the daily dashboard?

- A) Session channel (last non-direct)
- B) First touch only
- C) Data-driven only
- D) Random

4. What's an "assister"?

- A) Channel gaining credit under data-driven
- B) A developer

- C) A plugin
- D) A tag

5. What must every number show?

- A) Its author
- B) Its font
- C) Its colour
- D) Its attribution model

6. Why does blog look weak in last click?

- A) It's untracked
- B) It's slow
- C) It has no visitors
- D) It often starts, not finishes, journeys

7. Where are attribution paths?

- A) Engagement → Pages
- B) Admin → Streams
- C) Explore → Cohort
- D) Advertising → Attribution

8. Our other key events lookback?

- A) 7 days
- B) 1 day
- C) 1 year
- D) 90 days

9. How is backend channel known?

- A) Social DMs
- B) Portal context parameter (if approved), as totals
- C) Patient emails
- D) Guessing

10. In the cricket analogy, first touch is…

- A) The opener
- B) The finisher

- C) The crowd
- D) The umpire

Sources

1. Google Analytics Help, Attribution models (changes Nov 2023) ●
2. Google Analytics Help, Attribution settings ●

Answer key, Chapter 49: 1-B · 2-D · 3-A · 4-A · 5-D · 6-D · 7-D · 8-D · 9-B · 10-A

LEVEL 4 · CHAPTER 50

Chapter 50: Landing Page and Content Performance

Every page has a job: TOFU pages attract and teach, MOFU pages compare and reassure, BOFU pages convince and convert. This chapter scores each page against its own job.

> 💡 **Trivia**
>
> GA4 defines an engaged session as one lasting longer than 10 seconds, having a key event, or having two or more page views; engagement rate is engaged sessions divided by sessions. [1] ● Official (Google)
>
> Engagement rate replaced bounce rate as the headline quality metric; bounce rate is simply its opposite.

50.1 Scorecards by funnel stage

Stage	Primary job	Main KPIs	Secondary
TOFU (blog basics)	Attract, educate	Organic entrances, engagement rate, scroll 75% rate	Internal link CTR to MOFU/BOFU
MOFU (comparisons, cost, eligibility explainers)	Move closer	% to BOFU, FAQ opens, tool starts	Engagement time
BOFU (service, medication pages)	Convert	CTA click rate, portal exit rate	Phone/chat rate, plain exit rate
Tools	Engage, qualify	Completion rate, % to BOFU	Return rate
Trust (medical team, reviews, policies)	Reassure	Views from BOFU, % back to BOFU	Time

> 🍛 **Desi analogy: team ke players**
>
> Bowler ko runs se nahi, wickets se judge karte ho; batsman ko wickets se nahi. **Har page ko uske kaam se judge karo.**

50.2 Content performance table (monthly)

Column	Source
Page, page_type, funnel_stage, cluster	GA4 page_view params
GSC impressions, clicks, CTR, position	GSC bulk export
Organic entrances	GA4/BigQuery
Engagement rate, avg engagement time	GA4
Scroll 75% rate	scroll_depth ÷ views
Internal link CTR	internal_link_click ÷ views
% sessions reaching BOFU	BigQuery
CTA click rate	cta_* ÷ views
Portal exits (first touch)	Query E
Last updated, reviewer	CMS (publish_month param)

50.3 Content actions matrix

Pattern	Diagnosis	Action
High impressions, low CTR	Title/description weak or SERP features	Rewrite title/meta (Book Six)
High clicks, low engagement	Intent mismatch or slow page	Rework intro; check CWV
High engagement, low internal CTR	Missing/weak links	Add contextual links (Book Three)
High BOFU reach, low CTA	BOFU page friction	CTA/position test (Chapter 67)

Pattern	Diagnosis	Action
Declining impressions	Freshness or competition	Refresh with medical review
Low everything for 6 months	Not needed	Merge/redirect or noindex (with SEO lead)

50.4 Content decay check

Signal	Threshold (starting point)
Impressions down vs 3 months ago	> 20%
Position down	> 3 places
Last updated	> 12 months (YMYL)

50.5 Building it

Tool	How
GA4	Ongo Weight Loss collection "Pages by funnel stage" (Chapter 43)
BigQuery	Scheduled table `reporting.content_monthly` joining GSC + GA4
Looker	SEO & content dashboard (Chapter 62)

> 🟧 **Ongo Weight Loss example**
>
> "Semaglutide side effects" may have huge impressions and low portal exits, and that's fine: it's TOFU. Judge it on engagement and internal link CTR to the semaglutide page. Judging it on portal exits would wrongly mark a strong article as failing. 🔵

50.6 Do's and Don'ts

✅ Do	❌ Don't
Score pages by stage	Use one KPI for all pages

✅ Do	❌ Don't
Combine GSC and GA4	Look at GA4 alone
Act on patterns	Report without actions
Refresh YMYL content with reviewer	Edit medical claims casually

50.7 Pros and cons of stage scorecards

✅ Pros	❌ Cons
Fair judgement	Needs funnel_stage on every page
Clear actions	More columns
Aligns with Book Three	—

50.8 Where freshers go wrong in Chapter 50 topics

Mistake	Why it happens	Avoid it by	Fix it by
Deleting TOFU for "no conversions"	Last-click thinking	50.1	Stage KPIs
Chasing bounce rate	UA habit	Engagement rate	Retrain
Refresh without measurement	No baseline	Note date	Annotate

50.9 Thumb rules

> 👍 Judge the bowler by wickets.
>
> 👍 Impressions → clicks → engagement → links → CTA → portal.
>
> 👍 Every refresh gets a date note.

50.10 Notes to remember

> 📝 Engaged session: >10 s, key event, or 2+ views.
>
> 📝 Decay: impressions −20%, position −3, age >12 months.

Chapter 50 Glossary

Term	Simple meaning
Scorecard	Set of KPIs for a page type
Internal link CTR	Internal link clicks ÷ page views
Content decay	Gradual performance decline
Intent mismatch	Page doesn't match what searchers want
Refresh	Updating existing content

Chapter 50 Quiz

1. What's a TOFU page's main job?

- A) Take payments
- B) Collect emails
- C) Attract and educate
- D) Show policies

2. What's a BOFU KPI?

- A) Portal exit rate
- B) Followers
- C) Impressions only
- D) Scroll only

3. High impressions, low CTR means…

- A) Delete page
- B) Add more ads
- C) Page is perfect
- D) Title/description needs work

4. What replaced bounce rate as headline?

- A) Exit rate
- B) Followers
- C) Engagement rate
- D) Pageviews

5. High engagement, low internal CTR means...

- A) Slow server
- B) Wrong DNS
- C) No impressions
- D) Missing or weak internal links

6. When is content "decaying"?

- A) Impressions down >20% vs 3 months
- B) Never
- C) When a competitor posts
- D) Any one-day drop

7. Who must review YMYL refreshes?

- A) Social team only
- B) Nobody
- C) Medical reviewer
- D) Developer only

8. What's the right action for a page dead for 6 months?

- A) Ignore forever
- B) Duplicate it
- C) Merge/redirect or noindex with SEO lead
- D) Add keywords

9. Where's the monthly content table built?

- A) Clarity
- B) BigQuery scheduled table
- C) GTM
- D) Bing only

10. In the cricket analogy, judging pages by job means...

- A) Only captains
- B) Bowlers by wickets, batsmen by runs
- C) Nobody judged
- D) Everyone by runs

Sources

1. Google Analytics Help, Engagement rate and bounce rate 🟢
2. Ongo Weight Loss, Book Three content map, internal 🔵

Answer key, Chapter 50: 1-C · 2-A · 3-D · 4-C · 5-D · 6-A · 7-C · 8-C · 9-B · 10-B

LEVEL 4 · CHAPTER 51

Chapter 51: Internal Link Performance

You asked: "How many users clicked the link from my GLP-1 article to my Semaglutide page?" This chapter answers exactly that, and then ranks every internal link by how well it moves people toward the portal.

> 💡 **Trivia**
>
> GA4 doesn't track internal link clicks by default; Enhanced Measurement's "outbound clicks" only covers links to other domains. [1] 🟢 Official (Google)
>
> That's why Chapter 18 built `internal_link_click` with link_target and link_position.

51.1 The internal link event (recap)

Parameter	Example
`link_target`	`/services/weight-loss/glp-1-medications/semaglutide/`
`link_text`	"compounded semaglutide program" (trimmed)
`link_position`	body, sidebar, related, breadcrumb, nav, footer, toc
`page_location`	Source page (cleaned)
`link_type`	contextual, nav, cta-styled (if set)

51.2 Link metrics

Metric	Formula
Clicks	Count of internal_link_click for source → target

Metric	Formula
Link CTR	Clicks ÷ source page views
Downstream engagement	Engagement on target page in the same session
Downstream portal rate	Sessions with this click that later exit_to_portal ÷ sessions with this click

51.3 GA4 exploration: "GLP-1 article → Semaglutide page"

#	Step
1	Free form; rows: Page path (source), `link_target`, `link_position`
2	Values: Event count
3	Filter: Event name = internal_link_click; Page path contains `/blog/what-is-glp-1/`
4	Read the row with link_target = semaglutide page

51.4 BigQuery: downstream portal rate per link

```
WITH clicks AS (
  SELECT CONCAT(user_pseudo_id,'.',CAST((SELECT value.int_value FROM UNNEST(event_params)
WHERE key='ga_session_id') AS STRING)) AS session_key,
    event_timestamp AS click_ts,
    REGEXP_REPLACE((SELECT value.string_value FROM UNNEST(event_params) WHERE
key='page_location'), r'\?.*$','') AS from_page,
    (SELECT value.string_value FROM UNNEST(event_params) WHERE key='link_target') AS
to_page,
    (SELECT value.string_value FROM UNNEST(event_params) WHERE key='link_position') AS
position
  FROM `PROJECT.analytics_PROPERTY.events_*`
  WHERE _TABLE_SUFFIX BETWEEN '20261101' AND '20261130' AND
event_name='internal_link_click'
),
portal AS (
  SELECT CONCAT(user_pseudo_id,'.',CAST((SELECT value.int_value FROM UNNEST(event_params)
WHERE key='ga_session_id') AS STRING)) AS session_key,
    MIN(event_timestamp) AS portal_ts
  FROM `PROJECT.analytics_PROPERTY.events_*`
  WHERE _TABLE_SUFFIX BETWEEN '20261101' AND '20261130' AND event_name='exit_to_portal'
  GROUP BY session_key
)
SELECT from_page, to_page, position,
  COUNT(*) AS clicks,
  COUNTIF(p.portal_ts > c.click_ts) AS later_portal,
  ROUND(SAFE_DIVIDE(COUNTIF(p.portal_ts > c.click_ts), COUNT(*))*100,1) AS
portal_rate_pct
FROM clicks c LEFT JOIN portal p USING (session_key)
GROUP BY from_page, to_page, position
HAVING clicks >= 20
ORDER BY later_portal DESC;
```

> **Desi analogy: shortcut gali**
>
> Mohalle mein kuch galiyan seedha bazaar le jaati hain, kuch ghuma ke wapas. **Internal link report batata hai kaunsi gali bazaar tak pahunchati hai.**

51.5 Link position comparison

Position	Typical pattern	What to do
Body (contextual)	Highest intent	Write more contextual links in key paragraphs
Related articles	Exploration	Keep relevant, 3–5 items
Sidebar	Low CTR on mobile	Don't depend on it
Breadcrumb	Navigation	Fine
Nav/footer	Generic	Not a content signal

51.6 Hub-and-spoke health check

Check	Expectation
Every spoke links to its hub	Clicks recorded spoke → hub
Hub links to every spoke	Clicks hub → each spoke
TOFU → BOFU links exist in every cluster	At least one per article
Orphan pages	No internal link clicks into them (and no links in crawl)

> 🟧 **Ongo Weight Loss example**
>
> If body links from "GLP-1 side effects" to the semaglutide page have a 2% CTR but those clickers exit to the portal at 14%, while sidebar links have 0.3% CTR and 3%, the SEO team rewrites two more paragraphs in that cluster with contextual links. Measured, not guessed. 🔵

51.7 Do's and Don'ts

✅ Do	❌ Don't
Track position on every link	Track clicks without position

✅ Do	❌ Don't
Measure downstream portal rate	Stop at clicks
Use thresholds (20+ clicks)	Rank tiny samples
Feed results into Book Three linking plans	Keep insights in reports only

51.8 Pros and cons of internal link tracking

✅ Pros	❌ Cons
Proves which links move users	Developer setup needed
Guides content writing	More events
Validates hub-and-spoke	Position must be maintained

51.9 Where freshers go wrong in Chapter 51 topics

Mistake	Why it happens	Avoid it by	Fix it by
CTA buttons counted as links	Overlap	CTA exclusion (Chapter 18)	Fix selectors
Link targets with query strings	Raw hrefs	Clean target	Regex
"(not set)" position	Missing container attribute	data-link-position	Add attribute

51.10 Thumb rules

> 👍 Find the lanes that reach the bazaar.
>
> 👍 Clicks are good; clicks that lead on are better.
>
> 👍 Body links beat sidebars.

51.11 Notes to remember

📝 internal_link_click: link_target, link_text, link_position.

📝 Link CTR = clicks ÷ source page views.

Chapter 51 Glossary

Term	Simple meaning
Internal link	Link to another page on our sites
Link CTR	Link clicks ÷ page views
Downstream	What happens after the click
Orphan page	Page with no internal links pointing to it
Contextual link	Link inside body text

Chapter 51 Quiz

1. Does GA4 track internal links by default?

- A) Yes
- B) Only nav links
- C) Only on mobile
- D) No

2. Which parameter stores the destination?

- A) `link_target`
- B) `page_type`
- C) `cta_name`
- D) `metric_name`

3. What's link CTR?

- A) Clicks ÷ impressions
- B) Views ÷ users
- C) Clicks ÷ source page views
- D) Users ÷ sessions

4. What's downstream portal rate?

- A) GSC CTR
- B) Share of clicking sessions that later exit to portal
- C) Followers
- D) Server speed

5. Which position usually has highest intent?

- A) Footer
- B) Nav
- C) Body (contextual)
- D) Sidebar

6. What's an orphan page?

- A) No internal links pointing to it
- B) A portal page
- C) A page with many links
- D) A deleted page

7. Why set a 20-click threshold?

- A) No reason
- B) For speed
- C) Required by Google
- D) Avoid tiny-sample rankings

8. What causes "(not set)" link_position?

- A) Consent
- B) GSC delays
- C) Too many users
- D) Missing container attribute

9. What should results feed into?

- A) Paid ads only
- B) Book Three linking plans
- C) HR
- D) Nothing

10. In the gali analogy, the report shows...

- A) Which lanes reach the bazaar
- B) Street names only
- C) Shop prices
- D) Traffic police

Sources

1. Google Analytics Help, Enhanced measurement events 🟢
2. Ongo Weight Loss, Chapter 18 internal link tracking, internal 🔵

Answer key, Chapter 51: 1-D · 2-A · 3-C · 4-B · 5-C · 6-A · 7-D · 8-D · 9-B · 10-A

LEVEL 4 · CHAPTER 52

Chapter 52: CTA Performance

Not one generic `button_click`, but separate events for Get Started, Check Eligibility, Call Us, Chat With Us and more, each with a location. This chapter turns those events into decisions: which CTA, where, on which page type, moves people forward.

> 💡 **Trivia**
>
> GA4 allows up to 500 distinct event names per property (web), so naming every CTA as its own event is safe at our scale, as long as names follow a pattern. [1] ● Official (Google)
>
> Our pattern `cta_[group]_[action]` plus shared parameters keeps it tidy.

52.1 CTA metrics

Metric	Formula	Meaning
CTA clicks	Count of cta_*	Volume
CTA click rate	CTA clicks ÷ page views (of pages showing it)	Attraction
CTA → portal rate	Sessions with cta then exit_to_portal ÷ sessions with cta	Quality
Share by location	Clicks per cta_location ÷ total	Where clicks happen
Contact mix	phone / chat / email / portal shares	Preference

52.2 The CTA matrix (monthly)

cta_group	cta_action	cta_location	Page type	Clicks	Click rate	→ Portal
semaglutide	get_started	hero	medication			
semaglutide	get_started	pricing	medication			

cta_group	cta_action	cta_location	Page type	Clicks	Click rate	→ Portal
weight_loss	check_eligibility	sticky_mobile	service			
global	call_us	header	all			
blog	learn_more	inline	blog_article			

52.3 Getting it from GA4

#	Step
1	Exploration free form: rows `cta_name`, `cta_location`, `page_type`; values Event count
2	Filter Event name begins with `cta_`
3	Add segment "Portal exiters" (session scope) to compare
4	Or use the "CTA performance" report in the Ongo Weight Loss collection

52.4 BigQuery: CTA → portal rate

```
WITH cta AS (
  SELECT CONCAT(user_pseudo_id,'.',CAST((SELECT value.int_value FROM UNNEST(event_params)
WHERE key='ga_session_id') AS STRING)) AS session_key,
    event_name,
    (SELECT value.string_value FROM UNNEST(event_params) WHERE key='cta_location') AS
cta_location
  FROM `PROJECT.analytics_PROPERTY.events_*`
  WHERE _TABLE_SUFFIX BETWEEN '20261101' AND '20261130' AND
STARTS_WITH(event_name,'cta_')
),
portal AS (
  SELECT DISTINCT CONCAT(user_pseudo_id,'.',CAST((SELECT value.int_value FROM
UNNEST(event_params) WHERE key='ga_session_id') AS STRING)) AS session_key
  FROM `PROJECT.analytics_PROPERTY.events_*`
  WHERE _TABLE_SUFFIX BETWEEN '20261101' AND '20261130' AND event_name='exit_to_portal'
)
SELECT event_name, cta_location, COUNT(*) AS clicks,
  COUNT(DISTINCT c.session_key) AS sessions,
  COUNT(DISTINCT p.session_key) AS portal_sessions
FROM cta c LEFT JOIN portal p USING (session_key)
GROUP BY event_name, cta_location
ORDER BY clicks DESC;
```

> **Desi analogy: dukaan ke darwaaze**
>
> Badi dukaan ke chaar darwaaze hain. Kaunse darwaaze se zyada graahak andar aate hain aur kaunse se sirf jhaank ke chale jaate hain? **CTA report har darwaaze ka register hai.**

52.5 Reading patterns

Pattern	Likely meaning	Action
Hero CTA high clicks, low portal rate	Curious clickers, unclear next step	Clarify CTA copy and portal landing
Pricing CTA low clicks, high portal rate	Strong intent, low visibility	Move pricing higher
Sticky mobile CTA dominates	Mobile users prefer it	Keep; test copy

Pattern	Likely meaning	Action
Phone > portal on BOFU	Need for reassurance	Strengthen FAQs; staffing for calls
Blog CTAs ~0	Not relevant	Replace with contextual links

52.6 Compliance check on CTAs

Check	Rule
CTA copy	No guaranteed results; no brand drug names; "compounded" clear (Book Six)
Event params	No personal data; cta_name from our list only

> 🟧 **Ongo Weight Loss example**
>
> If the "Check eligibility" CTA has half the clicks of "Get started" but double the portal rate, the team tests "Check eligibility" as the primary hero CTA on the tirzepatide page (Chapter 67). 🔵

52.7 Do's and Don'ts

✅ Do	❌ Don't
Report by name × location × page type	Report one CTA total
Measure quality (→ portal)	Celebrate clicks alone
Keep a CTA register (Appendix A)	Let developers invent names
Test one change at a time	Change all CTAs together

52.8 Pros and cons of specific CTA events

✅ Pros	❌ Cons
Clear reports without filters	More event names
Location insight	Needs a register
Quality measurement	—

52.9 Where freshers go wrong in Chapter 52 topics

Mistake	Why it happens	Avoid it by	Fix it by
Click rate using all page views	Wrong denominator	Pages showing the CTA	Filter page_type
Duplicate CTA events	Click + link listeners	Chapter 17 exclusions	QA
New CTA without name	Rushed release	Register first	Add to Appendix A

52.10 Thumb rules

> 👍 Register every door.
>
> 👍 Clicks are interest; portal is intent.
>
> 👍 Location matters as much as copy.

52.11 Notes to remember

> 📝 cta_[group]_[action] + cta_name, cta_group, cta_location.
>
> 📝 CTA → portal rate is the quality KPI.

Chapter 52 Glossary

Term	Simple meaning
CTA	Call to action button/link
CTA click rate	CTA clicks ÷ views of pages showing it
CTA register	List of approved CTA names
Sticky CTA	Button fixed on screen while scrolling
Denominator	The number you divide by

Chapter 52 Quiz

1. What's the CTA quality KPI?

- A) Font size
- B) Colour
- C) Total clicks only
- D) CTA → portal rate

2. What's the event name pattern?

- A) Random
- B) `button_click`
- C) `cta_[group]_[action]`
- D) `click1`

3. Which parameter says where the CTA was?

- A) `link_target`
- B) `page_title`
- C) `cta_location`
- D) `metric_id`

4. Correct denominator for CTA click rate?

- A) Impressions
- B) Views of pages showing that CTA

- C) Users worldwide
- D) All site views

5. Hero CTA high clicks, low portal rate suggests...

- A) Too few pages
- B) Perfect CTA
- C) Broken GA4
- D) Unclear next step

6. What's GA4's event-name limit for web?

- A) 500
- B) 10
- C) Unlimited
- D) 5

7. Phone > portal on BOFU suggests...

- A) No visitors
- B) Broken phone
- C) Need for reassurance
- D) Spam

8. Where are approved CTA names kept?

- A) GSC
- B) Developer memory
- C) Bing
- D) Appendix A register

9. What compliance rule applies to CTA copy?

- A) Promise outcomes
- B) Name the pharmacy
- C) Use superlatives
- D) No guaranteed results or brand drug names

10. In the darwaaza analogy, the CTA report is...

- A) The cashier
- B) The godown

- C) Each door's register
- D) The shop sign

Sources

1. Google Analytics Help, Collection and configuration limits
2. Ongo Weight Loss, Chapter 17 CTA tracking, internal

Answer key, Chapter 52: 1-D · 2-C · 3-C · 4-B · 5-D · 6-A · 7-C · 8-D · 9-D · 10-C

LEVEL 4 · CHAPTER 53

Chapter 53: Search Performance Analysis

Search is our main growth engine. This chapter turns Search Console, Bing and GA4 data into a weekly SEO analysis: entity performance, cannibalisation, AI Overview effects, and striking-distance opportunities.

> 💡 **Trivia**
>
> Google counts an impression when a link to your site appears in search results; clicks that open the result count as clicks, and a result inside an AI Overview counts for position too. [1]
> 🟢 Official (Google)
>
> AI features change CTR without changing your visibility, so read CTR shifts carefully.

53.1 Weekly search analysis checklist

#	Check	Tool
1	Clicks, impressions, CTR, position vs previous 4 weeks	GSC / Looker
2	Entity groups (weight loss, GLP-1, semaglutide, tirzepatide, cost, eligibility)	BigQuery entity CASE
3	Top gaining/losing pages	GSC compare
4	Striking-distance list	Chapter 37 query
5	Cannibalisation check	Query → multiple URLs
6	Index coverage changes	GSC Pages
7	Bing clicks and AI citations	Bing WMT
8	Organic engagement and portal exits by landing cluster	GA4/BigQuery

53.2 Entity grouping (BigQuery)

```
SELECT
  CASE
    WHEN REGEXP_CONTAINS(query, r'tirzepatide') THEN 'tirzepatide'
    WHEN REGEXP_CONTAINS(query, r'semaglutide') THEN 'semaglutide'
    WHEN REGEXP_CONTAINS(query, r'glp-?1') THEN 'glp-1'
    WHEN REGEXP_CONTAINS(query, r'(cost|price|insurance)') THEN 'cost'
    WHEN REGEXP_CONTAINS(query, r'(weight loss|lose weight|obesity)') THEN 'weight-loss'
    ELSE 'other'
  END AS entity,
  SUM(clicks) AS clicks, SUM(impressions) AS impressions
FROM `PROJECT.searchconsole_main.searchdata_site_impression`
WHERE data_date >= DATE_SUB(CURRENT_DATE(), INTERVAL 28 DAY)
  AND NOT is_anonymized_query AND search_type='WEB'
GROUP BY entity
ORDER BY clicks DESC;
```

Brand queries ("ongo weight loss") get their own group: **branded vs non-branded** is a core split. 🔵

> 🍛 **Desi analogy: mandi ke bhaav**
>
> Mandi mein har sabzi ka bhaav alag chalta hai. "Total sabzi" ka bhaav kisi kaam ka nahi. **Entity groups har sabzi ka alag bhaav hain.**

53.3 Cannibalisation check

```
SELECT query, COUNT(DISTINCT url) AS urls, SUM(impressions) AS impressions,
  STRING_AGG(DISTINCT url, ' ; ') AS url_list
FROM `PROJECT.searchconsole_main.searchdata_url_impression`
WHERE data_date >= DATE_SUB(CURRENT_DATE(), INTERVAL 28 DAY)
  AND NOT is_anonymized_query AND search_type='WEB'
GROUP BY query
HAVING urls >= 2 AND impressions >= 200
ORDER BY impressions DESC;
```

Not every overlap is bad: a blog and a service page can both rank. It's a problem when positions swap week to week or both sit below 10. 🔵

53.4 Reading CTR changes

Pattern	Possible reason	Check
CTR drops, position stable	AI Overview / new SERP feature	Look at the live SERP (manual, view-only)
CTR drops, position drops	Ranking loss	Content/competitor review
Impressions spike, CTR falls	Ranking for broader queries	Query list
Clicks up, engagement down	Intent mismatch	Page intro

53.5 Branded vs non-branded

Metric	Why
Non-branded clicks	SEO growth (content working)
Branded clicks	Brand awareness (social, word of mouth)
Branded share	Business health trend

> 🟧 **Ongo Weight Loss example**
>
> Weekly SEO note format: "Non-branded clicks +9% (semaglutide +14%, tirzepatide −6%). Tirzepatide cost page CTR fell from 4.1% to 2.7% at stable position 4; AI Overview now shown. Action: add a clear cost summary table and FAQ; monitor 4 weeks." 🔵

53.6 Do's and Don'ts

✅ Do	❌ Don't
Group queries by entity	Read thousands of rows
Separate branded and non-branded	Mix them
Check SERPs before rewriting	Rewrite on CTR alone

✅ Do	❌ Don't
Note actions and dates	Forget what changed

53.7 Pros and cons of entity analysis

✅ Pros	❌ Cons
Matches site architecture	Regex upkeep
Clear priorities	Overlapping queries
Shareable	Anonymised queries missing

53.8 Where freshers go wrong in Chapter 53 topics

Mistake	Why it happens	Avoid it by	Fix it by
Panic over CTR drop	Ignoring SERP changes	53.4	Check SERP
Calling any overlap cannibalisation	Rule of thumb misuse	Thresholds	Review positions
Counting brand clicks as SEO wins	Mixing groups	53.5	Split

53.9 Thumb rules

> 👍 Each sabzi has its own bhaav.
>
> 👍 Look at the SERP before you rewrite.
>
> 👍 Branded and non-branded never in one number.

53.10 Notes to remember

📝 Weekly 8-step search checklist.

📝 Cannibalisation = 2+ URLs, swapping, both weak.

Chapter 53 Glossary

Term	Simple meaning
Entity group	Queries about the same topic/thing
Branded query	Search containing our brand
Cannibalisation	Our pages competing for one query
SERP feature	Special result types (AI Overview, FAQ, video)
Non-branded	Searches without our name

Chapter 53 Quiz

1. Why group queries by entity?

- A) No reason
- B) To match site architecture and prioritise
- C) To hide data
- D) For colours

2. CTR down, position stable usually means…

- A) Ranking loss
- B) GA4 broken
- C) A SERP feature like AI Overview
- D) Consent issue

3. Is every query overlap cannibalisation?

- A) Yes
- B) Only branded
- C) Always on mobile
- D) No

4. What do non-branded clicks show?

- A) Brand awareness only
- B) Email
- C) SEO content growth
- D) Ads

5. Should branded and non-branded be mixed?

- A) Only on Bing
- B) Only monthly
- C) No
- D) Yes

6. Which table has URL + query?

- A) events_*
- B) searchdata_url_impression
- C) searchdata_site_impression
- D) ExportLog

7. What should a weekly SEO note include?

- A) Only totals
- B) Change, reason, action, date
- C) Nothing
- D) Only screenshots

8. Clicks up, engagement down means...

- A) Server outage
- B) Bing issue
- C) Intent mismatch possible
- D) Perfect page

9. How should SERPs be checked?

- A) By clicking ads
- B) Not at all
- C) Manually, view-only
- D) By commenting

10. In the mandi analogy, entity groups are…

- A) The total mandi price
- B) The truck
- C) The weighing scale
- D) Each vegetable's own price

Sources

1. Google Search Console Help, Performance report metrics and AI features

---

Answer key, Chapter 53: 1-B · 2-C · 3-D · 4-C · 5-C · 6-B · 7-B · 8-C · 9-C · 10-D

LEVEL 4 · CHAPTER 54

Chapter 54: Analysing with Microsoft Clarity

GA4 tells you what happened; Clarity shows how it happened: rage clicks on a price table, a dead "Learn more" button, a hero image that hides the CTA on small phones. This chapter turns Clarity into a weekly UX fix list.

> 💡 **Trivia**
>
> Clarity's masking changes are not retroactive and can take up to an hour to apply; inputs and dropdowns are always masked regardless of mode. [1] 🟢 Official (Microsoft)
>
> Confirm Strict masking before reviewing recordings (Chapter 28).

54.1 What Clarity shows

Feature	Shows	Use
Dashboard	Sessions, dead clicks, rage clicks, excessive scrolling, quick backs, JavaScript errors	Weekly trend
Heatmaps (click, scroll, area)	Where people click and how far they scroll	Page design review
Recordings	Masked replays of sessions	Understand friction
Filters/segments	By page, device, source, smart events, custom tags	Focus
Insights / Copilot summaries	AI summaries of patterns (if enabled)	Starting point only 🟡
GA4 integration	Clarity dimension in GA4	Optional; review privacy first

54.2 Behaviour signals explained

Signal	Meaning	Usual cause
Dead click	Click with no response	Text that looks clickable
Rage click	Rapid repeated clicks	Broken or slow element
Excessive scrolling	Scrolling more than expected	Can't find info
Quick back	Leaves a page quickly and returns	Wrong link expectation
JS error	Script error in the session	Bugs

> 🪔 **Desi analogy: dukaan mein chhupa camera nahi, aaina**
>
> Clarity aaina hai jo dikhata hai graahak kahan atka: darwaaza bhaari tha, board padha nahi gaya. Kisi ka chehra nahi dikhta (masked). **Atkane waali jagah dikhaata hai, insaan nahi.**

54.3 Weekly Clarity routine (45 minutes)

#	Step
1	Dashboard, last 7 days vs previous: any spike in rage/dead clicks or JS errors?
2	Filter page type BOFU (URL contains `/services/`) → click heatmap mobile → note top 3 issues
3	Scroll heatmap of top 5 landing pages → where does 50% of the audience stop?
4	Recordings filtered: rage click = true, BOFU pages → watch 10
5	Recordings filtered: quick back on blog → watch 5
6	Log issues in the UX fix list (page, device, issue, evidence link, severity)
7	Share with UI/UX designer and developer

54.4 Custom tags (optional)

Developers can tag sessions with non-personal context:

```
// Only non-personal values
window.clarity?.('set', 'page_type', 'medication');
window.clarity?.('set', 'funnel_stage', 'bofu');
```

Never set tags with personal or health answers. 🔵

54.5 Clarity + GA4 together

Question	GA4	Clarity
Is the pricing CTA under-clicked?	Click rate low	Heatmap shows it below a large image on mobile
Why do people drop after FAQ?	faq_open high, exit after	Recording shows long FAQ answers pushing CTA away
Is a link broken?	404 events	Rage clicks on that link

> 🟧 **Ongo Weight Loss example**
>
> Clarity shows many dead clicks on the "compounded semaglutide" text in a comparison table: people expect it to open the medication page. Turning that text into a tracked internal link (Chapter 18) converts confusion into a measured path. 🔵

54.6 Do's and Don'ts

✅ Do	❌ Don't
Use Strict masking	Use Relaxed on a health site
Watch recordings in batches with filters	Watch random sessions
Log issues with evidence	Rely on memory
Keep Clarity off Zone C	Tag the portal

54.7 Pros and cons of Clarity

✅ Pros	❌ Cons
Free	Recordings are time-consuming
Visual evidence for design changes	Masking needs care
Behaviour signals automatic	Not a counting tool

54.8 Where freshers go wrong in Chapter 54 topics

Mistake	Why it happens	Avoid it by	Fix it by
Changing design from one recording	Vivid example	10+ recordings	Confirm in GA4
Ignoring device	Desktop heatmap only	Mobile first	Switch device
Personal data in tags	Convenience	54.4 rule	Remove tag

54.9 Thumb rules

👍 A mirror for friction, not a camera on people.

👍 Ten recordings, then a conclusion.

👍 Mobile heatmap first.

54.10 Notes to remember

📝 Signals: dead, rage, excessive scroll, quick back, JS errors.

📝 Weekly routine: 45 minutes, 7 steps.

Chapter 54 Glossary

Term	Simple meaning
Heatmap	Visual of clicks or scroll depth
Session recording	Masked replay of a visit
Dead click	Click with no response
Rage click	Repeated frustrated clicks
Quick back	Fast return after visiting a page

Chapter 54 Quiz

1. Are masking changes retroactive?

- A) Yes
- B) Only for heatmaps
- C) No
- D) Sometimes

2. What's a dead click?

- A) A link to the portal
- B) Click with no response
- C) A scroll
- D) Purchase

3. Which masking mode for us?

- A) None
- B) Strict
- C) Balanced always
- D) Relaxed

4. How many recordings before concluding?

- A) 0
- B) 1

- C) 1,000
- D) 10+

5. Which heatmap device first?

- A) Tablet only
- B) TV
- C) Mobile
- D) Desktop only

6. What may Clarity tags contain?

- A) Health answers
- B) Emails
- C) Names
- D) Non-personal context like page_type

7. Rage clicks usually mean...

- A) Ads
- B) Happy users
- C) Bots only
- D) Broken or slow element

8. Where must Clarity never run?

- A) Tools page
- B) Blog
- C) Home
- D) Zone C (portal)

9. What does GA4 add that Clarity doesn't?

- A) Recordings
- B) Rage clicks
- C) Heatmaps
- D) Reliable counts and funnels

10. In the aaina analogy, Clarity shows...

- A) Where customers get stuck, not who they are
- B) Customer faces

- C) Stock
- D) Prices

Sources

1. Microsoft Learn, Clarity masking
2. Microsoft Learn, Clarity dashboard and insights

Answer key, Chapter 54: 1-C · 2-B · 3-B · 4-D · 5-C · 6-D · 7-D · 8-D · 9-D · 10-A

LEVEL 4 · CHAPTER 55

Chapter 55: Data Quality: Can We Trust the Numbers?

A dashboard is only as good as the data under it. This chapter gives a weekly data-quality checklist, common data problems and how to spot them, and the rules for annotating and explaining gaps.

> 💡 **Trivia**
>
> GA4 can apply "data thresholds" that hide rows when user counts are small and certain identity signals are present, to protect user privacy. [1] ● Official (Google)
>
> Missing rows aren't always broken tracking; sometimes they're privacy protection.

55.1 The five qualities of good data

Quality	Question
Complete	Are all pages and events collecting?
Accurate	Do numbers match reality (e.g., backend portal starts)?
Consistent	Are names and values stable over time?
Clean	No spam, internal traffic, duplicates or personal data?
Timely	Is data arriving on schedule (exports, Sheets)?

55.2 Weekly data-quality checklist

#	Check	Pass rule
1	page_view count vs last 4 weeks	Within ±30% unless explained

#	Check	Pass rule
2	Pages with "(not set)" page_type	0 (new templates?)
3	Event names not in Appendix A	0
4	Parameter values not lowercase/unknown	0
5	Duplicate page_views (same page, same session, <1 s)	<1%
6	Internal traffic filter active	Office/team IPs excluded
7	Unwanted referrals (e.g., payment pages, child sites)	None
8	Spam/bot spikes (weird sources, 0-second sessions)	Investigate
9	BigQuery export arrived daily	ExportLog / tables present
10	GSC export arrived	ExportLog
11	`ops.daily_conversions` loaded	Yesterday's row present
12	Personal-data scan (Query H)	0 rows
13	GA4 portal exits vs backend portal starts	Ratio stable week to week

> 🪔 **Desi analogy: doodh mein paani**
>
> Doodh waala thoda paani mila de toh chai ka swaad badal jaata hai, aur roz check na karo toh pata bhi nahi chalta. **Data quality check roz ka lactometer hai.**

55.3 Common problems and fixes

Problem	How it shows	Fix
Missing page_view on new template	Page absent from reports	Add `PageMeta` (Chapter 15)
Double counting	Views ~2×	History-change setting off
Self-referrals	Own domain as referral	Configure domains / unwanted referrals
Internal traffic	Spikes on test days	Define internal traffic + filter active

Problem	How it shows	Fix
Bot traffic	Many 0-engagement sessions from one city/source	GA4 excludes known bots; add exclusions in reports; Clarity bot check
Consent drop	Sudden traffic drop with no GSC drop	CMP change; check consent rate
Sheet not updated	Social tiles empty	Owner reminder

55.4 Annotations

GA4 now supports **annotations** on charts (where available) and Looker allows text notes; also keep a **change log** sheet:

Date	Change	Owner	Expected effect
2026-11-12	New medication page template	Dev	page_views for medication pages
2026-11-20	CMP banner text change	Legal	Possible consent rate change

55.5 Reconciliation (GA4 vs backend)

Pair	Expected relation
exit_to_portal (GA4) vs portal sessions started (backend count)	GA4 lower (consent, blockers); ratio stable
GSC clicks vs GA4 organic Google sessions	GA4 lower or similar; trend matches
Social link clicks (platform) vs GA4 social sessions	GA4 lower; ratio stable

Rule: we watch **ratios over time**, not exact matches. A ratio change = investigate. 🔵

> 🟧 **Ongo Weight Loss example**
>
> If GA4 portal exits ÷ backend portal starts was 0.62 for six weeks and suddenly drops to 0.40, something broke on the website side (CTA tracking, consent banner) — not demand. The data-quality check catches it before the monthly report misleads anyone. 🔵

55.6 Do's and Don'ts

✅ Do	❌ Don't
Run the checklist weekly	Wait for someone to notice
Keep a change log	Rely on memory
Watch ratios	Chase exact matches
Explain known gaps in reports	Hide them

55.7 Pros and cons of formal data quality

✅ Pros	❌ Cons
Trustworthy reports	Weekly time cost
Early problem detection	Needs discipline
Fewer arguments about numbers	—

55.8 Where freshers go wrong in Chapter 55 topics

Mistake	Why it happens	Avoid it by	Fix it by
Reporting a drop as demand loss	No reconciliation	Ratios	Investigate tracking
Internal filter left in "Testing"	Default state	Activate	Change to Active
Ignoring thresholds	Unknown	55 trivia	Use BigQuery

55.9 Thumb rules

👍 Check the milk before making chai.

👍 Ratios over exact matches.

👍 Every change goes in the log.

55.10 Notes to remember

📝 13-point weekly data-quality checklist.

📝 Personal-data scan must be zero.

Chapter 55 Glossary

Term	Simple meaning
Data quality	How trustworthy data is
Reconciliation	Comparing two sources for consistency
Internal traffic	Visits from our own team
Annotation	Note on a chart marking a change
Change log	Record of website/tracking changes

Chapter 55 Quiz

1. What can hide rows in GA4 for privacy?

- A) GTM
- B) Bing
- C) Data thresholds
- D) Fonts

2. What do we compare GA4 and backend by?

- A) Exact match
- B) Ratios over time
- C) Names
- D) Emails

3. What does a sudden ratio change mean?

- A) Ignore
- B) Celebrate
- C) Investigate tracking
- D) Delete data

4. How many personal-data rows are acceptable?

- A) 100
- B) Zero
- C) Any
- D) Ten

5. What causes self-referrals?

- A) Too many pages
- B) Domains not configured
- C) GSC
- D) Slow site

6. What should internal traffic filter state be?

- A) Deleted
- B) Off
- C) Active
- D) Testing forever

7. What records website/tracking changes?

- A) Nothing
- B) GSC
- C) Clarity
- D) Change log

8. Double page_views usually mean...

- A) History-change page views on
- B) Consent
- C) Popular content
- D) Bots only

9. How often is the checklist run?

- A) Never
- B) Hourly
- C) Weekly
- D) Yearly

10. In the doodh analogy, the checklist is...

- A) The lactometer
- B) The cow
- C) The tea
- D) The sugar

Sources

1. Google Analytics Help, Data thresholds
2. Google Analytics Help, Define internal traffic

Answer key, Chapter 55: 1-C · 2-B · 3-C · 4-B · 5-B · 6-C · 7-D · 8-A · 9-C · 10-A

LEVEL 4 · CHAPTER 56

Chapter 56: Alerts and Anomaly Detection

You want continuous supervision. Nobody can stare at dashboards all day, so the data must tap you on the shoulder when something unusual happens: a traffic crash, a tracking break, a sudden spike, or a personal-data leak.

> 💡 **Trivia**
>
> GA4 lets you create custom insights that check conditions (hourly for some metrics, daily, weekly or monthly) and email chosen people when they're met. [1] ● Official (Google)
>
> Alerts are cheap; missed problems are expensive.

56.1 Alert layers

Layer	Tool	Speed
1 Tracking break	GA4 custom insight (page_view near zero)	Hourly/daily
2 Business anomaly	GA4 custom insights; Looker scheduled email	Daily
3 SEO problems	GSC email notices (manual actions, indexing)	As they happen
4 Speed/errors	Clarity JS errors trend; uptime monitor	Daily
5 Data pipeline	BigQuery scheduled query that emails/fails on missing data	Daily
6 Privacy	Weekly PII scan result	Weekly
7 Costs	Cloud budget alerts	On threshold

56.2 GA4 custom insights to create

Name	Condition	Frequency	Email
OWL – page_view crash	Event count page_view decreases by more than 50% vs previous period	Daily (hourly if available)	Team lead, developer
OWL – portal exits drop	Key event exit_to_portal decreases > 30%	Daily	Team lead
OWL – organic drop	Sessions, Organic Search, decrease > 30%	Daily	SEO lead
OWL – traffic spike	Sessions increase > 100%	Daily	Team lead (bot check)
OWL – 404 spike	page_not_found count increase > 100%	Daily	Developer

Where: Home → Insights → **View all insights** → **Create** (or Reports → Insights). 🟢

> 🪔 **Desi analogy: chaukidaar ki seeti**
>
> Raat ko chaukidaar seeti bajata hai jab kuch ajeeb dikhe. Tum so sakte ho kyunki seeti bajegi. **Alerts data ka chaukidaar hain.**

56.3 Looker Studio scheduled delivery

Setting	Value
Report	Daily monitoring dashboard (Chapter 60)
Share → **Schedule delivery**	Every day 9:00 am, recipients team lead + leads
Pages	Page 1 (summary)
Conditional (if available in your Looker version)	Alert when a scorecard crosses a threshold 🟡

56.4 BigQuery pipeline checks

```
-- Fails loudly if yesterday's GA4 table or ops row is missing
SELECT
  IF((SELECT COUNT(*) FROM `PROJECT.analytics_PROPERTY.INFORMATION_SCHEMA.TABLES`
      WHERE table_name = CONCAT('events_', FORMAT_DATE('%Y%m%d', DATE_SUB(CURRENT_DATE(),
INTERVAL 1 DAY)))) = 0,
     ERROR('GA4 export missing for yesterday'), 'ok') AS ga4_check,
  IF((SELECT COUNT(*) FROM `PROJECT.ops.daily_conversions`
      WHERE date = DATE_SUB(CURRENT_DATE(), INTERVAL 1 DAY)) = 0,
     ERROR('ops.daily_conversions missing yesterday'), 'ok') AS ops_check;
```

Schedule daily at 6 pm (after export arrives) with **email notification on failure**.

56.5 Anomaly thinking: is it real?

Question	Check
Is it everywhere or one page/channel?	Break down
Did we change something?	Change log
Does GSC/backend show the same?	Reconcile
Is it a holiday or news event?	Calendar
Is it bots?	Source, city, engagement time

56.6 Alert response playbook

Severity	Example	Response time	Action
P1	Personal data detected; page_view crash	Same hour	Stop source / roll back; inform lead and legal
P2	Portal exits −40%	Same day	Investigate tracking and site
P3	Organic −25% one day	Next day	Watch; check GSC
P4	Spike from referral spam	Weekly	Filter in reports

> 🟧 **Ongo Weight Loss example**
>
> The team lead's morning: 9:00 Looker email (2 minutes), GA4 insight emails if any, then decide. On most days there's nothing to do — and that calm is exactly what supervision should feel like. 🔵

56.7 Do's and Don'ts

✅ Do	❌ Don't
Alert on tracking breaks first	Only alert on business metrics
Set severity and owners	Send all alerts to everyone
Tune thresholds after a month	Leave noisy alerts on
Write what happened after each P1/P2	Fix silently

56.8 Pros and cons of alerts

✅ Pros	❌ Cons
Fast detection	Noise if thresholds poor
Less manual checking	Weekend coverage needed
Clear ownership	Setup time

56.9 Where freshers go wrong in Chapter 56 topics

Mistake	Why it happens	Avoid it by	Fix it by
Alert fatigue	Too sensitive	Tune thresholds	Raise thresholds
No owner	Group emails	Named owner	Assign

Mistake	Why it happens	Avoid it by	Fix it by
Missing export unnoticed	No pipeline check	56.4	Add check

56.10 Thumb rules

> 👍 Let the chaukidaar whistle, so you can sleep.
>
> 👍 Tracking alerts before business alerts.
>
> 👍 Every alert has an owner and a severity.

56.11 Notes to remember

> 📝 Five GA4 custom insights; daily Looker email; BigQuery pipeline check.
>
> 📝 P1–P4 response playbook.

Chapter 56 Glossary

Term	Simple meaning
Custom insight	GA4 alert condition with email
Anomaly	Unusual change in data
Alert fatigue	Ignoring alerts because there are too many
Severity	How urgent a problem is
Scheduled delivery	Automatic emailing of a Looker report

Chapter 56 Quiz

1. What creates GA4 email alerts?

- A) Custom insights
- B) Clarity
- C) GTM
- D) Explorations

2. Which alert comes first in priority?

- A) Tracking breaks
- B) Fonts
- C) Social likes
- D) Ads

3. What's P1?

- A) Small spike
- B) Personal data detected or tracking crash
- C) Referral spam
- D) Font change

4. What checks pipeline data?

- A) GSC sitemap
- B) Clarity heatmap
- C) Bing
- D) Scheduled BigQuery query with ERROR()

5. What causes alert fatigue?

- A) Owners
- B) Few alerts
- C) Too-sensitive thresholds
- D) Severity levels

6. First question for an anomaly?

- A) Everywhere or one page/channel?
- B) Ignore?
- C) Delete data?
- D) Who to blame?

7. When does the daily Looker email go?

- A) Midnight
- B) Never
- C) 9:00 am
- D) Monthly

8. What tells you if something changed on the site?

- A) GSC only
- B) Change log
- C) Nothing
- D) Weather

9. When should thresholds be tuned?

- A) Hourly
- B) Every year
- C) After about a month
- D) Never

10. In the chaukidaar analogy, alerts let you…

- A) Sell more
- B) Stay awake all night
- C) Lock the door
- D) Sleep while someone watches

Sources

1. Google Analytics Help, Create custom insights
2. Looker Studio Help, Schedule email delivery
3. Google Cloud, Scheduling queries

Answer key, Chapter 56: 1-A · 2-A · 3-B · 4-D · 5-C · 6-A · 7-C · 8-B · 9-C · 10-D

05

LEVEL 5

Looker Studio and Reporting

Looker basics, fast data sources, calculated fields, five dashboards, weekly and monthly reports, and KPI target ranges.

57 · Looker Studio Basics

58 · Data Sources, Quotas and Fast Dashboards

59 · Calculated Fields, Blends and Parameters

60 · The Daily Monitoring Dashboard

61 · The Executive Dashboard

62 · The SEO and Content Dashboard

63 · The Social and Channel Dashboard

64 · The Technical Health Dashboard

65 · Weekly and Monthly Reporting (Telling the Story)

66 · KPI Dictionary and Target Ranges

Chapter 57: Looker Studio Basics

Looker Studio is Google's free dashboard tool. It pulls GA4, Search Console, BigQuery and Sheets into one shareable report, which is exactly what daily supervision needs.

> 💡 **Trivia**
>
> Looker Studio (formerly Data Studio) is free; Looker Studio Pro adds team workspaces, asset management by organisation, and support for a fee. [1] 🟢 Official (Google)
>
> The free version is enough to start; Pro helps when ownership of many reports becomes messy.

57.1 Key concepts

Concept	Meaning
Report	The dashboard (pages, charts)
Data source	Connection to data (GA4 property, BigQuery table, Sheet) with field settings
Connector	Type of connection (GA4, BigQuery, Sheets, Search Console…)
Field	Dimension (text, date) or metric (number)
Chart	Table, scorecard, time series, bar, pie, geo, pivot…
Control	Filters the viewer can change (date range, drop-down)
Theme	Colours and fonts

57.2 Creating your first report (clicks)

#	Step
1	lookerstudio.google.com → **Create → Report** (company account)
2	Add data → **Google Analytics** → choose account → property → Add
3	Insert → **Scorecard** → metric Sessions
4	Insert → **Time series** → dimension Date, metric Sessions
5	Insert → **Table** → dimension Landing page, metrics Sessions, Engagement rate, Key events
6	Add a control → **Date range control** → default "Last 28 days"
7	Theme and layout → choose a simple theme; brand colours
8	File → rename "OWL – Practice"; Share → company domain viewers

> 🪔 **Desi analogy: dashboard gaadi ka**
>
> Gaadi ke dashboard mein speed, petrol, temperature ek nazar mein dikhte hain. Engine kholke nahi dekhna padta. **Looker Studio business ki gaadi ka dashboard hai.**

57.3 Ownership and sharing rules

Rule	Why
Reports owned by a **company account** (or Pro workspace)	Staff changes don't lose reports
Data source **credentials**: owner's credentials for shared dashboards	Viewers don't need GA4 access
Share as **Viewer** by default	Prevent accidental edits
Don't share publicly ("anyone with link")	Business data
Naming "OWL – [Dashboard] – v#"	Findability

57.4 Design principles

Principle	Practice
Question first	Each page answers one question
Top-left = most important	Scorecards first
Compare	Show change vs previous period
Few colours	One highlight colour
Labels and definitions	Info text on every KPI
Mobile check	View on phone (you read daily)

57.5 Our dashboards (built in Chapters 60–64)

Dashboard	Audience	Frequency
Daily monitoring	Team lead	Daily
Executive	Leadership	Weekly/monthly
SEO & content	SEO + writers	Weekly
Social & channel	Social team	Weekly
Technical health	Developer + tester	Weekly

> 🟧 **Ongo Weight Loss example**
>
> Build the practice report today with GA4 alone; it teaches the interface in an hour. The real dashboards then use BigQuery tables (Chapter 58) so they load fast and avoid GA4 quota errors. 🔵

57.6 Do's and Don'ts

✅ Do	❌ Don't
Own reports with a company account	Use personal Gmail
One question per page	Cram 30 charts
Add definitions	Assume everyone knows "engaged session"
Share as viewer	Give edit to all

57.7 Pros and cons of Looker Studio

✅ Pros	❌ Cons
Free	GA4 connector quotas
Many connectors	Can be slow with raw data
Easy sharing and scheduling	Limited advanced stats

57.8 Where freshers go wrong in Chapter 57 topics

Mistake	Why it happens	Avoid it by	Fix it by
Report breaks when owner leaves	Personal ownership	Company account	Transfer ownership
Viewers see "no access"	Viewer credentials set	Owner credentials	Change data source setting
Chart soup	No question	57.4	Redesign

57.9 Thumb rules

> 👍 A car dashboard, not the engine.
>
> 👍 One page, one question.
>
> 👍 Company owns, viewers view.

57.10 Notes to remember

> 📝 Report → data source → fields → charts → controls.
>
> 📝 Name "OWL – [Dashboard] – v#".

Chapter 57 Glossary

Term	Simple meaning
Looker Studio	Google's free dashboard tool
Data source	Saved connection with fields
Scorecard	Single big number chart
Control	Viewer filter like date range
Credentials	Whose access a data source uses

Chapter 57 Quiz

1. Is Looker Studio free?

- A) No
- B) Only for 30 days
- C) Only for Ads
- D) Yes (Pro is paid)

2. What should own reports?

- A) A personal Gmail
- B) A company account
- C) An intern
- D) Nobody

3. What credentials let viewers see data without GA4 access?

- A) Public link
- B) Viewer's credentials
- C) Owner's credentials
- D) None

4. What's a scorecard?

- A) A table
- B) A single big number
- C) A map
- D) A filter

5. What's the main design principle?

- A) Many colours
- B) No labels
- C) One question per page
- D) Maximum charts

6. Default sharing role?

- A) Public
- B) Viewer
- C) Editor for all
- D) Owner

7. Why use BigQuery tables for real dashboards?

- A) Faster and avoids GA4 quota errors
- B) Prettier
- C) No reason
- D) Required by law

8. Where does the most important KPI go?

- A) Hidden page
- B) Footer
- C) Bottom-right
- D) Top-left

9. What's a control?

- A) A GA4 tag
- B) A viewer filter like date range
- C) A server
- D) A CMP

10. In the gaadi analogy, Looker is…

- A) The driver
- B) The car's dashboard
- C) The petrol
- D) The engine

Sources

1. Google Cloud, Looker Studio and Looker Studio Pro
2. Looker Studio Help, Data credentials

Answer key, Chapter 57: 1-D · 2-B · 3-C · 4-B · 5-C · 6-B · 7-A · 8-D · 9-B · 10-B

LEVEL 5 · CHAPTER 58

Chapter 58: Data Sources, Quotas and Fast Dashboards

Dashboards that connect straight to GA4 often hit quota errors and load slowly. This chapter shows how to choose data sources, use extracts and BigQuery tables, and set freshness so daily dashboards are fast and reliable.

> 💡 **Trivia**
>
> The GA4 Data API that Looker Studio's GA4 connector uses has per-property quotas (tokens per hour and per day, and limits on concurrent requests). Busy dashboards can show "quota exceeded" errors. [1] 🟢 Official (Google) [2] 🟠 Industry
>
> The fix is architecture, not refreshing harder.

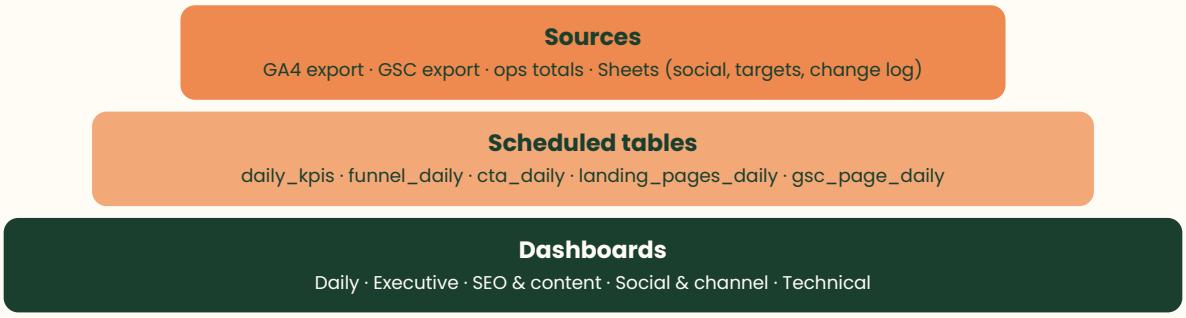
58.1 Source options

Source	Speed	Quota risk	Best for
GA4 connector (live)	Medium	High	Small practice reports
Extract data (snapshot of GA4/other)	Fast	Low	Simple dashboards; refreshes daily
BigQuery tables/views	Fast (small tables)	None (BigQuery costs)	All production dashboards
Google Sheets	Fast	None	Social data, targets, change log
Search Console connector	Medium	Low	Quick GSC views

58.2 Our production data model

BigQuery table (dataset `reporting`)	Built by	Grain
`daily_kpis`	Scheduled query 7:00 pm daily	date × site_id × channel
`landing_pages_daily`	Scheduled	date × landing_page × channel
`content_monthly`	Scheduled monthly	month × page
`cta_daily`	Scheduled	date × cta_name × cta_location × page_type
`internal_links_monthly`	Scheduled	month × from × to × position
`funnel_daily`	Scheduled	date × channel × step counts
`gsc_page_daily`	Scheduled	date × page
`cwv_monthly`	Scheduled (web-vitals + CrUX)	month × page_type × metric
`ops.daily_conversions`	Backend loader	date × channel_group

The Reporting Data Model



Dashboards read small tables, never raw events.

> **Desi analogy: dabbawala**
>
> Mumbai ke dabbawale har ghar se khaana uthaake, chhaant ke, sahi daftar tak pahunchaate hain. Har baar ghar se seedha daftar nahi bhaagte. **Scheduled tables dabbawale hain: data chhaant ke tayyar rakhte hain.**

58.3 Example scheduled table: daily_kpis

```
CREATE OR REPLACE TABLE `PROJECT.reporting.daily_kpis`
PARTITION BY date AS
WITH e AS (
  SELECT PARSE_DATE('%Y%m%d', event_date) AS date,
    (SELECT value.string_value FROM UNNEST(event_params) WHERE key='site_id') AS site_id,
    IFNULL(collected_traffic_source.manual_medium, '(none)') AS medium,
    user_pseudo_id, event_name,
    CONCAT(user_pseudo_id,'.',CAST((SELECT value.int_value FROM UNNEST(event_params)
WHERE key='ga_session_id') AS STRING)) AS session_key
  FROM `PROJECT.analytics_PROPERTY.events_*`
  WHERE _TABLE_SUFFIX >= FORMAT_DATE('%Y%m%d', DATE_SUB(CURRENT_DATE(), INTERVAL 400
DAY))
)
SELECT date, site_id, medium,
  COUNT(DISTINCT user_pseudo_id) AS users,
  COUNT(DISTINCT session_key) AS sessions,
  COUNTIF(event_name='page_view') AS page_views,
  COUNTIF(STARTS_WITH(event_name,'cta_')) AS cta_clicks,
  COUNTIF(event_name='internal_link_click') AS internal_link_clicks,
  COUNTIF(event_name='exit_to_portal') AS portal_exits,
  COUNTIF(event_name IN ('phone_click','chat_open')) AS contact_actions
FROM e
GROUP BY date, site_id, medium;
```

Note: for channel, use GA4's session attribution fields where available, or map source/medium to OWL channels with a CASE (Appendix G). Rebuilding 400 days nightly is fine at our size; switch to incremental `MERGE` when volumes grow. 🔵

58.4 Data freshness settings

Source	Setting	Our value
GA4 connector	Data freshness (1, 4 or 12 hours)	12 hours

Source	Setting	Our value
BigQuery	Data freshness (as low as 15 minutes)	12 hours (tables refresh daily)
Sheets	15 minutes default	Fine
Extracts	Auto-update daily/weekly/monthly	Daily

58.5 Avoiding GA4 quota errors

Fix	How
Move charts to BigQuery tables	Production rule
Use extracts for GA4-only reports	Data → Extract data
Fewer charts per page	6–10
Avoid high-cardinality dimensions on live GA4	Use BigQuery
Stagger scheduled emails	Not all at 9:00

> 🟧 **Ongo Weight Loss example**
>
> The daily dashboard reads only `daily_kpis`, `funnel_daily`, `cta_daily` and `ops.daily_conversions`: four small tables, a few thousand rows. It opens in two seconds on your phone, every morning, with zero GA4 quota errors. 🔵

58.6 Do's and Don'ts

✅ Do	❌ Don't
Build production on BigQuery tables	Connect 20 charts to live GA4
Partition tables by date	Query raw events from Looker
Document each table's owner	Leave mystery tables
Set sensible freshness	Set 15 min on everything

58.7 Pros and cons of the BigQuery layer

✅ Pros	❌ Cons
Fast, reliable	Setup and SQL
One definition of each KPI	Small storage/query cost
Joins backend and GSC	Needs monitoring

58.8 Where freshers go wrong in Chapter 58 topics

Mistake	Why it happens	Avoid it by	Fix it by
Quota errors every morning	Live GA4 charts	58.5	Move to BigQuery
Dashboard shows yesterday missing	Table refresh before export	Schedule 7 pm	Reschedule
Different numbers per dashboard	Different definitions	One table per KPI	Consolidate

58.9 Thumb rules

> 👍 Let the dabbawala sort before delivery.
>
> 👍 Dashboards read small tables, never raw events.
>
> 👍 One KPI, one definition, one table.

58.10 Notes to remember

> 📝 reporting.* tables refreshed by scheduled queries.
>
> 📝 GA4 connector freshness options: 1, 4, 12 hours.

Chapter 58 Glossary

Term	Simple meaning
Quota	Usage limit on an API
Extract	Saved snapshot of data in Looker
Data freshness	How often Looker refreshes cached data
Partitioned table	Table split by date for cheaper queries
Incremental load	Adding only new data instead of rebuilding

Chapter 58 Quiz

1. What causes GA4 connector errors?

- A) API quotas
- B) Fonts
- C) Clarity
- D) GSC

2. What's the production data source?

- A) Live GA4 for everything
- B) Email
- C) Screenshots
- D) BigQuery tables

3. What does an extract do?

- A) Sends emails
- B) Creates GA4 events
- C) Deletes data
- D) Saves a snapshot that refreshes on schedule

4. When should scheduled tables run?

- A) Every minute

- B) Never
- C) At midnight before export
- D) After the GA4 export arrives (evening)

5. How many charts per page?

- A) 6–10
- B) 1
- C) 100
- D) 50

6. Why partition by date?

- A) Cheaper, faster queries
- B) Required by GSC
- C) No reason
- D) For colours

7. What gives one definition per KPI?

- A) Random filters
- B) One table per KPI
- C) Many owners
- D) Many dashboards

8. GA4 connector freshness options?

- A) Weekly only
- B) None
- C) 1, 4 or 12 hours
- D) 1 minute only

9. Where does social data come from?

- A) GA4 only
- B) Google Sheets
- C) GSC
- D) Clarity

10. In the dabbawala analogy, scheduled tables…

- A) Drive the train

- B) Cook the food
- C) Eat the lunch
- D) Sort data before delivery

Sources

1. Google Analytics, Data API quotas
2. Swydo, How to solve Looker Studio quota errors
3. Looker Studio Help, Manage data freshness

Answer key, Chapter 58: 1-A · 2-D · 3-D · 4-D · 5-A · 6-A · 7-B · 8-C · 9-B · 10-D

LEVEL 5 · CHAPTER 59

Chapter 59: Calculated Fields, Blends and Parameters

Calculated fields create new metrics (rates, ratios, groupings); blends join two data sources; parameters let viewers change a value. With these three, one dashboard can answer many questions.

> 💡 **Trivia**
>
> A Looker Studio blend can join up to **5 data sources**, using left outer, right outer, inner, full outer or cross joins. [1] ● Official (Google)
>
> Blends are handy, but heavy blending is a sign the join belongs in BigQuery.

59.1 Calculated fields you'll use

Field	Formula	Type
Engagement rate	`SUM(engaged_sessions) / SUM(sessions)`	Percent
Portal exit rate	`SUM(portal_exits) / SUM(sessions)`	Percent
CTA click rate	`SUM(cta_clicks) / SUM(page_views)`	Percent
Lead rate (aggregated)	`SUM(leads) / SUM(sessions)`	Percent
Payment rate (aggregated)	`SUM(payments) / SUM(sessions)`	Percent
WoW change	Use chart **comparison date range** (no formula)	—
Page group	`CASE WHEN REGEXP_MATCH(page, '.*/blog/.*') THEN 'Blog' WHEN REGEXP_MATCH(page, '.*/services/.*') THEN 'Services' ELSE 'Other' END`	Text

Field	Formula	Type
Channel (lowercase)	`LOWER(medium)`	Text

Tip: compute ratios as **SUM ÷ SUM**, never the average of daily rates. 🔵

> 🪔 **Desi analogy: masala mix**
>
> Haldi, mirch, dhaniya alag-alag hain; garam masala unka mix. **Calculated field purane ingredients se naya masala banata hai.**

59.2 Blends

Blend	Left source	Right source	Join keys	Type
Social on-site vs on-platform	Sheet (date, platform)	`daily_kpis` by source	date + platform/source	Left outer
Targets vs actuals	`daily_kpis` (month)	Targets Sheet (month, KPI)	month	Left outer
GSC vs GA4 by page	`gsc_page_daily`	`landing_pages_daily`	date + path	Left outer

Steps: select a chart → Data → **Blend** → add sources → **Configure join** → choose keys and type → Save. 🟢

59.3 Parameters

Parameter	Use
`p_threshold` (number)	Viewer sets minimum sessions for tables
`p_metric` (text list: sessions, portal_exits, cta_clicks)	One chart switches metric via CASE
`p_target` (number)	What-if target lines

Example: `CASE p_metric WHEN 'portal_exits' THEN SUM(portal_exits) WHEN 'cta_clicks' THEN SUM(cta_clicks) ELSE SUM(sessions) END`. 🔵

59.4 Filters and controls

Control	Field
Date range	Default last 28 days, compare previous period
Site	site_id
Channel	OWL channel
Funnel stage	funnel_stage
Page group	calculated Page group

> 🟧 **Ongo Weight Loss example**
>
> A "Targets" Google Sheet (month, KPI, low, high) blended with actuals shows each KPI in green/amber/red against Chapter 66 target ranges. Changing a target is a Sheet edit, not a dashboard rebuild. 🔵

59.5 Do's and Don'ts

✅ Do	❌ Don't
SUM ÷ SUM for rates	Average daily rates
Keep blends simple (2–3 sources)	Blend 5 large sources
Name fields clearly	Leave "Calculated field 1"
Move complex joins to BigQuery	Fight slow blends

59.6 Pros and cons of blends

✅ Pros	❌ Cons
No SQL	Slow on big data

✅ Pros	❌ Cons
Quick combos	Join confusion
Viewer-friendly	Hard to debug

59.7 Where freshers go wrong in Chapter 59 topics

Mistake	Why it happens	Avoid it by	Fix it by
Rates over 100%	Wrong aggregation	SUM ÷ SUM	Fix formula
Blank blends	Key formats differ (dates as text)	Same types	Convert types
Duplicated totals	Wrong join type	Left outer	Change join

59.8 Thumb rules

> 👍 Mix masalas from good ingredients.
>
> 👍 SUM over SUM, always.
>
> 👍 Blend lightly; join heavily in BigQuery.

59.9 Notes to remember

> 📝 Blends: up to 5 sources; 5 join types.
>
> 📝 Targets live in a Sheet.

Chapter 59 Glossary

Term	Simple meaning
Calculated field	New field from a formula
Blend	Join of data sources in Looker
Join key	Shared field used to match rows
Parameter	Viewer-changeable value
Left outer join	Keep all left rows, match right where possible

Chapter 59 Quiz

1. How many sources can a blend join?

- A) 1
- B) Up to 5
- C) Unlimited
- D) 50

2. How should rates be calculated?

- A) Random
- B) MAX ÷ MIN
- C) SUM ÷ SUM
- D) Average of daily rates

3. What causes blank blends?

- A) Mismatched key formats
- B) Fonts
- C) Too few charts
- D) GSC

4. What's a parameter?

- A) A table
- B) A value viewers can change

- C) A GA4 event
- D) A tag

5. Where do targets live?

- A) A Google Sheet
- B) Developer's laptop
- C) GSC
- D) Clarity

6. What join type is usually right?

- A) Cross
- B) None
- C) Random
- D) Left outer

7. When should joins move to BigQuery?

- A) Never
- B) Always first
- C) When blends are complex or slow
- D) Only on Fridays

8. What does LOWER(medium) fix?

- A) Dates
- B) Consent
- C) Mixed-case channel values
- D) Speed

9. How do you show WoW change?

- A) Manual typing
- B) Comparison date range
- C) GSC
- D) Screenshots

10. In the masala analogy, a calculated field is…

- A) A new mix from existing ingredients
- B) The pot

- C) The stove
- D) A raw spice

Sources

1. Looker Studio Help, About data blending ●
2. Looker Studio Help, Calculated fields ●

Answer key, Chapter 59: 1-B · 2-C · 3-A · 4-B · 5-A · 6-D · 7-C · 8-C · 9-B · 10-A

LEVEL 5 · CHAPTER 60

Chapter 60: The Daily Monitoring Dashboard

This is your morning dashboard: one page, two minutes, read on a phone. It tells you whether the website is healthy, whether traffic and leads are normal, and whether anything needs action today.

> 💡 **Trivia**
>
> Looker Studio can email a report as a PDF on a schedule (daily, weekly, monthly) to people inside or outside your organisation. [1] 🟢 Official (Google)
>
> The dashboard comes to you; you don't have to remember to open it.

60.1 Layout (one page, phone-friendly)

Row	Tiles	Source
1 Health	Tracking status (page_views yesterday vs 4-week same-day avg); PII scan last result; export status	daily_kpis, checks
2 Traffic	Sessions, users, engaged rate (yesterday vs same day last week)	daily_kpis
3 Intent	BOFU sessions, CTA clicks, contact actions	daily_kpis, funnel_daily
4 Outcomes	Portal exits; leads; payments (aggregated backend, yesterday + 7-day)	daily_kpis, ops
5 Channels	Sessions and portal exits by OWL channel (bar)	daily_kpis
6 Trend	28-day time series: sessions, portal exits, payments	daily_kpis, ops
7 Movers	Top 5 landing pages up/down vs last week	landing_pages_daily
8 Notes	Change log entries in last 7 days	Sheet

> 🪔 **Desi analogy: subah ki chai aur akhbaar**
>
> Subah chai ke saath akhbaar ki headlines padhte ho, poora akhbaar nahi. Kuch badi khabar ho toh andar ka page kholte ho. **Daily dashboard headlines hai.**

60.2 Scorecard settings

Setting	Value
Date range	Yesterday (control default "Yesterday")
Comparison	Same day last week (custom: previous period shifted 7 days)
Conditional colour	Green if within target range, amber outside, red if beyond alert threshold
Info icon text	Definition + source + model

60.3 The two-minute reading routine

#	Look	If odd
1	Health row all green?	Red → developer (P1/P2)
2	Traffic within ±20% of same day last week?	Check channel row
3	Portal exits and payments normal?	Check funnel and CTA
4	Any channel unusual?	Open channel detail
5	Movers: any key page crashed?	GSC URL Inspection / site check
6	Notes: did we change something?	Expected effect?

60.4 Daily decisions it supports

Signal	Decision
Tracking broken	Hotfix today
Portal exits down, sessions normal	Check CTA/portal links, site errors

Signal	Decision
Payments down, portal exits normal	Portal/ops issue (not website); notify ops
Organic down, others normal	GSC check; algorithm news
Social spike	Find the post; repurpose

60.5 Build steps

#	Step
1	New report "OWL – Daily Monitor – v1"; page size 400×1600 (mobile portrait) or responsive layout if available
2	Add data sources: reporting.daily_kpis, funnel_daily, landing_pages_daily, ops.daily_conversions, Change log Sheet, Targets Sheet
3	Build rows 1–8 as in 60.1
4	Add date control default Yesterday
5	Share → Schedule delivery 9:00 am → team lead (+ leads)
6	Test on phone

> **🟧 Ongo Weight Loss example**
>
> A Tuesday reading: health green; sessions +6%; portal exits −31%; payments normal yesterday. Channel row: drop only on medication pages from Organic Search. Movers: tirzepatide page −45%. Two minutes in, you know exactly where to look — and you ask the developer whether yesterday's release touched that template. 🔵

60.6 Do's and Don'ts

✅ Do	❌ Don't
Keep it to one page	Add every metric

✅ Do	❌ Don't
Compare to same weekday	Compare to yesterday only
Show health first	Hide tracking status
Read daily, act weekly unless red	Change the site daily on noise

60.7 Pros and cons of a one-page daily view

✅ Pros	❌ Cons
Fast, habit-forming	Not deep
Catches breaks early	Needs drill-down elsewhere
Phone-friendly	—

60.8 Where freshers go wrong in Chapter 60 topics

Mistake	Why it happens	Avoid it by	Fix it by
Overreacting to daily noise	Small numbers	Thresholds and 7-day view	Wait for pattern
Dashboard empty before noon	Export timing	"Yesterday" uses tables refreshed previous evening	Schedule properly
No definitions	Rushing	Info icons	Add

60.9 Thumb rules

👍 Headlines with chai; full paper only if needed.

👍 Health row first, always.

👍 Same weekday comparisons.

60.10 Notes to remember

📝 Eight rows: health, traffic, intent, outcomes, channels, trend, movers, notes.

📝 9:00 am scheduled email.

Chapter 60 Glossary

Term	Simple meaning
Monitoring dashboard	Daily health and performance view
Mover	Page with big change
Conditional formatting	Colour based on value
Same-day comparison	Comparing to the same weekday last week
Drill-down	Going to more detail

Chapter 60 Quiz

1. How long should the daily read take?

- A) Two hours
- B) A week
- C) About two minutes
- D) A day

2. What row comes first?

- A) Social
- B) Health
- C) Movers
- D) Notes

3. What's the comparison?

- A) None
- B) Same day last week
- C) Yesterday vs today
- D) Last year only

4. Portal exits normal but payments down suggests...

- A) Social spike
- B) Portal/ops issue
- C) Organic loss
- D) Website tracking broken

5. How does the dashboard reach you?

- A) Scheduled 9:00 am email
- B) You guess
- C) Fax
- D) GSC email

6. Which sources feed it?

- A) Clarity only
- B) Bing only
- C) Raw GA4 live
- D) Small reporting tables and Sheets

7. What do info icons hold?

- A) Definition, source, model
- B) Nothing
- C) Jokes
- D) Passwords

8. When to act on amber?

- A) Never
- B) Weekly, unless pattern grows
- C) Immediately always
- D) Yearly

9. What's a mover?

- A) A page with a big change
- B) A CTA
- C) A developer
- D) A server

10. In the chai analogy, the daily dashboard is…

- A) The tea
- B) The headlines
- C) The hawker
- D) The whole newspaper

Sources

1. Looker Studio Help, Schedule email delivery ●

Answer key, Chapter 60: 1-C · 2-B · 3-B · 4-B · 5-A · 6-D · 7-A · 8-B · 9-A · 10-B

LEVEL 5 · CHAPTER 61

Chapter 61: The Executive Dashboard

Leadership doesn't need 40 charts; they need to know: Are we growing? Is the website turning visitors into paying patients? Which investments work? This dashboard answers those three questions on two pages.

> 💡 **Trivia**
>
> Research on dashboard design (e.g., Stephen Few's work) stresses that a dashboard should fit on a single screen and be readable "at a glance". [1] 🟡 Industry/expert
>
> Two pages is our compromise: one for results, one for drivers.

61.1 Page 1: Results

Tile	Metric	Period
Payments (aggregated)	Count, vs last month, vs target range	Month to date + last full month
Leads (sign-ups + eligibility complete, aggregated)	Count, vs last month	Same
Lead → payment rate	Payments ÷ leads (aggregate)	Same
Portal exits (website)	Count and rate	Same
Sessions	Count, % new	Same
Trend	13-month line: sessions, portal exits, leads, payments	Monthly

61.2 Page 2: Drivers

Tile	Metric
Channel mix	Sessions and payments (aggregated, by channel_group from backend context) by OWL channel
Content engine	Non-branded organic clicks, top 5 clusters by first-touch portal exits
Search visibility	Share of voice (Chapter 42), Bing AI citations
Site health	CWV pass rate (p75 good), uptime/JS errors trend
Social	Reach, social sessions, social share of leads
Notable changes	3 bullet notes (text box, written monthly)

> 🪔 **Desi analogy: dukaan ka galla aur bahi-khata**
>
> Seth ji ko roz ka poora bahi-khata nahi chahiye; galle mein kitna aaya aur kis maal se aaya, bas. **Executive dashboard galla aur uska kaaran hai.**

61.3 Rules for executive numbers

Rule	Why
Backend counts for leads/payments	Truth for business outcomes
Label attribution model	Avoid arguments
Show target ranges, not single targets	Honest uncertainty
Monthly, not daily	Strategic view
Three written notes	Numbers need meaning
No personal data, aggregates only	Privacy

61.4 Build

#	Step
1	Report "OWL – Executive – v1", 2 pages
2	Sources: `reporting.monthly_kpis` (scheduled monthly from daily tables), `ops.daily_conversions` (aggregated monthly), Targets Sheet, SoV Sheet, CWV table
3	Date control default "Last full month" with "Month to date" option
4	Scorecards with comparison and conditional colours
5	Share to leadership as Viewer; schedule 1st of month 10:00 am

> 🟧 **Ongo Weight Loss example**
>
> Three notes for October might read: "Payments +12% MoM, within target range. Organic GLP-1 cluster started 46% of portal exits (first touch). Tirzepatide page speed fix lifted mobile LCP to 'good'; CTA rate +18%." Leadership reads it in one minute and knows what worked. 🔵

61.5 Do's and Don'ts

✅ Do	❌ Don't
Lead with business outcomes	Lead with page views
Use backend counts	Use GA4 for payments
Write three notes	Let charts speak alone
Keep to two pages	Add exploratory charts

61.6 Pros and cons of a minimal exec view

✅ Pros	❌ Cons
Clear priorities	Hides detail (by design)
Fast reading	Needs good notes
Trusted numbers	Monthly lag

61.7 Where freshers go wrong in Chapter 61 topics

Mistake	Why it happens	Avoid it by	Fix it by
Vanity metrics on top	Easy to show	Outcomes first	Reorder
GA4 vs backend mismatch questions	Different sources	Labels, notes	Explain
No targets	Unknown	Chapter 66	Add ranges

61.8 Thumb rules

👍 Show the galla and why it filled.

👍 Backend for outcomes, GA4 for drivers.

👍 Three notes beat thirty charts.

61.9 Notes to remember

📝 Page 1 results, page 2 drivers.

📝 Monthly, scheduled on the 1st.

Chapter 61 Glossary

Term	Simple meaning
Executive dashboard	Leadership summary view
Driver	Factor that influences results
Target range	Acceptable low–high band for a KPI
MoM	Month over month
Vanity metric	Looks good, doesn't drive decisions

Chapter 61 Quiz

1. Where do payments come from?

- A) GA4 events
- B) Clarity
- C) Aggregated backend counts
- D) GSC

2. How many pages?

- A) Two
- B) Zero
- C) One hundred
- D) Twenty

3. What's on page 1?

- A) Developer notes
- B) Results
- C) Font tests
- D) Raw events

4. What's a vanity metric?

- A) Looks good, doesn't drive decisions
- B) Portal exits

- C) Payments
- D) Leads

5. Why target ranges?

- A) Look nicer
- B) Honest uncertainty
- C) Required by GA4
- D) No reason

6. How often is it delivered?

- A) Never
- B) Monthly on the 1st
- C) Hourly
- D) Yearly

7. What should accompany charts?

- A) Raw SQL
- B) Nothing
- C) Three written notes
- D) Emojis only

8. What metric leads page 1?

- A) Page views
- B) Scroll depth
- C) Followers
- D) Payments

9. Is personal data shown?

- A) Only emails
- B) Only names
- C) Yes
- D) No, aggregates only

10. In the seth ji analogy, the exec dashboard is...

- A) The signboard
- B) The whole bahi-khata

- C) The galla and why it filled
- D) The godown

Sources

1. Stephen Few, *Information Dashboard Design* (Analytics Press, 2006/2013) 🟡
2. Looker Studio Help, Schedule email delivery 🟢

Answer key, Chapter 61: 1-C · 2-A · 3-B · 4-A · 5-B · 6-B · 7-C · 8-D · 9-D · 10-C

LEVEL 5 · CHAPTER 62

Chapter 62: The SEO and Content Dashboard

This dashboard is the SEO and content team's weekly workspace: visibility by entity, page scorecards by funnel stage, internal-link flow and a ready-made worklist of pages to fix.

> 💡 **Trivia**
>
> Looker Studio supports cross-filtering: clicking a row in one chart can filter the other charts on the page. [1] ● Official (Google)
>
> One click on "semaglutide" turns the whole page into a semaglutide report.

62.1 Pages

Page	Question	Main charts
1 Visibility	Are we more visible?	Clicks/impressions trend; branded vs non-branded; entity table; Bing clicks; AI citations
2 Pages	Which pages perform their job?	Page table with funnel_stage scorecard columns (Chapter 50)
3 Opportunities	What should we fix this week?	Striking-distance list; low CTR high impressions; decaying pages
4 Internal links	Which links move people?	Link table with CTR and downstream portal rate; position comparison
5 Journeys	How do people move?	Funnel_daily stage bars; top entry → exit pairs; first-touch clusters
6 Health	Can Google crawl and index?	Index changes (manual Sheet from GSC), CWV by page_type, 404s

62.2 Controls

Control	Field
Date range	Last 28 days vs previous 28
Site	site_id
Entity/cluster	content_cluster / entity
Funnel stage	funnel_stage
Min impressions/sessions	Parameter p_threshold

> 🪔 **Desi analogy: khet ka naksha**
>
> Kisaan ke paas naksha hota hai: kaunsa khet achha ugaa raha, kahan paani kam, kahan keede. **SEO dashboard content ke khet ka naksha hai.**

62.3 Opportunities page logic

List	Rule	Action owner
Striking distance	Avg position 8–20, impressions ≥ 200 (28 days)	On-page SEO
Low CTR	Impressions ≥ 1,000, CTR below expected for position	Writer (titles/meta)
Decay	Impressions −20% vs 3 months, or position −3	Writer + medical reviewer
Weak links	Engagement good, internal link CTR < 1%	SEO + writer
Thin BOFU path	TOFU page with no internal clicks to BOFU	SEO

62.4 Weekly SEO meeting (30 minutes) using the dashboard

Minute	Agenda
0–5	Visibility page: headline changes
5–15	Opportunities: choose 5 tasks

Minute	Agenda
15–20	Internal links: one linking improvement
20–25	Health: any index or CWV issue
25–30	Assign owners and due dates; add to change log

> 🟧 **Ongo Weight Loss example**
>
> The opportunities page shows "compounded tirzepatide cost" at position 11 with 3,200 impressions. The writer adds a clear cost table and an FAQ, the SEO lead adds two contextual links from GLP-1 articles, and the change log records the date. Four weeks later, the same page shows whether it moved. 🔵

62.5 Do's and Don'ts

✅ Do	❌ Don't
Turn the dashboard into tasks	Just look at it
Use thresholds	List every tiny query
Include medical review for YMYL edits	Edit claims without review
Track outcomes 4 weeks later	Forget what changed

62.6 Pros and cons of a task-driven SEO dashboard

✅ Pros	❌ Cons
Weekly priorities	Needs GSC export and GA4 tables
Shared language	Rules need tuning
Measures impact	—

62.7 Where freshers go wrong in Chapter 62 topics

Mistake	Why it happens	Avoid it by	Fix it by
Fixing 30 pages at once	Enthusiasm	5 tasks/week	Prioritise
Ignoring branded split	Mixed totals	Page 1 split	Separate
No re-check	No calendar	4-week reminder	Add to log

62.8 Thumb rules

👍 The farmer's map shows where to water.

👍 Five tasks a week, measured.

👍 YMYL edits go through the reviewer.

62.9 Notes to remember

📝 Six pages: visibility, pages, opportunities, links, journeys, health.

📝 30-minute weekly SEO meeting.

Chapter 62 Glossary

Term	Simple meaning
Cross-filtering	Clicking one chart filters others
Opportunity list	Rule-based list of pages to improve
Expected CTR	Typical CTR for a position
Worklist	Assigned tasks from data
Entity	Topic such as semaglutide or GLP-1

Chapter 62 Quiz

1. What does cross-filtering do?

- A) Changes fonts
- B) Clicking a chart filters others
- C) Deletes data
- D) Sends email

2. Which page lists pages to fix?

- A) Health
- B) Visibility
- C) Journeys
- D) Opportunities

3. Striking distance rule?

- A) Position 100
- B) Position 1
- C) Position 8–20, impressions ≥ 200
- D) No impressions

4. How many tasks per week?

- A) 0
- B) About 5
- C) 500
- D) 50

5. Who reviews YMYL edits?

- A) Developer
- B) Nobody
- C) Medical reviewer
- D) Social team

6. When do we re-check results?

- A) Never
- B) About 4 weeks later

- C) Next hour
- D) Next year

7. What does the Internal links page show?

- A) Social followers
- B) DNS
- C) CTR and downstream portal rate per link
- D) Payments

8. Which control sets minimum volume?

- A) Parameter p_threshold
- B) Theme
- C) Font size
- D) Logo

9. How long is the weekly SEO meeting?

- A) A day
- B) 3 hours
- C) 30 minutes
- D) 5 minutes

10. In the khet analogy, the dashboard is…

- A) The market
- B) The seed
- C) The farmer's field map
- D) The tractor

Sources

1. Looker Studio Help, Cross-filtering 🟢
2. Ongo Weight Loss, Book Six editorial process, internal 🔵

Answer key, Chapter 62: 1-B · 2-D · 3-C · 4-B · 5-C · 6-B · 7-C · 8-A · 9-C · 10-C

LEVEL 5 · CHAPTER 63

Chapter 63: The Social and Channel Dashboard

Which channels bring people who move forward? This dashboard compares Organic Search, Organic Social, AI Assistants, Referral, Email and Direct on quality, not just volume, and puts on-platform social data beside on-site results.

> 💡 **Trivia**
>
> GA4 lets you create up to two custom channel groups on standard properties, in addition to the default group, and they apply retroactively to reports. [1] ● Official (Google)
>
> Our "OWL channels" group (Chapter 27) is one of those two.

63.1 Page 1: Channel quality

Column	Metric
Channel	OWL channels
Sessions	Count, share
Engaged rate	Engaged ÷ sessions
BOFU reach rate	Sessions reaching BOFU ÷ sessions
CTA rate	Sessions with CTA ÷ sessions
Portal exit rate	Sessions with exit_to_portal ÷ sessions
Leads / payments share	From ops.daily_conversions channel_group (if context param approved)

63.2 Page 2: Social on-platform + on-site

Section	Content
Platform scorecards	Followers, reach, engagement rate, link clicks per platform (Sheet)
On-site per platform	Sessions, engaged rate, portal exits (GA4 by utm_source)
Efficiency	Sessions per 1,000 reach; portal exits per 1,000 reach
Top content	utm_content ranked by portal exits and engaged sessions
Calendar	Posts per week vs sessions per week (blend)

> 🪔 **Desi analogy: alag-alag bus routes**
>
> Shehar mein kai bus routes hain. Kis route ki bus bhari aati hai, aur kis route ke yatri sach mein bazaar mein khareedari karte hain? **Channel dashboard har route ki quality batata hai.**

63.3 Page 3: AI and referrals

Tile	Source
AI Assistants sessions and portal rate	GA4 channel
Top AI referrers (chatgpt.com, perplexity.ai, copilot, gemini)	Session source
Bing AI citations	Bing WMT Sheet
Top referral domains	GA4 referral
Child-site → main-site flow	site_id and cross-domain sessions

63.4 Reading channel quality

Pattern	Meaning	Action
Social high volume, low BOFU reach	Entertaining, not intent	Add more educational posts linking to MOFU

Pattern	Meaning	Action
AI Assistants small but highest portal rate	High intent visitors	Strengthen pages AI cites
Referral spike, 0 engagement	Spam	Filter
Child site strong portal rate	Good funnel	Keep linking strategy

> 🟧 **Ongo Weight Loss example**
>
> If Instagram brings 3× more sessions than Pinterest but Pinterest visitors reach BOFU twice as often, the social team doesn't drop Instagram; it studies which Pinterest pins work and brings that educational style to Instagram carousels. 🔵

63.5 Do's and Don'ts

✅ Do	❌ Don't
Compare rates, not just volume	Rank channels by sessions only
Use the OWL channel group	Mix default and custom groups
Normalise by reach	Compare raw reach across platforms
Separate AI Assistants	Leave AI traffic in Referral

63.6 Pros and cons of quality-first channel views

✅ Pros	❌ Cons
Fair comparison	Small channels noisy
Guides effort	Needs thresholds
Combines platform + site	Manual Sheet upkeep

63.7 Where freshers go wrong in Chapter 63 topics

Mistake	Why it happens	Avoid it by	Fix it by
Declaring a channel "winner" from a week	Noise	28-day windows	Wait
AI traffic in Referral	Default grouping	OWL channels	Update regex list
Comparing on-platform metrics across apps	Different definitions	Trends per platform	Explain

63.8 Thumb rules

👍 Full buses aren't always shopping buses.

👍 Rates beat volume.

👍 Per 1,000 reach makes platforms comparable.

63.9 Notes to remember

📝 Three pages: channel quality, social, AI and referrals.

📝 Up to two custom channel groups; retroactive.

Chapter 63 Glossary

Term	Simple meaning
Channel group	Rules grouping sources into channels
BOFU reach rate	Share of sessions reaching BOFU pages
Normalise	Make numbers comparable (e.g., per 1,000)
Referral	Visit from another website link

Term	Simple meaning
AI Assistants channel	Our channel for AI chat tools

Chapter 63 Quiz

1. How many custom channel groups can standard properties create?

- A) Ten
- B) Unlimited
- C) Two
- D) Zero

2. Are custom channel groups retroactive?

- A) Only weekly
- B) No
- C) Yes
- D) Only for social

3. What's the fair way to compare platforms?

- A) Raw reach
- B) Followers only
- C) Per 1,000 reach
- D) Likes

4. AI Assistants with high portal rate means...

- A) Bots
- B) Broken tracking
- C) Spam
- D) High-intent visitors

5. Referral spike with zero engagement means...

- A) Great content
- B) More payments
- C) GSC error

- D) Likely spam

6. What window should channel decisions use?

- A) 28 days
- B) 1 day
- C) 10 years
- D) 1 hour

7. What does page 2 combine?

- A) Payments and passwords
- B) On-platform and on-site social data
- C) DNS and SSL
- D) GTM and Clarity

8. Which group should all reports use?

- A) OWL channels
- B) Default only
- C) None
- D) Both mixed

9. What's BOFU reach rate?

- A) Payments ÷ leads
- B) Clicks ÷ impressions
- C) BOFU pages ÷ all pages
- D) Sessions reaching BOFU ÷ sessions

10. In the bus analogy, the dashboard shows…

- A) Driver names
- B) Fuel prices
- C) Bus colours
- D) Which routes bring shoppers

Sources

1. Google Analytics Help, Custom channel groups

Answer key, Chapter 63: 1-C · 2-C · 3-C · 4-D · 5-D · 6-A · 7-B · 8-A · 9-D · 10-D

LEVEL 5 · CHAPTER 64

Chapter 64: The Technical Health Dashboard

Developers and testers need their own view: is tracking healthy, is the site fast, are there errors, broken links or index problems? This dashboard makes technical health visible every week.

> 💡 **Trivia**
>
> Google's Search Console Crawl Stats report shows total crawl requests, average response time and host status (robots.txt fetch, DNS, server connectivity) over the last 90 days. [1] 🟢 Official (Google)
>
> If Googlebot struggles, users probably do too.

64.1 Pages

Page	Tiles	Source
1 Tracking health	page_views/day; events not in dictionary; (not set) page_type count; duplicate page_view %; export status; PII scan result	BigQuery checks
2 Speed	CWV p75 by page_type (web-vitals); CrUX origin trend; slow pages list	cwv_monthly, CrUX
3 Errors	404 count and top missing paths; Clarity JS error trend; rage-click pages	GA4, Clarity Sheet
4 Index and crawl	GSC indexed vs not indexed (Sheet), crawl response codes, sitemap status	GSC (manual weekly Sheet)
5 Releases	Change log; event volume before/after each release	Sheet + daily_kpis

64.2 Tracking-health SQL (feeds page 1)

```
WITH e AS (
  SELECT PARSE_DATE('%Y%m%d', event_date) AS date, event_name,
    (SELECT value.string_value FROM UNNEST(event_params) WHERE key='page_type') AS
page_type
  FROM `PROJECT.analytics_PROPERTY.events_*`
  WHERE _TABLE_SUFFIX >= FORMAT_DATE('%Y%m%d', DATE_SUB(CURRENT_DATE(), INTERVAL 30 DAY))
)
SELECT e.date,
  COUNTIF(e.event_name='page_view') AS page_views,
  COUNTIF(e.event_name='page_view' AND e.page_type IS NULL) AS pv_missing_type,
  COUNTIF(d.event_name IS NULL) AS unknown_events
FROM e
LEFT JOIN `PROJECT.reporting.event_dictionary` d ON e.event_name = d.event_name
GROUP BY e.date ORDER BY e.date;
```

Note: GA4 automatic events (first_visit, session_start, user_engagement) must be in the dictionary table too. 🔵

`reporting.event_dictionary` is a one-column table loaded from Appendix A. 🔵

> 🍛 **Desi analogy: gaadi ki servicing**
>
> Gaadi chal rahi hai, par har 5,000 km pe oil, brake, tyre check hota hai. **Technical dashboard website ki servicing card hai.**

64.3 404 tracking

Item	Detail
Event	`page_not_found` with `requested_path` (path only, no query)
Report	Top missing paths + referring page (page_referrer)
Action	Redirect (301) to best page or fix the linking page

64.4 Release impact view

Before (7 days)	After (7 days)	Change
page_views/day		
CTA clicks/day		
portal exits/day		
JS errors/day		
p75 LCP (medication pages)		

64.5 Weekly technical check (developer + tester, 20 minutes)

#	Check
1	Page 1 all green
2	Any page_type with p75 LCP > 2.5 s or INP > 200 ms
3	Top 10 404s: fix or redirect
4	JS error spikes after releases
5	Index: new "not indexed" reasons
6	Log tickets

> 🟧 **Ongo Weight Loss example**
>
> After a release adds a new hero video to medication pages, the release view shows p75 LCP on mobile jumped from 2.1 s to 3.4 s and CTA clicks fell 15%. The developer lazy-loads the video poster image; the next week both recover. The dashboard made the cost of a design choice visible. 🔵

64.6 Do's and Don'ts

✅ Do	❌ Don't
Check after every release	Wait for complaints
Track 404 sources	Only count 404s
Measure release impact	Assume releases are neutral
Keep the event dictionary table updated	Let it drift

64.7 Pros and cons of a technical dashboard

✅ Pros	❌ Cons
Developers see impact	Manual GSC/Clarity Sheets
Catches regressions	Setup time
Shared priorities	—

64.8 Where freshers go wrong in Chapter 64 topics

Mistake	Why it happens	Avoid it by	Fix it by
404 path includes query strings with data	Raw URL	Path only	Clean variable
Ignoring INP	New metric	64.5 step 2	Add tile
No release dates	No log	Change log	Add

64.9 Thumb rules

👍 Service the car before it breaks.

👍 Every release gets a before/after.

👍 Fix the link, not just the 404.

64.10 Notes to remember

📝 Five pages: tracking, speed, errors, index, releases.

📝 20-minute weekly developer check.

Chapter 64 Glossary

Term	Simple meaning
404	Page not found error
301 redirect	Permanent redirect
Crawl stats	Googlebot request report
Regression	Something breaking after a change
Event dictionary table	List of approved event names

Chapter 64 Quiz

1. What period does Crawl Stats cover?

- A) Last 90 days
- B) Forever
- C) 1 hour
- D) 10 years

2. What does page 1 show?

- A) Tracking health
- B) Competitors
- C) Payments
- D) Social reach

3. What should 404 events contain?

- A) Path only
- B) Email
- C) Full URL with query
- D) Name

4. What fixes a 404 best?

- A) Fix the linking page or 301 redirect
- B) Delete GA4
- C) Ignore
- D) Add ads

5. What does the release view compare?

- A) Year vs year
- B) 7 days before vs after
- C) Nothing
- D) Hour vs hour

6. Which table lists approved events?

- A) searchconsole
- B) crux
- C) reporting.event_dictionary
- D) ops.daily_conversions

7. When is the technical check?

- A) Only on complaints
- B) Never
- C) Weekly and after releases
- D) Yearly

8. What's a regression?

- A) Something breaking after a change
- B) A chart type
- C) A GSC report
- D) A CTA

9. INP threshold for good?

- A) ≤ 5 s
- B) ≤ 200 ms
- C) ≤ 2.5 s
- D) ≤ 1 ms

10. In the servicing analogy, the dashboard is…

- A) The road
- B) The car's service card
- C) The driver
- D) The fuel

Sources

1. Google Search Console Help, Crawl Stats report
2. web.dev, Web Vitals

Answer key, Chapter 64: 1-A · 2-A · 3-A · 4-A · 5-B · 6-C · 7-C · 8-A · 9-B · 10-B

Chapter 65: Weekly and Monthly Reporting (Telling the Story)

Dashboards show numbers; reports explain them. This chapter gives the weekly and monthly report templates, the "What, So what, Now what" method, and how to write data findings anyone can act on.

> 💡 **Trivia**
>
> The "What? So what? Now what?" reflection model comes from Rolfe, Freshwater and Jasper's 2001 framework for reflective practice in health care. [1] 🟡 Academic
>
> A model born in health care fits a health brand's reporting nicely.

65.1 Report rhythm

Report	When	Length	Audience
Daily	9:00 am email (dashboard)	2 min read	Team lead
Weekly	Monday	1 page	Team
Monthly	1st–3rd working day	2–3 pages + exec dashboard	Leadership + team
Quarterly	First week of quarter	Deck/doc	Strategy

65.2 What, So what, Now what

Part	Question	Example
What	What happened? (fact + number + comparison)	"Portal exits from organic fell 18% WoW."

Part	Question	Example
So what	Why does it matter / why did it happen?	"The tirzepatide page dropped from position 4 to 7 after a competitor update; it drives 22% of organic portal exits."
Now what	What will we do, who, by when?	"Writer refreshes cost and safety sections by Thursday; SEO adds 3 links; re-check in 4 weeks."

> 🪔 **Desi analogy: doctor ki parchi**
>
> Doctor sirf report ke numbers nahi padhta; batata hai matlab kya hai aur dawai kya leni hai. **Report numbers + matlab + dawai hai.**

65.3 Weekly report template (1 page)

Section	Content
Headline	One sentence: the week in one line
3 wins	What, so what
3 concerns	What, so what, now what
Funnel snapshot	Sessions → BOFU → CTA → portal exits (WoW)
SEO	Non-branded clicks, top movers, tasks done
Social	Reach, sessions, portal exits per platform
UX (Clarity + user explorer)	3 observations
Data quality	Pass/fail of the 13 checks
Next week	5 tasks with owners

65.4 Monthly report template

Section	Content
Executive summary	5 bullets (results vs target ranges)

Section	Content
Results	Payments, leads, lead→payment, portal exits (backend + GA4)
Drivers	Channels, content clusters (first touch), search visibility, social
Experiments	What we tested, outcome (Chapter 67)
Benchmarks	Competitor table (Chapter 42)
Technical	CWV, errors, index
Risks and compliance	Privacy checks, consent rate (CMP), incidents
Plan	Next month's priorities

65.5 Writing rules for data findings

Rule	Bad	Good
Number + comparison	"Traffic went up"	"Sessions +12% vs last month (28,400 vs 25,350)"
Label the model	"Blog drove 40% of conversions"	"Blog started 40% of portal-exit journeys (first touch)"
Avoid false causation	"The new title caused +20% clicks"	"Clicks +20% after the title change; position also rose"
No personal data	"A user from Austin..."	"Users who opened the cost FAQ..."
Plain language	"INP p75 regression"	"Taps on mobile felt slower (INP 260 ms vs 180 ms)"

> 🟧 **Ongo Weight Loss example**
>
> Weekly headline: "Steady week: portal exits +4%, but mobile tirzepatide page slowed after Tuesday's release (fix shipped Friday)." In one line, the reader knows the state, the risk and the action. 🔵

65.6 Do's and Don'ts

✅ Do	❌ Don't
Lead with the headline	Start with methodology
Give comparisons	Give lonely numbers
End with actions and owners	End with "interesting"
Keep weekly to one page	Send 20-slide weeklies

65.7 Pros and cons of written reports

✅ Pros	❌ Cons
Meaning and decisions	Time to write
Shared memory	Can drift into opinions
Accountability	—

65.8 Where freshers go wrong in Chapter 65 topics

Mistake	Why it happens	Avoid it by	Fix it by
Data dump	Fear of missing things	Template	Cut to essentials
Claiming causation	Excitement	65.5 rule	Rephrase
No owners	Vague actions	Now what	Name owner/date

65.9 Thumb rules

> 👍 Numbers, meaning, medicine.
>
> 👍 Every concern ends with a "now what".
>
> 👍 One page weekly.

65.10 Notes to remember

> 📝 What, So what, Now what.
>
> 📝 Weekly Monday; monthly by the 3rd working day.

Chapter 65 Glossary

Term	Simple meaning
Headline	One-sentence summary
WoW / MoM	Week over week / month over month
Causation	One thing causing another
Correlation	Things changing together
Action item	Task with owner and date

Chapter 65 Quiz

1. What are the three parts of the method?

- A) Read, Write, Delete
- B) What, So what, Now what
- C) Who, When, Where
- D) Plan, Do, Stop

2. How long is the weekly report?

- A) Twenty pages
- B) One word
- C) No limit
- D) One page

3. What must every number have?

- A) A font
- B) A comparison
- C) A colour
- D) An emoji

4. Which is a good finding?

- A) "Numbers changed"
- B) "Interesting"
- C) "Sessions +12% vs last month (28,400 vs 25,350)"
- D) "Traffic went up"

5. What does "Now what" contain?

- A) A chart
- B) A joke
- C) Action, owner, date
- D) Raw data

6. When is the monthly report due?

- A) Never
- B) End of quarter
- C) Daily
- D) By the 3rd working day

7. What's wrong with "The title caused +20%"?

- A) Too short
- B) Claims causation without proof
- C) Uses numbers
- D) Nothing

8. Can reports mention individual users?

- A) Yes
- B) No
- C) Only by city
- D) Only by name

9. Where did the method originate?

- A) Reflective practice in health care
- B) Cricket
- C) Advertising
- D) Banking

10. In the parchi analogy, a report includes…

- A) Numbers, meaning and medicine
- B) Only the bill
- C) Only numbers
- D) Only medicine

Sources

1. Rolfe, Freshwater & Jasper, *Critical Reflection for Nursing and the Helping Professions* (2001)

Answer key, Chapter 65: 1-B · 2-D · 3-B · 4-C · 5-C · 6-D · 7-B · 8-B · 9-A · 10-A

LEVEL 5 · CHAPTER 66

Chapter 66: KPI Dictionary and Target Ranges

You asked for KPI targets. Here they are, as starting ranges with the honest caveat: until we have 90 days of clean data, targets are hypotheses. This chapter defines every KPI precisely and explains how to reset ranges from your own history.

> 💡 **Trivia**
>
> Public conversion benchmarks for healthcare websites vary widely between studies and almost none separate telehealth GLP-1 programs; vendor studies often report healthcare landing-page conversion rates in the low single digits. [1] 🟠 Industry (vendor studies)
>
> Treat outside benchmarks as rough; your own first 90 days set the real baseline.

66.1 How targets work here

Phase	Targets come from	Review
Days 0–90	Starting ranges below (🟠/🔵)	Monthly
After 90 days	Our median ± typical variation (e.g., 25th–75th percentile of weekly values)	Quarterly
After 12 months	Same month last year + growth goal	Quarterly

Format: every KPI has a **low** (below = investigate) and **high** (above = celebrate/verify) band. 🔵

> 🍛 **Desi analogy: BP ki normal range**
>
> Doctor BP ko ek number nahi, ek range se dekhte hain. Range se bahar gaya toh dhyaan dete hain. **KPI targets bhi range hain, ek number nahi.**

66.2 KPI dictionary (core)

KPI	Definition	Source	Starting range	Owner
Sessions	GA4 sessions (consented)	daily_kpis	Baseline ± 20% WoW	Team lead
Engagement rate	Engaged sessions ÷ sessions	GA4	55–70% site-wide 🟠	On-page SEO
TOFU → BOFU rate	TOFU-landing sessions reaching BOFU ÷ TOFU-landing sessions	BigQuery	3–8% 🔵	SEO + writers
Internal link CTR (TOFU)	internal_link_click ÷ views on TOFU pages	BigQuery	5–15% 🔵	SEO
BOFU CTA rate	Sessions with CTA on BOFU ÷ BOFU sessions	BigQuery	10–25% 🔵	UX + SEO
Portal exit rate (site)	Sessions with exit_to_portal ÷ sessions	GA4	2–6% 🔵	Team lead
Portal exit rate (BOFU sessions)	÷ BOFU sessions	BigQuery	10–25% 🔵	Team lead
Lead rate	Leads (backend, aggregated) ÷ sessions	ops + GA4	Set after 90 days	Team lead
Lead → payment rate	Payments ÷ leads (backend)	ops	Set after 90 days	Ops/lead
Non-branded organic clicks	GSC clicks excluding brand queries	GSC export	+5–15% MoM in growth phase 🔵	SEO
Share of voice (tracked set)	Semrush visibility %	Semrush	Upward trend	SEO
CWV pass rate	% page types with p75 good on all three	web-vitals	≥ 90% 🔵	Developer
p75 LCP mobile	From web-vitals	GA4/BQ	≤ 2.5 s 🟢	Developer
p75 INP mobile	From web-vitals	GA4/BQ	≤ 200 ms 🟢	Developer
CLS p75	From web-vitals	GA4/BQ	≤ 0.1 🟢	Developer

KPI	Definition	Source	Starting range	Owner
404 rate	page_not_found ÷ page_views	GA4	< 0.5% 🔵	Developer
Data-quality pass	Weekly checks passed ÷ 13	Checklist	13/13	Analytics owner
PII scan	Rows found	Query H	0 🟢	Analytics owner
Social portal exits per 1,000 reach	exit_to_portal (social) ÷ reach × 1,000	GA4 + Sheet	Set after 90 days	Social lead
Clarity rage-click sessions	% sessions with rage clicks	Clarity	< 3% 🟠	UX

66.3 Setting ranges from your own data (after 90 days)

```
SELECT
  APPROX_QUANTILES(rate, 100)[OFFSET(25)] AS low,
  APPROX_QUANTILES(rate, 100)[OFFSET(50)] AS median,
  APPROX_QUANTILES(rate, 100)[OFFSET(75)] AS high
FROM (
  SELECT DATE_TRUNC(date, WEEK) AS wk,
    SAFE_DIVIDE(SUM(portal_exits), SUM(sessions)) AS rate
  FROM `PROJECT.reporting.daily_kpis`
  WHERE date >= DATE_SUB(CURRENT_DATE(), INTERVAL 90 DAY)
  GROUP BY wk
);
```

66.4 KPI hygiene rules

Rule	Why
Every KPI has one definition and one table	No arguments
Every KPI has an owner	Accountability
Ranges, not single numbers	Natural variation
Separate website KPIs from business KPIs	Website can't control ops

Rule	Why
Review quarterly	Business changes

> 🟧 **Ongo Weight Loss example**
>
> If after 90 days the weekly portal exit rate's 25th–75th percentile is 2.8%–3.9%, that becomes the band. A week at 2.1% turns amber and triggers the Chapter 56 checks; a week at 4.6% triggers "verify, then learn what worked". 🔵

66.5 Do's and Don'ts

✅ Do	❌ Don't
Label starting ranges as hypotheses	Present them as industry truth
Reset after 90 days	Keep day-one guesses forever
Assign owners	Leave KPIs ownerless
Keep the dictionary in Appendix J	Define KPIs in chats

66.6 Pros and cons of range targets

✅ Pros	❌ Cons
Realistic	Less "motivational" than single goals
Fewer false alarms	Need data to compute
Data-driven	—

66.7 Where freshers go wrong in Chapter 66 topics

Mistake	Why it happens	Avoid it by	Fix it by
Copying a blog's "average conversion rate"	Search for a number	66.1	Use own history
KPI defined differently in two dashboards	Copy-paste	One table	Consolidate
Website blamed for ops issues	Mixed KPIs	Separate	Split

66.8 Thumb rules

👍 Like BP, KPIs live in a range.

👍 Your own 90 days beat any blog benchmark.

👍 One KPI, one definition, one owner.

66.9 Notes to remember

📝 Starting ranges until day 90; percentile bands after.

📝 CWV ranges are Google's official thresholds 🟢.

Chapter 66 Glossary

Term	Simple meaning
KPI	Key performance indicator
Target range	Low–high band for a KPI
Percentile	Value below which a % of data falls
Baseline	Starting reference level
Owner	Person accountable for a KPI

Chapter 66 Quiz

1. What are day-one targets?

- A) Google rules
- B) Legal requirements
- C) Fixed forever
- D) Hypotheses to reset after 90 days

2. How are ranges set after 90 days?

- A) Single best week
- B) 25th–75th percentile of weekly values
- C) Competitor claims
- D) Random

3. Which KPI ranges are official Google thresholds?

- A) Core Web Vitals
- B) Engagement rate
- C) Portal exit rate
- D) Social reach

4. What's the PII scan target?

- A) 10
- B) Any
- C) 0
- D) 100

5. Why separate website and business KPIs?

- A) Website can't control ops
- B) GSC requires it
- C) For colours
- D) No reason

6. Where is the KPI dictionary kept?

- A) Bing
- B) Clarity

- C) Appendix J
- D) Chat messages

7. What happens at amber?

- A) Ignore
- B) Investigate with Chapter 56 checks
- C) Delete data
- D) Celebrate

8. Who owns CWV KPIs?

- A) Writer
- B) Nobody
- C) Social lead
- D) Developer

9. How often are targets reviewed?

- A) Never
- B) Hourly
- C) Every 10 years
- D) Quarterly

10. In the BP analogy, KPI targets are…

- A) The medicine
- B) The doctor
- C) A normal range
- D) A single number

Sources

1. Unbounce, Conversion Benchmark Report 🟠
2. web.dev, Web Vitals thresholds 🟢

Answer key, Chapter 66: 1-D · 2-B · 3-A · 4-C · 5-A · 6-C · 7-B · 8-D · 9-D · 10-C

06

LEVEL 6

Operating

Experiments, governance, the 90-day roadmap and ZETA for analytics.

67 · Experiments and Measuring Change

68 · Governance and Documentation

69 · The 90-Day Implementation Roadmap

70 · The ZETA Model for Analytics and the Final Gate

LEVEL 6 · CHAPTER 67

Chapter 67: Experiments and Measuring Change

Data shows problems; experiments prove fixes. This chapter explains how to test CTA copy, page layouts and content changes properly, with A/B tests where traffic allows and careful before/after comparisons where it doesn't.

> 💡 **Trivia**
>
> Google Optimize, Google's free A/B testing tool, was shut down on September 30, 2023. Google now points users to third-party testing tools that integrate with GA4. [1] 🟢 Official (Google)
>
> Testing needs a separate tool now, or a developer feature-flag setup.

67.1 Two ways to measure change

Method	When	Strength
A/B test (randomised split)	Enough traffic on the page (rule of thumb: hundreds of key events per variant over 2–4 weeks)	Proves cause
Before/after with control	Low traffic; content refreshes; SEO changes	Suggests cause; weaker

67.2 A/B testing options

Tool	Type	Note
GrowthBook	Open source / cloud, feature flags + experiments	Works with GA4/BigQuery 🟡
PostHog	Product analytics + experiments	Check HIPAA/BAA terms before use 🟡

Tool	Type	Note
VWO / Optimizely / AB Tasty	Commercial testing	Check BAA and data handling 🟡
Next.js middleware + cookie split	Developer-built	Full control; send `experiment_id`, `variant` to GA4

Health caution: any testing tool that runs on Zone B pages must pass the same legal review as analytics (Chapter 4). 🔵

> 🪔 **Desi analogy: do chai ki dukaan**
>
> Ek hi galli mein do chai ki dukaanein, ek mein adrak waali chai, ek mein elaichi. Graahak bina chune kisi bhi dukaan pe jaayein, phir dekho kahan zyada log dobara aaye. **A/B test do dukaanon ka imaandaar muqabla hai.**

67.3 Experiment tracking in GA4

Item	Value
Event	`experiment_impression` (once per session per experiment)
Params	`experiment_id` (e.g., exp_2026_11_tirz_hero_cta), `variant` (control / b)
Custom dims	experiment_id, variant (event scope)
Analysis	BigQuery: portal exit rate by variant; GA4 exploration segment by variant

```
track('experiment_impression', { experiment_id: 'exp_2026_11_tirz_hero_cta', variant: 'b' });
```

67.4 Experiment card (fill before starting)

Field	Example
ID	exp_2026_11_tirz_hero_cta

Field	Example
Hypothesis	"Check eligibility" as the hero CTA will raise portal exit rate on the tirzepatide page because it feels lower-commitment
Primary metric	Portal exit rate (sessions)
Guardrail metrics	Engagement rate, phone_click rate, CTA compliance
Audience	All visitors to tirzepatide page
Split	50/50
Duration	Minimum 2 full weeks; until planned sample reached
Decision rule	Ship if variant wins with ≥ 95% confidence and guardrails ok
Owner	UX designer + on-page SEO

67.5 Before/after with control

Step	Detail
1	Pick changed pages (test group) and similar unchanged pages (control)
2	Measure 4 weeks before and 4 weeks after
3	Compare the change in test vs change in control (difference-in-differences)
4	Note seasonality, algorithm updates, releases

67.6 Significance (plain words)

Term	Meaning
Sample size	Visitors/key events needed to detect a change
Confidence	How sure we are the difference isn't luck
Peeking	Stopping early when it "looks good" — inflates false wins
MDE	Minimum detectable effect: smallest change worth finding

Use an online sample-size calculator before starting; don't stop early. 🔵

> 🟧 **Ongo Weight Loss example**
>
> With ~2,000 tirzepatide page sessions a month and a 15% portal rate, a test can detect a large change (e.g., 15% → 19%) in about a month, but not a tiny one. If traffic is too low, the team runs a before/after with the semaglutide page as control instead. 🔵

67.7 Do's and Don'ts

✅ Do	❌ Don't
Write the card before starting	Decide metrics after seeing results
Run full weeks	Stop on day 3
Use guardrails	Win on CTA clicks while calls collapse
Log results, even losses	Hide failed tests

67.8 Pros and cons of A/B tests

✅ Pros	❌ Cons
Proves cause	Needs traffic
Removes opinion fights	Tool and legal review
Builds learning library	Takes weeks

67.9 Where freshers go wrong in Chapter 67 topics

Mistake	Why it happens	Avoid it by	Fix it by
Peeking and stopping early	Excitement	Fixed duration	Re-run

Mistake	Why it happens	Avoid it by	Fix it by
Testing 5 changes at once	Impatience	One variable	Split tests
Flicker (original shows then swaps)	Client-side swap	Server-side split	Use middleware

67.10 Thumb rules

👍 Two chai shops, fair contest.

👍 Card first, test second, decision by rule.

👍 Losses are learning too.

67.11 Notes to remember

📝 Google Optimize ended Sept 30, 2023.

📝 experiment_impression with experiment_id and variant.

Chapter 67 Glossary

Term	Simple meaning
A/B test	Random split comparing two versions
Hypothesis	Testable prediction with a reason
Guardrail metric	Metric that must not get worse
Difference-in-differences	Compare change in test vs change in control
Peeking	Checking results early and stopping

Chapter 67 Quiz

1. When did Google Optimize shut down?

- A) 2030
- B) 2010
- C) Never
- D) September 30, 2023

2. What proves cause best?

- A) One user's recording
- B) A blog post
- C) Opinion
- D) Randomised A/B test

3. What's a guardrail metric?

- A) A font
- B) The main metric
- C) A server
- D) A metric that must not get worse

4. What's peeking?

- A) Running full weeks
- B) Using controls
- C) Writing hypotheses
- D) Stopping early when it looks good

5. Which event records experiment exposure?

- A) experiment_impression
- B) scroll
- C) page_view only
- D) faq_open

6. What do you use with low traffic?

- A) Nothing
- B) Guessing
- C) More colours

- D) Before/after with control

7. Minimum duration?

- A) Two full weeks
- B) One day
- C) One year always
- D) One hour

8. What needs legal review?

- A) Testing tools on Zone B pages
- B) Fonts
- C) Colours
- D) Logos

9. How many variables per test?

- A) Ten
- B) Five
- C) All
- D) One

10. In the chai analogy, an A/B test is…

- A) Closing a shop
- B) One shop only
- C) A fair contest between two shops
- D) Raising prices

Sources

1. Google, Sunset of Google Optimize 🟢
2. GrowthBook, Docs 🟡

Answer key, Chapter 67: 1-D · 2-D · 3-D · 4-D · 5-A · 6-D · 7-A · 8-A · 9-D · 10-C

LEVEL 6 · CHAPTER 68

Chapter 68: Governance and Documentation

Tracking decays without rules: new pages without metadata, events with random names, people who left still holding admin access. Governance is the set of habits that keeps the measurement system trustworthy for years.

> 💡 **Trivia**
>
> GA4 keeps a **change history** (Admin → Change history) of configuration changes, and GTM keeps every published **version** with who published it and when. [1] ● Official (Google)
>
> The tools record what changed; governance records why.

68.1 The governance pillars

Pillar	What it covers	Owner
Ownership	Who owns each tool, dashboard, KPI	Team lead
Access	Least privilege, quarterly review	Team lead
Naming	Events, params, tags, dashboards (Appendices A, D)	Analytics owner
Change control	Requests → build → QA → publish → log	Developer + tester
Documentation	Living docs: dictionary, data layer, SQL library	Analytics owner
Privacy	Zones, consent, PII scans, vendor BAAs	Team lead + legal
Training	This book, onboarding, practice	Team lead

68.2 Tracking change request (template)

Field	Example
Request	Track "Download meal guide" button on blog
Business question	Does the guide move readers to BOFU?
Event/params	cta_blog_download_guide; cta_name, cta_location
Privacy zone	A
PII risk	None (no email capture)
Dashboards affected	CTA performance
Owner/dev	—
QA steps	Chapter 30 steps 1, 6, 15
Published (GTM version)	v42, date

> 🪔 **Desi analogy: ghar ki chaabi ka register**
>
> Ghar ki chaabiyan kisko di, kab di, kab wapas li — ek register mein likhte ho. Naukar badla toh chaabi wapas. **Access governance chaabi ka register hai.**

68.3 Access review (quarterly)

Tool	Check
GA4	Users and roles; remove leavers; Admin limited to 2
GTM	Publish rights limited (team lead + senior dev)
BigQuery/Cloud	IAM roles; service accounts still needed
GSC / Bing	Owners/full users
Clarity	Team members
Looker	Report ownership, shared links

Tool	Check
CMP	Admins

68.4 Documentation set (where each lives)

Doc	Location	Update when
Event Dictionary (Appendix A)	Shared Sheet + BigQuery table	Any event change
Data layer reference (Appendix B)	Repo `/docs/analytics.md`	Code changes
GTM build sheet (Appendix D)	Shared Sheet	GTM changes
Custom dimensions register (Appendix F)	Shared Sheet	Registrations
SQL library (Appendix H)	BigQuery saved queries + repo	New queries
KPI dictionary (Appendix J)	Shared Sheet	Quarterly
Change log	Shared Sheet	Every release
Privacy decisions log	Legal folder	Every decision

68.5 Release rule for every website change

Gate	Question
1	Does the new page/template include `PageMeta` with all fields?
2	Are new CTAs/links using data attributes from the register?
3	Any new form? Privacy zone decided?
4	QA script run (Appendix K)?
5	Change log updated?

🟧 **Ongo Weight Loss example**

Because there are no existing docs, Week 1 of the roadmap (Chapter 69) creates the shared folder "OWL Analytics" with the eight documents above, each with an owner. From then on, "if it isn't in the dictionary, it doesn't ship" becomes the team rule. 🔵

68.6 Do's and Don'ts

✅ Do	❌ Don't
Keep living docs with owners	Write docs once and forget
Review access quarterly	Leave leavers with access
Use version notes in GTM	Publish "v12" with no description
Make the dictionary the gate	Let events ship undocumented

68.7 Pros and cons of strong governance

✅ Pros	❌ Cons
Reliable data for years	Some process overhead
Safer privacy	Slower ad-hoc changes
Easy onboarding	Needs discipline

68.8 Where freshers go wrong in Chapter 68 topics

Mistake	Why it happens	Avoid it by	Fix it by
Undocumented events	"Quick fix"	Dictionary gate	Document or remove
Shared logins	Convenience	Named accounts	Rotate and split

Mistake	Why it happens	Avoid it by	Fix it by
No version notes	Hurry	Template	Add retroactively

68.9 Thumb rules

👍 Keep the chaabi register.

👍 Not in the dictionary, not in production.

👍 Docs have owners, or they die.

68.10 Notes to remember

📝 Seven pillars: ownership, access, naming, change control, documentation, privacy, training.

📝 Quarterly access review.

Chapter 68 Glossary

Term	Simple meaning
Governance	Rules and habits keeping data reliable
Least privilege	Minimum access needed
Change control	Process for safe changes
Change history	GA4 record of admin changes
Living document	Kept up to date continuously

Chapter 68 Quiz

1. Where does GA4 record configuration changes?

- A) GSC
- B) Realtime
- C) Explore
- D) Admin → Change history

2. How often is access reviewed?

- A) Hourly
- B) Quarterly
- C) Every 10 years
- D) Never

3. What's the dictionary rule?

- A) Optional
- B) Only for big events
- C) Not in the dictionary, not in production
- D) Document later

4. Who should have GTM publish rights?

- A) Everyone
- B) Interns only
- C) Limited people (team lead + senior dev)
- D) Nobody

5. What does the change request capture?

- A) Question, event, zone, PII risk, QA
- B) Only the date
- C) Only colour
- D) Nothing

6. What is least privilege?

- A) Maximum access
- B) Minimum access needed
- C) Shared logins

- D) Public access

7. Where does the change log live?

- A) Shared Sheet
- B) Clarity
- C) Bing
- D) Someone's memory

8. What must every new template include?

- A) Nothing
- B) PageMeta with all fields
- C) Ads
- D) Random scripts

9. What should GTM versions have?

- A) Emojis only
- B) Descriptive notes
- C) Passwords
- D) No names

10. In the chaabi analogy, governance is…

- A) The lock
- B) The key register
- C) The house
- D) The guest

Sources

1. Google Analytics Help, Change history
2. Google Tag Manager Help, Versions

Answer key, Chapter 68: 1-D · 2-B · 3-C · 4-C · 5-A · 6-B · 7-A · 8-B · 9-B · 10-B

LEVEL 6 · CHAPTER 69

Chapter 69: The 90-Day Implementation Roadmap

You have nothing live yet, but you have a plan. This chapter turns the whole book into a week-by-week schedule: who does what, in what order, and how you'll know each stage is finished.

> 💡 **Trivia**
>
> GA4 does not reprocess historical data when you add custom dimensions, key events or channel groups later (channel groups are an exception and do apply retroactively). [1] ● Official (Google)
>
> That's why setup order matters: definitions first, traffic second.

69.1 Phases

Phase	Weeks	Goal
0 Decide	1–2	Legal zones, consent model, BAA vendor scope, owners
1 Foundation	2–4	GA4/GTM/CMP/Clarity configured, data layer coded, QA passed
2 Platforms	4–6	BigQuery exports (GA4 + GSC), Bing, backend aggregate table
3 Reporting	6–9	Reporting tables, daily + SEO dashboards, alerts
4 Analysis	9–12	First funnels, paths, content and link reports; baseline captured
5 Optimise	12+	Targets reset, experiments, monthly cycle

69.2 Week-by-week plan

Week	Tasks	Owner	Done when
1	Legal meeting: Zone A/B/C list, consent approach, portal context params, BAA vendor scope; create "OWL Analytics" folder and docs (Chapter 68)	Team lead + legal	Signed decision log
1	Access audit and roles (Chapter 10)	Team lead	All tools named accounts
2	GA4 property settings, retention 14 months, internal traffic, domains, OWL channels (Chapters 11, 27)	Analytics owner	Admin checklist (Appendix E)
2	CMP chosen and configured (Chapters 5, 29)	Team lead + dev	Banner on staging
2–3	Next.js: GTM install, consent default, PageMeta, track() helper, SmartLink, data attributes (Chapters 13–21)	Front-end dev	Code merged to staging
3	GTM container build per Appendix D	Analytics owner + dev	Preview shows all events
3	Custom dimensions and key events registered (Chapters 25–26)	Analytics owner	Register complete
3	Clarity Strict masking, exclusion of Zone C (Chapter 28)	Dev	Verified on staging
4	Full QA script (Chapter 30, Appendix K), PII scan, portal check	Tester	16/16 pass
4	**Go live** (publish GTM, deploy)	Team lead approves	Realtime shows events
4	BigQuery project, billing, budget; GA4 link (Chapters 31–32)	Analytics owner	Tables arriving
5	GSC domain properties, sitemaps, bulk export, GA4 link; 16-month API backfill (Chapters 36–38)	SEO lead + dev	Export log present
5	Bing import, IndexNow hook (Chapter 39)	SEO lead + dev	IndexNow submissions visible

Week	Tasks	Owner	Done when
5–6	Backend `ops.daily_conversions` loader (Chapter 24)	Back-end dev	Yesterday's row daily
6	web-vitals component (Chapter 40)	Dev	web_vitals events
6–7	Reporting tables and scheduled queries (Chapter 58, Appendix H)	Analytics owner	Tables refresh nightly
7	Daily monitoring dashboard + 9 am email (Chapter 60)	Analytics owner	You receive it
7	GA4 custom insights; pipeline check; budget alerts (Chapter 56)	Analytics owner	Test alert received
8	SEO & content dashboard (Chapter 62); social Sheet + dashboard (Chapters 41, 63)	SEO lead, social lead	Weekly meeting uses it
9	Technical dashboard (Chapter 64); executive dashboard (Chapter 61)	Dev, team lead	First monthly report
9–10	Standard explorations, funnels, paths (Chapters 44–46)	Analytics owner	Shared "OWL –" explorations
10	Benchmark setup: Semrush tracking, CrUX query (Chapter 42)	SEO lead	First benchmark table
11	Content, link and CTA reports reviewed; first worklists	SEO + writers + UX	Tasks assigned
12	90-day review: reset KPI ranges from data (Chapter 66); first experiment card (Chapter 67)	Team lead	KPI dictionary v2



Foundation before paint.

> **Desi analogy: ghar banana**
>
> Pehle naksha aur permission, phir neev, phir deewarein, phir bijli-paani, phir rang-rogan. Rang pehle karoge toh neev ke kaam mein sab kharab. **Roadmap ghar banane ka kram hai.**

69.3 Dependencies (what must come first)

Before this	You need
Any tracking on Zone B pages	Legal zone decision
GA4 events reaching reports with our dimensions	Custom dimensions registered
BigQuery path queries	PageMeta live + export running
Backend channel totals	Legal approval of ctx params
Dashboards	Reporting tables
KPI targets	90 days of clean data

69.4 Roles summary

Role	Main chapters
Team lead	1–10, 56, 60–61, 65–66, 68–70
Front-end developer	13–24, 28–30, 40, 67
Back-end developer	23–24, 37–39 (scripts), 58
On-page SEO / analytics owner	11–12, 16, 25–27, 31–38, 43–53, 55–59, 62
Off-page SEO	39, 42, 53
Content writer	50–51, 53, 62
UI/UX designer	17, 52, 54, 67
Social media (senior/junior)	27, 41, 63
Tester	30, 55, 64, Appendix K
Graphic designer / video editor	20 (video events), 41

> 🟧 **Ongo Weight Loss example**
>
> Week 4 go-live is the biggest moment: GTM publish, CMP banner live, Clarity masked, BigQuery linked the same day. Everything after that builds on clean data from day one — the advantage of starting with nothing implemented. 🔵

69.5 Do's and Don'ts

✅ Do	❌ Don't
Get legal decisions in week 1	Track first, ask later
Register definitions before go-live	Register after traffic
Link BigQuery on go-live day	Wait months
Hold a 90-day review	Let the setup drift

69.6 Pros and cons of a phased roadmap

✅ Pros	❌ Cons
Clear order and owners	Requires coordination
Clean data from day one	Dashboards come after week 6
Measurable progress	—

69.7 Where freshers go wrong in Chapter 69 topics

Mistake	Why it happens	Avoid it by	Fix it by
Building dashboards first	Visible output	Phases	Reorder
Skipping legal	"We'll fix later"	Week 1	Pause Zone B tracking
No owner per task	Group tasks	69.2 owners	Assign

69.8 Thumb rules

👍 Foundation before paint.

👍 Definitions before traffic.

👍 Day 90 is a review, not the finish line.

69.9 Notes to remember

📝 Go-live in week 4; dashboards by week 9; targets at day 90.

📝 Custom dimensions and key events aren't retroactive.

Chapter 69 Glossary

Term	Simple meaning
Roadmap	Ordered plan over time
Dependency	Task that must finish first
Go-live	Publishing to production
Phase	Group of related weeks/tasks
Retroactive	Applies to past data

Chapter 69 Quiz

1. What happens in week 1?

- A) Dashboards
- B) Targets
- C) Legal decisions and docs folder
- D) Experiments

2. When is go-live?

- A) Week 12
- B) Week 1
- C) Week 4
- D) Never

3. When should BigQuery be linked?

- A) Go-live day
- B) Never
- C) Week 12
- D) After a year

4. Are custom dimensions retroactive?

- A) Only Mondays
- B) Yes

- C) Sometimes
- D) No

5. What's needed before Zone B tracking?

- A) Legal zone decision
- B) A dashboard
- C) An ad campaign
- D) A logo

6. When are KPI targets reset?

- A) Day 90
- B) Never
- C) Day 1
- D) Year 5

7. Who builds the backend aggregate loader?

- A) Back-end developer
- B) Social lead
- C) Writer
- D) Designer

8. What must exist before dashboards?

- A) Fonts
- B) Reporting tables
- C) Videos
- D) Paid ads

9. Which grouping is retroactive?

- A) Key events
- B) Custom dimensions
- C) Channel groups
- D) Audiences

10. In the ghar analogy, the roadmap is…

- A) The furniture
- B) The order of building a house

- C) The address
- D) The paint colour

Sources

1. Google Analytics Help, Custom dimensions and metrics
2. Google Analytics Help, Custom channel groups

Answer key, Chapter 69: 1-C · 2-C · 3-A · 4-D · 5-A · 6-A · 7-A · 8-B · 9-C · 10-B

LEVEL 6 · CHAPTER 70

Chapter 70: The ZETA Model for Analytics and the Final Gate

Zero-Error Tolerance Approach: the analytics mistakes that must never happen on a health brand's website, because they can expose patients' information, break the law, or make the whole team trust wrong numbers.

> 💡 **Trivia**
>
> HHS's guidance on online tracking says regulated entities must ensure that all tracking technology vendors that receive PHI have signed a business associate agreement, and that a cookie banner is not a HIPAA authorization. [1] 🟢 Official (HHS)
>
> In analytics, one wrong tag can be a privacy incident.

70.1 The ZETA zones (analytics edition)

Zone	Meaning	Example
🟢 **Green**	Low risk, fix any time	A badly named exploration; a chart colour
🟠 **Amber**	Fix before the next report; log it	Event not in dictionary; (not set) page_type; missing annotation
🔴 **Red**	Stop immediately; report the same day	Personal or health data in GA4/Clarity; tags on the portal; consent ignored

70.2 The Analytics Red List: never, ever

#	Never do this	Why
1	Send **names, emails, phone numbers, DOB, addresses** or IDs to GA4, Clarity, Looker or any non-BAA tool	Privacy law; HIPAA risk
2	Send **health answers** (BMI value, weight, conditions, medications, eligibility answers) as event parameters	PHI/consumer health data
3	Put **GTM, GA4, Clarity or pixels on Zone C** (portal, intake, payment)	Authenticated pages = PHI
4	Track **Zone B** pages without a written legal decision	HHS guidance; state laws
5	Fire analytics tags **before or against consent**	State privacy laws; trust
6	Send **payments or leads as user-level records** to GA4 (Measurement Protocol with client ID)	Links health service to a person
7	Build **health-condition audiences** for advertising	Sensitive data; platform policies
8	Use **Clarity Relaxed masking** or unmask inputs	Recordings could show data
9	Share **user-level timelines or exports** outside the analytics team	Re-identification
10	Publish GTM changes **without Preview and QA**	Silent breaks and leaks
11	Present **numbers without a definition or model**, or estimates as facts	Wrong decisions
12	**Hide** a tracking mistake	Worse harm later

> **Desi analogy: doctor ki file**
>
> Clinic mein mareez ki file sirf doctor ke cabin mein rehti hai, reception ke notice board pe nahi lagti. **Red List yeh pakka karti hai ki file kabhi notice board pe na pahunche.**

70.3 Warning signs

You hear or think…	It usually means…
"Just send the email as a user ID so we can match"	Red List #1, #6
"Let's capture the BMI result to see who qualifies"	Red List #2
"Add GTM to the portal so we see the whole funnel"	Red List #3
"The banner slows things; fire GA4 anyway"	Red List #5
"Make an audience of diabetes article readers for ads"	Red List #7
"Unmask the form so we can see where they get stuck"	Red List #8
"It's a tiny tag, skip Preview"	Red List #10
"Similarweb says Ro gets 3 million; put that in the deck"	Red List #11

70.4 The final gate (MEASURE)

Before any tracking change ships, or any report goes out:

Check	Question
Minimum data	Do we collect only what the question needs?
Exclusion zones	Is Zone C untouched and Zone B legally approved?
Approved names	Are all events/params in the dictionary?
Safe values	Did the PII scan return zero?
User consent	Do all tags respect consent?
Reviewed	Did Preview + QA script pass?
Explained	Do reports state definitions, models and data limits?

70.5 If a Red List item happens

Step	Action	When
1	Stop the source (pause tag, roll back code)	Immediately
2	Tell the team lead and legal	Same day
3	Assess what data, where, how long	Same day
4	Delete per legal advice (GA4 data deletion, BigQuery, Clarity)	As advised
5	Fix root cause and add a QA check	Within the week
6	Log the incident and lesson	Within the week

> 🟧 **Ongo Weight Loss example**
>
> A developer adds a "Your BMI is 31.2" message to the tool page and, to be helpful, sends `bmi_value: 31.2` with tool_complete. The weekly PII scan doesn't flag numbers, but the tester's dictionary check does: `bmi_value` isn't an approved parameter. The tag is paused the same hour, the data deleted, and the dictionary check stays as a permanent gate. That's ZETA working. 🔵

70.6 Do's and Don'ts

✅ Do	❌ Don't
Run MEASURE before every change	Ship and hope
Report mistakes the same day	Wait for someone to find them
Keep the Red List visible	Assume everyone remembers
Learn from each incident	Blame people

70.7 Pros and cons of zero tolerance

✅ Pros	❌ Cons
Protects patients and the brand	Slower changes
Clear rules	Some useful ideas rejected
Trusted data	Needs training

70.8 Where freshers go wrong in Chapter 70 topics

Mistake	Why it happens	Avoid it by	Fix it by
Thinking "it's anonymous" because no name	Numbers + context can identify	Red List #2	Remove values
Assuming consent banner = HIPAA OK	Common myth	70 trivia	Legal zones
Quietly fixing a leak	Fear	#12	Report

70.9 Thumb rules

> 👍 The patient's file never goes on the notice board.
>
> 👍 MEASURE before you ship.
>
> 👍 The worst ZETA mistake is hiding a ZETA mistake.

70.10 Notes to remember

> 📝 12-item Analytics Red List.
>
> 📝 MEASURE: Minimum, Exclusion, Approved, Safe, User consent, Reviewed, Explained.

Chapter 70 Glossary

Term	Simple meaning
ZETA	Zero-Error Tolerance Approach
Red List	Mistakes that must never happen
MEASURE	Our final gate checklist
Incident	A privacy or data mistake that happened
Root cause	The real reason a problem occurred

Chapter 70 Quiz

1. What does ZETA stand for?

- A) Zero Effort Testing
- B) Zone Event Tracking Analytics
- C) Zonal Entry Tag Audit
- D) Zero-Error Tolerance Approach

2. Which is a Red List item?

- A) A chart colour
- B) A font
- C) An exploration name
- D) Tags on the portal

3. Is a cookie banner a HIPAA authorization?

- A) Yes
- B) No
- C) Only in Texas
- D) Sometimes

4. What's step 1 if a Red item happens?

- A) Ignore it
- B) Stop the source

- C) Write a report next month
- D) Delete GA4 entirely

5. What does M in MEASURE mean?

- A) Marketing
- B) Minimum data
- C) Mobile
- D) Monthly

6. Can BMI values go to GA4?

- A) No
- B) Yes
- C) Only on weekends
- D) Only rounded

7. Which Clarity masking is forbidden?

- A) None
- B) Default for inputs
- C) Strict
- D) Relaxed

8. Who is told the same day?

- A) Team lead and legal
- B) Competitors
- C) Nobody
- D) Social followers

9. What's the worst ZETA mistake?

- A) A slow chart
- B) A long meeting
- C) A typo
- D) Hiding a ZETA mistake

10. In the clinic analogy, the Red List keeps the file…

- A) In the waiting room
- B) Off the notice board

- C) In the parking lot
- D) On the notice board

Sources

1. U.S. HHS, Use of Online Tracking Technologies by HIPAA Covered Entities and Business Associates ●
2. Google Analytics Help, HIPAA and Google Analytics ●

Answer key, Chapter 70: 1-D · 2-D · 3-B · 4-B · 5-B · 6-A · 7-D · 8-A · 9-D · 10-B

Back Matter

Appendix A: Event Dictionary v2 (Full Reference)

The single source of truth for every event. If an event isn't here, it doesn't ship (Chapter 68). Also kept as a shared Sheet and as the BigQuery table `reporting.event_dictionary`.

A.1 Page and navigation events

Event	Trigger	Parameters	Key event	Zone
`page_view`	Data-layer `page_meta` on every route	page_location (clean), page_referrer (clean), page_title, site_id, page_type, page_group, funnel_stage, content_cluster, primary_entity, hub_parent, template, content_id, privacy_zone, author_role, publish_month, index_state	No	A, B (if approved)
`internal_link_click`	Click on a link to an own domain (not a CTA)	link_source, link_target, link_position, link_text, source_stage, target_stage, target_page_type	No	A, B
`page_not_found`	404 template renders	requested_path	No	A
`outbound_click_source` / `outbound_click_social` / `outbound_click_partner`	Click to another domain	link_domain, link_url_path, page_type	No	A

A.2 CTA and contact events

Event	Trigger	Parameters	Key event
`cta_[group]_[action]`	Element with `data-event` starting `cta_`	cta_name, cta_group, cta_location, cta_text, link_url_path	Optional (`cta_*_get_started`)
`exit_to_portal`	Any click leaving for the portal (`data-exit-portal="true"`)	source_page, cta_name, site_id, page_type, funnel_stage	✅ Primary
`phone_click`	`tel:` link	contact_location, page_type	✅
`email_click`	`mailto:` link	contact_location, page_type	No
`chat_open`	Chat widget opened	contact_location, page_type	✅

Approved CTA groups: weight_loss, semaglutide, tirzepatide, glp1, longevity, mens_health, womens_health, pricing, blog, tool, global. **Approved CTA actions:** get_started, check_eligibility, book_appointment, call_us, chat_with_us, email_us, learn_more, see_pricing, download_guide. **Approved locations:** hero, body, inline, pricing, sticky_mobile, nav, header, footer, sidebar, card, faq.

A.3 Engagement events

Event	Trigger	Parameters
`scroll_depth`	25/50/75/90% scrolled (once each per page)	percent_scrolled, page_type
`faq_open`	FAQ item opened	faq_id, page_type
`video_start` / `video_progress` / `video_complete`	Video played	video_title, video_percent, page_type
`web_vitals`	LCP/INP/CLS measured	metric_name, metric_value, metric_rating, metric_id

A.4 Tools, forms and experiments

Event	Trigger	Parameters	Never include
`tool_start_[tool]`	First input in a public tool	tool_name, page_type	Inputs
`tool_complete_[tool]`	Result shown	tool_name, page_type	Inputs, results, categories
`form_start_[form]`	First field focused (non-health forms only)	form_name, page_type	Field values
`form_submit_[form]`	Successful submit (non-health forms only)	form_name, page_type	Field values
`experiment_impression`	Visitor sees an experiment variant	experiment_id, variant	—
`consent_update` (optional)	Consent choice changed	consent_state	—

A.5 GA4 automatic events (also in the dictionary table)

`first_visit`, `session_start`, `user_engagement`, `scroll` (Enhanced measurement 90%), `click` (outbound), `file_download`, `video_*` (if Enhanced measurement video is on), `view_search_results` (if site search exists).

A.6 Retired or forbidden

Event	Status	Reason
`portal_signup`, `portal_payment` and any `portal_*`	**Forbidden in GA4**	Zone C (Chapter 4)
`button_click` (generic)	Retired	Replaced by specific CTA events
GA4 event `page_meta`	Retired	Now a data-layer event that becomes `page_view`

Appendix B: Data Layer Reference

B.1 Load order

```
1. window.dataLayer created + gtag('consent','default', …)   [inline, beforeInteractive]
2. GTM container loads; CMP template runs on Consent Initialization
3. page_meta pushed on every route (PageMeta component)
4. Interaction pushes (track() helper or data attributes → GTM)
```

B.2 `page_meta` fields

Field	Type	Allowed values	Source
site_id	string	ongo_main, wcr, child_[name]	Env variable per site
page_type	string	home, service, medication, pricing, how_it_works, blog_article, blog_hub, tool, eligibility_start, policy, about, medical_team, contributor, reviews, contact, not_found	Content map
page_group	string	weight_loss, longevity, mens_health, womens_health, general	Content map (silo)
funnel_stage	string	tofu, mofu, bofu, retention, none	Content map
content_cluster	string	lowercase snake_case cluster	Content map
primary_entity	string	semaglutide, tirzepatide, glp1, none	Content map
hub_parent	string	Parent hub slug	Content map
template	string	e.g., article_v2	Code
content_id	string	e.g., blog-0142	CMS
privacy_zone	string	A, B	Legal decision log
author_role	string	writer, clinician, none	CMS

Field	Type	Allowed values	Source
publish_month	string	YYYY-MM	CMS
index_state	string	index, noindex	SEO

B.3 Interaction push format

```
window.dataLayer.push({
  event: 'cta_semaglutide_get_started',
  cta_name: 'get_started',
  cta_group: 'semaglutide',
  cta_location: 'hero'
});
```

B.4 Data attributes (HTML)

Attribute	On	Example
`data-event`	CTAs	cta_semaglutide_get_started
`data-cta-name` / `data-cta-group` / `data-cta-location`	CTAs	get_started / semaglutide / hero
`data-exit-portal`	Links to the portal	true
`data-link-position`	Containers	body, nav, footer, related, breadcrumb, toc, card
`data-target-stage` / `data-target-type`	SmartLink	bofu / medication
`data-faq-id`	FAQ items	sema-nausea-q3
`data-clarity-mask`	Sensitive areas	True

B.5 Rules

Rule	Detail
No personal or health values	Ever (Chapter 70)

Rule	Detail
Lowercase, snake_case values	Consistent reporting
Text ≤ 100 characters	GA4 truncation
Paths, not full URLs	No query strings

Appendix C: Next.js Code Pack

All code is from Chapters 13–23, 40 and 67, collected in one place. Paths assume the App Router.

C.1 Root layout with consent default and GTM

```
// app/layout.tsx
import Script from 'next/script'
import { GoogleTagManager } from '@next/third-parties/google'
import { WebVitals } from '@/components/WebVitals'

const GTM_ID = process.env.NEXT_PUBLIC_GTM_ID!

export default function RootLayout({ children }: { children: React.ReactNode }) {
  return (
    <html lang="en">
      <head>
        <Script id="consent-default" strategy="beforeInteractive">
          {`window.dataLayer = window.dataLayer || [];
            function gtag(){dataLayer.push(arguments);}
            gtag('consent','default',{analytics_storage:'denied',ad_storage:'denied',
              ad_user_data:'denied',ad_personalization:'denied',wait_for_update:500});`}
        </Script>
      </head>
      <GoogleTagManager gtmId={GTM_ID} />
      <body>
        <WebVitals />
        {children}
      </body>
    </html>
  )
}
```

If the CMP template sets the defaults in GTM, remove the inline default (Chapter 29, step 10).

C.2 Types and helper

```
// types/analytics.d.ts
export {}
declare global { interface Window { dataLayer: Record<string, unknown>[]; clarity?:
(...a: unknown[]) => void } }

// lib/analytics.ts
export function track(event: string, params: Record<string, string | number | boolean> =
{}) {
  if (typeof window === 'undefined') return
  window.dataLayer = window.dataLayer || []
  window.dataLayer.push({ event, ...params })
}
```

C.3 PageMeta component

```
// components/PageMeta.tsx
'use client'
import { useEffect } from 'react'
import { usePathname } from 'next/navigation'
import type { PageMeta as Meta } from '@/lib/page-meta'
import { setVisitContext } from '@/lib/visit-context'

export function PageMeta(meta: Meta) {
  const pathname = usePathname()
  useEffect(() => {
    setVisitContext({ page_type: meta.page_type, content_cluster: meta.content_cluster })
    window.dataLayer = window.dataLayer || []
    window.dataLayer.push({
      event: 'page_meta',
      site_id: meta.site_id, page_type: meta.page_type, page_group: meta.page_group,
      funnel_stage: meta.funnel_stage, content_cluster: meta.content_cluster,
      primary_entity: meta.primary_entity, hub_parent: meta.hub_parent,
      template: meta.template, content_id: meta.content_id, privacy_zone:
meta.privacy_zone,
      author_role: meta.author_role ?? 'none', publish_month: meta.publish_month ??
'none',
      index_state: meta.index_state ?? 'index',
    })
  }, [pathname, meta.content_id])
  return null
}
```

C.4 CTA (data attributes) and portal link

```
import Link from 'next/link'
import { portalUrl } from '@/lib/visit-context'

<Link href={portalUrl(process.env.NEXT_PUBLIC_PORTAL_URL!)}
  data-event="cta_semaglutide_get_started" data-cta-name="get_started"
  data-cta-group="semaglutide" data-cta-location="hero" data-exit-portal="true">
  See if I'm eligible
</Link>
```

Use `portalUrl()` only if legal approved the context parameters (Chapter 23); otherwise link to the plain portal URL.

C.5 SmartLink and containers

```
// components/SmartLink.tsx
import Link from 'next/link'
import { getPageMeta } from '@/lib/content-map'
export function SmartLink({ href, children, linkId }: { href: string; children:
React.ReactNode; linkId?: string }) {
  const t = getPageMeta(href)
  return <Link href={href} data-target-stage={t?.funnel_stage ?? 'none'}
    data-target-type={t?.page_type ?? 'unknown'} data-link-id={linkId}>{children}</Link>
}
// Wrap regions: <article data-link-position="body">, <aside data-link-
position="related">, etc.
```

C.6 Scroll, FAQ, tool, 404

```
// hooks/useScrollDepth.ts — see Chapter 20 (fires 25/50/75/90 once per page)
// FAQ
<details data-faq-id="sema-nausea-q3"
  onToggle={(e) => (e.currentTarget as HTMLDetailsElement).open && track('faq_open', {
faq_id: 'sema-nausea-q3' })}>
// Tool: track('tool_start_bmi_calculator', { tool_name: 'bmi_calculator' }) on first
input;
//       track('tool_complete_bmi_calculator', { tool_name: 'bmi_calculator' }) on
result. NO values.
// 404 (app/not-found.tsx, client effect):
track('page_not_found', { requested_path: window.location.pathname })
```

C.7 Visit context (optional, legal approval required)

See Chapter 23 for `lib/visit-context.ts` (`setVisitContext`, `channelGroup`, `portalUrl`). It stores only the first page type/cluster and a channel label in `sessionStorage` and appends `ctx_ch`, `ctx_lp`, `ctx_site` to portal links.

C.8 Web vitals

```
// components/WebVitals.tsx
'use client'
import { useEffect } from 'react'
import { onLCP, onINP, onCLS } from 'web-vitals'
import { track } from '@/lib/analytics'
export function WebVitals() {
  useEffect(() => {
    const send = (m: { name: string; value: number; rating: string; id: string }) =>
      track('web_vitals', { metric_name: m.name,
        metric_value: Math.round(m.name === 'CLS' ? m.value * 1000 : m.value),
        metric_rating: m.rating, metric_id: m.id })
    onLCP(send); onINP(send); onCLS(send)
  }, [])
  return null
}
```

C.9 Experiments

```
track('experiment_impression', { experiment_id: 'exp_2026_11_tirz_hero_cta', variant: 'b'
})
```

C.10 Clarity (non-personal tags only)

```
window.clarity?.('set', 'page_type', meta.page_type)
window.clarity?.('set', 'funnel_stage', meta.funnel_stage)
```

C.11 Developer checklist per template

Check	Done
`<PageMeta>` with all 13 fields from the content map	☐
Link regions have `data-link-position`	☐

Check	Done
CTAs use registered `data-event` names	☐
Portal links have `data-exit-portal="true"`	☐
Sensitive areas have `data-clarity-mask="True"`	☐
No GTM/Clarity on Zone C routes	☐
QA script passed on staging	☐

Appendix D: GTM Build Sheet

D.1 Folders

01 Consent · 02 Google tag · 03 Page views · 04 CTAs · 05 Links · 06 Engagement · 07 Tools & forms · 08 Portal · 09 Clarity · 10 Utilities.

D.2 Variables

Name	Type	Setting
Const – GA4 ID	Lookup on Environment Name	Live → G-LIVE, Staging → G-STAGE
DLV – site_id … DLV – index_state (13)	Data Layer Variable v2	Default `(not set)`
DLV – cta_name / cta_group / cta_location	Data Layer Variable	For pushed CTA events
CJS – clean page_location	Custom JS	Keep utm_* only (Chapter 12)
CJS – clean page_referrer	Custom JS	Strip query
CJS – cta event	Custom JS	`closest('[data-event^="cta_"]')`
CJS – cta attr name / group / location	Custom JS	closest CTA's data attributes
CJS – click path	Custom JS	`new URL({{Click URL}}).pathname`

Name	Type	Setting
CJS – link position	Custom JS	`closest('[data-link-position]')`
CJS – link target stage / type	Custom JS	closest `a` data-target-*
RegEx Table – own domain	RegEx table on Click Hostname	Own-domains pattern below → true
Google tag: Event Settings – shared	Event settings variable	page_location, page_referrer, site_id, privacy_zone

Own-domains pattern (keep in this sheet; update when child sites are added):

```
^(www\.)?(ongoweightloss\.com|weightclinicsresults\.com)$
```

D.3 Triggers

Name	Type	Condition
Consent Init – All Pages	Consent Initialization	All pages
Init – All Pages	Initialization	All pages
CE – page_meta	Custom Event	Event name `page_meta`
Click – CTA	All Elements click	`CJS – cta event` matches RegEx `^cta_`
Click – Portal exit	All Elements click	closest `[data-exit-portal="true"]` exists
Click – Internal link	Just Links	Click Hostname is own domain AND not a CTA AND not portal
Click – Outbound	Just Links	Click Hostname not own domain AND not portal
Click – tel	Just Links	Click URL starts with `tel:`
Click – mailto	Just Links	Click URL starts with `mailto:`
CE – cta pushed	Custom Event	Event name matches RegEx `^cta_`

Name	Type	Condition
CE – tool / form / faq / scroll / web_vitals / experiment / 404	Custom Event	Event name matches each
CE – chat_open	Custom Event	From chat widget callback
Exception – Zone C	Page Hostname/Path	Portal hostnames or paths (blocking trigger on everything)

D.4 Tags

Name	Type	Trigger	Key settings
CMP – CookieYes/Cookiebot	Template	Consent Init – All Pages	Defaults per legal
GA4 – Google tag	Google tag	Init – All Pages	send_page_view false; shared settings
GA4 – Event – page_view	GA4 Event	CE – page_meta	Event name `page_view`; params from 13 DLVs
GA4 – Event – cta (click)	GA4 Event	Click – CTA	Event name `{{CJS – cta event}}`; cta params
GA4 – Event – cta (pushed)	GA4 Event	CE – cta pushed	Event name `{{Event}}`; DLV params
GA4 – Event – exit_to_portal	GA4 Event	Click – Portal exit	source_page, cta_name, page_type, funnel_stage
GA4 – Event – internal_link_click	GA4 Event	Click – Internal link	link params

Name	Type	Trigger	Key settings
GA4 – Event – outbound	GA4 Event	Click – Outbound	link_domain, link_url_path
GA4 – Event – phone_click / email_click / chat_open	GA4 Event	Respective triggers	contact_location
GA4 – Event – generic pushed	GA4 Event	CE – tool/form/faq/scroll/web_vitals/experiment/404	Event name `{{Event}}`; DLV params
Clarity – base	Custom HTML or template	Init – All Pages (Zone A/B approved)	Requires analytics consent; Zone C exception

All GA4 and Clarity tags: **Additional consent required: analytics_storage**; **blocking trigger: Exception – Zone C**.

D.5 Publishing

Version name `vNN – what – chapter/ticket`; description lists tags changed; Preview + Appendix K before publishing; export container JSON monthly as backup.

Appendix E: GA4 Admin Checklist

#	Setting	Value	☐
1	Account data sharing	Only what legal approves	☐
2	Property time zone / currency	Company reporting zone / USD	☐
3	Web stream	"Public sites", one G- ID for all public domains	☐
4	Enhanced measurement	Page views on (history changes OFF); scrolls, outbound, downloads, video on; form interactions OFF	☐
5	Data retention	14 months; reset on new activity ON	☐

#	Setting	Value	☐
6	Google signals	OFF	☐
7	Granular location/device data	Legal decides (recommend OFF)	☐
8	Ads personalisation	Disabled for all regions	☐
9	Internal traffic	Defined; filter Active	☐
10	Unwanted referrals	Payment processors, portal if needed	☐
11	Configure your domains	All public domains	☐
12	Session timeout	30 min (default)	☐
13	Reporting identity	Device-based	☐
14	Attribution settings	Data-driven; 30-day acquisition, 90-day other	☐
15	Custom dimensions	Appendix F registered	☐
16	Key events	exit_to_portal, phone_click, chat_open, form_submit_contact	☐
17	Channel group	"OWL channels" (Appendix G)	☐
18	Product links	Search Console (each property), BigQuery; no Ads link yet	☐
19	Access	Named users, least privilege (Chapter 10)	☐
20	Custom insights	Five alerts (Chapter 56)	☐
21	Report library	"Ongo Weight Loss" collection published	☐
22	Staging property	Separate property with its own G- ID	☐

Appendix F: Custom Dimensions Register

Priority	Display name	Parameter	Scope
1	Site	site_id	Event

Priority	Display name	Parameter	Scope
1	Page type	page_type	Event
1	Page group	page_group	Event
1	Funnel stage	funnel_stage	Event
1	Content cluster	content_cluster	Event
1	CTA name	cta_name	Event
1	CTA location	cta_location	Event
1	Link position	link_position	Event
1	Link target	link_target	Event
1	Source stage	source_stage	Event
1	Target stage	target_stage	Event
2	Primary entity	primary_entity	Event
2	Hub parent	hub_parent	Event
2	CTA group	cta_group	Event
2	Link source	link_source	Event
2	Link text	link_text	Event
2	Tool name	tool_name	Event
2	Form name	form_name	Event
2	Metric name	metric_name	Event
2	Metric rating	metric_rating	Event
2	Experiment ID	experiment_id	Event
2	Variant	variant	Event
3	Template	template	Event
3	Content ID	content_id	Event

Priority	Display name	Parameter	Scope
3	Author role	author_role	Event
3	Publish month	publish_month	Event
3	FAQ ID	faq_id	Event
3	Percent scrolled	percent_scrolled	Event
3	Requested path	requested_path	Event
3	Privacy zone	privacy_zone	Event
3	CTA text	cta_text	Event
3	Target page type	target_page_type	Event
3	Link domain	link_domain	Event
3	Contact location	contact_location	Event
3	Source page	source_page	Event
Metric	Metric value	metric_value	Event (custom metric)

Register before go-live; names are case-sensitive; not retroactive.

Appendix G: UTM and Channel Convention

G.1 When to use UTMs

Situation	UTM?
Search, organic referrals, typed visits	No
Any link inside our own sites (including child site → main site)	**Never**
Links we place outside (social, email, QR, partners, press)	Yes, per G.2

G.2 Values

Parameter	Format	Examples
utm_source	platform or partner	instagram, youtube, facebook, pinterest, linkedin, reddit, threads, newsletter, partner_[name], qr_[place]
utm_medium	channel type	social, email, qr, partner, referral
utm_campaign	pillar_yyyy_mm or event_yyyy	glp1_basics_2026_11
utm_content	format_topic_v# or placement	carousel_protein_v2, bio_link

Lowercase, underscores, no personal or health data, built in the shared UTM sheet.

G.3 "OWL channels" group (in rule order)

Channel	Rule
AI Assistants	Source matches chatgpt.com, perplexity.ai, copilot.microsoft.com, gemini.google.com, claude.ai (check referrer names monthly)
Email	Medium exactly email
QR	Medium exactly qr
Partner	Medium exactly partner
Organic Social	Default rule or medium exactly social
Organic Search	Default rule
Child Sites	Source matches child domains (only if journeys start there without cross-domain)
Referral	Default rule
Direct	Default rule

AI Assistants must sit **above** Referral so AI tools aren't counted as referrals.

G.4 BigQuery channel mapping (for scheduled tables)

```
CASE
  WHEN REGEXP_CONTAINS(LOWER(source), r'chatgpt|perplexity|copilot|gemini|claude') THEN
'ai_assistants'
  WHEN LOWER(medium) = 'email' THEN 'email'
  WHEN LOWER(medium) = 'qr' THEN 'qr'
  WHEN LOWER(medium) = 'partner' THEN 'partner'
  WHEN LOWER(medium) IN ('social','social-network','sm') OR
REGEXP_CONTAINS(LOWER(source),
r'instagram|facebook|youtube|pinterest|linkedin|reddit|tiktok|threads') THEN
'organic_social'
  WHEN LOWER(medium) = 'organic' THEN 'organic_search'
  WHEN LOWER(medium) = 'referral' THEN 'referral'
  WHEN source IS NULL OR source = '(direct)' THEN 'direct'
  ELSE 'other'
END AS channel_group
```

Appendix H: BigQuery SQL Library

Replace `PROJECT` and `PROPERTY`. All queries filter dates. Saved queries live in BigQuery (shared) and in the repo under `/analytics/sql/`.

#	Name	Chapter	Purpose
H1	events_by_name	34	Event counts by name
H2	users_sessions_daily	34	Users and sessions per day
H3	top_pages	34	Page views by page
H4	landing_pages	34	Entrances by landing page
H5	cta_by_location	34	CTA clicks by name and location
H6	v_page_views (view)	35	Clean page-view rows (base view)
H7	session_paths	35	Every session's path, entry → exit
H8	entry_exit_pairs	35	Entry and exit page per session
H9	next_page	35	Page-to-page flow

#	Name	Chapter	Purpose
H10	funnel_any_order	35	TOFU → MOFU → BOFU → portal
H11	first_touch_portal	35	Landing page credit for portal exits
H12	internal_link_flow	35	Link clicks by from/to/position
H13	ga4_backend_join	35	Portal exits vs backend totals by date
H14	pii_scan	35	Weekly personal-data scan (expect 0)
H15	gsc_daily	37	Clicks, impressions, CTR, position by day
H16	gsc_page_queries	37	Top queries per URL
H17	striking_distance	37	Positions 8–20 with impressions
H18	gsc_ga4_join	38	GSC clicks vs organic entrances by path
H19	crux_competitors	40	Origin CWV vs competitors
H20	web_vitals_p75	40	p75 by metric and page type
H21	link_downstream_portal	51	Portal rate after each internal link
H22	cta_to_portal	52	Portal sessions by CTA
H23	entity_groups	53	GSC by entity
H24	cannibalisation	53	Queries with 2+ URLs
H25	pipeline_check	56	ERROR() if exports missing
H26	daily_kpis (table)	58	Dashboard base table
H27	tracking_health	64	page_views, missing page_type, unknown events
H28	kpi_bands	66	25th/50th/75th percentile bands

H.1 Scheduled-query timetable

Time (property zone)	Query	Destination
18:00	H25 pipeline_check	— (email on failure)

Time (property zone)	Query	Destination
19:00	H26 daily_kpis	reporting.daily_kpis
19:10	funnel_daily, cta_daily, landing_pages_daily	reporting.*
19:20	gsc_page_daily	reporting.gsc_page_daily
Monday 07:00	H14 pii_scan	reporting.pii_scan_log
1st of month 06:00	content_monthly, internal_links_monthly, cwv_monthly	reporting.*

H.2 Snippets you'll reuse

```
-- one parameter
(SELECT value.string_value FROM UNNEST(event_params) WHERE key = 'page_type') AS
page_type
-- session key
CONCAT(user_pseudo_id, '.', CAST((SELECT value.int_value FROM UNNEST(event_params) WHERE
key = 'ga_session_id') AS STRING)) AS session_key
-- last 28 days
WHERE _TABLE_SUFFIX BETWEEN FORMAT_DATE('%Y%m%d', DATE_SUB(CURRENT_DATE(), INTERVAL 28
DAY))
                       AND FORMAT_DATE('%Y%m%d', DATE_SUB(CURRENT_DATE(), INTERVAL 1
DAY))
-- strip query string
REGEXP_REPLACE(page_location, r'\?.*$', '')
```

Appendix I: Looker Studio Dashboard Specs

Dashboard	Pages	Sources	Default range	Audience	Delivery
OWL – Daily Monitor	1 (8 rows)	daily_kpis, funnel_daily, landing_pages_daily, ops.daily_conversions, Change log, Targets	Yesterday vs same day last week	Team lead (+ leads)	Daily 09:00 email

Dashboard	Pages	Sources	Default range	Audience	Delivery
OWL – Executive	2	monthly_kpis, ops, Targets, SoV Sheet, cwv_monthly	Last full month	Leadership	1st of month 10:00
OWL – SEO & Content	6	gsc_page_daily, content_monthly, internal_links_monthly, funnel_daily, cwv_monthly, Index Sheet	Last 28 vs previous 28	SEO, writers	Monday 08:30
OWL – Social & Channel	3	daily_kpis, social Sheets, Bing AI Sheet	Last 28 days	Social team, team lead	Monday 08:45
OWL – Technical Health	5	tracking_health, cwv_monthly, 404 table, Clarity Sheet, Change log	Last 30 days	Developers, tester	Monday 09:15

I.1 Standard elements on every dashboard

Element	Rule
Title + version	"OWL – [Name] – v#"
Date control	Top right
Info icons	Definition, source, attribution model
Colours	Brand green for actuals, orange for highlights, grey for comparisons
Footer note	"Aggregated data. No personal or health information."
Owner	Named on page 1

I.2 Sheets layouts

Sheet	Columns
Social (one tab per platform)	date, platform, account, followers, posts, reach, impressions, engagements, link_clicks, video_views
Targets	kpi, month, low, high, owner

Sheet	Columns
Change log	date, change, owner, expected_effect, chapter_or_ticket
SoV	month, domain, share_of_voice, keywords_top3
Bing AI	month, citations, cited_pages, top_grounding_query_1, _2, _3
Index (GSC)	week, indexed, not_indexed, top_reason
Clarity	week, sessions, dead_click_pct, rage_click_pct, quick_back_pct, js_error_sessions

Appendix J: KPI Dictionary

KPI	Formula	Source	Starting range	Review	Owner
Sessions	GA4 sessions	daily_kpis	±20% vs same weekday	Daily	Team lead
Engagement rate	Engaged sessions ÷ sessions	GA4	55–70%	Weekly	On-page SEO
TOFU → BOFU rate	TOFU-landing sessions reaching BOFU ÷ TOFU-landing sessions	BigQuery	3–8%	Weekly	SEO + writers
Internal link CTR (TOFU)	internal_link_click ÷ TOFU views	BigQuery	5–15%	Monthly	SEO
BOFU CTA rate	BOFU sessions with CTA ÷ BOFU sessions	BigQuery	10–25%	Weekly	UX + SEO
Portal exit rate (site)	Sessions with exit_to_portal ÷ sessions	GA4	2–6%	Daily	Team lead
Portal exit rate (BOFU)	÷ BOFU sessions	BigQuery	10–25%	Weekly	Team lead

KPI	Formula	Source	Starting range	Review	Owner
Contact rate	phone_click + chat_open sessions ÷ sessions	GA4	Set at day 90	Weekly	Team lead
Leads	eligibility_completed + signups	ops	Set at day 90	Daily	Team lead
Lead rate	Leads ÷ sessions	ops + GA4	Set at day 90	Weekly	Team lead
Portal completion rate	Leads ÷ exit_to_portal	ops + GA4	Set at day 90	Weekly	Team lead + ops
Payments	Count	ops	Set at day 90	Daily	Team lead + ops
Lead → payment rate	Payments ÷ leads	ops	Set at day 90	Monthly	Ops
Non-branded organic clicks	GSC excluding brand	GSC export	+5–15% MoM	Weekly	SEO
Share of voice	Tracked-set visibility	Semrush	Upward trend	Monthly	SEO
Bing AI citations	Citations per month	Bing WMT	Upward trend	Monthly	SEO
CWV pass rate	Page types good on all three	web-vitals	≥ 90%	Weekly	Developer
p75 LCP / INP / CLS (mobile)	web-vitals p75	GA4/BQ	≤ 2.5 s / ≤ 200 ms / ≤ 0.1	Weekly	Developer
404 rate	page_not_found ÷ page_views	GA4	< 0.5%	Weekly	Developer
Social portal exits per 1,000 reach	Social exit_to_portal ÷ reach × 1,000	GA4 + Sheet	Set at day 90	Monthly	Social lead
Rage-click sessions	% sessions	Clarity	< 3%	Weekly	UX

KPI	Formula	Source	Starting range	Review	Owner
Data-quality pass	Checks passed ÷ 13	Checklist	13/13	Weekly	Analytics owner
PII scan rows	Count	H14	0	Weekly	Analytics owner

Starting ranges are hypotheses (🔵/🟠) except Core Web Vitals (🟢). Reset at day 90 with H28.

Appendix K: QA and Release Checklist

K.1 The 16-step QA script (every release)

#	Step	Pass	☐
1	Open GTM Preview on staging	Container connects	☐
2	Load home	1 page_view, correct metadata	☐
3	Navigate to a blog article (client-side)	1 page_view; funnel_stage correct	☐
4	Back / forward / refresh	1 page_view each	☐
5	Click a body internal link to a money page	internal_link_click with correct params	☐
6	Click a hero CTA	cta_*; no internal_link_click	☐
7	Click a portal CTA	cta_* + exit_to_portal; no `_gl` to portal	☐
8	Phone / chat / email	Correct events, no numbers/emails in params	☐
9	Scroll an article	scroll_depth 25/50/75/90 once each	☐
10	Open an FAQ, play a video	faq_open, video_*	☐
11	Use a tool	tool_start/complete; no values	☐
12	Visit a 404	page_view (not_found) + page_not_found (path only)	☐

#	Step	Pass	☐
13	Child site → main site	Same session; `_gl` present	☐
14	Consent: reject	No GA4/Clarity requests	☐
15	Personal-data scan	No `@`, phone patterns, `email=`, `name=` in any hit	☐
16	Portal pages	No GTM/GA4/Clarity requests	☐

K.2 Release gate (Chapter 68)

Gate	☐
New templates include PageMeta	☐
New CTAs/links registered in Appendix A	☐
New forms have a privacy-zone decision	☐
QA script passed (K.1)	☐
GTM version named and described	☐
Change log updated	☐
Next-morning check: daily dashboard health row green	☐

Appendix L: Privacy, HIPAA and Consent Checklist

#	Item	Owner	☐
1	Zone A / B / C page list written and signed by legal	Team lead + legal	☐
2	BAA vendor scope confirmed: which systems and data it covers	Team lead + legal	☐
3	No GTM, GA4, Clarity or pixels on Zone C (verified by K.1 step 16)	Developer + tester	☐
4	Zone B tracking: written decision (none / minimal / BAA tool only)	Legal	☐
5	CMP live on all public domains; Consent Mode v2; GPC honoured	Developer	☐

#	Item	Owner	☐
6	Consent defaults and region rules approved by legal	Legal	☐
7	All GA4/Clarity tags require analytics_storage	Analytics owner	☐
8	Clean page_location (only utm_*); no full URLs with queries	Analytics owner	☐
9	No health values in any event (tools, forms, FAQs)	Developer	☐
10	Clarity Strict masking; sensitive areas masked	Developer	☐
11	Google signals off; ads personalisation off; no Ads link	Analytics owner	☐
12	Backend counts aggregated; small-number rule applied	Back-end dev + legal	☐
13	Portal context params approved (or not used)	Legal	☐
14	Consumer health data privacy policy linked where state law requires (e.g., Washington MHMDA)	Legal	☐
15	Weekly PII scan returns 0	Analytics owner	☐
16	Incident playbook known (Chapter 70)	Everyone	☐
17	Quarterly access review done	Team lead	☐
18	Privacy decisions log updated	Team lead	☐

This checklist supports, and does not replace, legal advice.

Appendix M: GSC, Bing and Core Web Vitals Setup Checklists

M.1 Google Search Console

#	Step	☐
1	Domain property per public domain (DNS TXT)	☐
2	Users added (Restricted where possible)	☐
3	Sitemaps submitted per site	☐

#	Step	☐
4	GA4 link per property; Search Console collection published	☐
5	Bulk export: billing project, APIs, service account roles, dataset per property, same location as GA4	☐
6	One-time 16-month API backfill	☐
7	Weekly: Performance compare, Pages, CWV, Security/Manual actions	☐
8	Monthly: Links, Crawl stats	☐

M.2 Bing Webmaster Tools

#	Step	☐
1	Import from Google Search Console	☐
2	Users added	☐
3	IndexNow key file and publish hook	☐
4	Weekly: Search Performance; monthly: AI Performance, Backlinks, Site Scan, Recommendations	☐
5	Monthly export to Sheets (Bing AI tab)	☐

M.3 Core Web Vitals

#	Step	☐
1	WebVitals component live; web_vitals events in GA4	☐
2	Custom definitions: metric_name, metric_rating, metric_value	☐
3	H20 p75 query scheduled monthly into cwv_monthly	☐
4	H19 CrUX competitor query monthly	☐
5	CrUX Vis bookmarked for main origin and competitors	☐
6	GSC CWV report reviewed weekly	☐

Appendix N: Case Appendices

N1 · Scenario: traffic halves overnight

The 9:00 email shows sessions −52% and the health row red. Channel row: every channel down equally. Change log: CMP banner text updated yesterday. **Finding:** the new banner made "Reject" the default button style; consent rate fell. **Action:** legal-approved neutral design restored; note in monthly report that the gap is a measurement change, not demand. **Lesson:** check consent before blaming SEO.

N2 · Scenario: a developer wants GA4 on the portal

"We can't see payments, so let's add GTM to the portal." **ZETA Red List #3. Action:** decline; explain the aggregated backend feed (Chapter 24) delivers payments by channel without sending anything from the portal to Google. **Lesson:** outcomes travel as totals, not as tags.

N3 · Scenario: the BMI value leak

A tool update sends `bmi_value`. The tracking-health check flags an unknown parameter. **Action:** pause tag within the hour, delete data per legal, add a dictionary check to QA. **Lesson:** the dictionary is a privacy control, not just tidiness (Chapter 70).

N4 · Scenario: "the blog doesn't convert"

Last-click report shows blog with 4% of portal exits. First-touch Query E shows blog articles started 44% of portal-exit journeys. **Action:** report both, labelled; keep investing in TOFU. **Lesson:** label the model (Chapter 49).

N5 · Scenario: CTR drop on the tirzepatide cost page

Position steady at 4; CTR 4.1% → 2.7%. Manual SERP check: AI Overview added. **Action:** add a clear cost table and FAQ, monitor 4 weeks. **Lesson:** look at the SERP before rewriting (Chapter 53).

N6 · Scenario: child site traffic looks like referrals

weightclinicsresults.com shows as a referral source on the main site. **Finding:** child domain missing from "Configure your domains". **Action:** add it; QA step 13. **Lesson:** every new child site joins the domain list and the own-domains pattern (Appendix D).

N7 · Scenario: Looker quota errors every morning

The SEO dashboard was built on live GA4 with 30 charts. **Action:** move to reporting tables (Chapter 58). **Lesson:** dashboards read small tables.

N8 · Scenario: a winning test that wasn't

A CTA test "won" after 3 days, then reversed. **Finding:** peeking. **Action:** rerun for two full weeks with a written card. **Lesson:** decide the rule before the result (Chapter 67).

Appendix O: Practice Assignments by Level

Level 1 · Foundation

1. Draw the 360° loop for one real journey (search → blog → medication page → portal) and name the tool at each step.
2. Sort 20 of our page URLs into Zone A, B or C and write one reason for each (for legal review, not final).
3. Write the measurement plan row for "Is the GLP-1 cluster moving people to BOFU?" (Chapter 8 format).
4. Pass the Chapter 1–10 quizzes with 8/10 or better.

Level 2 · Setup and Collection

1. On staging, add `PageMeta` to one template and show the page_view in GTM Preview with all 13 fields.
2. Build the `Click – CTA` trigger and the CTA tag; show two different CTAs firing with correct cta_location.
3. Run the full 16-step QA script and record results in a sheet.
4. Find one way personal data could leak on our site and write the fix.

Level 3 · Data Platforms

1. In the BigQuery sandbox, run H1–H5 on one day of data and record bytes processed for each.
2. Create the `v_page_views` view and run H7 (paths) for one week; share the 10 most common paths.
3. Write a striking-distance list (H17) and choose three pages with reasons.
4. Run the CrUX competitor query for our origin and two competitors; summarise in three sentences.

Level 4 · Analysis

1. Build Funnel 1 (Chapter 45) and write one What / So what / Now what.
2. Build a backward path from exit_to_portal; name the top three preceding pages.

3. Review 10 user-explorer timelines of portal exiters and write three patterns (no individual details).
4. Do a 45-minute Clarity routine and log five UX issues with evidence.

Level 5 · Looker Studio and Reporting

1. Build the Daily Monitor from the reporting tables and schedule it to yourself.
2. Add a Targets Sheet and colour three KPIs by range.
3. Write a one-page weekly report using the template in Chapter 65.
4. Explain to a non-analyst, in five sentences, why GA4 portal exits and backend leads differ.

Level 6 · Operating

1. Write an experiment card for a CTA change on a medication page.
2. Run a quarterly access review and list changes made.
3. Update the 90-day roadmap with real owners and dates.
4. Teach the Analytics Red List to a teammate and have them pass Chapter 70's quiz.

Glossary (A to Z)

Term	Simple meaning	Ch.
301 redirect	Permanent redirect	64
360° measurement	Measuring the whole journey from source to conversion	1
404	Page not found error	64
Abandonment	Users who didn't reach the next step	45
A/B test	Random split comparing two versions	67
Access review	Regular check of who has access	10
Action item	Task with owner and date	65
Aggregated data	Totals (e.g., per day), not individual records	4
Aggregated table	Daily totals with no individual rows	24
AI assistant referral	Traffic from AI chat tools	27

Term	Simple meaning	Ch.
AI Assistants channel	Our channel for AI chat tools	63
AI citation	Our page referenced in an AI answer	39
Alert fatigue	Ignoring alerts because there are too many	56
Anchor text	The clickable words	18
Annotation	Note on a chart marking a change	55
Anomaly	Unusual change in data	56
Anonymised query	Rare query hidden for privacy	37
App Router	Next.js's routing system using the `app/` folder	13
Attribution	Rules for giving credit to touchpoints	49
Audience	Saved group of users in Admin	48
Authenticated page	A page you see only after logging in	3
Auto-Event Variable	GTM variable reading the clicked element	17
BAA	Business associate agreement: a HIPAA contract with a vendor	3
Backfill	Loading historical data after setup	38
Backward path	Path ending at a chosen event	46
Baseline	The starting state before changes	9
`beforeInteractive`	Load a script before the page becomes interactive	13
Benchmark	A reference point to compare against	42
Benchmarking	Comparing with competitors	6
BigQuery	Google's cloud data warehouse	31
Bing Webmaster Tools	Microsoft's free site-owner console	39
Blend	Looker Studio's join of two sources	38
BOFU reach rate	Share of sessions reaching BOFU pages	63

Term	Simple meaning	Ch.
Branded query	Search containing our brand	53
Budget alert	Email when spending reaches a threshold	32
Build-time test	A check that runs when the site is built	14
Bulk export	Daily copy of GSC data into BigQuery	37
Business associate	A vendor handling PHI for a covered entity	4
Bytes processed	Amount of data a query reads (cost driver)	31
Calculated field	New field from a formula	59
Cannibalisation	Our pages competing for one query	53
Canvas	Area where results appear	44
Cardinality	How many unique values a dimension has	25
Causation	One thing causing another	65
Change control	Process for safe changes	68
Change history	GA4 record of admin changes	68
Change log	Record of all tracking changes	10
Channel group	A broad traffic type (organic search, social…)	23
Clickstream	Panel data on browsing used for estimates	42
Client ID	An anonymous ID GA4 stores in the browser	1
Closed funnel	Must enter at step 1	45
`closest()`	Finds the nearest matching parent element	17
CMP	Consent management platform	5
Cohort	Group sharing a start date	48
Collection	A group of reports in GA4's library	38
Columnar storage	Data stored by column, so fewer columns cost less	34

Term	Simple meaning	Ch.
Completion rate	Completes ÷ starts	21
Conditional formatting	Colour based on value	60
Consent Initialization	The earliest GTM trigger, for consent defaults	5
Consent Mode	Google's way to pass consent choices to its tags	5
Consent overview	GTM screen showing each tag's consent settings	29
Consent update	The state after a choice	29
Consumer health data	Health-related data protected by state laws like Washington's	3
Contact location	Where the contact link sits	19
Container	A GTM installation for a site	12
Container export	A JSON backup of GTM	9
Content decay	Gradual performance decline	50
Content ID	A stable identifier for a piece of content	14
Contextual link	Link inside body text	51
Control	Viewer filter like date range	57
Conversion	For Ongo Weight Loss: a completed payment	1
Correlation	Things changing together	65
Counting method	Once per event or once per session	26
Crawl stats	Report on Googlebot's requests	36
Credentials	Whose access a data source uses	57
Cross-domain measurement	Treating visits across your domains as one journey	23
Cross-filtering	Clicking one chart filters others	62
CrUX	Chrome UX Report: real-user speed data from Chrome	40
CSP	Content Security Policy: rules for allowed scripts	13

Term	Simple meaning	Ch.
CTA	Call to action (button or link asking for a next step)	17
CTA click rate	CTA clicks ÷ views of pages showing it	52
CTA location	Where on the page the CTA sits	17
CTA register	List of approved CTA names	52
CTE	A named step created with `WITH`	34
CTR	Clicks divided by impressions	36
Custom dimension	A registered parameter usable in reports	25
Custom Event trigger	GTM trigger on a data layer event name	15
Custom insight	GA4 alert condition with email	56
Custom JavaScript variable	Small function returning a value	12
Custom metric	A registered numeric parameter	25
Daily export	Once-a-day full table of yesterday's events	32
Data attribute	An HTML attribute starting with `data-`	17
Data deletion request	GA4 feature to remove collected data	30
Data-driven attribution	Credit by machine-learned contribution	49
Data freshness	How often Looker refreshes cached data	58
Data layer	Where the website writes facts for GTM	2
Data Layer Variable	Reads a value from the data layer	12
Data quality	How trustworthy data is	55
Dataset	A folder of tables	31
Data source	Saved connection with fields	57
Data stream	A GA4 source (website or app)	9
Dead click	Click on something not clickable	28

Term	Simple meaning	Ch.
DebugView	GA4's real-time debugging screen	15
Default consent	The state before a visitor chooses	29
De-identified data	Data with identifiers removed	4
Denominator	The number you divide by	52
Dependency	Task that must finish first	69
Detail report	Table report with chosen dimensions/metrics	43
`<details>`	HTML element for expandable content	20
Difference-in-differences	Compare change in test vs change in control	67
Domain property	GSC property covering a whole domain	36
Downstream	What happens after the click	51
Drill-down	Going to more detail	60
Driver	Factor that influences results	61
Duplicate tag	The same tracking firing twice	9
Editorial link	A link written into content by a writer	18
Effective user ID	ID GA4 uses for a user	47
Engaged session	A meaningful visit by GA4's rules	7
Engagement	Signs that a visitor interacted meaningfully	20
Engagement rate	Engagements divided by reach	41
Enhanced measurement	Automatic events GA4 can collect	11
Entity	Topic such as semaglutide or GLP-1	62
Entity group	Queries about the same topic/thing	53
Entry page	The first page of a visit (landing page)	1
Environment	Live or staging version of GTM	12

Term	Simple meaning	Ch.
Event	Something that happened	7
Event dictionary	The master list of events and parameters	16
Event dictionary table	List of approved event names	64
Executive dashboard	Leadership summary view	61
Exit page	Last page of a session	46
Expected CTR	Typical CTR for a position	62
Exploration	Custom analysis workspace in GA4	44
ExportLog	Table recording each export	37
Extract	Saved snapshot of data in Looker	58
Field data	Measurements from real visitors	40
First touch	Credit to the first landing page	35
First user dimension	How a user first arrived	43
Funnel	Ordered steps toward a goal	8
Gap tool	A tool added for something the core can't do	6
`generate_lead`	GA4 recommended event for leads	22
GET submission	A form that puts values in the URL	21
`_gl`	GA4's linker parameter carrying IDs between domains	23
Go-live	Publishing to production	69
Google tag	The base tag that loads GA4	12
Governance	Rules and habits keeping data reliable	68
GPC	Global Privacy Control: a browser opt-out signal	5
Grounding query	Query the AI used to find our content	39
Guardrail metric	Metric that must not get worse	67

Term	Simple meaning	Ch.
HAVING	Filter after grouping	35
Headline	One-sentence summary	65
Heatmap	Visual of clicks or scrolls on a page	28
Hit	A single request sent to GA4	30
Hypothesis	Testable prediction with a reason	67
Impression	Our link appeared in results	36
Incident	A privacy or data mistake that happened	70
Incremental load	Adding only new data instead of rebuilding	58
IndexNow	Protocol to notify search engines of URL changes	39
Intent mismatch	Page doesn't match what searchers want	50
Internal link	A link to another page on our sites	18
Internal link CTR	Internal link clicks ÷ page views	50
Internal promotion	A banner or link promoting our own pages	27
Internal traffic	Visits from our own team	55
Internal traffic filter	Removes your team's visits	11
Join	Combining two tables on a shared column	38
Join key	Shared field used to match rows	59
JS API	A way for scripts to talk to an embedded player	20
Key event	An event marked as important for conversions	16
KPI	Key performance indicator	8
Lab data	Measurements from a simulated test	40
Landing page	First page of a session	46
Last non-direct click	Credit to last non-direct channel	49

Term	Simple meaning	Ch.
Lead	Someone who signed up or completed eligibility	1
LEAD()	Gets the next row's value	35
Lead rate	Leads ÷ portal exits	26
Lead signal	A public action showing intent	22
Leak	Personal data accidentally collected	9
Least privilege	Minimum access needed	68
Left outer join	Keep all left rows, match right where possible	59
Link CTR	Clicks on a link ÷ views of its page	18
Link-in-bio	Single link in a social profile	41
Link position	Where the link sits (body, nav, footer…)	18
Living document	Kept up to date continuously	68
Lookback window	How far back touchpoints count	49
Looker Studio	Google's free dashboard tool	57
Lookup table	GTM variable mapping inputs to outputs	19
`mailto:` link	A link that opens an email	19
Masking	Hiding content in recordings	21
MEASURE	Our final gate checklist	70
Measurement plan	Document linking goals to events and reports	8
Micro-conversion	A small step toward the goal	8
MoM	Month over month	61
Monitoring dashboard	Daily health and performance view	60
Mover	Page with big change	60
`next/script`	Next.js component for loading scripts	13

Term	Simple meaning	Ch.
Node	One step in a path	46
Non-branded	Searches without our name	53
Nonce	A one-time value allowing an inline script under CSP	13
Normalise	Make values the same format	38
Offline/backend conversion	A conversion recorded outside the browser	24
On-platform metric	Measured by the social app	41
On-site metric	Measured by GA4 on our website	41
Open funnel	Can enter at any step	45
Opportunity list	Rule-based list of pages to improve	62
Opt-in / opt-out	Agree first vs. allowed until refused	5
Origin	Protocol + host (e.g., https://www.site.com)	40
Orphan page	Page with no internal links pointing to it	51
(other)	GA4's grouping when too many values exist	25
Outbound click	A click to another website	19
Owner	Person responsible for a plan item	8
p75	Value that 75% of visits are at or better than	40
Page metadata	Facts about a page (type, stage, cluster)	14
Page-zone register	A list of templates with their zone and allowed tools	4
Parameter	Detail about an event	7
Partition	A slice of a table by date	37
Partitioned table	Table split by date for cheaper queries	58
Partner connector	Third-party Looker connector, usually paid	41
Path	Sequence of pages/events	46

Term	Simple meaning	Ch.
Payment rate	Payments ÷ leads	26
Peeking	Checking results early and stopping	67
Percentile	Value below which a % of data falls	66
Phase	Group of related weeks/tasks	69
PHI	Protected health information	3
Position tracking	Daily ranking checks for chosen keywords	42
POST submission	Sending form data without putting it in the URL	22
Preview mode	GTM's testing mode	30
Project ID	Unique name of a Cloud project	32
Property	A GA4 container for one business's data	11
Proxy metric	A stand-in signal for something harder to measure	26
Qualitative	Insight from examples, not counts	47
Quick back	Leaving a page and quickly returning	28
Quota	Usage limit on an API	58
RACI	Responsible, Accountable, Consulted, Informed	10
Rage click	Repeated frustrated clicks	28
Raw data	Every event, unsummarised	6
Reconciliation	Comparing numbers from two systems	24
Red List	Mistakes that must never happen	70
Referral	Visit from another website link	63
Referring domain	A unique website linking to us	42
Refresh	Updating existing content	50
Region	Where data is stored geographically	31

Term	Simple meaning	Ch.
Region default	Different defaults by location	29
Regression	Something that used to work breaking	30
Re-identification	Figuring out who a record belongs to	24
Repeated record	A list of structured items in one cell	33
Reporting identity	How GA4 recognises users	11
Reserved name	A name GA4 keeps for its own events	16
Result category	A label from a tool's output (not allowed in analytics)	21
Retention	% returning later	48
Retire	Stop using an event but keep its history notes	16
Retroactive	Applying to past data (custom dimensions don't)	25
Roadmap	Ordered plan over time	69
Root cause	The real reason a problem occurred	70
Route change	Moving to a new URL inside the app	15
Same-day comparison	Comparing to the same weekday last week	60
Sampling	Using part of the data to estimate results	44
Sandbox	Free BigQuery mode with limits	31
Scheduled delivery	Automatic emailing of a Looker report	56
Scheduled query	A query BigQuery runs automatically	34
Schema	The structure of a table	33
Scope	Whether a dimension describes an event, user or item	7
Scorecard	Set of KPIs for a page type	50
Scroll depth	How far down a page a visitor scrolled	20
Segment	Subset of data in explorations	48

Term	Simple meaning	Ch.
Segment overlap	Venn view of segments	48
Self-referral	Our own site appearing as a traffic source	27
SERP feature	Special result types (AI Overview, FAQ, video)	53
Server-side GTM	Tag manager running on your own server	6
Service account	A non-human account used by a program	24
Session	One visit; ends after 30 minutes inactive	2
Session dimension	How a session arrived	43
Session recording	Masked replay of a visit	28
sessionStorage	Browser storage cleared when the tab closes	23
Severity	How urgent a problem is	56
Share of voice	Our share of visibility on a keyword set	42
Site Explorer	Bing's crawled-URL browser	39
SPA navigation	Changing pages without a full reload	15
SQL	Language for asking databases questions	34
Stack	The set of tools used together	6
Stale value	A value left over from a previous page	14
Sticky CTA	Button fixed on screen while scrolling	52
Streaming export	Near-real-time intraday table	32
Striking distance	Ranking just off page one (8–20)	37
STRING_AGG	Joins values into one text	35
Success state	Confirmation that the server accepted the form	22
`_TABLE_SUFFIX`	The date part of a wildcard table name	33
Tag	Code that sends data to a tool	2

Term	Simple meaning	Ch.
Target range	Acceptable low–high band for a KPI	61
Technique	Type of exploration (free form, funnel…)	44
`tel:` link	A link that starts a phone call	19
Template	A ready-made GTM tag from the gallery	29
Threshold	A percentage mark (25, 50…)	20
Thresholding	GA4 hiding small rows for privacy	43
Timeline	Events in time order	47
Tool event	An event showing a tool was used	21
Tracking technology	Code that collects information about users' actions	3
Trended funnel	Funnel completion over time	45
Trigger	Rule for when a tag fires	2
Truncation	Cutting text that's too long	16
UNNEST	SQL function to open a list into rows	33
URL Inspection	Tool to check one URL's index status	36
`usePathname`	Next.js hook returning the current path	15
User explorer	GA4 tool showing one user's event timeline	47
User property	A user-scoped attribute	7
UTM	Tags on a link telling analytics its source	27
Vanity metric	Looks good, doesn't drive decisions	61
Variable	A value used by tags and triggers	2
Version	A saved, publishable snapshot of GTM	10
View	A saved query that acts like a table	34
Visit context	Broad, non-personal info about how a visit started	23

Term	Simple meaning	Ch.
Wildcard table	Querying many daily tables with `events_*`	33
Window function	Calculation over related rows (e.g., ROW_NUMBER)	35
Worklist	Assigned tasks from data	62
Workspace	A GTM draft area for one set of changes	10
WoW / MoM	Week over week / month over month	65
ZETA	Zero-Error Tolerance Approach	70
Zone	A category of pages with its own tracking rules	4

MCQ Answer Keys (all chapters)

Try every quiz first, then turn the page upside down to check.

Chapter	Answers
1	1-A · 2-B · 3-B · 4-B · 5-B · 6-C · 7-A · 8-A · 9-D · 10-B
2	1-B · 2-B · 3-A · 4-B · 5-D · 6-A · 7-D · 8-C · 9-B · 10-B
3	1-C · 2-C · 3-C · 4-D · 5-C · 6-D · 7-B · 8-D · 9-D · 10-B
4	1-B · 2-D · 3-D · 4-C · 5-A · 6-C · 7-C · 8-A · 9-B · 10-D
5	1-B · 2-D · 3-D · 4-D · 5-B · 6-C · 7-D · 8-B · 9-D · 10-D
6	1-C · 2-A · 3-A · 4-C · 5-C · 6-B · 7-C · 8-A · 9-B · 10-A
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Measure what matters. Protect who matters.

Collect only what a question needs. Keep the portal out of every tag. Label every number. Check the data before you trust it. Then turn it into one clear action, every day.

Edition: October 2026. Tools, limits and laws change; every source is labelled so you can recheck it.

Prepared by **Ashish Osten**, Team Lead, Ongo Weight Loss.

Ongo Weight Loss